Exact (Non-)Accelerations of Heavy-Ball Discretizations

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Abstract

Momentum methods with the same slow dynamics can have different worst-case rates on smooth strongly convex functions. We identify a half-step phase barrier in the gradient query that governs this difference. For momentum with extrapolation $\tau_\kappa \to \tau$, $\beta = 1-\theta/\sqrt{\kappa}+o(\kappa^{-1/2})$ and $hL \to y$, where $\kappa = L/\mu$ and $\theta, y > 0$, the exact uniform distance factor is $1 - c/\sqrt{\kappa} + o(\kappa^{-1/2})$, with $c = \min\{\theta/3, c_Q(\theta, y)\}$, when $\tau > 1/2$ and $y(1 + 2\tau) < 4$; c_Q is the slowest quadratic decay exponent. Quadratically accelerated families with $\limsup \tau_\kappa < 1/2$ have divergent trajectories on the full function class. Upper bounds use finite-lag certificates: nonnegative combinations of inequalities for values and gradients at finitely separated query points, together with comparisons to the minimizer. For any prescribed accuracy, spiral mixtures or quadratic trajectories of one finite-dimensional objective give matching lower bounds at all iteration horizons. We resolve the parameter layer near $\tau = 1/2$ and classify all tunings of this family attaining the optimal leading constant 1. The flow also bounds the leading constant of every second-order stationary linear family with the same slow parameters (θ, y) and fixed finite memory.

1. Introduction

Momentum methods can have the same continuous-time limit and different worst-case behavior. Polyak's heavy ball is optimal on strongly convex quadratics [34]. On smooth strongly convex functions it can fail to converge [17, 23]. Nesterov's method accelerates on this whole class [30], and the triple momentum method (TMM) attains the optimal asymptotic distance factor $1 - \kappa^{-1/2}$, where $\kappa = L/\mu$ [10, 48]. These methods belong to the family $\mathcal{M}$,

$$\tilde{x}_k = x_k + \tau(x_k - x_{k-1}), \qquad x_{k+1} = x_k + \beta(x_k - x_{k-1}) - h\nabla f(\tilde{x}_k) : \tag{1}$$

heavy ball has $\tau = 0$, Nesterov's method has $\tau = \beta$, and TMM has $\tau \to 1/2$ as $\kappa \to \infty$. We determine how extrapolation controls acceleration on the full class.

Slow dynamics and the half-step rule. Accelerated tunings have $1-\beta \asymp \kappa^{-1/2}$. For families with limits

$$\beta = 1 - \theta\kappa^{-1/2} + o(\kappa^{-1/2}), \qquad hL \to y, \tag{2}$$

we measure performance by the constant c in the uniform distance factor $\rho_{\mathcal{F}} = 1-c\kappa^{-1/2}+o(\kappa^{-1/2})$. For squared distances the leading constant is $2c$. The slow equation is the heavy-ball flow $\ddot{x}+\theta\dot{x}+\nabla F(x) = 0$ with a y-strongly convex potential. We call (θ, y) its *slow data*: damping θ and strong convexity y of the scaled flow. Over all finite smoothness bounds, its worst distance decay exponent is

$$c_{\mathcal{F}}(\theta, y) = \min\{\theta/3, c_Q(\theta, y)\},$$

where c_Q is the slowest quadratic exponent. This exponent bounds the leading constant of every second-order stationary linear family with the same slow data and fixed finite memory (Definition 5.1 and Theorem 5.2). The family $\mathcal{M}$ attains it when $\tau > 1/2$ and $y(1+2\tau) < 4$ (Theorem 2.6); acceleration on the full class requires $\liminf_{\kappa\to\infty} \tau_\kappa \geq 1/2$ (Theorem 6.2). We call $\tau = 1/2$ the *half-step wall*.

The half step has a spectral explanation. On a Fourier mode with $\omega \neq 0$,

$$1 - e^{i\omega} = \frac{2\sin(\omega/2)}{\omega}\, e^{i\omega/2}(-i\omega).$$

Replacing a derivative by a difference adds half a step of phase. For limit methods of $\mathcal{M}$, the low-frequency lead that compensates for this shift is τ. For larger memory, the full frequency-dependent phase enters the rate (Section 5.4 and Figure 3). Upper bounds use finite-lag certificates: nonnegative combinations of inequalities for values and gradients at finitely separated queries and comparisons to the minimizer (Definition 4.6). We match them with trajectories generated by one objective.

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Contributions. We extend the spectral characterizations for heavy ball [32] and standard Nesterov [33] to general stationary linear methods. Finite-dimensional $C^{1,1}$ witnesses establish distance sharpness for the flow estimates of [2] over the union of finite smoothness classes. A trace identity complements certificate-based method synthesis [24]: quadratic stability and a nonnegative limiting lag-one symbol force $y \le 2$, with equality precisely for TMM's limiting transfer function.

The standard Nesterov tuning has $c = 2/3$; its quadratic-optimal tuning has full-class constant $4/(3\sqrt{3})$. The best full-class constant within the Nesterov family is $\sqrt{2/3}$, attained by $h = 4/(3L)$, $\tau = \beta = 1 - \sqrt{6/\kappa}$ (Corollaries 6.6 and 6.7). The optimum $c = 1$ over $\mathcal{M}$ requires $(\sqrt{\kappa}(1-\beta), hL, \tau) \to (3, 2, 1/2)$. We call $(hL, \tau) = (2, 1/2)$ the *TMM corner*. In its natural scale, $1/2 - \tau = a\kappa^{-1/2} + o(\kappa^{-1/2})$, $2 - hL = b\kappa^{-1/2} + o(\kappa^{-1/2})$, it is attained exactly when $\theta = 3$, $a \le 5/4$ and $2a + b \ge 7/2$: a wedge with TMM at its vertex (Figure 5). We also classify approaches at arbitrary rates. For some suboptimal corner tunings, integer rotation periods in the lower-bound construction leave a narrow arithmetic gap between the available bounds (Theorem 7.2 and Section 9).

Related work: flows and exact rates. Continuous-time momentum models go back to Polyak [34]; for Nesterov's convex method see [43]. High-resolution equations, Hessian-driven damping and semi-implicit discretizations expose the gradient corrections separating the methods [1, 29, 41]; Lyapunov analyses connect these flows to accelerated schemes [38, 53]. For heavy-ball flow, Aujol, Dossal and Rondepierre proved function-value bounds on quasi-strongly convex functions and sharpness using non-Lipschitz gradients [2]; see [3] for the Łojasiewicz setting. Theorem 3.1 gives the sharp distance exponent using $C^{1,1}$ witnesses. For Nesterov's standard tuning, explicit integral quadratic constraint (IQC) bounds appear in [37]; Gannot obtained the constant $2/3$ using the Jury–Lee multiplier [13], and Pokutta and Stellato proved its sharpness by mixtures of planar spirals [33]. Our formula recovers it and identifies the best asymptotic tuning within the Nesterov family. For heavy ball, planar cycles establish non-acceleration [15, 17]. The one-dimensional question of [16] was settled in [12]; local acceleration with controlled initialization and a gradient-descent warm-up is studied in [25]. Spectral realizations characterize the exact rate, separation and one-exponent principle in [32]. Theorem 4.7 extends this framework to general stationary linear methods; Lemmas C.8 to C.10 give finite-support and atom-count results under explicit nondegeneracy conditions. Hernanz [18] identifies the optimal nonaccelerated asymptotic constant 8 in dimension at least two.

Related work: certificates, synthesis and optimal methods. The family (1) includes robust momentum, which interpolates between TMM and gradient descent [8], and is a standard synthesis template [23, 28, 39]. Rates can be certified by IQCs and dissipativity [19, 23, 27], Popov and Jury–Lee multipliers [20, 35], and O'Shea–Zames–Falb multipliers [6, 31, 42, 52, 54, 55]. Exactness in dimension-free lifted settings and limitations for a single plant are studied in [21, 22]; periodic counterexamples and nonlinear rates beyond linearizations appear in [40, 56]. Performance estimation and automated Lyapunov analysis compute tight bounds at fixed parameters [11, 45–47]; lifting and interpolation connect these descriptions [49]. Lessard and Seiler derive memory-independent synthesis bounds for a fixed IQC, with a certificate using consecutive-query data recovering TMM's rate [24]. Our trace identity adds an explicit step formula and rigidity of its equality case. TMM and the information-theoretic exact method attain the optimal asymptotic factor [10, 44, 48]. Individual asymptotic root factors under additional local regularity are studied in [7, 50]. We use the uniform factor, with the supremum over functions taken before the asymptotic limit (Remark 2.4). For broader context on acceleration see [9].

Organization. Section 2 states the main rate for $\mathcal{M}$. Sections 3 and 4 develop the flow witnesses and spectral closure; Section 5 gives the ceiling, half-step rule and trace formula for general memory. Sections 6 and 7 resolve the wall layer and the TMM corner, and Section 8 treats finite condition numbers. Proofs and detailed algebra are collected in the appendices.

Formal verification. The mathematical results of this article have formal proofs in Lean. The package is available at `PolynomialRealizationsProgram/nesterov`[3].

2. The Momentum Family and Its Main Rate

Notation and scope. Let H be a real Hilbert space and let $\mathcal{F}_{\mu,L}(H)$, $0 < \mu < L$, consist of differentiable μ-strongly convex functions with L-Lipschitz gradient; thus *smooth* means $C^{1,1}$. We write $x^\star$ for the minimizer, $\kappa = L/\mu$, and $\mathbb{N} = \{1, 2, \dots\}$. We identify a real plane with $\mathbb{C}$, using $\langle a, b\rangle = \mathrm{Re}(a\bar{b})$. Frequencies are $\omega \in [0, \pi]$, and $\xi = 1-\cos\omega \in [0, 2]$. Bars over complex quantities mean conjugation; bars over real spectral quantities mean averages with respect to the specified probability measure.

2.1. Quadratic stability and the rate convention

For (1) set $y_\mu = h\mu$, $y = hL$, $N_\tau(z) = (1+\tau)z - \tau$, and

$$\chi_\lambda(z) = (z-1)(z-\beta) + h\lambda N_\tau(z).$$

This is the characteristic polynomial on $\lambda\|x\|^2/2$. A *tuning* is a triple (β, τ, h), with $\beta \in [0, 1)$, $\tau \geq 0$, $h > 0$. A tuning is *stable* when $\rho_q := \max_{\lambda\in[\mu,L]} \max\{|z| : \chi_\lambda(z) = 0\} < 1$, so stability refers to quadratics.

Lemma 2.1. *A tuning of $\mathcal{M}$ is stable if and only if $y(1+2\tau) < 2(1+\beta)$.*

For $\mathcal{M}$ the state is $S_k = (x_{k-1} - x^\star, x_k - x^\star)$. The following convention also applies to the states of stationary methods defined in Definition 4.3.

Definition 2.2. For $n \geq 0$, let $\Gamma(n) = \sup\|S_{k+n}\|/\|S_k\|$, where the supremum ranges over real Hilbert spaces H, functions $f \in \mathcal{F}_{\mu,L}(H)$, trajectories and indices $k \geq 0$ with $S_k \neq 0$. The *worst-case rate* is $\rho_{\mathcal{F}} = \lim_{n\to\infty} \Gamma(n)^{1/n}$. For $\mathcal{M}$ we also use the *rate from rest*,

$$\rho^0_{\mathcal{F}} = \limsup_{N\to\infty} \left(\sup \frac{\|x_N - x^\star\|}{\|x_0 - x^\star\|} \right)^{1/N},$$

where the supremum ranges over $d \geq 1$, $f \in \mathcal{F}_{\mu,L}(\mathbb{R}^d)$ and $x_0 \neq x^\star$, with $x_{-1} = x_0$.

Lemma 2.3. *The envelope Γ is finite and submultiplicative, and*

$$\rho_{\mathcal{F}} = \lim_{n\to\infty} \Gamma(n)^{1/n} = \inf_{n\geq 1} \Gamma(n)^{1/n} \geq \rho_q.$$

For $\mathcal{M}$, $\rho^0_{\mathcal{F}} \leq \rho_{\mathcal{F}}$.

Remark 2.4 (Uniform rates and smooth subclasses). The supremum in $\Gamma(n)$ is taken *before* the limit in n. For any fixed stationary linear method and finite n, restricting that supremum to finite-dimensional C^∞ objectives in $\mathcal{F}_{\mu,L}$ preserves its value: smooth convolution preserves μ, L and approximates every finite trajectory (Appendix A). Individual asymptotic root factors, as in [7, 50], place the limit inside the supremum and describe a different rate convention. Distance and squared-distance factors have respective first-order constants c and $2c$.

2.2. Accelerated scaling and the main theorem

In the accelerated scaling we normalize $L = 1$ (replace f by f/L and h by hL), write $\mu = \varepsilon^2 = 1/\kappa$, and consider families

$$\beta = 1 - \theta_\varepsilon\varepsilon, \quad h = y_\varepsilon, \quad \tau = \tau_\varepsilon, \qquad (\theta_\varepsilon, y_\varepsilon, \tau_\varepsilon) \to (\theta, y, \tau), \quad \theta, y > 0,\ \tau \geq 0, \tag{3}$$

stable for all small ε. Let $\zeta_\pm$ be the roots of $\zeta^2 - \theta\zeta + y$, ordered as $\zeta_- \leq \zeta_+$ when real, and set

$$c_Q(\theta, y) = \begin{cases} \theta/2, & \theta^2 \leq 4y, \\ (\theta - \sqrt{\theta^2 - 4y})/2, & \theta^2 > 4y. \end{cases} \tag{4}$$

[3] `https://github.com/dmivilensky/PolynomialRealizationsProgram/tree/main/nesterov`

Lemma 2.5 (Quadratic constant)**.** *Under* (3), $\limsup_{\kappa\to\infty}\sqrt{\kappa}\,(1-\rho_q) \le c_Q(\theta,y)$, *and if* $y(1+2\tau)<4$, *then* $\sqrt{\kappa}\,(1-\rho_q)\to c_Q(\theta,y)$.

The function c_Q is continuous, $c_Q \le \theta/2$ and $c_Q \le \sqrt{y}$, with equality in both at critical damping $\theta = 2\sqrt{y}$.

A family is *accelerated* on a function class if its worst-case rate is at most $1-c\varepsilon$ for some $c>0$ and all small ε. Acceleration on $\mathcal{F}_{\mu,L}$ implies acceleration on its quadratic subclass.

In asymptotic statements, (θ,y,τ) denotes the limit of $(\theta_\varepsilon,y_\varepsilon,\tau_\varepsilon)$. Finite-tuning identities use β,τ,h and $y=hL$. Put

$$c_{\mathcal{F}}(\theta,y) = \min\{\theta/3, c_Q(\theta,y)\}. \tag{5}$$

Spiral mixtures and quadratic modes supply the two branches of this bound.

Theorem 2.6 (Exact constant)**.** *Consider a family* (3) *of stable tunings of* $\mathcal{M}$.

(a) (Lower bound, every $\tau \ge 0$.) *For every* $\delta>0$ *there are* $J=J(\theta,y,\delta)\in\mathbb{N}$ *and* $\varepsilon_0>0$ *such that for every* $\varepsilon\in(0,\varepsilon_0)$ *there exist* $f\in\mathcal{F}_{\varepsilon^2,1}(\mathbb{R}^{2J+1})$ *and* $x_0\neq x^\star$ *with, from rest,* $\|x_N-x^\star\| \ge c_0\, e^{-(c_{\mathcal{F}}+\delta)\varepsilon N}\|x_0-x^\star\|$ *for all* $N\ge 0$, *where* $c_0=c_0(\varepsilon,\delta)>0$ *is independent of* N.

(b) (Upper bound beyond the wall.) *Let* $\tau>\frac{1}{2}$ *and* $y(1+2\tau)<4$. *Then for every* $c\in(0,c_{\mathcal{F}}(\theta,y))$ *there is* $\varepsilon_0>0$ *such that* $\rho_{\mathcal{F}}<1-c\varepsilon$ *for* $\varepsilon\in(0,\varepsilon_0)$*; this bound is certified by the lag-one criterion (Proposition 4.9).*

(c) *Consequently, under the hypotheses of (b),*

$$\lim_{\kappa\to\infty}\sqrt{\kappa}\,(1-\rho_{\mathcal{F}}) = \lim_{\kappa\to\infty}\sqrt{\kappa}\,(1-\rho^0_{\mathcal{F}}) = c_{\mathcal{F}}(\theta,y) = \min\big(\tfrac{\theta}{3},\ c_Q(\theta,y)\big).$$

Every accelerated family satisfies $\liminf \tau_\varepsilon \ge 1/2$ (Theorem 6.2); TMM reaches this limit from below. The boundary layer and the stability boundary have their own scales (Sections 6.3 and 7).

Designing a target constant. For $c>0$, the condition $c_{\mathcal{F}}(\theta,y)\ge c$ is equivalent to

$$\theta \ge 3c, \qquad y \ge c\theta - c^2.$$

Indeed, under $\theta\ge 3c$, the quadratic condition $c_Q\ge c$ is equivalent to $c^2-\theta c+y\ge 0$. For $\tau>1/2$ and $y(1+2\tau)<4$, these are also the level sets of the method's constant, by Theorem 2.6(c). The smallest step compatible with a target constant is $y=2c^2$, at $\theta=3c$. Combining this with $y(1+2\tau)\le 4$ explains the fixed-extrapolation ceiling $\sqrt{2/(1+2\tau)}$ (Corollary E.11).

2.3. The quadratic price of extrapolation

Theorem 2.7 (Price of extrapolation)**.** *Every tuning of* $\mathcal{M}$ *with* $\rho_q<1$ *satisfies*

$$\rho_q \ \ge\ \frac{\sqrt{\kappa_\tau}-1}{\sqrt{\kappa_\tau}+1}, \qquad \kappa_\tau := (1+2\tau)\kappa - 2\tau.$$

Equality holds exactly at Polyak's tuning,

$$\tau=0, \qquad \beta=\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^2, \qquad h=\frac{4}{(\sqrt{L}+\sqrt{\mu})^2}.$$

Extrapolation therefore costs at least the replacement of κ by κ_τ, asymptotically a factor $1+2\tau$. Queries also admit a gradient-correction realization, including quasi-hyperbolic momentum; Remark 4.4 gives the precise parameter translation and initialization convention.

3. The Heavy-Ball Flow

In the accelerated scaling (3), the slow time $t=\varepsilon k$ gives the heavy-ball equation

$$\ddot{x}+\theta\dot{x}+\nabla F(x)=0, \qquad F=\tfrac{y}{\varepsilon^2}f\in\mathcal{F}_{y,y\kappa}.$$

Its smoothness bound grows with κ. We determine the worst decay exponent over all finite smoothness bounds and transfer sampled spiral witnesses to discrete time by interpolation (Lemma D.5).

Let H be a real Hilbert space and $m > 0$. Let $\mathcal{F}_m(H)$ consist of differentiable m-strongly convex functions whose gradient is Lipschitz on bounded sets; thus $\bigcup_{L>m} \mathcal{F}_{m,L}(H) \subseteq \mathcal{F}_m(H)$. Each has a unique minimizer, which we normalize to 0, with $f(0) = 0$. The convex remainder $f_m(x) = f(x) - \frac{m}{2}\|x\|^2$ satisfies $f_m(0) = 0$ and $\nabla f_m(0) = 0$, so

$$f_m(x) \geq 0, \qquad \langle \nabla f_m(x), x\rangle - f_m(x) \geq 0 \qquad (x \in H), \tag{6}$$

by convexity between x and 0. For $\theta > 0$ consider

$$\ddot{x} + \theta\dot{x} + \nabla f(x) = 0. \tag{7}$$

Every initial position and velocity determine a unique solution for all $t \geq 0$: the vector field is Lipschitz on bounded sets, and the energy $f(x) + \frac{1}{2}\|\dot{x}\|^2$ keeps the orbit bounded. For a nonzero solution define the state decay exponent by $\ell(x) = \liminf_{t\to\infty} -\frac{1}{t}\log(\|x(t)\| + \|\dot{x}(t)\|)$. With $c_Q(\theta, m)$ as in (4) (with y replaced by m), put

$$\Delta_c = c^2 - \theta c + m, \qquad c_{\mathcal{F}}(\theta, m) = \min(\tfrac{\theta}{3},\ c_Q(\theta, m)). \tag{8}$$

For $f(x) = \frac{m}{2}\|x\|^2$, the slowest state decay exponent is $c_Q(\theta, m)$.

3.1. Exact decay

Theorem 3.1 (Exact decay of the heavy-ball flow)**.** *Let $\theta, m > 0$.*

(a) (Lyapunov identity.) *For $c \in \mathbb{R}$ put*

$$V_c = f(x) - \tfrac{c(\theta-c)}{2}\|x\|^2 + \tfrac{1}{2}\|\dot{x} + (\theta - c)x\|^2 = f_m(x) + \tfrac{\Delta_c}{2}\|x\|^2 + \tfrac{1}{2}\|\dot{x} + (\theta - c)x\|^2.$$

Along every solution of (7),

$$\frac{d}{dt}(e^{2ct}V_c) = -e^{2ct}\Big[(\theta - 2c)\Delta_c\|x\|^2 + (\theta - c)(\langle \nabla f_m(x), x\rangle - f_m(x)) + (\theta - 3c)\, f_m(x)\Big]. \tag{9}$$

If $0 < c \leq \theta/3$ and $\Delta_c > 0$, the right-hand side is nonpositive, and $\|x(t)\|^2 + \|\dot{x}(t)\|^2 \leq K_c\, e^{-2ct}V_c(0)$ with K_c depending only on (θ, m, c).

(b) (Quadratic witness.) *For $f(x) = \frac{m}{2}x^2$ on $\mathbb{R}$, some solution has $\ell = c_Q(\theta, m)$.*

(c) (Spiral witnesses.) *For every $c \in (\theta/3, c_Q(\theta, m))$ there are $J \in \mathbb{N}$, $L < \infty$, $f \in \mathcal{F}_{m,L}(\mathbb{R}^{2J})$ and a solution of (7) with $\|x(t)\| = e^{-ct}\|x(0)\|$ and $\|\dot{x}(t)\| = e^{-ct}\|\dot{x}(0)\|$ for all $t \geq 0$.*

(d) (Exact rate.) *Over all real Hilbert spaces H, all $f \in \mathcal{F}_m(H)$ and all nonzero solutions,*

$$\inf \ell(x) = c_{\mathcal{F}}(\theta, m).$$

The same infimum holds with $f \in \bigcup_{L>m} \mathcal{F}_{m,L}(H)$. On both objective classes, the energy $f(x) + \frac{1}{2}\|\dot{x}\|^2$ and the gap $f(x)$ have logarithmic decay exponents $\liminf_{t\to\infty} -t^{-1}\log(\cdot)$ with infimum $2c_{\mathcal{F}}(\theta, m)$, using $-\log 0 = +\infty$. If $\theta < 3\sqrt{m/2}$, every solution satisfies $\|x(t)\| + \|\dot{x}(t)\| = O(e^{-\theta t/3})$.

The function-value bounds in (d) are due to Aujol, Dossal and Rondepierre [2], who proved them for quasi-strongly convex C^1 functions. Their energy at $\theta \leq 3\sqrt{m/2}$ is $V_{\theta/3} = f + \frac{1}{2}\|\dot{x} + \frac{2\theta}{3}x\|^2 - \frac{\theta^2}{9}\|x\|^2$. For $\theta < 3\sqrt{m/2}$, they proved sharpness of the gap exponent $2\theta/3$ using one-dimensional examples with non-Lipschitz gradients, and improved the bound for each fixed Lipschitz constant L. Theorem 3.1(c) supplies finite-dimensional $C^{1,1}$ witnesses that are sharp for distances. Their dimension may depend on the exponent, and L grows as $c \downarrow \theta/3$.

The coefficient $\theta/3$ comes from the two convexity inequalities in (6). After the chain rule cancels the gradient–velocity term, a homogeneous remainder with $\langle \nabla f_m(x), x\rangle = r f_m(x)$ has coefficient $r\theta - (r+2)c$ inside the brackets in (9). This coefficient vanishes at $c = r\theta/(r+2)$: at $\theta/2$ for quadratic profiles and $\theta/3$ for conical profiles, whose degree 1 is the minimum allowed by convexity. The smooth witnesses in (c) have nearly conical profiles along their orbits (Section 3.2).

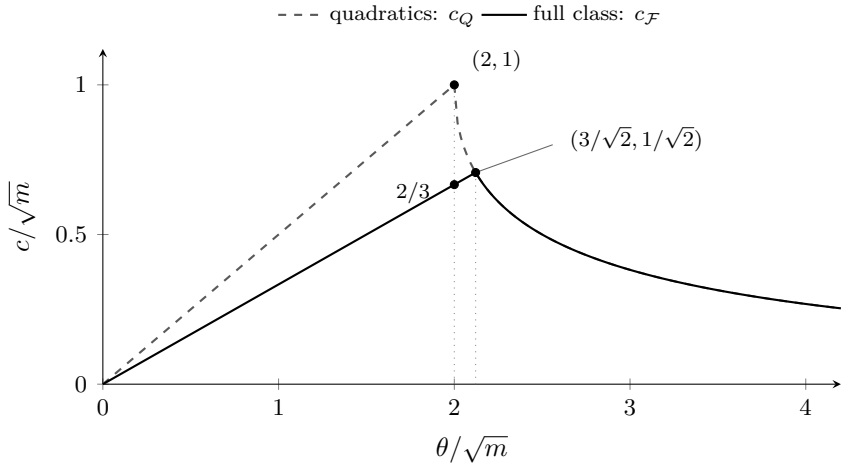


Figure 1: Normalized flow exponents (Theorem 3.1 and Corollary 3.2). The marked maxima occur at $\theta/\sqrt{m} = 2$ for quadratics and $3/\sqrt{2}$ for the full class. At critical damping, the full-class value is 2/3, corresponding to standard Nesterov (Corollary 6.6(i)).

Corollary 3.2 (The price of nonlinearity)**.** *For $m > 0$,*

$$\max_{\theta>0} c_{\mathcal{F}}(\theta, m) = \sqrt{m/2}, \qquad \max_{\theta>0} c_Q(\theta, m) = \sqrt{m}.$$

The maxima are attained uniquely at $\theta = 3\sqrt{m/2}$ and $\theta = 2\sqrt{m}$, respectively; at the first, $c_Q = \theta/3$. Every pair $(c_Q, c_{\mathcal{F}})(\theta, m)$ is dominated componentwise by a point of $\{(c_Q(\vartheta, m), \vartheta/3) : 2\sqrt{m} \le \vartheta \le 3\sqrt{m/2}\}$.

The full-class optimum is $1/\sqrt{2}$ times the quadratic optimum and uses $3/(2\sqrt{2}) \approx 1.06$ times as much friction. At critical damping the full-class exponent is $2\sqrt{m}/3$. The arc in Corollary 3.2 records the trade-off between the two exponents (Figure 1); discrete tunings also face stability constraints.

3.2. Why the worst case mixes spirals

The nonquadratic witnesses are orthogonal sums of spirals,

$$x(t) = \big(\sqrt{\nu_j}\, e^{(-c+i\gamma_j)t}\big)_{j=1}^{J}, \qquad \nu_j \ge 0, \quad \sum_j \nu_j = 1.$$

The flow fixes the gradient along each orbit. Existence of a strongly convex potential realizing these data is characterized by the interpolation program $\mathcal{P}_c$ in Lemma B.2. Its minimizer constraint is

$$(3c - \theta) \sum_j \nu_j \gamma_j^2 \ge c\Delta_c.$$

For $c < c_Q$, $\Delta_c > 0$, so it forces $c > \theta/3$, matching the coefficient $\theta - 3c$ in the Lyapunov identity. The other constraints test convexity between every pair of orbit points, in both time directions.

For $\theta/3 < c < c_Q$, the equally weighted frequencies $\gamma_j = \gamma_0(1 + (j - \frac{1}{2})/J)$ satisfy all constraints with a uniform margin when γ_0 and then J are large (Lemma B.4); Lemma B.3 supplies a $C^{1,1}$ potential. As the frequencies grow, $\langle \nabla f_m(x), x\rangle / f_m(x) \to 2c/(\theta - c)$. This limit tends to 1 as $c \downarrow \theta/3$, giving a nearly conical profile. The frequency spread separates the components' first full turns (Figure 2); Proposition 3.3 shows why one spiral fails near the limiting exponent.

Proposition 3.3 (Single-spiral obstruction)**.** *Let $\theta/3 < c_Q(\theta, m)$. There is $c_\sharp \in (\theta/3, c_Q)$ such that for $\theta/3 < c \le c_\sharp$ no one-atom measure satisfies (34)–(35). Consequently, for such c, no $f \in \mathcal{F}_m(H)$ has a solution of (7) of the form $x(t) = e^{-ct}(a\cos\gamma t + b\sin\gamma t)$ with $a \perp b$, $\|a\| = \|b\| > 0$ and $\gamma \ge 0$.*

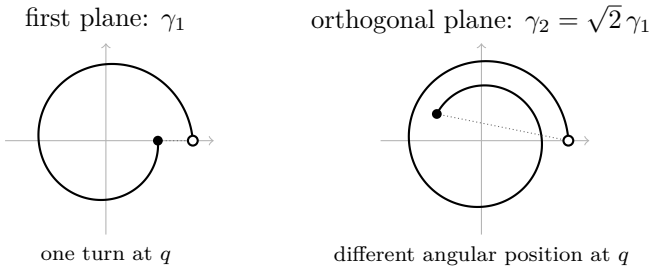


Figure 2: Schematic desynchronization in two orthogonal projections. At the common lag $q = 2\pi/\gamma_1$, the first returns to its starting ray and the second has a different angular position. Open and filled markers show times 0 and q; dotted chords join them.

Near $c = \theta/3$, the minimizer constraint forces a single spiral to wind rapidly, violating convexity at its first return to the starting ray (Remark B.5).

We now characterize discrete self-similar trajectories and the certificates that exclude larger growth factors.

4. Spectral Closure for Stationary Linear Methods

We characterize orbits generated by one smooth strongly convex function. Interpolation reduces this question to pairwise inequalities. For orthogonal rotation–dilation modes, these inequalities depend only on the lag; squared amplitudes define a spectral measure, and one scalar encodes the function values. The resulting spectral problem and its finite-lag certificates determine the same exact rate (Theorem 4.7).

4.1. Normalization and interpolation

Translate the minimizer to 0, set $f(0) = 0$, and define

$$\psi(e) = \tfrac{2}{L-\mu}\big(\tfrac{L+\mu}{4}\|e\|^2 - f(e)\big), \qquad \nabla f(e) = \tfrac{L+\mu}{2}\, e - \tfrac{L-\mu}{2}\,\nabla\psi(e). \tag{10}$$

Let $\mathcal{N}(H)$ be the set of differentiable $\psi : H \to \mathbb{R}$ with $\psi(0) = 0$, $\nabla\psi(0) = 0$ and 1-Lipschitz gradient.

Lemma 4.1 ([32, Lemma A.1])**.** *The map $f \mapsto \psi$ is a bijection from $\{f \in \mathcal{F}_{\mu,L}(H) : x^\star = 0,\ f(0) = 0\}$ onto $\mathcal{N}(H)$.*

The lemma is stated in [32] for $\mathbb{R}^d$; its proof (cocoercivity of the gradient of a convex $(L-\mu)$-smooth function [45]) is dimension-free. Every $\psi \in \mathcal{N}(H)$ satisfies $\|\nabla\psi(e)\| \le \|e\|$ and $|\psi(e)| \le \frac{1}{2}\|e\|^2$. For data $(\tilde{x}_i, \sigma_i, \psi_i)_{i\in I}$ in $H \times H \times \mathbb{R}$ put

$$T_{ij} = \psi_j - \psi_i - \tfrac{1}{2}\langle \sigma_i + \sigma_j, \tilde{x}_j - \tilde{x}_i\rangle, \qquad R_{ij} = \tfrac{1}{4}\big(\|\tilde{x}_j - \tilde{x}_i\|^2 - \|\sigma_j - \sigma_i\|^2\big). \tag{11}$$

Theorem 4.2 ([32, Theorem A.2])**.** *Suppose $(\tilde{x}_\star, \sigma_\star, \psi_\star) = (0, 0, 0)$ for some $\star \in I$. There is a function $\psi \in \mathcal{N}(H)$ interpolating all data, $\nabla\psi(\tilde{x}_i) = \sigma_i$, $\psi(\tilde{x}_i) = \psi_i$, if and only if $|T_{ij}| \le R_{ij}$ for every $i, j \in I$.*

Theorem 4.2 is Azagra and Mudarra's extension theorem for convex $C^{1,1}$ functions on Hilbert spaces [4, Theorem 1.3], expressed in the normalization ψ. It applies to arbitrary index sets and hence to infinite trajectories. On finite sets it is equivalent to the conditions of Taylor, Hendrickx and Glineur [45].

4.2. Stationary linear methods

Definition 4.3. Let $D \ge 1$, and let $u(z) = \sum_{j=0}^{D-1} u_j z^j \not\equiv 0$ and $v(z) = \sum_{j=0}^{D} v_j z^j$, $v_D \ne 0$, be real polynomials with $u(1) = 1$ and $v(1) = 0$. A *trajectory* of the stationary linear method (u, v) on $f \in \mathcal{F}_{\mu,L}(H)$ is a sequence $(w_k)_{k\ge 0} \subset H$ with

$$\tilde{x}_k := \sum_{j=0}^{D-1} u_j w_{k+j}, \qquad \sum_{j=0}^{D} v_j w_{k+j} = \nabla f(\tilde{x}_k) \qquad (k \ge 0). \tag{12}$$

We call w_k the *latent iterates* and $\tilde{x}_k$ the *query points*, where gradients are evaluated. Its *state* is $S_k = (w_k - x^\star, \dots, w_{k+D-1} - x^\star) \in H^D$; (12) determines S_{k+1} from S_k.

The identities $u(1) = 1$, $v(1) = 0$ preserve the minimizer as a fixed point; henceforth $x^\star = 0$. Every recurrence with constant scalar coefficients $x_{k+1} = \sum_{j<D} b_j x_{k-j} - h\nabla f(\sum_{j<D} a_j x_{k-j})$ with $\sum_j a_j = \sum_j b_j = 1$ fits this definition with $w_k = x_{k-D+1}$. Common roots of u and v represent modes that evolve independently of f. In particular, (1) is the case $D = 2$, $w_k = x_{k-1}$,

$$u(z) = N_\tau(z) = (1+\tau)z - \tau, \qquad v(z) = -\tfrac{1}{h}(z-1)(z-\beta), \tag{13}$$

and u, v are coprime unless $\beta = \tau/(1+\tau)$. In normalized form (10) the method reads $\sigma_k := \nabla\psi(\tilde{x}_k) = \sum_j p_j w_{k+j}$ with

$$p(z) = \tfrac{2}{L-\mu}\big(\tfrac{L+\mu}{2}\,u(z) - v(z)\big), \tag{14}$$

a polynomial of degree D, coprime to u whenever u, v are coprime. The *quadratic rate* and the *middle-curvature radius* are

$$\rho_q = \max_{\lambda\in[\mu,L]} \max\{|z| : v(z) = \lambda u(z)\}, \qquad r_c = \max\{|z| : p(z) = 0\} \le \rho_q.$$

Thus r_c is the root radius at curvature $(L+\mu)/2$. Moreover $\rho_q > 0$: otherwise both $v - \mu u$ and $v - Lu$ would be $v_D z^D$, forcing $u \equiv 0$. We set $\log 0 = -\infty$.

For $\mathcal{M}$, the endpoint polynomials χ_μ, χ_L give explicit normalized symbols. Indeed, (13) yields $u - p = -2\chi_\mu/(h(L-\mu))$ and $u + p = 2\chi_L/(h(L-\mu))$, hence for every $z \in \mathbb{C}$

$$|u|^2 - |p|^2 - 2i\,\mathrm{Im}(p\bar{u}) = (u-p)(\bar{u}+\bar{p}) = -\frac{4\,\chi_\mu(z)\overline{\chi_L(z)}}{h^2(L-\mu)^2}, \tag{15}$$

and r_c is the largest root modulus of $\chi_{(L+\mu)/2}$. On the unit circle the phase of $\chi_\mu\overline{\chi_L}$ is controlled by

$$\mathrm{Im}\big(\chi_\mu(e^{i\omega})\overline{\chi_L(e^{i\omega})}\big) = (y - y_\mu)\sin\omega\,\big[(1-\beta) + 2\tau\xi\big], \qquad \xi = 1 - \cos\omega, \tag{16}$$

which is positive on $(0, \pi)$ (Appendix A).

Remark 4.4 (Two-step methods). Every scalar stationary method with $D = 2$ has the algebraic form (1):

$$u = (1+\tau)z - \tau, \quad v = -h^{-1}(z-1)(z-\beta), \qquad \tau = -u(0), \quad h = -1/v_2, \quad \beta = v(0)/v_2.$$

The tunings studied here satisfy $\beta \in [0, 1)$, $\tau \ge 0$, $h > 0$. The queries obey the gradient-correction recurrence

$$\tilde{x}_{k+1} = \tilde{x}_k + \beta(\tilde{x}_k - \tilde{x}_{k-1}) - h_1\nabla f(\tilde{x}_k) + h_2\nabla f(\tilde{x}_{k-1}), \qquad h_1 = h(1+\tau),\ h_2 = h\tau,$$

Conversely, $h_1 > h_2 \ge 0$ gives the query representation (1) with $h = h_1 - h_2$, $\tau = h_2/(h_1 - h_2)$. For $\beta \ne z_\tau := \tau/(1+\tau)$, the two state descriptions have the same worst-case rate. Let $\rho^0_{\tilde{x}}$ denote the from-rest rate of the queries: replace x_N with $\tilde{x}_N$ in the numerator of the from-rest formula in Definition 2.2. Then

$$\rho^0_{\tilde{x}} \le \rho^0_{\mathcal{F}} \le \max\{\rho^0_{\tilde{x}}, z_\tau\}.$$

The two rates coincide above z_τ, including every accelerated regime (Appendix A).

For example, the quasi-hyperbolic momentum (QHM) method of Ma and Yarats [26], $d_k = \beta d_{k-1} + (1-\beta)\nabla f(x_k)$, $x_{k+1} = x_k - \alpha[(1-\nu_{\mathrm{qh}})\nabla f(x_k) + \nu_{\mathrm{qh}} d_k]$ with $\nu_{\mathrm{qh}} \in [0, 1]$, has the query representation (1) with

$$h = \alpha(1-\beta), \qquad \tau = \tau_{\mathrm{qh}} := \frac{(1-\nu_{\mathrm{qh}})\beta}{1-\beta};$$

$\nu_{\mathrm{qh}} = 1$ gives heavy ball and $\nu_{\mathrm{qh}} = \beta$ gives Nesterov's method. With the usual initialization $d_{-1} = 0$, its iterates are exactly the queries of (1) started from rest at x_0 (Appendix A). The query-rate comparison transfers the results below to QHM with this initialization.

4.3. Spectra and certificates

Fix a stationary linear method (u, v) with normalized polynomial p from (14). For $z = e^{t+i\omega}$ the *wave* $w_k = z^k$ rotates by ω and dilates by e^t per step; its queries are $\tilde{x}_k = u(z)z^k$, and (12) requires the normalized gradients $\sigma_k = p(z)z^k$. Put

$$g(t,\omega) = |u(z)|^2 - |p(z)|^2, \qquad s(t,\omega) = \operatorname{Im}(p(z)\overline{u(z)}), \tag{17}$$

$$a_n = g\,(\cosh nt - \cos n\omega), \qquad b_n = 2s \sin n\omega \qquad (n \geq 1). \tag{18}$$

For waves with a common exponent t, normalized squared amplitudes define a probability measure ν on $[0, \pi]$, called the *spectrum*. Each atom represents one planar wave component. One scalar C encodes the function values; interpolation with the minimizer bounds it by $|C| \leq \bar{g}$. This scalar is essential for sharp boundary-layer witnesses.

Definition 4.5. A pair (ν, C) is *feasible at* t if

$$\begin{gathered} |C| \leq \bar{g}, \\ \ell_n := \bar{a}_n - \bar{b}_n + C \sinh nt \geq 0, \qquad \upsilon_n := \bar{a}_n + \bar{b}_n - C \sinh nt \geq 0 \qquad (n \geq 1). \end{gathered} \tag{19}$$

The numbers ℓ_n, υ_n are the lower and upper *lag-n slacks*, for queries n steps apart; $\bar{g} \pm C \geq 0$ are the two minimizer inequalities. We call $|C| < \bar{g}$ *strict minimizer slack*. A lag n is *active* when $\ell_n = 0$ or $\upsilon_n = 0$. Let $\mathcal{T}$ be the set of $t \in \mathbb{R}$ at which some pair is feasible.

Definition 4.6. A *strict certificate at* t consists of nonnegative, finitely supported weights α_n, β_n, numbers $\gamma_\pm \geq 0$ and $\epsilon > 0$ such that

$$\begin{gathered} \Phi_t(\omega) := \sum_n \{\alpha_n(a_n - b_n) + \beta_n(a_n + b_n)\} + (\gamma_+ + \gamma_-)g \leq -\epsilon \quad (0 \leq \omega \leq \pi), \\ \sum_n (\alpha_n - \beta_n) \sinh nt + \gamma_+ - \gamma_- = 0. \end{gathered} \tag{20}$$

The inequalities (19) are the interpolation conditions between queries n steps apart and with the minimizer (Lemma C.1). A certificate combines them with nonnegative weights that cancel the function values. It is a finite-lag Lyapunov argument, or a discrete-time O'Shea–Zames–Falb multiplier [31, 32, 52, 54]. For heavy ball, $u(z) = z$, and (17) reduces, up to the positive factor $|z|^2$, to the quantities of [32, eq. (12) and Definition 2.4].

At $t = 0$ the pair (g, s) defines a phase: where $g - 2is \neq 0$, let $\Theta(\omega) \in (-\pi, \pi]$ be given by

$$g(0,\omega) - 2is(0,\omega) = \left|g(0,\omega) - 2is(0,\omega)\right| e^{-i\Theta(\omega)}. \tag{21}$$

For $\omega > 0$, the *spectral phase lead* is $\Theta(\omega)/\omega$, measured in steps. For a limit method of $\mathcal{M}$, this lead tends to τ at low frequencies (Sections 5.2 and 5.4).

4.4. The closure theorem

Theorem 4.7 (Spectral closure)**.** *Let (u, v) be a stationary linear method.*

(a) *$\mathcal{T}$ is nonempty, closed and bounded above, and $T := \max \mathcal{T} \geq \log \rho_q$.*

(b) *$\rho_{\mathcal{F}} = e^T$. Every $t > T$ admits a strict certificate, and feasibility at t excludes one.*

(c) (One exponent suffices.) *If a strict certificate exists at some t with $e^t > r_c$, then $\rho_{\mathcal{F}} < e^t$. Consequently $\mathcal{T} \cap (\log r_c, \infty) = (\log r_c, T]$.*

(d) *Every pair (ν, C) feasible at t is realized: there are $\psi \in \mathcal{N}(L^2(\nu; \mathbb{C}))$ and a trajectory with $S_{k+1} = e^t U S_k$ for an orthogonal U, hence $\|S_N\| = e^{Nt}\|S_0\|$ for all N; if ν has J atoms, then $L^2(\nu; \mathbb{C}) \cong \mathbb{R}^{2J}$.*

(e) *If $T = \log \rho_q$, a quadratic on $\mathbb{R}^2$ attains the worst case. If $T > \log \rho_q$ and $T \neq 0$, every pair feasible at T has $|C| < \bar{g}$ and finitely many active lags. Fix any such pair and let M be its largest active lag. Every pair feasible at T is then carried by at most $(D + M)/2$ points of $(0, \pi)$, and the worst case is attained on $\mathbb{R}^{2J}$ with $J \leq (D + M)/2$.*

Thus finite-lag certificates approach the exact rate from above, and each feasible spectrum generates one self-similar trajectory. Part (e) gives finite-dimensional attainment at the spectral boundary under its stated hypotheses. The interpolation, separation and one-exponent arguments for quadratically stable heavy ball are those of [32, Theorems 2.5, 3.2 and Lemmas A.3–A.7]; finite support, strict minimizer slack and inactive lag tails were proved there for Polyak's tuning [32, Lemma A.10]. For standard Nesterov, a linear program over spirals gives the exact squared-distance factor [33, Theorem 4.1]. The latent polynomials extend the characterization to arbitrary fixed finite memory and give the atom bound in (e). Finite-dimensional attainment at $T = 0 > \log \rho_q$ remains open (Section 9).

4.5. Consequences for the class $\mathcal{M}$

For (1) we use Theorem 4.7 with (13). By (15), g and s are explicit in χ_μ and χ_L, and on the unit circle (16) gives $s > 0$ on $(0, \pi)$. Hence, for a stable tuning, the phase (21) satisfies $\Theta(\omega) \in (0, \pi]$ for $\omega \in (0, \pi]$: it is the angle from $-\chi_\mu(e^{i\omega})$ to $\chi_L(e^{i\omega})$ (for heavy ball this is [32, Definition 3.1]).

Corollary 4.8 (Worst cases from rest)**.** *Let a tuning of $\mathcal{M}$ be stable, and let (ν, C) be feasible at t, with J atoms, and with $|C| < \bar{g}$ if $t > 0$. There are $f \in \mathcal{F}_{\mu,L}(\mathbb{R}^{2J+1})$ and $x_0 \neq x^\star$ such that the trajectory with $x_{-1} = x_0$ satisfies $\|x_N - x^\star\| \geq c_0 e^{Nt} \|x_0 - x^\star\|$ for all $N \geq 0$, with $c_0 > 0$. In particular, if either $T = \log \rho_q$ or $T > \log \rho_q$ and $T \neq 0$, then $\rho^0_{\mathcal{F}} = \rho_{\mathcal{F}}$. The worst case from rest is attained by a quadratic on $\mathbb{R}$ in the first case, and by one function in a fixed dimension in the second.*

Proposition 4.9 (Lag-one criterion)**.** *Let a tuning of $\mathcal{M}$ be stable.*

(a) *If $\operatorname{Re}[(1 - z)\chi_\mu(z)\overline{\chi_L(z)}] > 0$ on $|z| = r$ for some $r \in (r_c, 1)$, then $\rho_{\mathcal{F}} < r$.*
(b) *If $\operatorname{Re}[(1 - z)\chi_\mu(z)\overline{\chi_L(z)}] > 0$ on $|z| = 1$, $z \neq 1$, then $\rho_{\mathcal{F}} < 1$.*
(c) *More generally, if $\operatorname{Re}[\chi_\mu(z)\overline{\chi_L(z)}\,((1-z) + \lambda(1-\bar{z}^2))] > 0$ on $|z| = 1$, $z \neq 1$, for some $\lambda \geq 0$, then $\rho_{\mathcal{F}} < 1$.*

We call $\operatorname{Re}[(1 - z)\chi_\mu\overline{\chi_L}]$ the *lag-one symbol* and Proposition 4.9 the *lag-one criterion*. The corresponding *lag-one certificate* combines $\ell_1 \geq 0$ with a minimizer constraint to cancel C. Its Jury–Lee multiplier $1 - z$ [20] underlies rate bounds in [8, 13, 23] and is exact at the convergence threshold of Polyak's heavy ball [5]. On each circle its symbol is a cubic in ξ (Lemma E.1); Sections 6.2 and 6.3 describe when the criterion is exact.

For $r \in (r_c, 1]$, a strict certificate at $t = \log r$ yields a Lyapunov function (Proposition C.15). For lag one, it is a quadratic form in two iterates and a gradient plus a multiple of the function gap at the query, as in [8, 19, 23, 48]. Longer certificates use a finite window of query values and gradients. For a stable tuning, persistence in the exponent (Lemma C.5) turns a strict certificate at $t = 0$ into a contraction with some $r < 1$. Lower bounds come from the realizations of Corollary 4.8 and periodic rotating orbits (Proposition 4.10).

Proposition 4.10 (Phase locking)**.** *Let a tuning of $\mathcal{M}$ be stable, let $K \geq 3$, $1 \leq j < K/2$, $\gcd(j, K) = 1$ and $\omega_0 = 2\pi j/K$.*

(a) *If $\Theta(\omega_0) \leq \pi/K$, there are $f \in \mathcal{F}_{\mu,L}(\mathbb{R}^3)$ and x_0 such that, from rest, $\|x_N - x^\star\| \geq c_0 \|x_0 - x^\star\| > 0$ for all N; the trajectory converges to a K-periodic orbit that rotates by ω_0 in a plane.*
(b) *If $\Theta(\omega_0) < \pi/K$, then $\rho^0_{\mathcal{F}} > 1$.*

We call this periodic rotation *phase locking*, and the corresponding parameter regions *K-tongues*. For heavy ball the proposition is [32, Theorem 3.3(ii)], and these regions are the cycles of [17].

The flow supplies slowly rotating witnesses; the discrete symbol identifies the phase loss that determines whether the flow rate is attained.

5. The Half-Step Rule for Fixed Finite Memory

We now allow a fixed finite number of momentum terms, past queries and gradient corrections. With the slow dynamics fixed, the flow still bounds the rate. The limiting spectral phase governs attainment, and a trace identity relates the step to the lag-one symbol.

For a stationary method (u, v), write

$$\phi_\lambda(z) = \lambda u(z) - v(z).$$

This is the characteristic polynomial on $\lambda\|\cdot\|^2/2$: a wave $w_k = z^k$ is a trajectory exactly when $\phi_\lambda(z) = 0$. For $\mathcal{M}$, $\phi_\lambda = \chi_\lambda/h$. By (14), $u - p = -\frac{2}{L-\mu}\phi_\mu$ and $u + p = \frac{2}{L-\mu}\phi_L$, so that, as in (15),

$$g - 2is = (u - p)\overline{(u + p)} = -\frac{4\,\phi_\mu(z)\overline{\phi_L(z)}}{(L-\mu)^2}. \tag{22}$$

The proof of Proposition 4.9(a) uses this factorization and the one-exponent principle in Theorem 4.7(c). It therefore applies to every stationary method with ϕ_λ replacing χ_λ; we use the same terms *lag-one symbol*, *certificate* and *criterion*.

5.1. A ceiling for every memory

We keep the normalization $L = 1$, $\mu = \varepsilon^2$.

Definition 5.1 (Second-order families)**.** Let $D \geq 2$ and, for $\varepsilon \in (0, \varepsilon_0)$, let $(u_\varepsilon, v_\varepsilon)$ be stationary linear methods with memory D whose coefficients extend continuously to $\varepsilon = 0$, with $\deg v_0 = D$. The family is *second-order with slow data* (θ, y), $\theta, y > 0$, if

(i) $v_0(z) = (z-1)^2 q_0(z)$ with $q_0(1) = -1/y$;

(ii) the root β_ε of v_ε near 1 other than 1 satisfies $(1 - \beta_\varepsilon)/\varepsilon \to \theta$.

It is of class C^k if the coefficients are C^k functions of $\varepsilon \in [0, \varepsilon_0)$.

The limiting quotient $(z-1)q_0$ has a simple root at 1, so for small ε,

$$v_\varepsilon = (z-1)(z-\beta_\varepsilon)q_\varepsilon, \qquad \theta_\varepsilon := \frac{1-\beta_\varepsilon}{\varepsilon} \to \theta, \qquad y_\varepsilon := -\frac{1}{q_\varepsilon(1)} \to y.$$

Here β_ε is real, inherits the coefficient regularity, and $q_\varepsilon \to q_0$. For $\mathcal{M}$, $q = -1/h$, and the families (3) are second-order with slow data (θ, y). On a slow wave $z = e^{-\varepsilon\varpi}$ every such method sees the complex curvature $v_\varepsilon(z)/u_\varepsilon(z) = \frac{\varepsilon^2}{y_\varepsilon}\varpi(\theta_\varepsilon - \varpi) + O(\varepsilon^3)$ (Lemma D.1); after the change $F = \frac{y}{\varepsilon^2}f$ this is the curvature $\lambda' = \varpi(\theta - \varpi)$ at which $e^{-\varpi t}$ solves $\ddot{x} + \theta\dot{x} + \lambda' x = 0$. So every second-order family with slow data (θ, y) discretizes the same slow flow as the tunings (3) of $\mathcal{M}$.

Theorem 5.2 (Ceiling)**.** *For every second-order family with slow data (θ, y),*

$$\limsup_{\kappa\to\infty} \sqrt{\kappa}\,(1 - \rho_{\mathcal{F}}) \leq c_{\mathcal{F}}(\theta, y) = \min\big(\tfrac{\theta}{3}, c_Q(\theta, y)\big).$$

If $\theta/3 < c < c_Q(\theta, y)$, then for all small ε a spectrum with J atoms, J independent of ε, is feasible at $t = -c\varepsilon$, and the method has a trajectory in $\mathbb{R}^{2J}$ with $S_{k+1} = e^{-c\varepsilon}US_k$, U orthogonal.

Sampled flow spirals give the same ceiling for every family with these slow data and any fixed finite memory.

Corollary 5.3 (Consequences of the ceiling)**.** *Consider second-order families with slow data (θ, y).*

(i) $\limsup \sqrt{\kappa}(1 - \rho_{\mathcal{F}}) \leq \sqrt{y/2}$*, and the pair formed by* $\limsup \sqrt{\kappa}(1 - \rho_q)$ *and* $\limsup \sqrt{\kappa}(1 - \rho_{\mathcal{F}})$ *is dominated by a point of the arc of Corollary 3.2 with $m = y$.*

(ii) *At critical damping, $\theta = 2\sqrt{y}$, the constant is at most $\frac{2}{3}\sqrt{y}$. For any fixed finite memory, the slow data $(2, 1)$ of standard Nesterov and $(\frac{4}{\sqrt{3}}, \frac{4}{3})$ of its quadratic-optimal tuning therefore give ceilings $\frac{2}{3}$ and $\frac{4}{3\sqrt{3}}$, respectively (Corollary 6.6(i), (ii)).*

(iii) *A family with* $\liminf \sqrt{\kappa}(1 - \rho_{\mathcal{F}}) \geq 1$ *has $\theta \geq 3$ and $y \geq \theta - 1 \geq 2$.*

5.2. The limiting phase

For the *limit method* (u_0, v_0), corresponding to $\varepsilon = \mu = 0$, $L = 1$, write $\phi_\lambda = \lambda u_0 - v_0$. Thus the subscript in ϕ_0, ϕ_1 denotes curvature. Its lag-one symbol and *effective extrapolation* are

$$\mathcal{S}_0(\omega) = \mathrm{Re}\Big[(1 - e^{i\omega})\,\phi_0(e^{i\omega})\,\overline{\phi_1(e^{i\omega})}\Big], \qquad \tau_{\mathrm{eff}} = u_0'(1) - 1 + y\,q_0'(1),$$

with $\phi_0 = -v_0$ and $\phi_1 = u_0 - v_0$. The number τ_{eff} depends only on the transfer function of the limit method:

$$y\,\frac{v_0(z)}{u_0(z)} = -(z-1)^2\big[1 - (1+\tau_{\mathrm{eff}})(z-1) + O((z-1)^2)\big].$$

Thus multiplying u, v by a common real polynomial r with $r(1) = 1$ preserves τ_{eff}. For $\mathcal{M}$, $u_0 = N_\tau$ and $q_0 = -1/y$, so $\tau_{\mathrm{eff}} = \tau$.

Lemma 5.4 (Low frequencies of the limit method)**.** *As $\omega \to 0$,*

$$\phi_0\overline{\phi_1}\,(e^{i\omega}) = -\frac{\omega^2}{y} + i\,\frac{\tau_{\mathrm{eff}}}{y}\,\omega^3 + O(\omega^4), \qquad \mathcal{S}_0(\omega) = \frac{2\tau_{\mathrm{eff}} - 1}{2y}\,\omega^4 + O(\omega^6).$$

5.3. The half-step loss

On a Fourier mode with $\omega \neq 0$, the discrete difference and continuous derivative satisfy

$$1 - e^{i\omega} = \frac{2\sin(\omega/2)}{\omega}\,e^{i\omega/2}(-i\omega). \tag{23}$$

The factor $e^{i\omega/2}$ adds half a step of phase, which the spectral lead must compensate. The flow Lyapunov function in Theorem 3.1 is associated with a Popov-type multiplier [35]; $1 - z$ is its discrete Jury–Lee counterpart [20]. Theorem 5.5 turns this phase relation into a uniform rate theorem under explicit regularity and quadratic-root conditions.

Phase lead. Let Θ_0 be the phase (21) of the limit method ($t = 0$, $\mu = 0$, $L = 1$). By (22) it is defined where $\phi_0\phi_1 \neq 0$ and given by $-\phi_0\overline{\phi_1} = |\phi_0\phi_1|\,e^{-i\Theta_0}$, and by Lemma 5.4, $\Theta_0(\omega)/\omega \to \tau_{\mathrm{eff}}$ as $\omega \to 0$. Thus τ_{eff} is the low-frequency spectral phase lead. For $\mathcal{M}$, the phase is $\Theta_0(\omega) = \arg\big(y - (2 - y\tau)\xi + i\,y\tau\sin\omega\big)$. Two interpolation facts make the phase useful. At $\omega \in (0, \pi)$, the lag-one symbol is a positive multiple of $\sin(\Theta(\omega) - \omega/2)$ (Lemma C.2). For stable tunings of $\mathcal{M}$, $\Theta \in (0, \pi]$, so positivity means $\Theta > \omega/2$. For a general stationary method, the sine criterion is the branch-independent statement. Second, a one-atom spectrum at $2\pi/K$ is feasible at $t = 0$ exactly when $|\Theta(2\pi/K)| \leq \pi/K$; for $2\pi j/K$, $j \geq 2$, the threshold remains π/K (Lemma C.14).

The low-frequency wall is therefore $\tau_{\mathrm{eff}} = 1/2$. If $\mathcal{S}_0 \not\equiv 0$, write $\mathcal{S}_0(\omega) = s_{2\mathfrak{p}}\omega^{2\mathfrak{p}} + O(\omega^{2\mathfrak{p}+2})$. By Lemma 5.4, $\mathfrak{p} \geq 2$, and $\mathfrak{p} = 2$ exactly when $\tau_{\mathrm{eff}} \neq 1/2$.

5.4. Faithful discretization

A second-order family with slow data (θ, y) is *faithful to its slow flow* if $\lim_{\kappa\to\infty} \sqrt{\kappa}(1 - \rho_{\mathcal{F}}) = c_{\mathcal{F}}(\theta, y)$.

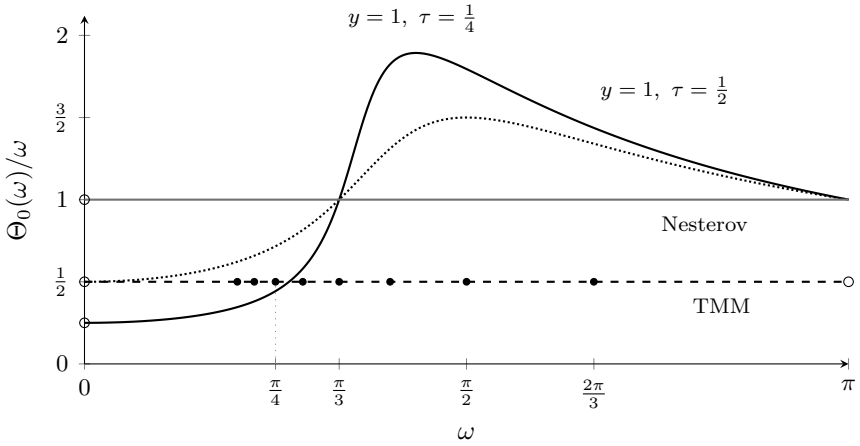


Figure 3: Spectral phase lead of limit methods of $\mathcal{M}$, in steps. The horizontal half-step line is TMM; the lag-one symbol is positive above it. Dots at $\omega = 2\pi/K$, $3 \le K \le 10$, mark the one-atom thresholds. For the solid curve, strict passage below a threshold occurs exactly for $K \ge 8$, starting at $\omega = \pi/4$. Open markers denote endpoint limits.

Theorem 5.5 (Faithful discretization and the half-step rule)**.** *Let a second-order family have slow data (θ, y), limit method (u_0, v_0), effective extrapolation τ_{eff}, phase Θ_0 and, if $\mathcal{S}_0 \not\equiv 0$, order $2\mathfrak{p}$ and leading coefficient $s_{2\mathfrak{p}}$ as above.*

(a) (Beyond the wall.) *If $\mathcal{S}_0 > 0$ on $(0, \pi]$, the roots of $u_0 - 2v_0$ lie in the open unit disk, and the family is of class $C^{2\mathfrak{p}}$, then*

$$\lim_{\kappa \to \infty} \sqrt{\kappa}\,(1 - \rho_{\mathcal{F}}) = c_{\mathcal{F}}(\theta, y),$$

and for every $c < c_{\mathcal{F}}(\theta, y)$ the bound $\rho_{\mathcal{F}} < 1 - c\varepsilon$, for small ε, is certified by the lag-one criterion.

(b) (Below the wall.) *If $|\tau_{\text{eff}}| < \frac{1}{2}$, or if $\tau_{\text{eff}} = \frac{1}{2}$ and $s_{2\mathfrak{p}} < 0$, there is an integer $K_0 \ge 3$ such that $|\Theta_0(2\pi/K)| < \pi/K$ for every $K \ge K_0$, and for each such K and all small ε the one-atom spectrum at frequency $2\pi/K$ is feasible at some $t > 0$.*

(c) (Behind the iterate.) *If $\tau_{\text{eff}} < 0$, then $\liminf_{\kappa \to \infty} \rho_q > 1$: the method diverges already on a quadratic on $\mathbb{R}$.*

(d) (Fast resonances.) *If $\phi_0\phi_1(e^{i\omega_0}) \ne 0$ and $|\Theta_0(\omega_0)| < \pi/K$ at $\omega_0 = 2\pi j/K$ for some $K \ge 3$ and $1 \le j < K/2$ with $\gcd(j, K) = 1$, then for all small ε the one-atom spectrum at frequency ω_0 is feasible at some $t > 0$.*

In (b)–(d), $\rho_{\mathcal{F}} > 1$: the method has growing trajectories on some $f \in \mathcal{F}_{\mu,L}(\mathbb{R}^2)$ for all small ε. These parts require coefficient continuity; part (a) uses the stated higher regularity.

Corollary 5.6 (Necessary lead)**.** *Every second-order family that is stable on quadratics for all small ε has $\tau_{\text{eff}} \ge 0$, and every second-order family with $\rho_{\mathcal{F}} < 1$ for all small ε has $\tau_{\text{eff}} \ge \frac{1}{2}$.*

In (a), $u_0 - 2v_0 = 2\phi_{1/2}$ controls the fast response at middle curvature. For $\tau_{\text{eff}} > 1/2$, the regularity hypothesis is C^4; at $\tau_{\text{eff}} = 1/2$, the quartic term of $\mathcal{S}_0$ vanishes and the first nonzero term fixes the required order. Parts (b), (c) detect low-frequency divergence under coefficient continuity, while (d) detects fixed-frequency resonances.

Remark 5.7 (Extremal leads). For the limit of standard Nesterov, $(y, \tau) = (1, 1)$, $\Theta_0(\omega) = \omega$ for $0 < \omega \le \pi$; for the limit of TMM, $(y, \tau) = (2, \frac{1}{2})$, $\Theta_0(\omega) = \omega/2$ for $0 < \omega < \pi$. For TMM, $\phi_0\phi_1(e^{i\pi}) = 0$ and $\mathcal{S}_0 \equiv 0$ on the entire circle. For every limit method of $\mathcal{M}$, $\Theta_0(\omega)/\omega \to \tau$ as $\omega \to 0$.

The slow part of the lag-one symbol is the flow identity: in the slow variables $\omega = \varepsilon\gamma$, the leading part of the symbol on the circle $|z| = 1 - c\varepsilon$ is $\frac{\varepsilon^3}{y}\big(c\Delta_c + (\theta - 3c)\gamma^2\big)$, with $\Delta_c = c^2 - \theta c + y$ (Lemma D.6),

which is the slack $(3c-\theta)\gamma^2 - c\Delta_c$ of the minimizer constraint (35) for one atom, with the opposite sign. For $0 < c < c_Q(\theta, y)$, we have $\Delta_c > 0$: the slow symbol is positive for every γ exactly when $c \le \theta/3$, which is also exactly when the minimizer constraint (35) excludes every flow atom. The certificate is the discrete form of (9).

5.5. A trace formula for the step

The slow step $y = -1/q_0(1)$, equal to $\lim hL$ for $\mathcal{M}$, satisfies an exact trace formula in the limiting lag-one symbol. Recall $\xi = 1 - \cos\omega$.

Theorem 5.8 (Trace formula)**.** *Let (u_0, v_0) be a stationary linear method with $v_0 = (z-1)^2 q_0$ and $q_0(1) = -1/y < 0$, and let $\phi_0 = -v_0$, $\phi_1 = u_0 - v_0$ and $\mathcal{S}_0$ be as above.*

(a) *If the roots of ϕ_1 lie in the open unit disk, except possibly a simple root at -1, then*

$$\frac{2}{y} - 1 = \frac{1}{2\pi} \int_0^\pi \frac{(2-\xi)\,\mathcal{S}_0(\omega)}{\xi^2\,|\phi_1(e^{i\omega})|^2}\, d\omega, \tag{24}$$

with a bounded integrand, and $\mathcal{S}_0 \equiv 0$ if and only if $v_0/u_0 = -(z-1)^2/(3z-1)$.

(b) *If the roots of $\lambda u_0 - v_0$ lie in the closed unit disk for all λ in an interval $(\lambda_0, 1)$, $0 \le \lambda_0 < 1$, and $\mathcal{S}_0 \ge 0$ on $(0, \pi)$, then $y \le 2$, with equality if and only if $v_0/u_0 = -(z-1)^2/(3z-1)$.*

Why equality is rigid. For the root hypothesis in (a), set

$$Q(z) = -\frac{z(1+z)q_0(z)}{\phi_1(z)}.$$

It is analytic outside the unit disk, with $Q(\infty) = 1$ and $Q(1) = 2/y$. On the circle, its imaginary part is minus $\mathcal{S}_0$ times a positive weight. The Schwarz formula gives (24). If $\mathcal{S}_0 \equiv 0$, the imaginary part vanishes on the boundary; analyticity and $Q(\infty) = 1$ force $Q \equiv 1$, hence $u_0 = (1-3z)q_0$ and $v_0/u_0 = -(z-1)^2/(3z-1)$. For part (b), cancel common factors and apply the same formula at curvature λ: $0 \le Q_\lambda(x) - 1 \le 2/(\lambda y) - 1$ for every real $x > 1$. If $y = 2$, letting $\lambda \uparrow 1$ gives $Q_1(x) = 1$ wherever defined, and hence the same polynomial identity.

For TMM's limit, $(y, \tau) = (2, \frac{1}{2})$ in (13), $\mathcal{S}_0 \equiv 0$ and $\tau_{\text{eff}} = \frac{1}{2}$. The integrand in (24) tends to $4(2\tau_{\text{eff}} - 1)/y$ as $\omega \to 0$. For a second-order family stable on quadratics, the quadratic-root hypothesis of (b) holds (proof of Corollary 5.9); hence a step $y > 2$ requires $\mathcal{S}_0 < 0$ somewhere.

Corollary 5.9 (The price of a faithful lead)**.** *Consider a second-order family with slow data (θ, y), stable on quadratics for small ε, whose limit method has $\mathcal{S}_0 \ge 0$ on $(0, \pi)$.*

(i) *$y \le 2$, hence $\limsup \sqrt{\kappa}(1 - \rho_{\mathcal{F}}) \le 1$ and $\limsup \sqrt{\kappa}(1 - \rho_q) \le \sqrt{2}$. The quadratic constant is at most $1/\sqrt{2}$ times Polyak's optimal constant 2.*

(ii) *If $\liminf \sqrt{\kappa}(1 - \rho_{\mathcal{F}}) \ge 1$, then $(\theta, y) = (3, 2)$, and the limit method has the transfer function $-(z-1)^2/(3z-1)$ of the limit of TMM; in particular $\tau_{\text{eff}} = \frac{1}{2}$ and $\mathcal{S}_0 \equiv 0$.*

(iii) *Under the hypotheses of Theorem 5.5(a), $y < 2$, and the constant is $c_{\mathcal{F}}(\theta, y) \le \sqrt{y/2} < 1$.*

For $\mathcal{M}$ the condition $\mathcal{S}_0 \ge 0$ is $\tau \ge \frac{1}{2}$, $y(1+2\tau) \le 4$ (Proposition 6.1(b)), and (i), (ii) reduce to the bounds behind Corollary E.11(ii) and (v). Under the nonnegative-symbol hypothesis, optimality fixes the limiting transfer function for every fixed finite memory. Extending the nonnegative-symbol condition from low to all frequencies is an open question (Section 9).

6. Two-Step Rates: The Wall, Boundary and Tunings

For $\mathcal{M}$, $\tau_{\text{eff}} = \tau$, and the lag-one symbol is cubic in $\xi = 1 - \cos\omega$. This gives explicit rate formulas, including the layer near $\tau = 1/2$. Section 7 treats the joint limit $(y, \tau) = (2, \frac{1}{2})$, including all tunings with constant 1 and the arithmetic gaps for suboptimal constants. Throughout this section, $L = 1$, $\mu = \varepsilon^2$, and $\Delta_c = c^2 - \theta c + y$.

6.1. The lag-one polynomial and the wall

Proposition 6.1 (Lag-one polynomial)**.**

(a) *For a tuning of $\mathcal{M}$ with $0 < y_\mu < y$ and $\omega \in [0, \pi]$,*

$$\mathrm{Re}[(1 - e^{i\omega})\chi_\mu(e^{i\omega})\overline{\chi_L(e^{i\omega})}] = \xi\, L_\tau(\xi), \qquad L_\tau(\xi) = \mathsf{A}_0 + \mathsf{A}_1\xi + \mathsf{A}_2\xi^2.$$

The coefficients are the polynomials in (β, τ, y_μ, y) given in (51); in particular $\mathsf{A}_0 = 2(1-\beta)(y - y_\mu) + y_\mu y$, and at $\beta = 1$, $y_\mu = 0$, $L_\tau(\xi) = 2y(2\tau - 1)\xi + 4(1 - \tau y)\xi^2$. For stable tunings $L_\tau(0) > 0$ and $L_\tau(2) = \chi_\mu(-1)\chi_L(-1) > 0$, and for $\omega \in (0, \pi]$, $\Theta(\omega) > \omega/2$ if and only if $L_\tau(\xi) > 0$, and $\Theta(\omega) = \omega/2$ if and only if $L_\tau(\xi) = 0$. If $L_\tau > 0$ on $(0, 2]$, then $\rho_{\mathcal{F}} < 1$.

(b) *For a family (3) of $\mathcal{M}$, $\tau_{\mathrm{eff}} = \tau$, and the lag-one symbol of the limit method is $\mathcal{S}_0 = \frac{\xi^2}{y^2}(2y(2\tau - 1) + 4(1 - \tau y)\xi)$. It is positive on $(0, \pi]$ if and only if $\tau \geq \frac{1}{2}$ and $y(1 + 2\tau) < 4$, and nonnegative if and only if $\tau \geq \frac{1}{2}$ and $y(1 + 2\tau) \leq 4$; if it is positive, the roots of $u_0 - 2v_0$ lie in the open unit disk.*

At $t = 0$, positivity of this quadratic on $[0, 2]$ gives the lag-one criterion; a negative value at $\xi = 1 - \cos(2\pi/K)$ gives phase locking with $j = 1$ (Proposition 4.10). In the notation of Section 5.4, the symbol in (b) has $\mathfrak{p} = 2$ for $\tau > \frac{1}{2}$, while at $\tau = \frac{1}{2}$ and $y < 2$, $\mathcal{S}_0 = \frac{2(2-y)}{y^2}\xi^3$ has $\mathfrak{p} = 3$.

Theorem 6.2 (The wall $\tau = \frac{1}{2}$)**.** *Let stable tunings $(\beta_\kappa, \tau_\kappa, h_\kappa)$ of $\mathcal{M}$, $\tau_\kappa \geq 0$, be accelerated on quadratics, with $\limsup_{\kappa\to\infty} \tau_\kappa < \frac{1}{2}$. Then for all large κ there is an integer $K \geq 6$ with $\Theta(2\pi/K) < \pi/K$; the method has trajectories from rest that approach K-periodic orbits in $\mathbb{R}^3$, and $\rho^0_{\mathcal{F}} > 1$. Consequently every family of stable tunings that is accelerated on $\mathcal{F}_{\mu,L}$ has $\liminf_{\kappa\to\infty} \tau_\kappa \geq \frac{1}{2}$.*

At $\tau = 0$, every heavy-ball tuning that accelerates on quadratics has periodic orbits on $\mathcal{F}_{\mu,L}$ for all large κ, strengthening the non-acceleration result of [17]. TMM approaches the wall from below: $\tau_\kappa < \frac{1}{2}$ and $\tau_\kappa \to \frac{1}{2}$ (Corollary 7.1). The leading low-frequency coefficient of the limiting lag-one symbol changes sign at $\tau = \frac{1}{2}$ (Proposition 6.1(b)), as predicted by the half-step rule; the resulting resonances occur at intermediate frequencies.

For fixed $\tau < \frac{1}{2}$, the resonant band extends from frequencies of order $\kappa^{-1/4}$ to a fixed frequency; the number and the maximal size of its integer periods are of order $\kappa^{1/4}$. Proposition E.5 gives the precise band and leading constant.

6.2. The flow constant and the stability boundary

The spiral lower witnesses in Theorem 2.6 are sampled from the flow. In the slow variables $\omega = \varepsilon\gamma$, $t = -c\varepsilon$, $q = n\varepsilon$, the lag constraints have the flow kernels with uniformly convergent coefficients on bounded frequency sets. Their common strict margin survives sampling. For every feasible slow spectrum, the value constant is determined up to $O(\varepsilon^3)$ (Lemmas D.3 to D.5).

Proposition 6.3 (Single-slow-frequency obstruction)**.** *Consider a family (3) with $\theta/3 < c_Q(\theta, y)$, and let $c_\sharp$ be as in Proposition 3.3 with $m = y$. For every $\gamma_{\max} > 0$ there is $\varepsilon_1 > 0$ such that for $\varepsilon \in (0, \varepsilon_1)$ and $\theta/3 < c \leq c_\sharp$, no pair $(\delta_{\varepsilon\gamma}, C)$ with $0 \leq \gamma \leq \gamma_{\max}$ is feasible at $t = -c\varepsilon$.*

The frequency range in this proposition is $\omega = O(\varepsilon)$.

The stability boundary. Several natural tunings, among them the tuning of Nesterov's method that is optimal on quadratics (Corollary 6.6(ii)), sit on the stability boundary $y(1 + 2\tau) = 4$ in the limit. For such families measure the stability margin by

$$\varsigma_\varepsilon = \frac{4 - y_\varepsilon(1 + 2\tau_\varepsilon)}{\varepsilon}, \qquad \text{so that} \qquad \chi_L(-1) = 2(1 + \beta) - y_\varepsilon(1 + 2\tau_\varepsilon) = (\varsigma_\varepsilon - 2\theta_\varepsilon)\,\varepsilon \tag{25}$$

exactly. Thus the tunings are stable for small ε if $\varsigma_\varepsilon \to \varsigma \in (2\theta, \infty]$, and unstable if $\varsigma < 2\theta$. For $\tau \geq \frac{1}{2}$ and $\varsigma \in (2\theta, \infty]$, χ_L has a real root close to -1: a sign-alternating mode at curvature L. Its coefficient

c_π complements the slow quadratic coefficient $c_Q(\theta, y)$ from (4):

$$c_\pi = \frac{(\varsigma - 2\theta)(1+2\tau)}{4\tau} \in (0, \infty], \tag{26}$$

and we put $c_\pi = \infty$ when $y(1+2\tau) < 4$.

Proposition 6.4 (On the stability boundary)**.** *Consider a family* (3) *of stable tunings of* $\mathcal{M}$ *with* $\tau > \frac{1}{2}$, $y(1+2\tau) = 4$ *and* $\varsigma_\varepsilon \to \varsigma \in (2\theta, \infty]$. *Then for every* $c \in (0, \min(c_{\mathcal{F}}(\theta, y), c_\pi))$ *there is* $\varepsilon_0 > 0$ *such that* $\rho_{\mathcal{F}} < 1 - c\varepsilon$ *for* $\varepsilon \in (0, \varepsilon_0)$, *certified by the lag-one criterion, and*

$$\lim_{\kappa\to\infty} \sqrt{\kappa}\,(1 - \rho_{\mathcal{F}}) = \lim_{\kappa\to\infty} \sqrt{\kappa}\,(1 - \rho^0_{\mathcal{F}}) = \min(\tfrac{\theta}{3},\ c_Q(\theta, y),\ c_\pi).$$

The term c_π *is realized from rest by the quadratic* $\frac{L}{2}\|x\|^2$ *on* $\mathbb{R}$.

The alternating mode affects the leading constant when the stability margin is of order $\kappa^{-1/2}$. For $\tau > \frac{1}{2}$, at $\varsigma = 2\theta$, stable tunings converge by Theorem 8.1 with constant 0, since $\rho_q \geq 1 - o(\varepsilon)$ (Lemma E.8(b)). At $(\tau, y) = (\frac{1}{2}, 2)$, the limiting lag-one symbol vanishes identically. For $\tau = \frac{1}{2}$, $y < 2$, its nonzero sixth-order term determines the wall layer.

6.3. The boundary layer of the wall

Theorems 2.6 and 6.2 leave open the families with $\tau_\varepsilon \to \frac{1}{2}$. The transition occurs at $\frac{1}{2} - \tau_\varepsilon \asymp \kappa^{-1/4}$. For a family (3) with $\tau = \frac{1}{2}$, $y < 2$, put

$$A_\varepsilon = (\tfrac{1}{2} - \tau_\varepsilon)\varepsilon^{-1/2}, \qquad \text{i.e.} \quad \tau_\varepsilon = \tfrac{1}{2} - A_\varepsilon \kappa^{-1/4},$$

and, with $A_+ = \max(A, 0)$,

$$A_* = \sqrt{\frac{\theta(2-y)}{y}}, \qquad \Lambda(A) = \theta - \frac{yA_+^2}{2-y}, \qquad c_{\text{layer}}(A) = \min\Big(c_Q(\theta, y), \frac{\Lambda(A)}{3}\Big). \tag{27}$$

Theorem 6.5 (The boundary layer)**.** *Consider a family* (3) *of stable tunings of* $\mathcal{M}$ *with* $\tau = \frac{1}{2}$ *and* $y < 2$.

(a) (Upper bound.) *If* $A_\varepsilon \to A \in [-\infty, A_*)$, *then for every* $c < c_{\text{layer}}(A)$ *there is* $\varepsilon_0 > 0$ *such that* $\rho_{\mathcal{F}} < 1 - c\varepsilon$ *for* $\varepsilon \in (0, \varepsilon_0)$, *certified by the lag-one criterion.*

(b) (Lower bound.) *If* $A_\varepsilon \to A \in (0, \infty)$, *then for every* $c > \Lambda(A)_+/3$ *and all small* ε *there are* $d \in \mathbb{N}$, $f \in \mathcal{F}_{\varepsilon^2,1}(\mathbb{R}^d)$ *and* $x_0 \neq x^\star$ *with, from rest,* $\|x_N - x^\star\| \geq c_0(1 - c\varepsilon)^N \|x_0 - x^\star\|$ *for all* $N \geq 0$, *where* $c_0 = c_0(\varepsilon) > 0$ *is fixed across all horizons* N.

(c) (Divergence.) *If* $\liminf A_\varepsilon > A_*$, *then for all small* ε *there is an integer* $K \geq 3$ *with* $\Theta(2\pi/K) < \pi/K$: *the method has trajectories from rest that approach* K*-periodic orbits in* $\mathbb{R}^3$, *and* $\rho^0_{\mathcal{F}} > 1$. *If* $A_\varepsilon \to A < \infty$, *these* K *are of order* $\varepsilon^{-1/4}$.

Consequently, if $A_\varepsilon \to A \in [-\infty, A_*)$, *then*

$$\lim_{\kappa\to\infty} \sqrt{\kappa}\,(1 - \rho_{\mathcal{F}}) = \lim_{\kappa\to\infty} \sqrt{\kappa}\,(1 - \rho^0_{\mathcal{F}}) = c_{\text{layer}}(A),$$

and if $A_\varepsilon \to A \in [A_*, \infty]$ *the family is not accelerated on* $\mathcal{F}_{\mu,L}$. *A family with* $y < 2$ *and* $\tau_\varepsilon \to \frac{1}{2}$ *is accelerated on* $\mathcal{F}_{\mu,L}$ *if and only if* $\limsup A_\varepsilon < A_*$.

The wall layer has width $A_*\kappa^{-1/4}$. Its constant equals the flow constant $c_{\mathcal{F}}(\theta, y)$ whenever $\Lambda(A) \geq 3c_{\mathcal{F}}(\theta, y)$, in particular for $A \leq 0$, and then decreases to 0 at $A = A_*$ (Figure 4, right). The lag-one certificate is exact throughout the accelerated layer. The mechanism is the lag-one polynomial at the scale $\xi \asymp \kappa^{-1/4}$, i.e. at frequencies $\omega \asymp \kappa^{-1/8}$: for the lag-one symbol $H_r = \mathrm{Re}[(1-z)\chi_\mu(z)\overline{\chi_L(z)}]$, $z = re^{i\omega}$ (Proposition 4.9 and Lemma E.1), with $\xi = \varepsilon^{1/2}\varphi$ and $r = 1 - c\varepsilon$,

$$\frac{H_r}{\xi} = \varepsilon\big[2y(\theta - 3c) - 4Ay\varphi + 2(2-y)\varphi^2\big] + o(\varepsilon).$$

Regime	Parameter scale	Relevant frequency	Rotation or cycle scale
Sampled flow	fixed slow data	$\omega \asymp \kappa^{-1/2}$	rotation time $\kappa^{1/2}$
Below the wall	fixed $\tau < 1/2$	lower edge $\kappa^{-1/4}$	maximal $K \asymp \kappa^{1/4}$
Wall layer, $y < 2$	$1/2 - \tau_\varepsilon \asymp \kappa^{-1/4}$	$\omega \asymp \kappa^{-1/8}$	$K \asymp \kappa^{1/8}$
TMM corner	$1/2 - \tau_\varepsilon,\ 2 - y_\varepsilon = O(\kappa^{-1/2})$	fixed $2\pi/K$ also matter	integer K leaves a gap

Table 1: Scales of the witnesses. In the first row, rotation time is the angular period of a decaying spiral component.

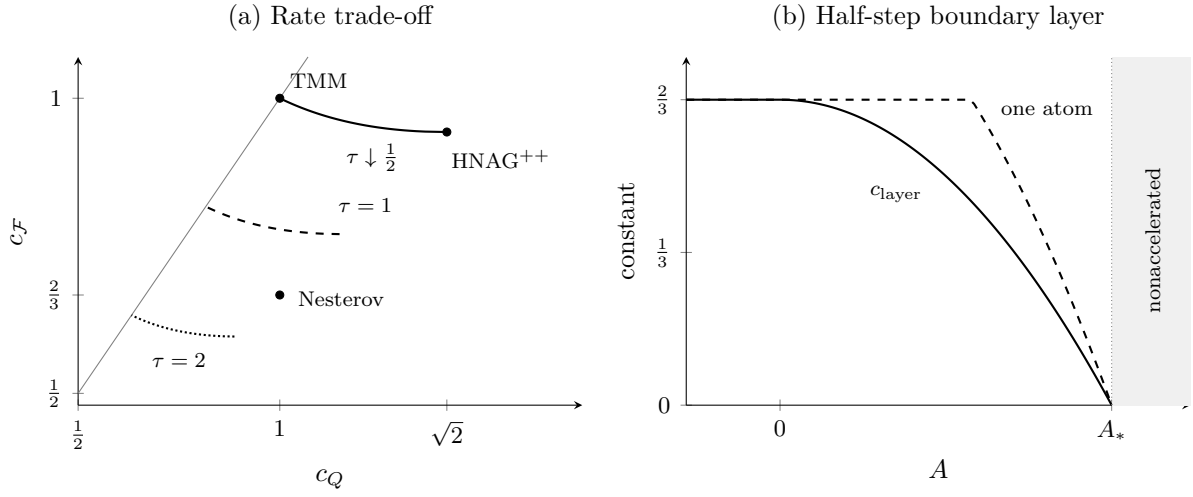


Figure 4: Left: the Pareto arcs $\mathcal{C}_\tau$ of Corollary E.11(iv), with the diagonal $c_{\mathcal{F}} = c_Q$, TMM, standard Nesterov and HNAG^{++} [7] (parameters in Corollary 6.6(v)). Right: for $\theta = 2$, $y = 1$, the exact wall-layer constant $c_{\text{layer}}(A) = (2 - A_+^2)/3$ on $A < A_* = \sqrt{2}$ (solid), compared with the one-atom bound $\min(\frac{2}{3}, \Lambda(A)_+/2)$ (dashed). The shaded region is nonaccelerated (Theorem 6.5).

Minimizing the bracket over $\varphi \geq 0$ gives $3c = \Lambda(A)$: the minimum occurs at a double root for $A > 0$, and at $\varphi = 0$ for $A \leq 0$. At $c = 0$, this gives $A = A_*$. The stabilizing term corresponds to $s_6\omega^6$ in the limiting symbol, with $s_6 = (2 - y)/(4y^2)$ (Proposition 6.1(b)). The resonant periods $\kappa^{1/8}$ interpolate between the periods up to order $\kappa^{1/4}$ at fixed $\tau < \frac{1}{2}$ (Proposition E.5) and the absence of resonances beyond the wall.

The witnesses in (b) spread frequencies over a smooth band around $2\pi/K$, $K \asymp \kappa^{1/8}$, desynchronizing the returns of the planar spirals (Lemma C.17). At these frequencies the gradient has a half-step lead up to relative error $O(\kappa^{-1/4})$. The spectra asymptotically saturate the lower lag-one and minimizer constraints, with C approaching its minimizer bound, and retain strict slack at the other lags. This matches the lag-one rate. Each witness is finite-dimensional; the number of spirals may grow with κ. The one-atom spectra of Lemma E.2, with $C = 0$, give the bound $\min(c_{\mathcal{F}}, \Lambda(A)_+/2)$ (Remark E.12).

The scales behind the different witnesses are summarized in Table 1. For sufficiently smooth families, Theorem 5.5 recovers Theorem 2.6(c) and the layer case $A = 0$; the direct two-step analysis also covers nonsmooth parameter families and trajectories from rest.

6.4. Known tunings and ceilings

We apply the rate formulas to standard tunings and optimize at fixed extrapolation.

Corollary 6.6 (Known tunings)**.**

(i) (Nesterov's method.) *The tuning* $h = 1/L$, $\tau = \beta = \frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}$ *has* $(\theta_\varepsilon, y_\varepsilon, \tau_\varepsilon) \to (2, 1, 1)$, $c_Q = 1$ *and*

$$\rho_{\mathcal{F}} = 1 - \tfrac{2}{3}\kappa^{-1/2} + o(\kappa^{-1/2}),$$

and the same for $\rho^0_{\mathcal{F}}$; for squared distances the factor is $1 - \frac{4}{3}\kappa^{-1/2} + o(\kappa^{-1/2})$.

(ii) (Nesterov's method, quadratic-optimal tuning.) *The tuning $h = 4/(3L + \mu)$, $\tau = \beta = (\sqrt{3\kappa+1} - 2)/(\sqrt{3\kappa+1} + 2)$, which is optimal for Nesterov's method on quadratics with rate $1 - 2/\sqrt{3\kappa+1}$ [23, Proposition 1], has $(\theta_\varepsilon, y_\varepsilon, \tau_\varepsilon) \to (\frac{4}{\sqrt{3}}, \frac{4}{3}, 1)$, on the stability boundary, with margin $\varsigma = \frac{32}{3\sqrt{3}}$ and $c_\pi = c_Q = \frac{2}{\sqrt{3}}$. On $\mathcal{F}_{\mu,L}$,*

$$\rho_{\mathcal{F}} = 1 - \frac{4}{3\sqrt{3}}\,\kappa^{-1/2} + o(\kappa^{-1/2}), \qquad \frac{4}{3\sqrt{3}} \approx 0.770,$$

and the same for $\rho^0_{\mathcal{F}}$.

(iii) (Quasi-hyperbolic momentum.) *For the method of [26] in the form of Remark 4.4, with step $\alpha > 0$, discount $\beta \in [0, 1)$, weight $\nu_{\mathrm{qh}} \in [0, 1]$ and τ_{qh} as there, rates refer to its iterates from the usual initialization $d_{-1} = 0$.*

(a) *If a family of tunings accelerates on quadratics, then $1 - \nu_{\mathrm{qh}} = O(\kappa^{-1/2})$; in particular a fixed $\nu_{\mathrm{qh}} < 1$ with $\beta \to 1$ does not accelerate. If it accelerates on $\mathcal{F}_{\mu,L}$, then moreover $\liminf \tau_{\mathrm{qh}} \geq \frac{1}{2}$.*

(b) *If $1 - \beta = \theta\kappa^{-1/2}$, $\alpha L(1 - \beta) \to y$ and $\tau_{\mathrm{qh}} \to \tau > \frac{1}{2}$ with $y(1 + 2\tau) < 4$, the constant is $\min(\theta/3, c_Q(\theta, y))$.*

(iv) (Over-relaxed heavy ball.) *The method $x_{k+1} = x_k + \beta(x_k - x_{k-1}) - \gamma(2\nabla f(x_k) - \nabla f(x_{k-1}))$ with $\gamma = 1/(\sqrt{L} + \sqrt{\mu})^2$ and $\beta = L\gamma$ has $\rho_{\mathcal{F}} = 1 - \frac{2}{3}\kappa^{-1/2} + o(\kappa^{-1/2})$.*

(v) (HNAG^{++}.) *Put $\eta = \sqrt{2/\kappa}$. The gradient-correction form of HNAG^{++} [7] has*

$$\beta = \frac{1}{(1+\eta)^2}, \qquad hL = \frac{2+\eta}{(1+\eta)^2}, \qquad \tau = \frac{1}{2+\eta}.$$

For arbitrary initial states, its uniform distance factor satisfies $\rho_{\mathcal{F}} = 1 - \frac{2\sqrt{2}}{3}\kappa^{-1/2} + o(\kappa^{-1/2})$, and its quadratic factor satisfies $\rho_q = 1 - \sqrt{2}\kappa^{-1/2} + o(\kappa^{-1/2})$.

Part (i) gives $c_{\mathcal{F}}(2, 1) = 2/3$, recovering Pokutta and Stellato [33] and Gannot [13]. Both Nesterov tunings are critically damped in the slow flow; their quadratic constant is $3/2$ times the full-class constant. In (ii), the quadratic endpoint modes balance at $c_Q = c_\pi = 2/\sqrt{3}$, and the spiral term determines the slower full-class rate. Squared-distance constants are twice the distance constants.

For the default quasi-hyperbolic weight $\nu_{\mathrm{qh}} = 0.7$ [26], Corollary 8.2 gives the finite-condition-number threshold. The gradient correction in (iv) is exactly $\tau = 1$ under the same translation, so the over-relaxed method of Wei and Chen [51] shares the slow data and the constant of standard Nesterov. For HNAG^{++}, the squared-distance coefficient is therefore $4\sqrt{2}/3$, improving the global coefficient $\sqrt{2}$ in [7, Theorem 5.1]. Their faster coefficient $2\sqrt{2}$ for the auxiliary sequence y_k is an individual asymptotic estimate under local asymptotic symmetry: the two first-order Taylor remainders of f between x and $x^\star$ have ratio tending to 1 as $x \to x^\star$ [7, Theorem 5.2]; Remark 2.4 describes the uniform rate convention. Optimizing the damping for the full class gives the following tuning.

Corollary 6.7 (Asymptotically optimal Nesterov tuning)**.** *For $\kappa \geq 6$, the tuning*

$$h = \frac{4}{3L}, \qquad \tau = \beta = 1 - \sqrt{\frac{6}{\kappa}}$$

is stable, converges linearly on $\mathcal{F}_{\mu,L}$, and satisfies

$$\rho_{\mathcal{F}} = 1 - \sqrt{\frac{2}{3}}\,\kappa^{-1/2} + o(\kappa^{-1/2}), \qquad \rho^0_{\mathcal{F}} = 1 - \sqrt{\frac{2}{3}}\,\kappa^{-1/2} + o(\kappa^{-1/2}).$$

The constant $\sqrt{2/3}$ is maximal among families with $\tau = \beta$ on $\mathcal{F}_{\mu,L}$.

Nesterov tuning	θ	y	Full-class constant c
Standard	2	1	2/3
Quadratic-optimal	$4/\sqrt{3}$	4/3	$4/(3\sqrt{3})$
Class-optimal asymptotically	$\sqrt{6}$	4/3	$\sqrt{2/3}$

Table 2: Asymptotic distance constants within $\tau = \beta$, so $\tau \to 1$ (Corollaries 6.6 and 6.7). Optimizing on quadratics and on the full function class gives different damping parameters.

Table 2 compares the three asymptotic Nesterov tunings; Figure 7 shows their geometry.

The target-rate inequalities in Section 2.2 give the fixed-extrapolation ceiling. Thus the supremum at fixed $\tau > 1/2$ is $\sqrt{2/(1+2\tau)}$, reached on the stability boundary when the sign-alternating mode is no slower. Every family accelerated on $\mathcal{F}_{\mu,L}$ loses at least a factor $\sqrt{2}$ in its quadratic constant relative to Polyak's optimum. Corollary E.11 gives the Pareto arcs and the necessary limit $(\theta, y, \tau) = (3, 2, 1/2)$ for constant 1.

At TMM, $y_\varepsilon \to 2$, so the stabilizing wall-layer coefficient vanishes. The joint limit $A_\varepsilon \to 0$, $y_\varepsilon \to 2$ motivates the corner scale below.

7. The Corner of the Triple Momentum Method

The optimal constant 1 forces $(\theta_\varepsilon, y_\varepsilon, \tau_\varepsilon) \to (3, 2, \frac{1}{2})$ (Corollary E.11(v)). We classify all approaches that attain it, first in the natural corner scale and then at arbitrary rates of approach. The lag-one certificate tests all frequencies; the lower-bound construction spreads spectra around $2\pi/K$ for integers $K \geq 3$. For some suboptimal tunings, the resulting bounds leave an *arithmetic gap*.

We begin with TMM's exact finite-κ certificate.

7.1. An exact certificate at TMM

For TMM, let $\varrho = 1 - \kappa^{-1/2}$. Its endpoint polynomials factor as

$$\chi_\mu(z) = (z - \varrho)(z - \varrho^2), \qquad \chi_L(z) = z(z + \varrho).$$

The lag-one symbol vanishes on $|z| = \varrho$ and is strictly positive on every circle $\varrho < r < 1$. Letting $r \downarrow \varrho$ gives the exact rate. Theorem F.1 gives the full factorization and the exact rate for the robust momentum family of [8], including TMM; the rate itself is already known from [8, 48].

For robust momentum with design rate $\varrho = 1 - c\kappa^{-1/2}$, $0 < c < 1$, the limiting data are $(\theta, y, \tau) = (3c, 2c^2, 1/(2c^2))$. They lie on the balance curve $\theta = 3\sqrt{y/2}$, where the quadratic and spiral bounds agree.

Corollary 7.1 (TMM)**.** *TMM [48] is the tuning* (52) *at* $\varrho = 1 - \kappa^{-1/2}$*. It has* $(\theta_\varepsilon, y_\varepsilon, \tau_\varepsilon) \to (3, 2, \frac{1}{2})$*, with* $2 - hL = \kappa^{-1/2}$ *and* $\frac{1}{2} - \tau = \frac{5}{4}\kappa^{-1/2}(1 + O(\kappa^{-1/2}))$*, so that* $\tau < \frac{1}{2}$ *for every* κ*, and* $\rho_{\mathcal{F}} = \rho^0_{\mathcal{F}} = \rho_q = 1 - \kappa^{-1/2}$ *exactly (the upper bound is [48, Theorem 1]). By [10] this rate, and the constant* $1 = c_{\mathcal{F}}(3, 2) = c_Q(3, 2)$*, are optimal on* $\mathcal{F}_{\mu,L}$ *among deterministic first-order methods with value and gradient access. The HNAG*$^+$ *tuning of [7, Theorem 4.1] has the same gradient-correction parameters.*

7.2. The corner

By Corollary 7.1, TMM approaches the corner along $2 - y_\varepsilon = \varepsilon$ and $\frac{1}{2} - \tau_\varepsilon = \frac{5}{4}\varepsilon(1 + o(1))$, at the same scale $\varepsilon = \kappa^{-1/2}$ in both coordinates. The natural scaling of the corner is

$$\beta = 1 - \theta_\varepsilon \varepsilon, \qquad hL = y_\varepsilon = 2 - b_\varepsilon \varepsilon, \qquad \tau_\varepsilon = \tfrac{1}{2} - a_\varepsilon \varepsilon, \qquad (\theta_\varepsilon, a_\varepsilon, b_\varepsilon) \to (\theta, a, b), \tag{28}$$

with $\theta > 0$, $a, b \in \mathbb{R}$. Expanding $y_\varepsilon(1 + 2\tau_\varepsilon) < 2(1 + \beta)$ shows that these tunings are stable for small ε if $2a + b > \theta$ and unstable if $2a + b < \theta$; we assume $2a + b > \theta$. Put

$$\begin{aligned} &\mathsf{N}(\xi) = 4\theta + 2(\theta - 4a)\,\xi + 2(4a + b - 2\theta)\,\xi^2, \qquad \mathsf{q}(\xi) = \frac{\mathsf{N}(\xi)}{4(3-\xi)}, \qquad c_\pi = 2(2a + b - \theta), \\ &c_{\mathrm{lag}} = \min_{0 \le \xi \le 2} \mathsf{q}(\xi), \qquad c_{\mathrm{disc}} = \min\{\mathsf{q}(\xi) :\ \xi \in \{0, 2\} \cup \{\xi_K : K \ge 3\}\}, \qquad \xi_K = 1 - \cos\tfrac{2\pi}{K}. \end{aligned} \tag{29}$$

Here c_π is (26) at $\tau = \frac{1}{2}$: the margin (25) of (28) tends to $\varsigma = 4a + 2b$. We call q the *lag-one ratio*. Its values at the ends of $[0, 2]$ are $\mathsf{q}(0) = \theta/3$ and $\mathsf{q}(2) = c_\pi$, so $c_{\mathrm{lag}} \le c_{\mathrm{disc}} \le \min(\theta/3, c_\pi)$.

Theorem 7.2 (The corner)**.** *Consider a family* (28) *with* $2a + b > \theta$.

(a) (Upper bound.) *For every* c *with* $0 < c < \min(c_Q(\theta, 2), c_{\mathrm{lag}})$ *there is* $\varepsilon_0 > 0$ *such that* $\rho_{\mathcal{F}} < 1 - c\varepsilon$ *for* $\varepsilon \in (0, \varepsilon_0)$*, certified by the lag-one criterion.*

(b) (Lower bound.) *For every* $c > \max\bigl(0, \min(c_Q(\theta, 2), c_{\mathrm{disc}})\bigr)$ *and all small* ε*,* $\rho^0_{\mathcal{F}} > 1 - c\varepsilon$*.*

(c) *Consequently, if* $c_{\mathrm{lag}} > 0$*,*

$$\min\bigl(c_Q(\theta, 2), c_{\mathrm{lag}}\bigr) \le \liminf_{\kappa \to \infty} \sqrt{\kappa}\,(1 - \rho_{\mathcal{F}}) \le \limsup_{\kappa \to \infty} \sqrt{\kappa}\,(1 - \rho^0_{\mathcal{F}}) \le \min\bigl(c_Q(\theta, 2), c_{\mathrm{disc}}\bigr),$$

and the constant equals $\min(c_Q(\theta, 2), c_{\mathrm{lag}})$ *whenever* q *attains its minimum over* $[0, 2]$ *at* 0*, at* 2 *or at a point* ξ_K*, or* $c_Q(\theta, 2) \le c_{\mathrm{lag}}$*. In particular, with* $c_\angle = \min(\theta/3, c_Q(\theta, 2), c_\pi)$*, the constant equals* $c_\angle$ *whenever* $\mathsf{N}(\xi) \ge 4c_\angle(3 - \xi)$ *on* $[0, 2]$*, for instance if* $4a + b \le 2\theta$*.*

(d) (Optimal tunings.) *The constant equals* 1 *if and only if* $\theta = 3$*,* $a \le \frac{5}{4}$ *and* $2a + b \ge \frac{7}{2}$*.*

The lower bounds in (b) have three sources, all realized from rest: flow or quadratic witnesses for $\min(\theta/3, c_Q(\theta, 2))$ (Theorem 2.6(a)); the sign-alternating mode of $\frac{L}{2}\|x\|^2$ for c_π; and spectra spread around $2\pi/K$ for $\mathsf{q}(\xi_K)$ (Lemma C.17).

Remark 7.3 (Explicit regions). Let $\theta \le 3$, so that $\min(\theta/3, c_Q(\theta, 2)) = \theta/3$. By Theorem 7.2(c), the constant equals $\theta/3$ if $a \le \frac{5\theta}{12}$ and $2a + b \ge \frac{7\theta}{6}$, and it equals $c_\pi = 2(2a + b - \theta) < \theta/3$ if $2a + b < \frac{7\theta}{6}$ and $5a + 2b \le \frac{11\theta}{4}$ (Appendix F).

TMM is the vertex $(\theta, a, b) = (3, \frac{5}{4}, 1)$ of the optimal wedge $\{a \le \frac{5}{4},\ 2a + b \ge \frac{7}{2}\}$ in (d) (Figure 5). It has the largest admissible deficit a and, at that deficit, the largest leading step $y_\varepsilon = 2 - b\varepsilon + o(\varepsilon)$. The flow and both quadratic endpoint constants coincide: $\theta/3 = c_Q(3, 2) = c_\pi = 1$. Also,

$$\mathsf{N}(\xi) = 12 - 4\xi = 4(3 - \xi),$$

so $\mathsf{q} \equiv 1$: to first order in ε the lag-one certificate at rate $1 - \varepsilon$ is tight at every frequency simultaneously. For $\theta = 3$,

$$\mathsf{N}(\xi) - 4(3 - \xi) = 2\xi\bigl[(5 - 4a) + (4a + b - 6)\xi\bigr],$$

an affine bracket whose values at $\xi = 0$ and $\xi = 2$ are $5 - 4a$ and $2(2a + b) - 7$: inside the wedge $\mathsf{q} \ge 1$ on $[0, 2]$, and beyond its edge $a = \frac{5}{4}$ the ratio dips below 1 at low frequencies, where the points ξ_K accumulate. The limit method of TMM, $(y, \tau) = (2, \frac{1}{2})$, leads the iterate by half a step for $0 < \omega < \pi$ (Remark 5.7). Theorem 7.2 recovers TMM's constant 1 [48], and Theorem F.1 gives its rate $1 - \kappa^{-1/2}$ at every κ, where the design-circle symbol vanishes identically. The optimal set also contains families with $\tau_\varepsilon \ge \frac{1}{2}$: these include $a < 0$, $b \ge \frac{7}{2} - 2a$, and exact-wall families with $a = 0$, $b \ge \frac{7}{2}$. Remark 7.3 gives the corresponding wedge and alternating-mode region for $\theta \le 3$.

The spread spectra for $\mathsf{q}(\xi_K)$ use the boundary-layer construction (Section 6.3); the limit method has a half-step lead on $(0, \pi)$. A gap between the bounds in (c) requires a minimum of q strictly between consecutive $\xi_{K+1} < \xi_K$, below the quadratic contribution. This occurs for some tunings with $\theta = 3$ and $a > \frac{5}{4}$, where (b) bounds every limiting leading coefficient strictly below 1. For instance, for $\theta = 3$ and $(a, b) = (2, 1)$, $\mathsf{q}(\xi) = (6\xi^2 - 10\xi + 12)/(4(3 - \xi))$ is minimal at $\xi = 3 - \sqrt{6}$, strictly between $\xi_6 = \frac{1}{2}$ and ξ_5, and every limiting leading coefficient lies in $[3\sqrt{6} - \frac{13}{2}, \frac{17}{20}] \approx [0.8485, 0.85]$ (Figure 5, right; see also Section 9).

Corollary 7.4 (The arc at $\tau = \frac{1}{2}$)**.**

(i) *The tunings* $\tau = \frac{1}{2}$, $\beta = 1 - 2\sqrt{2}\,\kappa^{-1/2}$, $hL = 2 - 4\kappa^{-1/2}$ *have constant* $\frac{2\sqrt{2}}{3}$ *on* $\mathcal{F}_{\mu,L}$ *and* $\sqrt{2}$ *on quadratics; so the bound* $\sqrt{2}$ *of Corollary E.11(ii) is attained.*

(ii) *Every point of the arc* $\mathcal{C}_{1/2}$ *of Corollary E.11(iv), from* $(\sqrt{2}, \frac{2\sqrt{2}}{3})$ *to TMM's point* $(1, 1)$*, is attained by a family* (28) *with* $a = 0$ *and* $b = 2\theta$*.*

7.3. Optimal tunings

Theorem 7.2(d) describes optimal families (28), with $\frac{1}{2} - \tau_\varepsilon$ and $2 - y_\varepsilon$ both $O(\kappa^{-1/2})$. By Corollary E.11(v), every family with constant 1 approaches the corner, possibly at other rates. For a family (3) with $(\theta_\varepsilon, y_\varepsilon, \tau_\varepsilon) \to (\theta, 2, \frac{1}{2})$ put

$$\alpha_\varepsilon = \frac{(\frac{1}{2} - \tau_\varepsilon)\, y_\varepsilon}{2\varepsilon}, \qquad \varsigma_\varepsilon = \frac{4 - y_\varepsilon(1 + 2\tau_\varepsilon)}{\varepsilon}, \qquad \mathsf{N}_\varepsilon(\xi) = 4\theta_\varepsilon + 2(\theta_\varepsilon - 4\alpha_\varepsilon)\xi + (\varsigma_\varepsilon + 4\alpha_\varepsilon - 4\theta_\varepsilon)\xi^2. \tag{30}$$

The tunings are stable if and only if $\varsigma_\varepsilon > 2\theta_\varepsilon$, and for (28), $\alpha_\varepsilon \to a$, $\varsigma_\varepsilon \to 4a + 2b$ and $\mathsf{N}_\varepsilon \to \mathsf{N}$. The criterion below allows unbounded α_ε and ς_ε.

Theorem 7.5 (Optimal tunings of $\mathcal{M}$)**.** *A family of stable tunings of* $\mathcal{M}$ *satisfies* $\lim_{\kappa\to\infty} \sqrt{\kappa}(1 - \rho_{\mathcal{F}}) = 1$ *if and only if, in the notation of* (3) *and* (30)*,*

$$(\theta_\varepsilon, y_\varepsilon, \tau_\varepsilon) \to (3, 2, \tfrac{1}{2}), \qquad \min_{0\le\xi\le 2} \frac{\mathsf{N}_\varepsilon(\xi)}{4(3 - \xi)} \to 1.$$

Under the first condition, put $d_\varepsilon = (4\alpha_\varepsilon - 5)_+$*. The second condition is equivalent to*

$$\liminf_{\varepsilon\to 0} \varsigma_\varepsilon \ge 7, \qquad \frac{d_\varepsilon^2}{\varsigma_\varepsilon + 5 + d_\varepsilon} \to 0.$$

The denominator is positive by stability. In this case also $\sqrt{\kappa}(1 - \rho_{\mathcal{F}}^0) \to 1$*.*

For the families (28), the criterion reduces to $\theta = 3$, $4a + 2b \ge 7$, $a \le \frac{5}{4}$, recovering Theorem 7.2(d). If $b_\varepsilon = (2 - y_\varepsilon)/\varepsilon \to \infty$ and α_ε stays bounded, then the exact identity $\varsigma_\varepsilon = 2b_\varepsilon + 4\alpha_\varepsilon$ gives $\varsigma_\varepsilon \to \infty$, and the ratio condition holds: any fixed multiple of $\kappa^{-1/2}$ for the deficit $\frac{1}{2} - \tau_\varepsilon$ is compatible with constant 1. The edge $a = \frac{5}{4}$ and TMM's vertex therefore belong to the scale $2 - y_\varepsilon = O(\kappa^{-1/2})$. The proof combines the expansion of the lag-one symbol near the corner, valid for arbitrary α_ε and ς_ε, with three lower-bound mechanisms: the sign-alternating mode when $\liminf \varsigma_\varepsilon < 7$, spread spectra (Lemma C.17) at frequencies where the lag-one ratio dips below 1, and, when this dip is deep, planar periodic orbits (Proposition 4.10).

8. Finite Condition Numbers

At fixed κ, the polynomial L_τ gives explicit convergence regions. Call a stable tuning *lag-one convergent* if $L_\tau > 0$ on $(0, 2]$. The periodic tongues of Proposition 4.10 and Remark C.16 lie outside this region. The key facts are that L_τ is affine in y_μ and equals $|\chi_L(e^{i\omega})|^2$ at $y_\mu = y$.

Theorem 8.1 (Tongue-free extrapolations)**.** *Let* $\beta \in [0, 1)$*,* $\tau \ge 0$*, and*

$$\tau_3(\beta) = \frac{\beta^2 + 3\beta - 1}{6}, \qquad \tau_*(\beta) = \frac{\beta^2 + 4\beta - 1}{8}.$$

(a) *If* $\tau \ge \tau_3(\beta)$*, every stable tuning* (β, τ, h)*, for every* $0 < \mu < L$*, converges:* $\rho_{\mathcal{F}} < 1$*. It is certified by the lag-two certificate with multiplier* $(1 - z) + \lambda(1 - \bar{z}^2)$*, with* $\lambda = 0$ *(the lag-one certificate) if* $\tau \ge \tau_*(\beta)$ *and* $\lambda = \frac{\beta - 2\tau}{1 + 2\beta - \beta^2 + 2\tau(1 + 3\beta)}$ *if* $\tau_3(\beta) \le \tau < \tau_*(\beta)$*.*

(b) *If* $\tau < \tau_3(\beta)$*, there are* $0 < \mu < L$ *and* $h > 0$ *such that* (β, τ, h) *is stable and lies in the* 3*-tongue,* $\Theta(2\pi/3) < \pi/3$*: the method has trajectories from rest approaching* 3*-periodic orbits, and* $\rho_{\mathcal{F}}^0 > 1$*.*

(c) *If* $\tau < \tau_*(\beta)$*, there are* $0 < \mu < L$ *and* $h > 0$ *such that* (β, τ, h) *is stable and not lag-one convergent.*

The 3-tongue sets the exact threshold for convergence of every stable tuning. All stable tunings admit lag-one certificates precisely when $\tau \ge \tau_*(\beta) = \tau_3(\beta) + (1 - \beta^2)/24$. Between the thresholds,

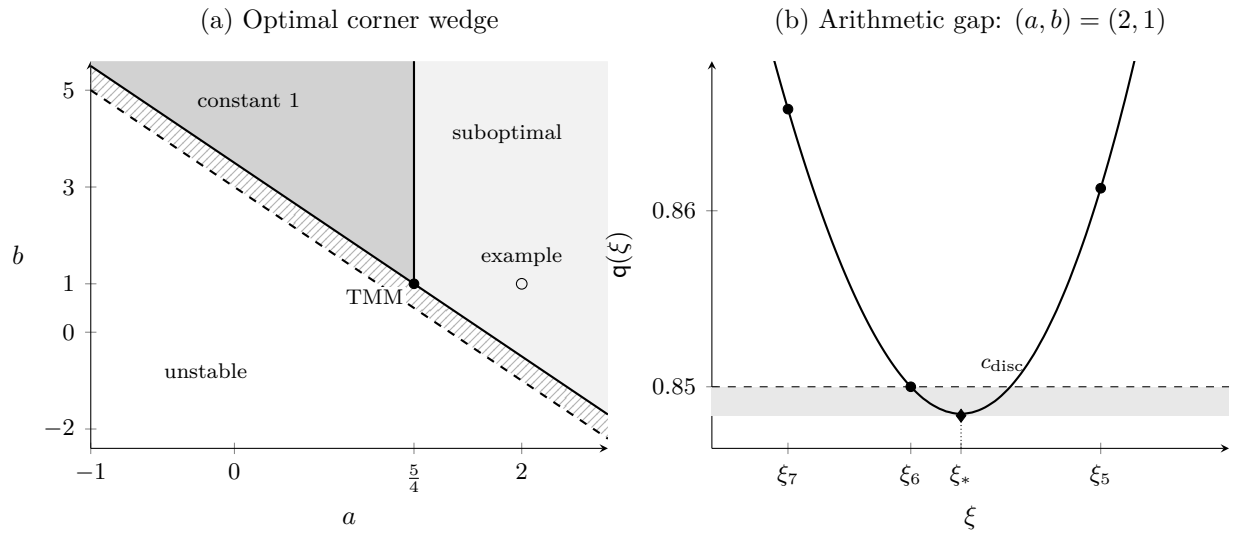


Figure 5: The corner at $\theta = 3$. Left: the dark wedge $a \le 5/4$, $2a + b \ge 7/2$ has constant 1, with TMM at its vertex; the dashed line is the stability boundary $2a + b = 3$, and the hatched strip has $c_\pi < 1$. Right: for the marked example, the continuous minimum $c_{\mathrm{lag}} = 3\sqrt{6} - 13/2$ at $\xi_* = 3 - \sqrt{6}$ (diamond) lies between ξ_6 and ξ_5; the resonant minimum is $c_{\mathrm{disc}} = 17/20$ (filled point at ξ_6). The shaded band is the gap between the certificate and witness bounds.

negative lag-one values occur only in $(2\pi/3, \pi)$; adding the upper lag-two constraint gives a strict certificate. Since $\tau_3 < \tau_* < \frac{1}{2}$ and $\tau_*(\beta) \le \beta$ on $[0, 1)$, every stable tuning with $\tau \ge \frac{1}{2}$, and every stable tuning of the Nesterov family $\tau = \beta$, converges linearly on $\mathcal{F}_{\mu,L}$, whatever h, μ and L; this agrees with the automated search of [15], which found cycles of Nesterov's method on smooth convex functions only on the boundary of its stability region. For heavy ball ($\tau = 0$), every stable tuning converges at every condition number exactly when $\beta \le (\sqrt{13} - 3)/2 \approx 0.303$; the lag-one threshold is $\beta \le \sqrt{5} - 2 \approx 0.236$. The lag-two certificate covers the interval between them. Earlier sufficient conditions depend on h, μ, L [14]. Heavy-ball 3-cycles are described by [17, Theorem 3.5], and each rotation-number tongue by [32, Corollary D.11(b)]; see also [15]. For $\mathcal{M}$, all such tongues are absent exactly when $\tau \ge \tau_3(\beta)$. Remark 4.4 transfers this threshold to QHM.

Corollary 8.2 (Quasi-hyperbolic momentum)**.** *For the method of [26] in the form of Remark 4.4, with fixed discount $\beta \in [0, 1)$ and weight $\nu_{\mathrm{qh}} \in [0, 1]$, and with the usual initialization $d_{-1} = 0$, every stable choice of the step α converges linearly on $\mathcal{F}_{\mu,L}$ for all $0 < \mu < L$ if and only if $\tau_{\mathrm{qh}} \ge (\beta^2 + 3\beta - 1)/6$.*

Proposition 8.3 (Monotonicity in the condition number)**.** *Fix a tuning (β, τ, h) of $\mathcal{M}$ and $L > 0$ with $hL(1 + 2\tau) < 2(1 + \beta)$. The set of $\mu \in (0, L)$ for which $\rho_{\mathcal{F}} < 1$ is an interval (μ_∂, L), possibly $(0, L)$, and it contains the interval (μ_*, L) on which the tuning is lag-one convergent.*

Thus uniform linear convergence holds below a critical condition number. For every fixed stationary method, class inclusion $\mathcal{F}_{\mu',L} \subseteq \mathcal{F}_{\mu,L}$, $\mu' \ge \mu$, makes the rate nonincreasing in μ. Strict-certificate persistence makes the convergence interval open. For heavy ball, the interval statement also appears in [32, Theorem 3.12].

Both thresholds approach the half-step wall as $\beta \to 1$, with $\tau_3(1 - \theta\varepsilon) = \frac{1}{2} - \frac{5\theta}{6}\varepsilon + O(\varepsilon^2)$. In the corner scale, stable margins reaching the 3-tongue exist for $a > 5\theta/6$ and are excluded for $a < 5\theta/6$. At $\theta = 3$, the threshold $a = 5/2$ is twice TMM's deficit.

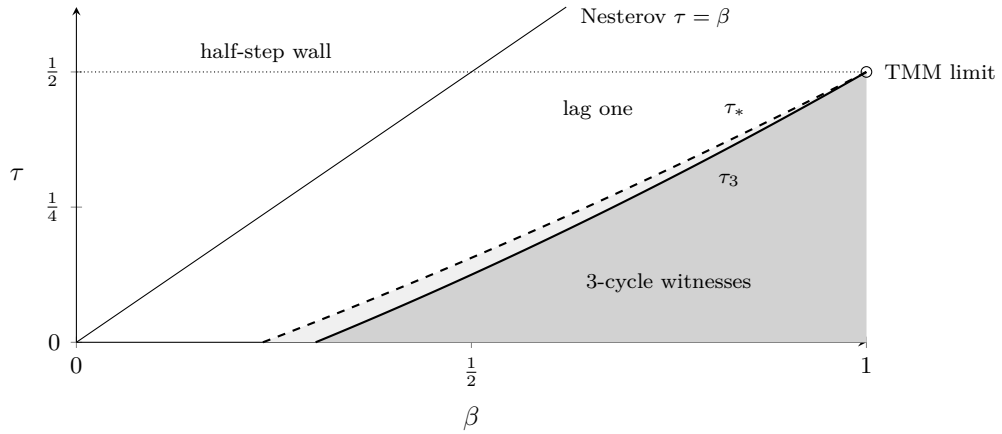


Figure 6: Regions for convergence of every stable tuning at fixed (β, τ) (Theorem 8.1). Lag one suffices for $\tau \geq \tau_*$ (dashed), and lag two covers $\tau_3 \leq \tau < \tau_*$, with τ_3 solid. For $\tau < \tau_3$, some stable choices of h, μ, L admit 3-cycle witnesses. The open endpoint $(1, \frac{1}{2})$ is the limit of TMM's parameters.

9. Conclusions

The flow ceiling $\min(\theta/3, c_Q)$ organizes the discrete rates. For $\mathcal{M}$, we determine the constants for $\tau > 1/2$, $y(1 + 2\tau) < 4$ and in the wall layer, and classify every family attaining the optimal leading constant 1. For fixed finite memory, positivity of the limiting lag-one symbol gives faithful discretization under the stated regularity and root conditions. Quadratic stability and nonnegativity of that symbol force $y \leq 2$, with equality fixing TMM's limiting transfer function. Spectral closure connects these rates to trajectories generated by one objective and to finite-lag certificates. Four directions remain open.

- *The arithmetic gap at the corner.* The bounds $\min(c_Q, c_{\text{lag}})$ and $\min(c_Q, c_{\text{disc}})$ can differ when the lag-one ratio q is minimized between consecutive points $\xi_{K+1} < \xi_K$. Spectra concentrated in the corresponding frequency gap violate the leading-order lag-$K+1$ constraint. After radial reweighting, the limiting interpolation constraints at the corner define the cone of finite nonnegative measures ν on $(0, \pi]$ satisfying $\int \sin \frac{n\omega}{2} \sin \frac{(n\mp1)\omega}{2}\, d\nu \geq 0$ for every $n \geq 1$; its Dirac measures are precisely those at $2\pi/K$, $K \geq 2$. Does linear-fractional optimization over this cone give the exact constant in the gaps, and how does it depend on the ordering of rational frequencies with bounded denominators?
- *Memory $D \geq 3$.* Does acceleration of a second-order family force $\mathcal{S}_0 \geq 0$ at every frequency? The unresolved case has a negative symbol somewhere and $|\Theta_0(2\pi j/K)| \geq \pi/K$ at every rational frequency where the phase is defined. An affirmative answer would extend $y \leq 2$ and TMM rigidity in Corollary 5.9 to every accelerated second-order family. How does the constant depend on the approach to a zero-symbol limit, $\mathcal{S}_0 \equiv 0$?
- *The size of the worst case.* For $\theta/3 < c_Q$, must the number of spirals in Theorem 2.6(a) grow as $\delta \to 0$? Can the dimension in Theorem 4.7(e) be bounded by the memory alone? At $T = 0 > \log \rho_q$, can the attained worst case in $L^2(\nu; \mathbb{C})$ always be realized by a finite spectrum?
- *Periodic and time-varying methods.* Which parts of the flow ceiling, phase rule and optimal-constant classification extend to these methods?

Use of AI tools

The authors used Claude Opus 5.5 and ChatGPT-6.1 Sol to assist with exposition, formal proof verification and literature checks. The authors verified the statements and take responsibility for the results and accompanying code.

References

[1] H. Attouch, Z. Chbani, J. Fadili, and H. Riahi. First-order optimization algorithms via inertial systems with Hessian driven damping. *Math. Program.*, 193(1):113–155, 2022. doi:10.1007/s10107-020-01591-1.

[2] J.-F. Aujol, C. Dossal, and A. Rondepierre. Convergence rates of the heavy ball method for quasi-strongly convex optimization. *SIAM J. Optim.*, 32(3):1817–1842, 2022. doi:10.1137/21M1403990.

[3] J.-F. Aujol, C. Dossal, and A. Rondepierre. Convergence rates of the heavy-ball method under the Łojasiewicz property. *Math. Program.*, 198(1):195–254, 2023. doi:10.1007/s10107-022-01770-2.

[4] D. Azagra and C. Mudarra. An extension theorem for convex functions of class $C^{1,1}$ on Hilbert spaces. *J. Math. Anal. Appl.*, 446(2):1167–1182, 2017. doi:10.1016/j.jmaa.2016.09.015.

[5] A. Badithela and P. Seiler. Analysis of the heavy-ball algorithm using integral quadratic constraints. In *Proc. 2019 American Control Conference (ACC)*, pages 4081–4085, 2019. doi:10.23919/ACC.2019.8814459.

[6] J. Carrasco, M. C. Turner, and W. P. Heath. Zames–Falb multipliers for absolute stability: from O'Shea's contribution to convex searches. *Eur. J. Control*, 28:1–19, 2016. doi:10.1016/j.ejcon.2015.10.003.

[7] L. Chen and Z. Xu. HNAG^{++}: An accelerated gradient method with a refined asymptotic rate for strongly convex optimization. *J. Optim. Theory Appl.*, 211(2):30, 2026. doi:10.1007/s10957-026-03131-x.

[8] S. Cyrus, B. Hu, B. Van Scoy, and L. Lessard. A robust accelerated optimization algorithm for strongly convex functions. In *Proc. 2018 Annual American Control Conference (ACC)*, pages 1376–1381, 2018. doi:10.23919/ACC.2018.8430824.

[9] A. d'Aspremont, D. Scieur, and A. B. Taylor. Acceleration methods. *Found. Trends Optim.*, 5(1–2):1–245, 2021. doi:10.1561/2400000036.

[10] Y. Drori and A. B. Taylor. On the oracle complexity of smooth strongly convex minimization. *J. Complexity*, 68:101590, 2022. doi:10.1016/j.jco.2021.101590.

[11] Y. Drori and M. Teboulle. Performance of first-order methods for smooth convex minimization: a novel approach. *Math. Program.*, 145(1–2):451–482, 2014. doi:10.1007/s10107-013-0653-0.

[12] K. Fotopoulos, M. Helcig, K. Deck, and J. Alsbach. Non-acceleration of the heavy-ball method in 1D. Preprint, arXiv:2610.25104, 2026.

[13] O. Gannot. A frequency-domain analysis of inexact gradient methods. *Math. Program.*, 194(1–2):975–1016, 2022. doi:10.1007/s10107-021-01665-8.

[14] E. Ghadimi, H. R. Feyzmahdavian, and M. Johansson. Global convergence of the heavy-ball method for convex optimization. In *Proc. 2015 European Control Conference (ECC)*, pages 310–315, 2015. doi:10.1109/ECC.2015.7330562.

[15] B. Goujaud, A. Dieuleveut, and A. B. Taylor. Counter-examples in first-order optimization: a constructive approach. *IEEE Control Syst. Lett.*, 7:2485–2490, 2023. doi:10.1109/LCSYS.2023.3286277.

[16] B. Goujaud, A. B. Taylor, and A. Dieuleveut. Open problem: two riddles in heavy-ball dynamics. Preprint, arXiv:2502.19916, 2025.

[17] B. Goujaud, A. B. Taylor, and A. Dieuleveut. Provable non-accelerations of the heavy-ball method. *Math. Program.*, online first, 2025. doi:10.1007/s10107-025-02269-2.

[18] M. Hernanz. The optimal rate constant of heavy-ball momentum on smooth strongly convex functions. Preprint, 16 October 2026. `https://hernanz.io/papers/heavy-ball-optimal-rate/`.

[19] B. Hu and L. Lessard. Dissipativity theory for Nesterov's accelerated method. In *Proc. 34th International Conference on Machine Learning (ICML)*, volume 70 of *PMLR*, pages 1549–1557, 2017. `https://proceedings.mlr.press/v70/hu17a.html`.

[20] E. I. Jury and B. W. Lee. On the stability of a certain class of nonlinear sampled-data systems. *IEEE Trans. Automat. Control*, 9(1):51–61, 1964. doi:10.1109/TAC.1964.1105622.

[21] A. Kharitenko. Existence of stable Lur'e systems for which the O'Shea–Zames–Falb stability test fails. Preprint, arXiv:2607.23599, 2026.

[22] A. Kharitenko and C. W. Scherer. On the exactness of a stability test for discrete-time Lur'e systems with slope-restricted nonlinearities. *IEEE Trans. Automat. Control*, 69(7):4851–4858, 2024. doi:10.1109/TAC.2024.3362859.

[23] L. Lessard, B. Recht, and A. Packard. Analysis and design of optimization algorithms via integral quadratic constraints. *SIAM J. Optim.*, 26(1):57–95, 2016. doi:10.1137/15M1009597.

[24] L. Lessard and P. Seiler. Direct synthesis of iterative algorithms with bounds on achievable worst-case convergence rate. In *Proc. 2020 American Control Conference (ACC)*, pages 119–125, 2020. doi:10.23919/ACC45564.2020.9147401.

[25] K. Liu, Q. Wu, Z. Sun, M. Chen, S. Li, and B. Qu. Sharp initialization guarantees for the heavy-ball method. Preprint, arXiv:2610.10369, 2026.

[26] J. Ma and D. Yarats. Quasi-hyperbolic momentum and Adam for deep learning. In *Proc. 7th International Conference on Learning Representations (ICLR)*, 2019. arXiv:1810.06801.

[27] A. Megretski and A. Rantzer. System analysis via integral quadratic constraints. *IEEE Trans. Automat. Control*, 42(6):819–830, 1997. doi:10.1109/9.587335.

[28] S. Michalowsky, C. W. Scherer, and C. Ebenbauer. Robust and structure exploiting optimisation algorithms: an integral quadratic constraint approach. *Internat. J. Control*, 94(11):2956–2979, 2021. doi:10.1080/00207179.2020.1745286.

[29] M. Muehlebach and M. I. Jordan. A dynamical systems perspective on Nesterov acceleration. In *Proc. 36th International Conference on Machine Learning (ICML)*, volume 97 of *PMLR*, pages 4656–4662, 2019. `https://proceedings.mlr.press/v97/muehlebach19a.html`.

[30] Y. Nesterov. *Lectures on Convex Optimization*, volume 137 of *Springer Optimization and Its Applications*. Springer, Cham, 2nd edition, 2018. doi:10.1007/978-3-319-91578-4.

[31] R. O'Shea and M. Younis. A frequency-time domain stability criterion for sampled-data systems. *IEEE Trans. Automat. Control*, 12(6):719–724, 1967. doi:10.1109/TAC.1967.1098747.

[32] D. Pasechnyuk-Vilensky and M. Takáč. Two answers on heavy-ball dynamics. Preprint, arXiv:2610.09912v2, 2026.

[33] S. Pokutta and B. Stellato. The exact worst-case asymptotic rate of Nesterov acceleration for strongly convex optimization. Preprint, arXiv:2610.15125, 2026.

[34] B. T. Polyak. Some methods of speeding up the convergence of iteration methods. *USSR Comput. Math. Math. Phys.*, 4(5):1–17, 1964. doi:10.1016/0041-5553(64)90137-5.

[35] V. M. Popov. On absolute stability of non-linear automatic control systems. *Avtomat. i Telemekh.*, 22(8):961–979, 1961 (in Russian). `https://www.mathnet.ru/eng/at12357`.

[36] A. Rantzer. On the Kalman–Yakubovich–Popov lemma. *Systems Control Lett.*, 28(1):7–10, 1996. doi:10.1016/0167-6911(95)00063-1.

[37] S. Safavi, B. Joshi, G. França, and J. Bento. An explicit convergence rate for Nesterov's method from SDP. In *Proc. 2018 IEEE International Symposium on Information Theory (ISIT)*, pages 1560–1564, 2018. doi:10.1109/ISIT.2018.8437794.

[38] J. M. Sanz-Serna and K. C. Zygalakis. The connections between Lyapunov functions for some optimization algorithms and differential equations. *SIAM J. Numer. Anal.*, 59(3):1542–1565, 2021. doi:10.1137/20M1364138.

[39] C. W. Scherer and C. Ebenbauer. Convex synthesis of accelerated gradient algorithms. *SIAM J. Control Optim.*, 59(6):4615–4645, 2021. doi:10.1137/21M1398598.

[40] P. Seiler and J. Carrasco. Construction of periodic counterexamples to the discrete-time Kalman conjecture. *IEEE Control Syst. Lett.*, 5(4):1291–1296, 2021. doi:10.1109/LCSYS.2020.3033443.

[41] B. Shi, S. S. Du, M. I. Jordan, and W. J. Su. Understanding the acceleration phenomenon via high-resolution differential equations. *Math. Program.*, 195(1–2):79–148, 2022. doi:10.1007/s10107-021-01681-8.

[42] L. Su, P. Seiler, J. Carrasco, and S. Z. Khong. On the necessity and sufficiency of discrete-time O'Shea–Zames–Falb multipliers. *Automatica*, 150:110872, 2023. doi:10.1016/j.automatica.2023.110872.

[43] W. J. Su, S. Boyd, and E. J. Candès. A differential equation for modeling Nesterov's accelerated gradient method: theory and insights. *J. Mach. Learn. Res.*, 17(153):1–43, 2016. `https://jmlr.org/papers/v17/15-084.html`.

[44] A. B. Taylor and Y. Drori. An optimal gradient method for smooth strongly convex minimization. *Math. Program.*, 199(1–2):557–594, 2023. doi:10.1007/s10107-022-01839-y.

[45] A. B. Taylor, J. M. Hendrickx, and F. Glineur. Smooth strongly convex interpolation and exact worst-case performance of first-order methods. *Math. Program.*, 161(1–2):307–345, 2017. doi:10.1007/s10107-016-1009-3.

[46] A. B. Taylor, B. Van Scoy, and L. Lessard. Lyapunov functions for first-order methods: tight automated convergence guarantees. In *Proc. 35th International Conference on Machine Learning (ICML)*, volume 80 of *PMLR*, pages 4897–4906, 2018. `https://proceedings.mlr.press/v80/taylor18a.html`.

[47] M. Upadhyaya, S. Banert, A. B. Taylor, and P. Giselsson. Automated tight Lyapunov analysis for first-order methods. *Math. Program.*, 209(1–2):133–170, 2025. doi:10.1007/s10107-024-02061-8.

[48] B. Van Scoy, R. A. Freeman, and K. M. Lynch. The fastest known globally convergent first-order method for minimizing strongly convex functions. *IEEE Control Syst. Lett.*, 2(1):49–54, 2018. doi:10.1109/LCSYS.2017.2722406.

[49] B. Van Scoy and L. Lessard. Absolute stability via lifting and interpolation. In *Proc. 61st IEEE Conference on Decision and Control (CDC)*, pages 6217–6223, 2022. doi:10.1109/CDC51059.2022.9993272.

[50] B. Van Scoy and L. Lessard. The fastest known first-order method for minimizing twice continuously differentiable smooth strongly convex functions. *IEEE Control Syst. Lett.*, 9:655–660, 2025. doi:10.1109/LCSYS.2025.3576681.

[51] J. Wei and L. Chen. Accelerated over-relaxation heavy-ball method: achieving global accelerated convergence with broad generalization. In *Proc. 13th International Conference on Learning Representations (ICLR)*, 2025. arXiv:2406.09772.

[52] J. C. Willems and R. W. Brockett. Some new rearrangement inequalities having application in stability analysis. *IEEE Trans. Automat. Control*, 13(5):539–549, 1968. doi:10.1109/TAC.1968.1098999.

[53] A. C. Wilson, B. Recht, and M. I. Jordan. A Lyapunov analysis of accelerated methods in optimization. *J. Mach. Learn. Res.*, 22(113):1–34, 2021. `https://jmlr.org/papers/v22/20-195.html`.

[54] G. Zames and P. L. Falb. Stability conditions for systems with monotone and slope-restricted nonlinearities. *SIAM J. Control*, 6(1):89–108, 1968. doi:10.1137/0306007.

[55] J. Zhang, J. Carrasco, and W. P. Heath. Duality bounds for discrete-time Zames–Falb multipliers. *IEEE Trans. Automat. Control*, 67(7):3521–3528, 2022. doi:10.1109/TAC.2021.3095418.

[56] J. Zhang, P. Seiler, and J. Carrasco. Zames–Falb multipliers for convergence rate: motivating example and convex searches. *Internat. J. Control*, 95(3):821–829, 2022. doi:10.1080/00207179.2020.1823484.

A. Elementary Rate and Quadratic Facts

Proof of Remark 2.4. Fix a stationary method (u, v) and a finite horizon $n \geq 1$. Translate the minimizer to 0 and restart at the state used in the ratio defining $\Gamma(n)$. For a trajectory in an arbitrary real Hilbert space, let

$$V = \operatorname{span}\{w_0, \dots, w_{D-1}, \nabla f(\tilde{x}_0), \dots, \nabla f(\tilde{x}_{n-1})\}.$$

This space is finite-dimensional. Since $v_D \neq 0$, the recursion (12) shows inductively that $w_0, \dots, w_{n+D-1}$ and all queries used to compute them lie in V. The restriction $f|_V$ belongs to $\mathcal{F}_{\mu,L}(V)$, has minimizer 0, and its gradient at each of these queries equals the original gradient. Thus the restriction preserves the finite trajectory and its ratio of state norms.

Identify V with $\mathbb{R}^d$. Let $\eta \geq 0$ be a smooth compactly supported probability density on the unit ball, and set $f_\delta(x) = \int f(x - \delta z)\eta(z)\, dz$. Integrating the strong-convexity inequality and the gradient Lipschitz bound gives $f_\delta \in C^\infty \cap \mathcal{F}_{\mu,L}(\mathbb{R}^d)$, with $\sup_x \|\nabla f_\delta(x) - \nabla f(x)\| \leq L\delta$. If a_δ is its minimizer, strong monotonicity gives $\|a_\delta\| \leq \|\nabla f_\delta(0)\|/\mu \leq L\delta/\mu$. Hence the recentered function $g_\delta(x) = f_\delta(x + a_\delta) - f_\delta(a_\delta)$ has minimizer 0, belongs to the same smooth class, and satisfies

$$\sup_x \|\nabla g_\delta(x) - \nabla f(x)\| \leq L\delta(1 + L/\mu) \longrightarrow 0.$$

Run the method on g_δ from the same initial coordinates $w_0, \dots, w_{D-1}$ of (12). Induction in the explicit recursion (12), using $v_D \neq 0$ and the common Lipschitz bound, shows convergence of every iterate up to w_{n+D-1}. The initial state is unchanged and nonzero, so the state-norm ratio converges to the original ratio. Every ratio in $\Gamma(n)$ is therefore approximated by finite-dimensional C^∞ objectives; the reverse inequality follows by class inclusion. The case $n = 0$ is immediate. Equality at every finite horizon gives equality of the corresponding uniform rates. For $\mathcal{M}$, the same approximation preserves the initialization $x_{-1} = x_0$, so it also applies to each finite-horizon supremum defining $\rho^0_{\mathcal{F}}$. □

Proof of Lemma 2.3. Center at $x^\star = 0$. The explicit recursion (12) and $\|\nabla f(x)\| \le L\|x\|$ give

$$\|w_{k+D}\| \le \frac{\sum_{j<D} |v_j| + L \sum_{j<D} |u_j|}{|v_D|} \|S_k\|.$$

Thus $\Gamma(n) \le C_\Gamma^n < \infty$, with C_Γ independent of the objective and dimension. Restarting at S_m gives $\|S_{m+n}\| \le \Gamma(n)\|S_m\| \le \Gamma(n)\Gamma(m)\|S_0\|$, including zero states, which are absorbing. Taking suprema proves submultiplicativity; the multiplicative Fekete lemma gives the root limit and infimum formula.

For a root $v(z) = \lambda u(z)$, $\lambda \in [\mu, L]$, the real planar quadratic $f(w) = \lambda|w|^2/2$ has the trajectory $w_k = z^k$, hence $\|S_n\| = |z|^n\|S_0\|$. Taking the largest root gives $\Gamma(n) \ge \rho_q^n$. For $\mathcal{M}$ started from rest, $\|S_0\| = \sqrt{2}\|x_0\|$, so $\|x_N\| \le \sqrt{2}\Gamma(N)\|x_0\|$; taking root limits gives $\rho^0_{\mathcal{F}} \le \rho_{\mathcal{F}}$. □

Proof of Lemma 2.1. A real monic quadratic $z^2 - Pz + Q$ has both roots in the open unit disk if and only if $|Q| < 1$, $1 + Q - P > 0$ and $1 + Q + P > 0$ (the Schur–Cohn, or Jury, conditions). For χ_λ we have $P = 1 + \beta - h\lambda(1 + \tau)$ and $Q = \beta - h\lambda\tau \le \beta < 1$. Further $1 + Q - P = \chi_\lambda(1) = h\lambda > 0$, and $1 + Q + P = \chi_\lambda(-1) = 2(1 + \beta) - h\lambda(1 + 2\tau)$, which decreases in λ; it is positive for all $\lambda \in [\mu, L]$ if and only if $y(1 + 2\tau) < 2(1 + \beta)$. Under this condition also $h\lambda\tau \le y\tau < 1 + \beta$, i.e. $Q > -1$. □

Proof of identities (15) *and* (16). Substitution of (13) in (14) gives

$$u - p = -\frac{2\chi_\mu}{h(L - \mu)}, \qquad u + p = \frac{2\chi_L}{h(L - \mu)}.$$

Their product $(u-p)(\bar{u}+\bar{p}) = |u|^2 - |p|^2 - 2i\,\mathrm{Im}(p\bar{u})$ gives (15); the roots of p are those of $(\chi_\mu + \chi_L)/2 = \chi_{(L+\mu)/2}$. For $\Omega = (z - 1)(z - \beta)$ and $N = N_\tau$, $\mathrm{Im}(\chi_\mu \overline{\chi_L}) = (y - y_\mu)\,\mathrm{Im}(\Omega\bar{N})$. On $|z| = 1$, expansion with $z\bar{z} = 1$ gives $\mathrm{Im}(\Omega\bar{N}) = \sin\omega[(1 - \beta) + 2\tau\xi]$, proving (16). □

Proof of Theorem 2.7. Let $\rho = \rho_q$, and let w_1, w_2 and v_1, v_2 be the roots of χ_μ and χ_L. Then $h\mu = \chi_\mu(1) = (1 - w_1)(1 - w_2) \ge (1 - \rho)^2$: for real roots both factors are at least $1 - \rho$, for complex ones $|1 - w_1|^2 \ge (1 - |w_1|)^2$. Similarly $\chi_L(-1) = (1 + v_1)(1 + v_2) \ge (1 - \rho)^2$, and $\beta - h\mu\tau = w_1 w_2 \le \rho^2$. Since $\chi_L(-1) = 2(1 + \beta) - hL(1 + 2\tau)$, $hL(1 + 2\tau) \le 2(1 + \beta) - (1 - \rho)^2 \le (1 + \rho)^2 + 2h\mu\tau$. Dividing by $h\mu \ge (1 - \rho)^2$ gives $\kappa(1 + 2\tau) \le (\frac{1+\rho}{1-\rho})^2 + 2\tau$, i.e. the claim. Equality forces equality in each step: $w_1 = w_2 = \rho$ (for complex roots $|1 - w| > 1 - |w|$), $v_1 = v_2 = -\rho$, so $\beta - h\mu\tau = \rho^2 = \beta - hL\tau$ and $\tau = 0$; then $\chi_\mu = (z - \rho)^2$ and $\chi_L = (z + \rho)^2$ give $\beta = \rho^2$, $h\mu = (1 - \rho)^2$, $hL = (1 + \rho)^2$, which is Polyak's tuning. □

Proof of Lemma 2.5. In this proof write $\beta = \beta_\varepsilon$, $\tau = \tau_\varepsilon$, and let $P_\ell(z) = (z - 1)(z - \beta) + \ell N_\tau(z)$, so $\chi_\lambda = P_{h\lambda}$ and $h\lambda \in [y_\varepsilon\varepsilon^2, y_\varepsilon]$. (i) At $\lambda = \mu$, $\chi_\mu(1-\varepsilon\zeta) = \varepsilon^2(\zeta^2 - \theta'\zeta + y_\varepsilon)$ exactly, with $\theta' = \theta_\varepsilon + y_\varepsilon(1+\tau_\varepsilon)\varepsilon$, because $(z - 1)(z - \beta) = \varepsilon^2\zeta(\zeta - \theta_\varepsilon)$ and $N_\tau(1 - \varepsilon\zeta) = 1 - (1 + \tau)\varepsilon\zeta$. So the roots at μ are $1 - \varepsilon\zeta$, ζ a root of $\zeta^2 - \theta'\zeta + y_\varepsilon$; if real, the largest is $1 - \varepsilon\zeta_-(\theta', y_\varepsilon)$, and if complex both have modulus $\sqrt{\beta - y_\mu\tau} = 1 - \frac{\theta_\varepsilon}{2}\varepsilon + O(\varepsilon^2)$. In both cases the largest root modulus at μ is $1 - \varepsilon c_Q(\theta', y_\varepsilon) + O(\varepsilon^2)$ (if the roots are real, $\zeta_-(\theta', y_\varepsilon) \le \theta'/2$), so $\rho_q \ge 1 - \varepsilon c_Q(\theta', y_\varepsilon) + O(\varepsilon^2)$; as c_Q is continuous, $\theta' \to \theta$ and $y_\varepsilon \to y$, this proves the first claim. For the second claim, assume the strict inequality in the statement. (ii) Complex roots at any ℓ have modulus $\sqrt{\beta - \ell\tau} \le \sqrt{\beta} = 1 - \frac{\theta_\varepsilon}{2}\varepsilon + O(\varepsilon^2)$. (iii) The discriminant of P_ℓ is a convex quadratic in ℓ; its smaller zero is $\ell_a = O(\varepsilon^2)$ and its larger zero is $4/(1 + \tau_\varepsilon)^2 + o(1)$.

On $[y_\varepsilon\varepsilon^2, \ell_a]$ (empty if the roots at μ are complex) the roots are real and positive, the larger root z_+ exceeds $\tau/(1+\tau)$ (it is close to 1), so $N_\tau(z_+) > 0$ and, in the interior, $dz_+/d\ell = -N_\tau(z_+)/P'_\ell(z_+) < 0$; by continuity the maximum over this interval is at $\ell = y_\varepsilon\varepsilon^2$. (iv) Fix $\ell_* > 0$ below the larger zero of the discriminant. On $[\ell_a, \ell_*]$ the roots are complex. On $[\ell_*, y_\varepsilon]$, real roots w satisfy $|w| < 1$ and $P_\ell(1) = \ell \geq \ell_*$, $P_\ell(-1) \geq 2(1+\beta) - y_\varepsilon(1+2\tau) \geq \delta_0 > 0$ for small ε (by this strict inequality); since $P_\ell(\pm 1) = (1 \mp w)(1 \mp w') \leq 2(1 \mp w)$ for the other root $w' \in [-1, 1]$, real roots lie in $[-1+\delta_1, 1-\delta_1]$, $\delta_1 = \frac{1}{2}\min(\ell_*, \delta_0)$. Combining: the root moduli on $[\ell_*, y_\varepsilon]$ are at most $1-\delta_1$ or $\sqrt{\beta}$; on $[\ell_a, \ell_*]$ they are at most $\sqrt{\beta}$; and on $[y_\varepsilon\varepsilon^2, \ell_a]$ they are at most $1 - \varepsilon\zeta_-(\theta', y_\varepsilon)$, attained at $\ell = y_\varepsilon\varepsilon^2$. If the roots at μ are complex, $\rho_q = \sqrt{\beta - y_\mu\tau} = 1 - \frac{\theta_\varepsilon}{2}\varepsilon + O(\varepsilon^2)$; if they are real, $\zeta_-(\theta', y_\varepsilon) \leq \theta'/2$ and $\rho_q = 1 - \varepsilon\zeta_-(\theta', y_\varepsilon) + O(\varepsilon^2)$. In both cases $1 - \rho_q = \varepsilon c_Q(\theta', y_\varepsilon) + O(\varepsilon^2)$, and c_Q is continuous, with $\theta' \to \theta$ and $y_\varepsilon \to y$. □

Proof of the claims in Remark 4.4. Applying the operator N_τ to (1) shows that the queries satisfy the gradient-correction recursion. Conversely, if $h_1 > h_2 \geq 0$, $h = h_1 - h_2$, $\tau = h_2/(h_1 - h_2)$ and $\beta \neq \tau/(1+\tau)$, each trajectory of the gradient-correction method is the query sequence of exactly one trajectory of (1). The two descriptions have the same rates. Indeed, $(\tilde{x}_{k-1},\ \tilde{x}_k + (1+\tau)h\nabla f(\tilde{x}_{k-1}))$ is the image of (x_{k-2}, x_{k-1}) under a fixed linear map with determinant $(1+\tau)\beta - \tau$; as $\|\nabla f(x)\| \leq L\|x\|$, the states (x_{k-2}, x_{k-1}) and $(\tilde{x}_{k-1}, \tilde{x}_k)$ bound each other up to factors independent of f when $\beta \neq \tau/(1+\tau)$, and the worst-case rate of the queries, defined as in Definition 2.2 with the state $(\tilde{x}_{k-1}, \tilde{x}_k)$, equals $\rho_{\mathcal{F}}$. From rest, $x_{-1} = x_0$, the queries start with $\tilde{x}_0 = x_0$ and $\tilde{x}_1 = \tilde{x}_0 - h_1\nabla f(\tilde{x}_0)$, and with $z_\tau = \tau/(1+\tau) < 1$, inverting $\tilde{x}_k = (1+\tau)x_k - \tau x_{k-1}$ gives

$$x_k = \frac{1}{1+\tau}\sum_{j=0}^{k} z_\tau^j\, \tilde{x}_{k-j} + z_\tau^{k+1} x_{-1};$$

hence $\rho_{\tilde{x}}^0 \leq \rho_{\mathcal{F}}^0 \leq \max(\rho_{\tilde{x}}^0, z_\tau)$. The two rates are equal whenever they exceed z_τ, in particular in every accelerated regime, where the rates tend to 1 while τ stays bounded: by Theorem 2.7, $c\kappa^{-1/2} \leq 1 - \rho_q \leq 2\kappa_\tau^{-1/2}$, and $\kappa_\tau \geq (1+2\tau)(\kappa - 1)$.

Quasi-hyperbolic momentum satisfies the gradient-correction recursion for $k \geq 1$ with $h_1 = \alpha(1 - \nu_{\mathrm{qh}}\beta)$ and $h_2 = \alpha(1-\nu_{\mathrm{qh}})\beta$. It therefore has the query representation (1) with $h = \alpha(1-\beta)$ and $\tau = \tau_{\mathrm{qh}}$. With $d_{-1} = 0$ its first step is $x_1 = x_0 - h_1\nabla f(x_0)$, so its iterates are exactly the queries of (1) started from rest at x_0. □

B. Proofs for Section 3

We use the following level-set and scaling identities for c_Q and $c_{\mathcal{F}}$.

Lemma B.1 (The quadratic constant)**.** *Let $\theta, m > 0$ and $\Delta_c = c^2 - \theta c + m$.*

(a) *For $c \in \mathbb{R}$, $c < c_Q(\theta, m)$ if and only if $c < \theta/2$ and $\Delta_c > 0$.*

(b) *For $\alpha > 0$, $c_Q(\alpha\theta, \alpha^2 m) = \alpha\, c_Q(\theta, m)$ and $c_{\mathcal{F}}(\alpha\theta, \alpha^2 m) = \alpha\, c_{\mathcal{F}}(\theta, m)$.*

Proof. (a) If $\theta^2 < 4m$, then $\Delta_c > 0$ for every c and $c_Q = \theta/2$. Otherwise Δ_c has roots $\zeta_- \leq \theta/2 \leq \zeta_+$, so $\Delta_c > 0$ together with $c < \theta/2$ is equivalent to $c < \zeta_- = c_Q$. (b) Rescaling (θ, m, c) to $(\alpha\theta, \alpha^2 m, \alpha c)$ preserves the two inequalities in (a), and therefore rescales c_Q by α. The same holds for $c_{\mathcal{F}} = \min(\theta/3, c_Q)$. □

The witnesses of Theorem 3.1(c) are orthogonal sums of self-similar spirals. For $c > 0$ and $\gamma \geq 0$ put

$$\mathcal{A}_\gamma(q) = \cosh cq - \cos\gamma q \geq 0, \qquad \mathcal{B}_\gamma(q) = \frac{\gamma \sinh cq}{c} - \sin\gamma q \geq 0 \qquad (q > 0). \tag{31}$$

Let $\nu = \sum_{j=1}^{J} \nu_j \delta_{\gamma_j}$ be a probability measure on $[0,\infty)$, let $z_j = -c + i\gamma_j$, and in $\mathbb{C}^J \cong \mathbb{R}^{2J}$ let

$$x(t) = \big(\sqrt{\nu_j}\, e^{z_j t}\big)_j, \qquad G(t) = \big(w_j \sqrt{\nu_j}\, e^{z_j t}\big)_j, \qquad w_j = -z_j^2 - \theta z_j - m, \tag{32}$$

$$\eta(t) = e^{-2ct} \sum_j \nu_j \eta_j, \qquad \eta_j = \frac{(\theta - c)\gamma_j^2 - c\Delta_c}{2c}. \tag{33}$$

Then $\ddot{x} + \theta \dot{x} + mx + G = 0$ and $\dot{\eta} = \langle G, \dot{x} \rangle$: the triples $(x(t), G(t), \eta(t))$ are the candidate values of $(x, \nabla f_m(x), f_m(x))$ along an orbit of (7). Consider the program $\mathcal{P}_c$ for such measures:

$$\textstyle\sum_j \nu_j \big[(\gamma_j^2 - \Delta_c)\mathcal{A}_{\gamma_j}(q) \mp (\theta - 2c)\gamma_j \mathcal{B}_{\gamma_j}(q)\big] \geq 0 \qquad (q > 0), \tag{34}$$

$$(3c - \theta) \textstyle\sum_j \nu_j \gamma_j^2 \geq c\, \Delta_c. \tag{35}$$

Lemma B.2 (Spirals of the flow)**.** *For $t \geq 0$ and $q > 0$,*

$$\eta(t+q) - \eta(t) - \langle G(t), x(t+q) - x(t) \rangle = e^{-2ct-cq} \textstyle\sum_j \nu_j \big[(\gamma_j^2 - \Delta_c)\mathcal{A}_{\gamma_j}(q) - (\theta - 2c)\gamma_j \mathcal{B}_{\gamma_j}(q)\big],$$
$$\eta(t) - \eta(t+q) - \langle G(t+q), x(t) - x(t+q) \rangle = e^{-2ct-cq} \textstyle\sum_j \nu_j \big[(\gamma_j^2 - \Delta_c)\mathcal{A}_{\gamma_j}(q) + (\theta - 2c)\gamma_j \mathcal{B}_{\gamma_j}(q)\big],$$
$$\|x(t+q) - x(t)\|^2 = 2e^{-2ct-cq} \textstyle\sum_j \nu_j \mathcal{A}_{\gamma_j}(q),$$

and $\langle G(t), x(t) \rangle - \eta(t) = e^{-2ct} \sum_j \nu_j \big((3c - \theta)\gamma_j^2 - c\Delta_c\big)/(2c)$. Hence, if $c \leq \theta/2$, there is a convex f_m on $\mathbb{C}^J$ with $f_m(0) = 0$, $0 \in \partial f_m(0)$, $f_m(x(t)) = \eta(t)$ and $G(t) \in \partial f_m(x(t))$ for all $t \geq 0$ if and only if ν satisfies (34)–(35)*.*

Proof of Lemma B.2. Orthogonality makes every quantity linear in the weights ν_j, so it suffices to take one atom. Write $z = -c + i\gamma$ and $w = -z^2 - \theta z - m = (\gamma^2 - \Delta_c) + i(2c - \theta)\gamma$. Then

$$\langle G(t), \dot{x}(t) \rangle = e^{-2ct} \operatorname{Re}(w\bar{z}) = e^{-2ct}\big(c\Delta_c + (c - \theta)\gamma^2\big) = \dot{\eta}(t).$$

For the forward lag, multiply $\eta(t+q) - \eta(t) - \langle G(t), x(t+q) - x(t) \rangle$ by e^{2ct+cq}. The result is

$$-2\eta_1 \sinh cq + (\gamma^2 - \Delta_c)(e^{cq} - \cos \gamma q) - (2c - \theta)\gamma \sin \gamma q = (\gamma^2 - \Delta_c)\mathcal{A}_\gamma(q) - (\theta - 2c)\gamma \mathcal{B}_\gamma(q),$$

using $2c\eta_1 = (\theta - c)\gamma^2 - c\Delta_c$. Reversing the pair changes the sign of the $\mathcal{B}_\gamma$-term. The distance identity follows from $|e^{zq} - 1|^2 = 2e^{-cq}(\cosh cq - \cos \gamma q)$; subtracting $\eta(t)$ from $\langle G(t), x(t) \rangle = e^{-2ct}(\gamma^2 - \Delta_c)$ gives the last identity.

Convex interpolation of values and subgradients is equivalent to $\eta_b \geq \eta_a + \langle G_a, x_b - x_a \rangle$ for every pair. Sufficiency follows by taking the supremum of these affine supports; here the orbit data are bounded, so the supremum is finite everywhere. Along the orbit, these inequalities are the two lag inequalities just computed; with the minimizer datum they add $\eta \geq 0$ and $\langle G, x \rangle - \eta \geq 0$. The latter is (35), and for $c \leq \theta/2$ it implies the former because

$$\eta(t) - \big(\langle G(t), x(t) \rangle - \eta(t)\big) = e^{-2ct} \frac{\theta - 2c}{c} \sum_j \nu_j \gamma_j^2 \geq 0.$$

□

Lemma B.3 (Smooth extension with a margin)**.** *Let $\{(x_a, G_a, \eta_a)\}_{a \in I}$ be data in $H \times H \times \mathbb{R}$ containing $(0,0,0)$, and suppose that for all $a, b \in I$, with $\Delta x = x_b - x_a$,*

$$\eta_b - \eta_a - \langle G_a, \Delta x \rangle \geq \epsilon_0 \|\Delta x\|^2, \qquad \|G_b - G_a\| \leq K \|\Delta x\|,$$

for some $\epsilon_0 > 0$, $K < \infty$. Then there is a convex $\tilde{f}_m \in C^{1,1}(H)$ with $\operatorname{Lip}(\nabla \tilde{f}_m) \leq K^2/(2\epsilon_0)$, $\tilde{f}_m(x_a) = \eta_a$ and $\nabla \tilde{f}_m(x_a) = G_a$ for all a, and, for every $m > 0$, $f = \tilde{f}_m + \frac{m}{2}\|\cdot\|^2$ is m-strongly convex with $\operatorname{Lip}(\nabla f) \leq m + K^2/(2\epsilon_0)$. In particular it belongs to $\mathcal{F}_{m,L}(H)$ for every $L > m$ at least this large.

Proof. If $K \le 0$, the gradient data are constant, and adding the two margin inequalities for a pair forces $x_a = x_b$; since the data contain $(0,0,0)$, the zero function is an interpolant. Suppose $K > 0$. By [4, Theorem 1.3], a convex $C^{1,1}$ interpolant with $\mathrm{Lip}(\nabla \tilde f_m) \le M$ exists if and only if $\eta_b \ge \eta_a + \langle G_a, x_b - x_a\rangle + \frac{1}{2M}\|G_b - G_a\|^2$ for all a, b. With $M = K^2/(2\epsilon_0)$ the last term is at most $\epsilon_0\|\Delta x\|^2$. $\square$

Lemma B.4 (Spread spectra)**.** *Let $\theta, m > 0$, $\theta/3 < c < c_Q(\theta, m)$, $\Delta_c = c^2 - \theta c + m$, and $W_\gamma = (\gamma^2 - \Delta_c)\mathcal{A}_\gamma - (\theta - 2c)\gamma\mathcal{B}_\gamma$. There are $\gamma_0 \ge 1$, $J \in \mathbb{N}$ and $\epsilon_* \in (0, 1]$ such that, for $\gamma_i = \gamma_0\lambda_i$, $\lambda_i = 1 + (i - \frac12)/J$, $\nu_i = 1/J$ $(1 \le i \le J)$, and all $q > 0$,*

$$\textstyle\sum_i \nu_i W_{\gamma_i}(q) \ge \epsilon_* \sum_i \nu_i(\mathcal{A}_{\gamma_i}(q) + \mathcal{B}_{\gamma_i}(q)), \tag{36}$$

$$\textstyle\sum_i \nu_i[(\gamma_i^2 - \Delta_c)\mathcal{A}_{\gamma_i} + (\theta - 2c)\gamma_i\mathcal{B}_{\gamma_i}] \ge \epsilon_* \sum_i \nu_i(\mathcal{A}_{\gamma_i} + \mathcal{B}_{\gamma_i}), \tag{37}$$

$$(3c - \theta)\textstyle\sum_i \nu_i\gamma_i^2 - c\Delta_c \ge 1. \tag{38}$$

Proof. We separate short lags, a compact band where the spread suppresses returns, and long lags where hyperbolic growth dominates. The frequency scale γ_0 is chosen first, and the quadrature size J last. Write $c = \theta/3 + \delta$, $\delta > 0$. By Lemma B.1(a), $\Delta_c > 0$ and $c < \theta/2$, so $\theta - 2c = c - 3\delta > 0$. Put $a = 3\delta/c$, $\mathcal{D}(q) = e^{-cq} + a\sinh cq$ and $\mathcal{D}_\delta = \min(e^{-1}, a\sinh 1) \le \inf \mathcal{D}$. The identity

$$W_\gamma(q) = \gamma^2(e^{-cq} - \cos\gamma q) + a\gamma^2\sinh cq - \Delta_c\mathcal{A}_\gamma(q) + (c - 3\delta)\gamma\sin\gamma q \tag{39}$$

follows from $\cosh x - \frac{c-3\delta}{c}\sinh x = e^{-x} + a\sinh x$. All atoms satisfy $\gamma_0 \le \gamma \le 2\gamma_0$. Let $\mathcal{U}(x) = \int_1^2 \lambda^2(1 - \cos\lambda x)\,d\lambda$, $\mathcal{V}(x) = \int_1^2 \lambda^2\cos\lambda x\,d\lambda$, and $\mathcal{U}_J$, $\mathcal{V}_J$ their midpoint sums $\frac1J\sum_i \lambda_i^2(1-\cos\lambda_i x)$, $\frac1J\sum_i \lambda_i^2\cos\lambda_i x$; then $|\mathcal{U} - \mathcal{U}_J|, |\mathcal{V} - \mathcal{V}_J| \le (1 + x)^2/(6J^2)$ (the second derivatives in λ are bounded by $4(1 + x)^2$) and, integrating by parts once, $|\mathcal{V}(x)| \le 8/x$. Fix $q_I = \frac12$, $q_{II} = \max(1, 32/\mathcal{D}_\delta)$, $\mathcal{U}_{\min} = \min_{[q_I, q_{II}]}\mathcal{U} > 0$ and $q_{III} = c^{-1}\,\mathrm{arsinh}(2/a)$.

Regime I: $q \le q_I/\gamma_0$. Assume $\gamma_0 \ge \max(2c, \sqrt{12\Delta_c})$. Then $\gamma q \le 1$, $cq \le \frac14$, and $1 - \cos x \ge \frac{x^2}{2}(1 - \frac{x^2}{12})$, $\cosh x - 1 \ge \frac{x^2}{2}$, $\sinh x - x \le \frac{x^3}{6}\cosh x$ give $\mathcal{A}_\gamma \ge \frac{11}{24}(c^2 + \gamma^2)q^2$ and $\mathcal{B}_\gamma \le \frac13\gamma(c^2 + \gamma^2)q^3$, while $\mathcal{A}_\gamma, \mathcal{B}_\gamma \le (c^2 + \gamma^2)q^2$. Using $\gamma^2 - \Delta_c \ge \frac{11}{12}\gamma^2$ and $(c - 3\delta)q/3 \le c/(6\gamma_0) \le \frac{1}{12}$,

$$W_\gamma(q) \ge (c^2 + \gamma^2)q^2[\tfrac{11}{12}\cdot\tfrac{11}{24} - \tfrac{1}{12}]\gamma^2 \ge \tfrac13\gamma^2(c^2 + \gamma^2)q^2 \ge \tfrac{\gamma_0^2}{6}(\mathcal{A}_\gamma + \mathcal{B}_\gamma)$$

atom by atom.

Regime II: $q_I/\gamma_0 \le q \le q_{II}/\gamma_0$, $x = \gamma_0 q$. Assume $\gamma_0 \ge cq_{II}$. By (39), using $\gamma^2(1 - e^{-cq}) \le 4\gamma_0^2 cq$, $\mathcal{A} \le 3$ and $|\gamma\sin\gamma q| \le 2\gamma_0$, $\sum_i \nu_i W_{\gamma_i} \ge \gamma_0^2\mathcal{U}_J(x) - 4c\gamma_0 x - 3\Delta_c - 2c\gamma_0$. With J so large that $(1 + q_{II})^2/(6J^2) \le \mathcal{U}_{\min}/2$ and γ_0 so large that $\gamma_0^2\mathcal{U}_{\min}/4 \ge (4cq_{II} + 2c)\gamma_0 + 3\Delta_c$, this is at least $\gamma_0^2\mathcal{U}_{\min}/4$, while $\sum_i \nu_i(\mathcal{A} + \mathcal{B}) \le 4 + 4q_{II}$ (as $\gamma\sinh(cq)/c \le 2\gamma_0 q\cosh 1 \le 4x$).

Regime III: $q \ge q_{II}/\gamma_0$. By (39), $\sum_i \nu_i W_{\gamma_i} = \gamma_0^2[m_2\mathcal{D}(q) - \mathcal{V}_J(\gamma_0 q)] - \Delta_c\sum_i \nu_i\mathcal{A}_{\gamma_i} + (c - 3\delta)\sum_i \nu_i\gamma_i\sin\gamma_i q$, $m_2 = \frac1J\sum_i \lambda_i^2 \in [1, 4]$. If $q \ge q_{III}$, then $\mathcal{D} \ge 2$ and $|\mathcal{V}_J| \le m_2$, so $m_2\mathcal{D} - \mathcal{V}_J \ge m_2\mathcal{D}/2 \ge \mathcal{D}/2$. If $q_{II}/\gamma_0 \le q \le q_{III}$, take J with $(1 + \gamma_0 q_{III})^2/(6J^2) \le \mathcal{D}_\delta/4$; then $|\mathcal{V}_J(\gamma_0 q)| \le 8/q_{II} + \mathcal{D}_\delta/4 \le \mathcal{D}_\delta/2 \le \mathcal{D}/2$, and again $m_2\mathcal{D} - \mathcal{V}_J \ge \mathcal{D}/2$. Since $\cosh x \le 1 + \sinh x$, $\mathcal{D} \ge \mathcal{D}_\delta$ and $\mathcal{D} \ge a\sinh cq$, $\Delta_c\sum_i \nu_i\mathcal{A}_{\gamma_i} + 2c\gamma_0 \le (2\Delta_c + 2c\gamma_0) + \Delta_c\sinh cq \le [(2\Delta_c + 2c\gamma_0)/\mathcal{D}_\delta + \Delta_c/a]\mathcal{D} \le \frac{\gamma_0^2}{4}\mathcal{D}$ for γ_0 large; so $\sum_i \nu_i W_{\gamma_i} \ge \frac{\gamma_0^2}{4}\mathcal{D}$, while $\sum_i \nu_i(\mathcal{A} + \mathcal{B}) \le 3 + (1 + 2\gamma_0/c)\sinh cq \le [3/\mathcal{D}_\delta + (1 + 2\gamma_0/c)/a]\mathcal{D}$.

Put

$$K_3 = \frac{3}{\mathcal{D}_\delta} + \frac{1 + 2\gamma_0/c}{a}, \qquad \epsilon_* = \min\left\{1, \frac{\gamma_0^2}{6}, \frac{\gamma_0^2\mathcal{U}_{\min}}{16(1 + q_{II})}, \frac{\gamma_0^2}{4K_3}, (\theta - 2c)\gamma_0\right\} > 0.$$

All lower bounds on γ_0 above are independent of J. Impose also $\gamma_0^2 \ge \max\{\Delta_c + 1, (c\Delta_c + 1)/(3\delta)\}$, then choose J to satisfy the two midpoint-error bounds. The three regimes prove (36) with this common ϵ_*. For (37), each atom has $\gamma^2 - \Delta_c \ge 1$ and $(\theta - 2c)\gamma \ge (\theta - 2c)\gamma_0$, while $\mathcal{A}, \mathcal{B} \ge 0$. Finally, $(3c - \theta)\sum_i \nu_i\gamma_i^2 - c\Delta_c = 3\delta\gamma_0^2 m_2 - c\Delta_c \ge 1$, proving (38). $\square$

Proof of Theorem 3.1. (a) Put $v = \dot{x}$, $G = \nabla f_m(x)$, and $a = \theta - c$. Since $\dot{v} = -\theta v - mx - G$,

$$\dot{V}_c = \langle G, v\rangle + \Delta_c\langle x, v\rangle + \langle v + ax, -cv - mx - G\rangle.$$

The gradient–velocity terms cancel. Using $\Delta_c = m - ca$ and adding $2cV_c$ yields

$$\dot{V}_c + 2cV_c = -(\theta - 2c)\Delta_c\|x\|^2 - (\theta - c)(\langle G, x\rangle - f_m) - (\theta - 3c)f_m,$$

which is (9). Under the stated conditions all three terms on the right are nonpositive by (6). Moreover

$$V_c \geq \tfrac{\Delta_c}{2}\|x\|^2 + \tfrac{1}{2}\|v + (\theta - c)x\|^2 \geq K_c^{-1}(\|x\|^2 + \|v\|^2),$$

with K_c depending only on (θ, m, c), proving (a).

(b) The solutions on $\frac{m}{2}x^2$ are combinations of e^{st}, $s^2 + \theta s + m = 0$, with te^{st} at a double root. The root with the larger real part has real part $-c_Q(\theta, m)$, and the solution with this mode alone (or its real part) has $\ell = c_Q(\theta, m)$.

(c) Let $\theta/3 < c < c_Q(\theta, m)$. Lemma B.4 gives $\gamma_0 \geq 1$, J, $\epsilon_* \in (0, 1]$ and atoms $\gamma_j \in [\gamma_0, 2\gamma_0]$ with equal weights such that, for all $q > 0$,

$$\textstyle\sum_j \nu_j[(\gamma_j^2 - \Delta_c)\mathcal{A}_{\gamma_j} \mp (\theta - 2c)\gamma_j\mathcal{B}_{\gamma_j}](q) \geq \epsilon_* \sum_j \nu_j\mathcal{A}_{\gamma_j}(q), \qquad (3c - \theta)\textstyle\sum_j \nu_j\gamma_j^2 - c\Delta_c \geq 1,$$

where we dropped the terms $\epsilon_*\mathcal{B}_{\gamma_j} \geq 0$. Use these atoms in (32)–(33). By Lemma B.2, for two times $t < t + q$ both lag quantities are at least $e^{-2ct-cq}\epsilon_* \sum_j \nu_j\mathcal{A}_{\gamma_j}(q) = \frac{\epsilon_*}{2}\|\Delta x\|^2$. For a time and the minimizer, $\langle G(t), x(t)\rangle - \eta(t) \geq \frac{1}{2c}e^{-2ct} = \frac{1}{2c}\|x(t)\|^2$, and $\eta(t) \geq \langle G(t), x(t)\rangle - \eta(t)$, as in the proof of Lemma B.2, since $c < c_Q \leq \theta/2$. Moreover $\|G(t') - G(t)\| \leq K\|x(t') - x(t)\|$ and $\|G(t)\| \leq K\|x(t)\|$ with $K = \max_j |w_j|$, since G acts on each coordinate as multiplication by w_j. Lemma B.3 with $\epsilon_0 = \min(\frac{\epsilon_*}{2}, \frac{1}{2c})$ gives $f \in \mathcal{F}_{m,L}(\mathbb{R}^{2J})$ with $\nabla f(x(t)) = mx(t) + G(t)$ for $t \geq 0$. Since $\ddot{x} + \theta\dot{x} + mx + G = 0$, the curve x solves (7), and $\|x(t)\| = e^{-ct}$, $\|\dot{x}(t)\|^2 = e^{-2ct}\sum_j \nu_j|z_j|^2$.

(d) Let $0 < c < c_{\mathcal{F}}(\theta, m)$. Then $c < \theta/3$, and $\Delta_c > 0$ by Lemma B.1(a), as $c < c_Q$. So (a) gives $\ell \geq c$ for every solution, and $\inf \ell \geq c_{\mathcal{F}}$. Conversely (b) gives $\ell = c_Q$, and (c) gives $\ell = c$ for every $c \in (\theta/3, c_Q)$; hence $\inf \ell \leq c_{\mathcal{F}}$, and the witnesses lie in $\bigcup_L \mathcal{F}_{m,L}$. For the energy and the gap: by (a), $f(x) = f_m + \frac{m}{2}\|x\|^2 \leq (1 + \frac{m}{\Delta_c})V_c$, and $\frac{1}{2}\|\dot{x}\|^2 \leq K_c e^{-2ct}V_c(0)$; along the spirals $f(x(t)) = e^{-2ct}(\sum_j \nu_j\eta_j + \frac{m}{2})$ and the energy is a fixed multiple of e^{-2ct}; on the quadratic $\frac{m}{2}x^2$ the slowest mode gives $2c_Q$. If $\theta < 3\sqrt{m/2}$, then $\Delta_{\theta/3} = m - \frac{2}{9}\theta^2 > 0$, and (a) applies with $c = \theta/3$. □

Proof of Corollary 3.2. If $\theta \leq 2\sqrt{m}$, then $c_Q = \theta/2 \leq \sqrt{m}$ and $c_{\mathcal{F}} = \theta/3 \leq \frac{2}{3}\sqrt{m} < \sqrt{m/2}$. On $\theta \geq 2\sqrt{m}$, $c_Q = \zeta_- = 2m/(\theta + \sqrt{\theta^2 - 4m})$ decreases from $\sqrt{m}$ and $\theta/3$ increases; they are equal if and only if $\frac{\theta^2}{9} - \frac{\theta^2}{3} + m = 0$, i.e. $\theta = 3\sqrt{m/2}$, with common value $\sqrt{m/2}$. This gives both maxima. For the domination: if $\theta \leq 2\sqrt{m}$, $(c_Q, c_{\mathcal{F}}) = (\theta/2, \theta/3)$ is dominated by $(\sqrt{m}, \frac{2}{3}\sqrt{m})$, the endpoint $\theta = 2\sqrt{m}$ of the arc; for $2\sqrt{m} \leq \theta \leq 3\sqrt{m/2}$ the point $(\zeta_-, \theta/3)$ lies on the arc; for $\theta \geq 3\sqrt{m/2}$, $c_Q = c_{\mathcal{F}} = \zeta_- \leq \sqrt{m/2}$, dominated by the other endpoint $(\sqrt{m/2}, \sqrt{m/2})$. □

Proof of Proposition 3.3. Take $c_\sharp \in (\theta/3, c_Q)$; then every $c \in (\theta/3, c_\sharp]$ has $c < c_Q$, hence $\Delta_c > 0$ and $\theta - 2c > 0$ by Lemma B.1(a). The atom $\gamma = 0$ violates (35) because $\Delta_c > 0$. For $\gamma > 0$, (35) gives $\gamma^2 \geq c\Delta_c/(3c - \theta)$. At $q = 2\pi/\gamma$ the inequality with the minus sign in (34) reads $(\gamma^2 - \Delta_c)(\cosh v - 1) \geq (\theta - 2c)\gamma^2 \sinh v/c$, $v = 2\pi c/\gamma$; it fails if $\gamma^2 \leq \Delta_c$, and otherwise, since $(\cosh v - 1)/\sinh v = \tanh(v/2) \leq v/2$, it forces $\pi c/\gamma \geq (\theta - 2c)/c$, i.e. $\gamma \leq \pi c^2/(\theta - 2c)$. As $c \downarrow \theta/3$, $c\Delta_c/(3c - \theta) \to \infty$ (because $\Delta_{\theta/3} > 0$ when $\theta/3 < c_Q$, by Lemma B.1(a)) while $\pi c^2/(\theta - 2c) \to \pi\theta/3$; choose $c_\sharp$ so that the first bound exceeds the square of the second on $(\theta/3, c_\sharp]$.

For the consequence, suppose such a solution exists. Replacing f by $f(r\,\cdot)/r^2 \in \mathcal{F}_m(H)$ and x by x/r, $r = \|a\|$, we may assume $\|a\| = \|b\| = 1$, and we identify the plane P spanned by a, b with $\mathbb{C}$ so that $x(t) = e^{zt}$, $z = -c + i\gamma$. Then $\nabla f_m(x(t)) = -\ddot{x} - \theta\dot{x} - mx = G(t)$ with $\nu = \delta_\gamma$ in (32); it lies in P, so it is also the gradient of the restriction of f_m to P, a convex function with $f_m(0) = 0$ and

$\nabla f_m(0) = 0$. Since $\frac{d}{dt} f_m(x(t)) = \langle G, \dot{x}\rangle = \dot{\eta}$ and both $f_m(x(t))$ and $\eta(t)$ tend to 0, $f_m(x(t)) = \eta(t)$. As $c \le \theta/2$, Lemma B.2 shows that δ_γ satisfies (34)–(35), a contradiction. □

Remark B.5 (The single-spiral cone). Let $0 < c < \theta/2$. For one atom $\gamma > 0$ with $\gamma^2 > \Delta_c$, put $\hat{\gamma} = \gamma/c$ and $\tan\phi = (\theta - 2c)\gamma/(\gamma^2 - \Delta_c)$, the (unsigned) angle between G and x. The inequality with the minus sign in (34) and (35) read

$$\tan\phi \le \Upsilon(\hat{\gamma}) := \inf_{x>0} \frac{\cosh x - \cos\hat{\gamma}x}{\hat{\gamma}\sinh x - \sin\hat{\gamma}x}, \qquad \hat{\gamma}\tan\phi \le 1.$$

A spiral with gradient twist ϕ and winding ratio $\hat{\gamma}$ is admissible if and only if both hold. When $\theta/3 < c_Q$, approaching $c = \theta/3$ from above requires $\hat{\gamma} \to \infty$ by the second inequality, while the first fails at the first return $q = 2\pi/\gamma$ of a fast winding. Indeed, for one atom the inequality with the minus sign in (34), divided by $(\gamma^2 - \Delta_c)\mathcal{B}_\gamma(q)$, with $x = cq$, reads $(\cosh x - \cos\hat{\gamma}x)/(\hat{\gamma}\sinh x - \sin\hat{\gamma}x) \ge \tan\phi$ for all $x > 0$, and (35) reads $(\theta - 2c)\gamma^2 \le c(\gamma^2 - \Delta_c)$, i.e. $\hat{\gamma}\tan\phi \le 1$; the inequality with the plus sign holds because $\mathcal{A}_\gamma, \mathcal{B}_\gamma \ge 0$.

C. Proofs for Section 4

Throughout this appendix, $x^\star = 0$. We compute the wave interpolation conditions, prove duality and the one-exponent principle, and construct the realizations. The first computation extends [32, Lemma A.3] to arbitrary polynomial amplitudes u, p.

Lemma C.1 (Exact spectral interpolation)**.** *Let ν be a probability measure on $[0,\pi]$, $C \in \mathbb{R}$, $t \in \mathbb{R}$, and let $H_\nu = L^2([0,\pi],\nu;\mathbb{C})$ with $\langle v, v'\rangle = \mathrm{Re}\int v\,\overline{v'}\,d\nu$. For $z = e^{t+i\omega}$ and complex polynomials (or constants) u, p, put $w_k(\omega) = z^k$, $\tilde{x}_k = u(z)z^k$, $\sigma_k = p(z)z^k$ and $\psi_k = \frac{1}{4}(C + 2\int \mathrm{Re}(p\bar{u})\,d\nu)e^{2kt}$. Then, with g, s, a_n, b_n as in (17)–(18), for $j \ge 0$, $n \ge 1$,*

$$R_{j,j+n} \pm T_{j,j+n} = \tfrac{1}{2}e^{(2j+n)t}(\bar{a}_n \mp \bar{b}_n \pm C\sinh nt), \qquad R_{j\star} \pm T_{\star j} = \tfrac{1}{4}e^{2jt}(\bar{g} \pm C). \tag{40}$$

Hence the data $(\tilde{x}_k, \sigma_k, \psi_k)_{k\ge 0}$ together with $(0,0,0)$ are interpolated by some $\psi \in \mathcal{N}(H_\nu)$ if and only if (ν, C) is feasible at t.

Proof. All expressions are ν-averages of the one-frequency expressions, so first take $\nu = \delta_\omega$, and put $m = j + n$. The identities

$$|z^n - 1|^2 = 2e^{nt}(\cosh nt - \cos n\omega), \qquad (1 + z^n)(\bar{z}^n - 1) = (e^{2nt} - 1) - 2ie^{nt}\sin n\omega$$

give

$$R_{jm} = \tfrac{1}{2}e^{(2j+n)t}a_n, \qquad \langle \sigma_j + \sigma_m, \tilde{x}_m - \tilde{x}_j\rangle = e^{2jt}[\mathrm{Re}(p\bar{u})(e^{2nt} - 1) + 2se^{nt}\sin n\omega].$$

Subtracting half of the second expression from $\psi_m - \psi_j = \frac{1}{4}(C + 2\,\mathrm{Re}(p\bar{u}))e^{2jt}(e^{2nt} - 1)$ cancels $\mathrm{Re}(p\bar{u})$ and yields

$$T_{jm} = \tfrac{1}{2}e^{(2j+n)t}(C\sinh nt - b_n).$$

For the minimizer pair, $R_{j\star} = \frac{1}{4}ge^{2jt}$ and $T_{\star j} = \psi_j - \frac{1}{2}\langle \sigma_j, \tilde{x}_j\rangle = \frac{1}{4}Ce^{2jt}$. Averaging proves (40). Since R is symmetric and T antisymmetric, nonnegativity of these slacks is exactly the pairwise interpolation condition of Theorem 4.2, including the minimizer pairs. This proves the equivalence with feasibility. □

Lemma C.2 (Phase of the lag-one symbol)**.** *Let (u, v) be a stationary linear method, $t \in \mathbb{R}$ and $\omega \in (0,\pi]$, and let $(g, s) = (g, s)(t, \omega) \ne (0,0)$, $g - 2is = Re^{-i\Theta}$ with $R > 0$. Then*

$$\mathrm{Re}[(1 - e^{i\omega})(g - 2is)] = 2R\sin\tfrac{\omega}{2}\,\sin\Big(\frac{\omega - 2\Theta}{2}\Big), \qquad \mathrm{Re}[(1 - e^{-i\omega})(g - 2is)] = 2R\sin\tfrac{\omega}{2}\,\sin\Big(\frac{\omega + 2\Theta}{2}\Big). \tag{41}$$

In particular, at $t = 0$ the lag-one symbol $\mathrm{Re}[(1 - e^{i\omega})\phi_\mu\overline{\phi_L}]$ is a positive multiple of $\sin(\Theta(\omega) - \frac{\omega}{2})$.

Proof. Since $1-e^{\pm i\omega} = 2\sin\frac{\omega}{2}\, e^{\pm i(\omega-\pi)/2}$, the left-hand sides are $2R\sin\frac{\omega}{2}\cos(\pm\frac{\omega-\pi}{2}-\Theta)$, and $\cos(\frac{\omega-\pi}{2}-\Theta) = \sin(\frac{\omega}{2}-\Theta)$, $\cos(\frac{\pi-\omega}{2}-\Theta) = \sin(\frac{\omega}{2}+\Theta)$. At $t=0$, $g-2is = -\frac{4}{(L-\mu)^2}\phi_\mu\overline{\phi_L}$ by (22) (for $\mathcal{M}$, (15)), so the lag-one symbol is $\frac{(L-\mu)^2}{2}R\sin\frac{\omega}{2}\sin(\Theta-\frac{\omega}{2})$. □

Lemma C.3 (Compactness)**.** *For every compact interval $I \subset \mathbb{R}$, the set of (t,ν,C) with $t \in I$ and (ν,C) feasible at t is compact (weak topology on ν).*

Proof. Put $G = \max_{I\times[0,\pi]}|g|$. Feasibility gives $|C| \le G$, so the set lies in the compact product $I \times \mathcal{P}([0,\pi]) \times [-G,G]$, where $\mathcal{P}([0,\pi])$ denotes probability measures with the weak topology. Every constraint is closed: its integrand is jointly continuous and hence varies uniformly with t, while integration is continuous under weak convergence of probability measures. □

Lemma C.4 (Duality)**.** (19) *is infeasible at t if and only if a strict certificate at t exists.*

Proof. Apply the compact-moment separation argument of [32, Theorem 2.5(a)]. By Lemma C.3, infeasibility is witnessed by finitely many constraints, including the two minimizer constraints. Finite-dimensional strict separation of their moment vectors gives nonnegative weights whose coefficient of C is zero and whose combined integrand is uniformly negative. These are exactly the two conditions in (20). Conversely, integrating a strict certificate against a feasible measure would give $0 \le \int \Phi_t\, d\nu < 0$. □

Lemma C.5 (Strict certificates persist)**.** *Let $(\alpha_n,\beta_n,\gamma_\pm)$ be a strict certificate at t, with margin ϵ. There is $\delta > 0$ such that, for any method and curvature interval with symbols g', s' at t',*

$$|t'-t| \le \delta, \qquad \sup_{\omega\in[0,\pi]} \max\{|g'(t',\omega)-g(t,\omega)|, |s'(t',\omega)-s(t,\omega)|\} \le \delta$$

imply a strict certificate with the same lag weights and margin $\epsilon/2$. Put $e = \sum_n(\alpha_n-\beta_n)(\sinh nt' - \sinh nt)$. Increase γ_- by e if $e \ge 0$, and γ_+ by $-e$ otherwise. For $t' = t = 0$, both minimizer weights are unchanged.

Proof. With finitely many fixed lag weights, the symbol is continuous in (t,g,s), uniformly in ω. The residual e defined in the statement tends to zero as $t' \to t$. The prescribed correction restores cancellation and adds $|e|g'$ to the symbol, also tending uniformly to zero. For sufficiently small δ, the strict margin is therefore at least $\epsilon/2$. At $t' = t = 0$, $e = 0$. □

Lemma C.6 (Certificate IQC)**.** *Let a strict certificate at t be given, put $r = e^t$, and define*

$$\mathsf{A}(\omega) = \sum_n(\alpha_n+\beta_n)(\cosh nt - \cos n\omega) + \gamma_+ + \gamma_-,$$

$$\mathsf{B}(\omega) = \sum_n(\alpha_n-\beta_n)\sin n\omega, \qquad \Pi(\omega) = \begin{pmatrix} \mathsf{A} & i\mathsf{B} \\ -i\mathsf{B} & -\mathsf{A}\end{pmatrix}.$$

(a) *For every $\psi \in \mathcal{N}(H)$ and sequence with $\check{x}_k = r^{-k}\tilde{x}_k \in \ell^2$, put $\check{\sigma}_k = r^{-k}\nabla\psi(\tilde{x}_k)$, and extend both sequences by zero to negative times. Then*

$$\mathcal{J}(\check{x},\check{\sigma}) := \frac{1}{2\pi}\int_{-\pi}^{\pi}[\hat{x};\hat{\sigma}]^*\Pi(\omega)[\hat{x};\hat{\sigma}]\, d\omega \ge 0, \qquad \hat{x}(\omega) = \sum_k \check{x}_k e^{-ik\omega},$$

The Fourier series are understood in L^2, with the same convention for $\hat{\sigma}$. (b) *For any $\mathsf{u},\mathsf{p} \in \mathbb{C}$, $[\mathsf{u};\mathsf{p}]^*\Pi(\omega)[\mathsf{u};\mathsf{p}]$ equals the certificate symbol computed with $g = |\mathsf{u}|^2 - |\mathsf{p}|^2$, $s = \mathrm{Im}(\mathsf{p}\bar{\mathsf{u}})$ and signed frequency ω. In particular, for the real method polynomials,*

$$[u(re^{i\omega});p(re^{i\omega})]^*\Pi(\omega)[u(re^{i\omega});p(re^{i\omega})] = \Phi_t(|\omega|) \qquad (|\omega| \le \pi).$$

Proof. Use the interpolation-to-IQC construction of [32, Lemma D.4]. With $M = \max(\{n : \alpha_n + \beta_n > 0\} \cup \{0\})$, extend $\tilde{x}_k, \sigma_k, \psi_k$ by zero for $k < 0$. The interpolation inequalities give

$$\mathcal{Q} = \sum_{k \geq -M} r^{-2k} \Big[\sum_{n \leq M} 2r^{-n} \{\alpha_n (R+T)_{k,k+n} + \beta_n (R-T)_{k,k+n}\} + 4\gamma_+ (R+T)_{\star k} + 4\gamma_- (R-T)_{\star k} \Big] \geq 0.$$

This sum converges absolutely by $\|\sigma_k\| \leq \|\tilde{x}_k\|$, $|\psi_k| \leq \frac{1}{2}\|\tilde{x}_k\|^2$ and $\check{x} \in \ell^2$. The coefficient of each ψ_m, including the zero-extension terms, is

$$4r^{-2m} \Big[\sum_n (\alpha_n - \beta_n) \sinh nt + \gamma_+ - \gamma_- \Big] = 0.$$

On harmonics $(\check{x}_k, \check{\sigma}_k) = (\mathsf{u}, \mathsf{p}) e^{ik\omega}$, expansion of the remaining finite-lag form gives $\mathsf{A}(|\mathsf{u}|^2 - |\mathsf{p}|^2) - 2\mathsf{B}\,\mathrm{Im}(\mathsf{p}\bar{\mathsf{u}}) = [\mathsf{u}; \mathsf{p}]^* \Pi [\mathsf{u}; \mathsf{p}]$. Parseval gives $\mathcal{Q} = \mathcal{J}$, proving both parts. For real u, p, the lag terms are even in ω, giving the last identity. □

Lemma C.7 (Homotopy)**.** *Let $\mathcal{G}$ be convolution with a real impulse response $\mathsf{h} = (\mathsf{h}_n)_{n \geq 0}$, $\sum_{n \geq 0} |\mathsf{h}_n| < \infty$, $\mathsf{h}_0 = 0$, acting coordinatewise on $\mathbb{R}^d$-valued sequences, and let G be its transfer function. This includes every real rational transfer function whose poles lie in the open unit disk and whose numerator has lower degree than its denominator (stable and strictly proper). Let Δ be a causal map on all one-sided sequences satisfying $\|P_N \Delta v\| \leq \|P_N v\|$, where P_N retains the indices $0, \dots, N-1$. Let Π be a bounded measurable Hermitian 2×2 matrix function, and suppose $\mathcal{J}(v, \vartheta \Delta v) \geq 0$ for every $v \in \ell^2$, $\vartheta \in [0,1]$, and*

$$[G(e^{i\omega}); 1]^* \Pi(\omega) [G(e^{i\omega}); 1] \leq -\epsilon' < 0 \qquad (-\pi \leq \omega \leq \pi).$$

Then $v = \mathcal{G}\vartheta\Delta v + v^0$ has a unique causal solution and $\|v\|_{\ell^2} \leq c_* \|v^0\|_{\ell^2}$ for every $\vartheta \in [0,1]$, $v^0 \in \ell^2$. The gain c_* depends only on G, $\sup\|\Pi\|$ and ϵ', and is independent of d, Δ and ϑ.

Proof. Use the discrete-time IQC continuation argument [32, Lemma D.5], following [27, Theorem 1]. The zero direct term gives a unique causal solution. Parseval and the strict frequency inequality give an a priori gain $\|v\| \leq C\|v^0\|$ for square-summable solutions, uniformly in ϑ. For continuation from ϑ_0 to $\vartheta_0 + \delta$, truncation and causality give

$$\|P_N v\| \leq C\|v^0\| + C\delta\|\mathcal{G}\|\|P_N \Delta v\| \leq C\|v^0\| + \tfrac{1}{2}\|P_N v\|$$

when $\delta \leq (2C\|\mathcal{G}\|)^{-1}$. This proves square summability; the a priori estimate restores the same gain C. Finitely many steps cover $[0,1]$. The scalar transfer function acts coordinatewise, so every constant is independent of d. □

Proof of Theorem 4.7(c). Let a strict certificate exist at t, and let $r = e^t > r_c$.

Reduction to finite dimension. The compression argument in the proof of Remark 2.4 preserves each finite trajectory and its state norms. It therefore suffices to obtain a bound in $\mathbb{R}^d$ uniform in d.

Feedback form. Fix $H = \mathbb{R}^d$, $\psi \in \mathcal{N}(H)$ and a trajectory with initial state S_0. Write $w = w^\sharp + w^0$, where w^0 solves $\sum_j p_j w^0_{k+j} = 0$ with initial state S_0, and $w^\sharp$ solves $\sum_j p_j w^\sharp_{k+j} = \sigma_k$ with zero initial state. Then $(\tilde{x}_k) = \mathcal{G}(\sigma_k) + (\tilde{x}^0_k)$, where $\tilde{x}^0_k = \sum_j u_j w^0_{k+j}$ is a free response and $\mathcal{G}$ is the causal convolution with transfer function $u(z)/p(z)$. After the scaling $\check{x}_k = r^{-k}\tilde{x}_k$, $\check{\sigma}_k = r^{-k}\sigma_k$, the transfer function becomes $G_t(z) = u(rz)/p(rz)$. It is stable, because the roots of p have modulus at most $r_c < r$, and strictly proper, because $\deg u < \deg p$. The free response $\check{x}^0$ is a combination of r^{-k}-scaled modes of p, so $\|\check{x}^0\|_{\ell^2} \leq c_1\|S_0\|$. The map $\Delta : \check{x} \mapsto (r^{-k}\nabla\psi(r^k\check{x}_k))_k$ is causal, memoryless, 1-Lipschitz in each step, and $\Delta 0 = 0$; moreover $\vartheta\Delta$ is the same map for $\vartheta\psi \in \mathcal{N}(H)$. Hence Lemma C.6(a) gives $\mathcal{J}(\check{x}, \vartheta\Delta\check{x}) \geq 0$ for all $\check{x} \in \ell^2$ and $\vartheta \in [0,1]$. By Lemma C.6(b) with $\mathsf{u} = u(re^{i\omega})$, $\mathsf{p} = p(re^{i\omega})$,

$$[G_t; 1]^* \Pi [G_t; 1] = \frac{\Phi_t(|\omega|)}{|p(re^{i\omega})|^2} \leq -\frac{\epsilon}{\max_{|z|=r} |p|^2}.$$

An ℓ^2 bound. Lemma C.7 establishes square summability of the causal feedback solution. With $\vartheta = 1$ and $v^0 = \check{x}^0$, it yields $\|\check{x}\| \le c_*\|\check{x}^0\| \le c_* c_1 \|S_0\|$. Then $\|\check{\sigma}\| \le \|\check{x}\|$, and since $w^\sharp$ is the output of the stable filter $1/p(rz)$ applied to $\check{\sigma}$,

$$\sum_{k\ge 0} r^{-2k}\|w_k\|^2 \le C_0\|S_0\|^2, \tag{42}$$

with C_0 independent of H, ψ and the trajectory.

From ℓ^2 to a rate. Put $\mathsf{E}_k = \sum_{j\ge k} r^{-2j}\|w_j\|^2$. Apply (42) after restarting at S_k:

$$\mathsf{E}_k \le C_0 r^{-2k}\|S_k\|^2 \le C'(\mathsf{E}_k - \mathsf{E}_{k+D}), \qquad C' = C_0 \max(1, r^{2(D-1)}),$$

because $r^{-2k}\|S_k\|^2 = \sum_{j<D} r^{-2k}\|w_{k+j}\|^2 \le \max(1, r^{2(D-1)})(\mathsf{E}_k - \mathsf{E}_{k+D})$. Increasing C_0 if necessary, assume $C' > 1$. Then $\mathsf{E}_{k+D} \le \rho_D \mathsf{E}_k$ with $\rho_D = 1 - 1/C' < 1$. Hence $\mathsf{E}_{mD} \le \rho_D^m \mathsf{E}_0 \le \rho_D^m C_0 \|S_0\|^2$, and $\|S_N\|^2 \le \max(1, r^{2(D-1)}) r^{2N} \mathsf{E}_N \le C'' r^{2N} \rho_D^{\lfloor N/D \rfloor}\|S_0\|^2$. Thus $\Gamma(N) \le (C'')^{1/2} r^N \rho_D^{(N/D-1)/2}$ and $\rho_{\mathcal{F}} \le r\rho_D^{1/(2D)} < r$. The second statement of (c) is proved after (b). □

Proof of Theorem 4.7(d). By Lemma C.1 there is $\psi \in \mathcal{N}(H_\nu)$ with $\nabla\psi(\tilde{x}_k) = \sigma_k$ for all $k \ge 0$. Since $\sigma_k = p(z)z^k = \sum_j p_j w_{k+j}$, this is the normalized form of (12) (proof of Lemma 2.3), and by Lemma 4.1, (w_k) is a trajectory on the corresponding $f \in \mathcal{F}_{\mu,L}(H_\nu)$. Multiplication by $e^{i\omega}$ is an orthogonal map U of H_ν^D, and $S_{k+1} = e^t U S_k$, so $\|S_N\| = e^{Nt}\|S_0\|$. If ν has J atoms, $H_\nu \cong \mathbb{C}^J \cong \mathbb{R}^{2J}$. □

Proof of Theorem 4.7(a), (b) and the end of (c). (a) Let $|z| = \rho_q$ with $v(z) = \lambda u(z)$, $\lambda \in [\mu, L]$; replacing z by $\bar{z}$ we may assume $\omega = \arg z \in [0, \pi]$. If $u(z) \ne 0$, then $p(z) = m\,u(z)$ with $m = \frac{L+\mu-2\lambda}{L-\mu} \in [-1, 1]$; if $u(z) = 0$, then also $v(z) = p(z) = 0$. In both cases $g \ge 0$ and $s = 0$, so for $\nu = \delta_\omega$, $C = 0$, one has $a_n \ge 0$, $b_n = 0$, and the pair is feasible at $\log\rho_q$. Thus $\log\rho_q \in \mathcal{T}$. If $t \in \mathcal{T}$, part (d) gives $\Gamma(n) \ge e^{nt}$, so $t \le \log C_\Gamma$ by the proof of Lemma 2.3. By Lemma C.3, $\mathcal{T} \cap I$ is compact for every compact I; hence $\mathcal{T}$ is closed, and it is bounded above. (b) By (d), $\rho_{\mathcal{F}} \ge e^T$. Let $t > T$. Then $t \notin \mathcal{T}$, and Lemma C.4 gives a strict certificate at t; since $e^t > e^T \ge \rho_q \ge r_c$, part (c) gives $\rho_{\mathcal{F}} < e^t$. Letting $t \downarrow T$, $\rho_{\mathcal{F}} \le e^T$. At $t \in \mathcal{T}$ a strict certificate is excluded by Lemma C.4. (c), second statement: if $\log r_c < t \le T$ and $t \notin \mathcal{T}$, a strict certificate at t would give $\rho_{\mathcal{F}} < e^t \le e^T$, contradicting (b). Hence $(\log r_c, T] \subset \mathcal{T}$, and the reverse inclusion holds since $T = \max\mathcal{T}$. □

Normalized lag functionals. For part (e), normalize the lag functionals to obtain a limit as $n \to \infty$. For $t \ne 0$ let $N_n(t) = \sinh(n|t|)/\sinh|t|$ and $\mathsf{L}_n = (a_n - b_n)/N_n$, $\mathsf{U}_n = (a_n + b_n)/N_n$. Then

$$\ell_n/N_n = \bar{\mathsf{L}}_n + \operatorname{sgn}(t)\sinh|t|\,C, \qquad \upsilon_n/N_n = \bar{\mathsf{U}}_n - \operatorname{sgn}(t)\sinh|t|\,C,$$

so the coefficient of C is independent of n. As $n \to \infty$, uniformly for $\omega \in [0, \pi]$ and $|t|$ in compact subsets of $(0, \infty)$, $\cosh(nt)/N_n \to \sinh|t|$ while $\cos n\omega/N_n, \sin n\omega/N_n \to 0$; hence $\mathsf{L}_n, \mathsf{U}_n \to g\sinh|t|$, and the limit constraints are the minimizer inequalities $\bar{g} \pm C \ge 0$ multiplied by $\sinh|t|$. Let $\mathcal{I} = \{\mathsf{L}, \mathsf{U}\} \times (\mathbb{N} \cup \{\infty\})$, where $\mathbb{N}\cup\{\infty\}$ is the one-point compactification. For $i = (\mathsf{L}, n)$, put $\phi_i = \mathsf{L}_n$, $\varpi_i = \operatorname{sgn}(t)\sinh|t|$; for $i = (\mathsf{U}, n)$, put $\phi_i = \mathsf{U}_n$, $\varpi_i = -\operatorname{sgn}(t)\sinh|t|$. Thus constraint i is $\int \phi_i(t,\omega)\,d\nu + \varpi_i(t)C \ge 0$, including the two minimizer constraints at infinity. The two distinct labels have the same limiting function, $\phi_{(\mathsf{L},\infty)} = \phi_{(\mathsf{U},\infty)}$. The functionals and coefficients are jointly continuous for $t \ne 0$.

Lemma C.8 (Finite support)**.** *Let $T = \max\mathcal{T} \ne 0$ and $T > \log\rho_q$. There are a probability measure Λ on $\mathcal{I}$ and a function $\Phi_T = \int \phi_i(T,\cdot)\,d\Lambda(i) \le 0$ on $[0,\pi]$, analytic in a complex neighbourhood of $[0,\pi]$, with $\Phi_T(0) < 0$, $\Phi_T(\pi) < 0$, such that every pair feasible at T is carried by $\{\Phi_T = 0\}$, a finite subset of $(0,\pi)$, and Λ is carried by constraints active for every pair feasible at T.*

Proof. A limiting certificate. Choose $t_k \downarrow T$ with $0 < t_k - T < |T|/2$. Part (b) gives strict certificates at t_k. Written in the normalized functionals and divided by their total weight, they are probability measures Λ_k on $\mathcal{I}$ whose coefficient of C vanishes and with $\int \phi_i(t_k,\omega)\,d\Lambda_k(i) < 0$ for all ω. A weak

limit point Λ exists by compactness of $\mathcal{I}$; by joint continuity, $\Phi_T(\omega) = \int \phi_i(T,\omega)\, d\Lambda(i) \le 0$, and the coefficient of C still vanishes.

Support. For (ν, C) feasible at T, integrate against ν and use cancellation of the C-coefficient: $\int \Phi_T\, d\nu = \int \mathrm{slack}_i\, d\Lambda(i) \ge 0$, where $\mathrm{slack}_i \ge 0$ is the normalized slack of constraint i. Since $\Phi_T \le 0$, both sides vanish: ν is carried by $\{\Phi_T = 0\}$, and Λ by the constraints with zero slack.

Analyticity. The functions $g(T,\cdot)$, $s(T,\cdot)$ are trigonometric polynomials. In the strip $|\operatorname{Im}\omega| < |T|/2$, $|\cos n\omega|, |\sin n\omega| \le \cosh(n|T|/2)$, and $\cosh(n|T|/2)/N_n(T) \to 0$ geometrically, so the normalized functionals are analytic and uniformly bounded there and converge uniformly on compact subsets. Hence Φ_T is analytic in the strip.

Endpoints. At $\omega = 0$, $z = e^T$ is real, $s = 0$ and $g(T,0) = u(e^T)^2 - p(e^T)^2$. If $u(e^T) \ne 0$ and $g(T,0) \ge 0$, then $p(e^T) = m\, u(e^T)$ with $m \in [-1,1]$, i.e. $v(e^T) = \lambda' u(e^T)$ with $\lambda' = \frac{L+\mu}{2} - \frac{L-\mu}{2} m \in [\mu, L]$, so $e^T \le \rho_q$, a contradiction. If $u(e^T) = 0 = p(e^T)$, then also $v(e^T) = 0$, making e^T a root of $v - \lambda u$ for every $\lambda \in [\mu, L]$, again contrary to $e^T > \rho_q$. Thus $u(e^T) = 0$ implies $g(T,0) = -p(e^T)^2 < 0$. In all cases $g(T,0) < 0$; every normalized functional is a positive multiple of g at $\omega = 0$ (there $b_n = 0$ and $\cosh nT - 1 > 0$), and $\Phi_T(0) < 0$. The same argument at $z = -e^T$ gives $\Phi_T(\pi) < 0$. A nonzero analytic function has finitely many zeros in $[0,\pi]$. □

Lemma C.9 (Strict minimizer slack)**.** *Under the hypotheses of Lemma C.8, every pair feasible at T has $|C| < \bar{g}$.*

Proof. By Lemma C.8, $\nu = \sum_{j \le J} \nu_j \delta_{\omega_j}$ with distinct $\omega_j \in (0,\pi)$. Put $F_\pm(n) = \sum_j \nu_j (g_j \cos n\omega_j \pm 2 s_j \sin n\omega_j)$. Then $\ell_n = \bar{g}\cosh nT - F_+(n) + C\sinh nT$ and $\upsilon_n = \bar{g}\cosh nT - F_-(n) - C\sinh nT$. Suppose $|C| = \bar{g}$. If $C = -\operatorname{sgn}(T)\bar{g}$, then $\ell_n \ge 0$ reads $F_+(n) \le \bar{g} e^{-n|T|}$; if $C = \operatorname{sgn}(T)\bar{g}$, then $\upsilon_n \ge 0$ reads $F_-(n) \le \bar{g} e^{-n|T|}$. In either case write $F = F_\pm = \operatorname{Re}\sum_j c_j e^{in\omega_j}$ with $c_j = \nu_j(g_j \mp 2 i s_j)$.

Each $c_j \ne 0$: $g_j = s_j = 0$ means $|u| = |p|$ and $\operatorname{Im}(p\bar{u}) = 0$ at z_j, i.e. $p\bar{u} = \pm|u|^2$. If $u(z_j) \ne 0$, this gives $p/u = \pm 1$; if $u(z_j) = 0$, then $p(z_j) = v(z_j) = 0$. In either case $v(z_j) = \mu u(z_j)$ or $v(z_j) = L u(z_j)$, a quadratic mode with $|z_j| = e^T > \rho_q$, which is impossible.

Distinct $\omega_j \in (0,\pi)$ make every nonconstant harmonic in F and F^2 nonzero modulo 2π. Thus the Cesàro means of F tend to 0, and those of F^2 tend to $\frac{1}{2}\sum_j |c_j|^2 > 0$. On the other hand, $F \le \bar{g} e^{-n|T|}$ makes the Cesàro means of the positive part F^+ tend to 0; then so do those of $F^- = F^+ - F$ and of $|F| = F^+ + F^-$, and since $|F| \le K$, the means of $F^2 \le K|F|$ tend to 0. This contradiction proves $|C| < \bar{g}$. □

Lemma C.10 (Atom count)**.** *Under the hypotheses of Lemma C.8, every pair feasible at T has finitely many active lags; if M is the largest active lag of one of them, every pair feasible at T is carried by at most $\frac{1}{2}(D+M)$ points.*

Proof. With $\eta = \bar{g} - |C| > 0$ (Lemma C.9) and $K = \max|g| + 2\max|s|$, $\ell_n, \upsilon_n \ge \bar{g}\cosh nT - |C|\sinh n|T| - K \ge \eta\cosh nT - K > 0$ for large n, so only finitely many lags are active. The probability measure Λ of Lemma C.8 is carried by active constraints, and strict minimizer slack excludes its two infinity labels. Hence every feasible pair has at least one active finite lag. Fix one such pair and let M be its largest active lag. Then Λ is carried by $\{(\mathsf{L}, n), (\mathsf{U}, n) : n \le M\}$. Since $\deg u < D$ and $\deg p = D$, $g(T,\cdot)$ is an even and $s(T,\cdot)$ an odd trigonometric polynomial of degree at most D. So a_n and b_n are even trigonometric polynomials of degree at most $D+n$, and $\Phi_T = P(\cos\omega)$ with a polynomial P of degree at most $D+M$. Now $P \le 0$ on $[-1,1]$, $P \not\equiv 0$ since $\Phi_T(0) < 0$, and every zero of P in $(-1,1)$ is a local maximum, hence of even multiplicity. So P has at most $\frac{1}{2}(D+M)$ zeros in $(-1,1)$, and by Lemma C.8 every feasible pair is carried by them. □

Proof of Theorem 4.7(e). If $T = \log\rho_q$, choose a maximizing root $v(z) = \lambda u(z)$, $\lambda \in [\mu, L]$. The wave $w_k = z^k$ is a trajectory on the quadratic $\frac{\lambda}{2}\|\cdot\|^2$ on $\mathbb{C} \cong \mathbb{R}^2$, and its state norm grows exactly by e^T at each step. If $T > \log\rho_q$ and $T \ne 0$, Lemmas C.8 to C.10 show that every pair feasible at T has

$|C| < \bar{g}$, finitely many active lags, and at most $\frac{1}{2}(D+M)$ atoms in $(0,\pi)$; part (d) realizes it on $\mathbb{R}^{2J}$, $J \le \frac{1}{2}(D+M)$. □

Lemma C.11 (Start from rest in $\mathcal{M}$)**.** *Let a tuning of $\mathcal{M}$ be stable, and let (ν, C) be feasible at t, with $\nu = \sum_{j=1}^{J} \nu_j \delta_{\omega_j}$, $\nu_j > 0$, and $|C| < \bar{g}$ if $t > 0$. Then there are $f \in \mathcal{F}_{\mu,L}(\mathbb{C}^J \oplus \mathbb{R})$ and $x_0 \neq 0$ such that the trajectory of (1) with $x_{-1} = x_0$ satisfies $\|x_N\| \ge c_0 e^{Nt}\|x_0\|$ for all $N \ge 0$, with $c_0 > 0$. The trajectory $(x_k)_{k\ge 0}$ is the orthogonal sum of the waves $x_k^{(j)} = \sqrt{\nu_j} z_j^{k+1}$, $z_j = e^{t+i\omega_j}$, and a sequence tending to 0 in the last coordinate.*

Proof. Following [32, Lemma A.8], prescribe the trajectory and its query data in latent coordinates. We verify their interpolation by a normalized objective; Lemma 4.1 then gives f.

Waves. Let $x_k^w = (\sqrt{\nu_j} z_j^{k+1})_j \in \mathbb{C}^J$ for $k \ge -1$, $z_j = e^{t+i\omega_j}$. The latent sequence $w_k = x_{k-1}^w$ is the wave z^k of Lemma C.1 in $H_\nu \cong \mathbb{C}^J$ (with the weights ν_j absorbed into the amplitudes). Hence the queries $\tilde{x}_k^w = (1+\tau)x_k^w - \tau x_{k-1}^w = (\sqrt{\nu_j} u(z_j) z_j^k)_j$, the normalized gradients $\sigma_k^w = (\sqrt{\nu_j} p(z_j) z_j^k)_j$ and the values ψ_k^w of Lemma C.1, $k \ge 0$, together with $(0,0,0)$, satisfy $|T| \le R$ for all pairs, and the waves satisfy (1) with these gradients.

Extra coordinate. For $\lambda \in (\mu, L)$ let $P_{-1} = P_0 = 1$, $Q_k = (1+\tau)P_k - \tau P_{k-1}$ and $P_{k+1} = (1+\beta)P_k - \beta P_{k-1} - h\lambda Q_k$: this is the run of (1) from rest on the scalar quadratic $\frac{\lambda}{2}v^2$, with iterates P_k and queries Q_k. By induction Q_k is a polynomial in λ of degree k with leading coefficient $(-h)^k(1+\tau)^k \neq 0$, so $Q_k(\lambda) = 1$ has finitely many roots for each $k \ge 1$; choose λ outside this countable set. By stability $Q_k \to 0$, so $(1-Q_k)^2 \ge \frac{1}{4}$ for large k, and $\eta_0 = \inf_{k\ge 1}(1-Q_k)^2 > 0$. In normalized form the quadratic $\frac{\lambda}{2}v^2$ is $\psi(v) = \frac{m_0}{2}v^2$ with $m_0 = \frac{L+\mu-2\lambda}{L-\mu} \in (-1,1)$. For data of a quadratic, $T_{ik} = 0$ for every pair, and $R_{ik} = \frac{1-m_0^2}{4}(v_i - v_k)^2 \ge 0$.

Trajectory. Put $x_{-1} = x_0 = (x_0^w, Y)$ with $Y > 0$ to be chosen. Then $\tilde{x}_0 = x_0$. At $\tilde{x}_0$ prescribe the gradient $((x_0^w - x_1^w)/h,\ \lambda Y)$ and the normalized value $\psi_0 = \frac{1}{2}\langle \sigma_0, \tilde{x}_0\rangle$, where σ_0 is the normalized gradient. Then $x_1 = x_0 - h\nabla f(\tilde{x}_0) = (x_1^w, (1-h\lambda)Y) = (x_1^w, P_1 Y)$. For $k \ge 1$, x_{k-1} and x_k have wave parts x_{k-1}^w, x_k^w, so $\tilde{x}_k = (\tilde{x}_k^w, Q_k Y)$; prescribe the gradient (wave gradient at $\tilde{x}_k^w$, $\lambda Q_k Y$) and the normalized value $\psi_k = \psi_k^w + \frac{m_0}{2}Q_k^2 Y^2$. By induction $x_k = (x_k^w, P_k Y)$ for all $k \ge 0$.

Interpolation. The data split orthogonally into the wave coordinates and the extra coordinate, and R and T are sums of the two contributions. In the extra coordinate, $T = 0$ and $R \ge 0$ for every pair, since $Q_0 = 1$ and the extra part of ψ_0 is $m_0 Y^2/2$. In the wave coordinates, pairs indexed in $\{\star, 1, 2, \ldots\}$ satisfy $|T| \le R$ by the first step. For the pairs $(0,k)$, $k \ge 1$, and $(\star, 0)$, the separately prescribed index-0 datum may cause a wave deficit $\max(0, |T| - R)$. This deficit is bounded uniformly in k. For $t \le 0$ all data are bounded. For $t > 0$, expand R_{0k} and T_{0k} around $R_{k\star}$ and $T_{\star k}$: $\|\tilde{x}_k - \tilde{x}_0\|^2 = \|\tilde{x}_k\|^2 - 2\langle \tilde{x}_k, \tilde{x}_0\rangle + \|\tilde{x}_0\|^2$ and similarly for σ, and $T_{0k} = T_{\star k} - T_{\star 0} - \frac{1}{2}\langle \sigma_0, \tilde{x}_k\rangle + \frac{1}{2}\langle \sigma_k, \tilde{x}_0\rangle$; the cross terms are $O(e^{kt})$ and the rest $O(1)$. By (40), $R_{k\star} - |T_{\star k}| = \frac{1}{4}(\bar{g} - |C|)e^{2kt}$, so $R_{0k} - |T_{0k}| \ge \frac{1}{4}(\bar{g} - |C|)e^{2kt} - O(e^{kt}) - O(1)$, which is bounded below because $\bar{g} - |C| > 0$. The extra coordinate adds $\frac{1-m_0^2}{4}(1-Q_k)^2 Y^2 \ge \frac{1-m_0^2}{4}\eta_0 Y^2$ to R_{0k} and $\frac{1-m_0^2}{4}Y^2$ to $R_{0\star}$, and nothing to T. One large Y makes all pairs interpolable, and Theorem 4.2 and Lemma 4.1 give ψ and f whose gradients at the queries are the prescribed ones; the trajectory of (1) on f from $x_{-1} = x_0$ is the constructed one.

Finally $\|x_N\| \ge \|x_N^w\| = (\sum_j \nu_j |z_j|^{2(N+1)})^{1/2} = e^{(N+1)t}$ and $\|x_0\| = (e^{2t} + Y^2)^{1/2}$, so $c_0 = e^t(e^{2t} + Y^2)^{-1/2}$. □

Lemma C.12 (From rest on a quadratic)**.** *Let (β, τ, h) be a tuning of $\mathcal{M}$, $\lambda > 0$, and let $\rho(\chi_\lambda)$ be the largest modulus of a root of χ_λ. The run of (1) from rest on $\frac{\lambda}{2}x^2$ on $\mathbb{R}$, with $x_{-1} = x_0 \neq 0$, satisfies $\limsup_{N\to\infty} |x_N|^{1/N} = \rho(\chi_\lambda)$. In particular $\rho_{\mathcal{F}}^0 \ge \rho(\chi_\lambda)$ for every $\lambda \in [\mu, L]$. If $\beta \neq \tau/(1+\tau)$, the queries $\tilde{x}_N = (1+\tau)x_N - \tau x_{N-1}$ of this run also satisfy $\limsup_{N\to\infty} |\tilde{x}_N|^{1/N} = \rho(\chi_\lambda)$.*

Proof. Normalize $x_0 = 1$, and write $\chi_\lambda(z) = z^2 - Sz + Q = (z - w_1)(z - w_2)$. The recurrence and $x_{-1} = x_0 = 1$ give the generating function

$$X(\zeta) := \sum_{N\geq 0} x_N \zeta^N = \frac{1 - Q\zeta}{1 - S\zeta + Q\zeta^2}.$$

A pole at $\zeta = 1/w_j$, $w_j \neq 0$, cannot cancel: the numerator there is $1 - w_{3-j} \neq 0$, since $\chi_\lambda(1) = h\lambda > 0$. The Cauchy–Hadamard formula therefore gives $\limsup_N |x_N|^{1/N} = \max_j |w_j|$. If both roots vanish, the series is a polynomial and the same conclusion holds with value zero. This proves the lower bound on $\rho^0_{\mathcal{F}}$.

For the queries,

$$\sum_{N\geq 0} \tilde{x}_N \zeta^N = ((1 + \tau) - \tau\zeta)X(\zeta) - \tau.$$

At a nonzero mode w_j, its multiplier is $N_\tau(w_j)/w_j$. This vanishes only if $w_j = \tau/(1 + \tau)$; substituting in χ_λ then gives $\beta = \tau/(1 + \tau)$. Under the stated exclusion no pole cancels, proving the query claim as well. □

Proof of Corollary 4.8. The first statement is Lemma C.11. If $T > \log \rho_q$ and $T \neq 0$, Theorem 4.7(e) provides a pair feasible at T with finitely many atoms and $|C| < \bar{g}$. Lemma C.11 realizes it from rest, so $\rho^0_{\mathcal{F}} \geq e^T = \rho_{\mathcal{F}} \geq \rho^0_{\mathcal{F}}$ by Lemma 2.3. If $T = \log \rho_q$, let ρ_q be attained by a root of χ_λ, $\lambda \in [\mu, L]$; Lemma C.12 gives $\rho^0_{\mathcal{F}} \geq \rho_q = e^T = \rho_{\mathcal{F}} \geq \rho^0_{\mathcal{F}}$, the worst case from rest being attained by $\frac{\lambda}{2}x^2$ on $\mathbb{R}$. □

Proof of Proposition 4.9. Let $r = e^t \leq 1$. Take $\alpha_1 = r$, all other α_n and all β_n equal to 0, $\gamma_- = 0$ and $\gamma_+ = -r \sinh t \geq 0$. The second condition of (20) holds: $\alpha_1 \sinh t + \gamma_+ = 0$. Since $r(\cosh t - \sinh t) = re^{-t} = 1$,

$$\Phi_t = r\big[g(\cosh t - \cos\omega) - 2s \sin\omega\big] - r \sinh t\, g = g(1 - r\cos\omega) - 2rs \sin\omega = \mathrm{Re}\big[(g - 2is)(1 - z)\big],$$

with $z = re^{i\omega}$, and by (15)

$$\Phi_t = -\frac{4\,\mathrm{Re}[(1 - z)\chi_\mu(z)\overline{\chi_L(z)}]}{h^2(L - \mu)^2}.$$

(a) The right-hand side is continuous and negative on the circle $|z| = r$, so $\Phi_t \leq -\epsilon$ for some $\epsilon > 0$, and Theorem 4.7(c) applies since $r > r_c$. (b) is (c) with $\lambda = 0$. (c) At $t = 0$ take in addition $\beta_2 = \lambda$; the second condition of (20) still holds since $\sinh 0 = 0$. At $t = 0$, $a_2 + b_2 = g(1 - \cos 2\omega) + 2s \sin 2\omega = \mathrm{Re}[(g - 2is)(1 - e^{-2i\omega})]$, so the symbol becomes

$$\Phi_0 = -\frac{4\,\mathrm{Re}[\chi_\mu \overline{\chi_L}\,((1 - z) + \lambda(1 - \bar{z}^2))]}{h^2(L - \mu)^2}, \qquad z = e^{i\omega}.$$

By hypothesis $\Phi_0 < 0$ on $(0, \pi]$ and $\Phi_0(0) = 0$. To obtain strict negativity on the full circle, add $\eta > 0$ to both γ_+ and γ_-; this keeps the second condition and adds $2\eta g$ to the symbol. At $z = 1$, $u(1) = 1$ and $p(1) = \frac{L+\mu}{L-\mu} > 1$, so $g(0, 0) < 0$, and $g \leq -g_1 < 0$ on some $[0, \omega_1]$, where $\Phi_0 \leq 0$; on $[\omega_1, \pi]$, $\Phi_0 \leq -\epsilon_1 < 0$. For $\eta \leq \epsilon_1/(4 \max |g|)$ the new symbol is at most $-\min(2\eta g_1, \epsilon_1/2) < 0$. Stability gives $r_c < 1$, and Theorem 4.7(c) applies at $t = 0$. □

Lemma C.13 (Separation of quadratic modes)**.** *Let a tuning of $\mathcal{M}$ and $r \in (0, 1)$ satisfy* $\mathrm{Re}[(1 - z)\chi_\mu(z)\overline{\chi_L(z)}] > 0$ *on $|z| = r$. Then no χ_λ, $\lambda \in [\mu, L]$, has a root on $|z| = r$. If moreover the roots of χ_μ lie in $|z| < r$, then $\rho_q < r$; in particular the tuning is stable, $r_c < r$, and $\rho_{\mathcal{F}} < r$.*

Proof. As χ_λ is affine in λ, $\chi_\lambda = (1 - \vartheta)\chi_\mu + \vartheta\chi_L$ with $\vartheta = (\lambda - \mu)/(L - \mu) \in [0, 1]$. Suppose $\chi_\lambda(z) = 0$ with $|z| = r$. If $\vartheta = 0$ or $\vartheta = 1$, then $\chi_\mu(z)\overline{\chi_L(z)} = 0$. Otherwise $\chi_\mu(z) = -\frac{\vartheta}{1-\vartheta}\chi_L(z)$, so $\chi_\mu\overline{\chi_L} = -\frac{\vartheta}{1-\vartheta}|\chi_L|^2 \leq 0$ at z, and $\mathrm{Re}[(1 - z)\chi_\mu\overline{\chi_L}] \leq 0$ because $\mathrm{Re}(1 - z) \geq 1 - r > 0$. Both contradict

the hypothesis. The roots of the monic quadratics χ_λ depend continuously on $\lambda \in [\mu, L]$ and never meet $|z| = r$; if those of χ_μ lie in $|z| < r$, so do those of every χ_λ. Hence $\rho_q < r < 1$, and $r_c \le \rho_q$ since r_c is the largest root modulus of $\chi_{(L+\mu)/2}$. Proposition 4.9(a) gives $\rho_{\mathcal{F}} < r$. □

Lemma C.14 (One-atom spectra)**.** *Let (u, v) be a stationary linear method, $K \ge 3$, $1 \le j < K/2$, $\gcd(j, K) = 1$ and $\omega_0 = 2\pi j/K$. For $t \in \mathbb{R}$ with $(g, s)(t, \omega_0) \ne (0, 0)$ write $g - 2is = |g - 2is|e^{-i\Theta^{(t)}}$ at (t, ω_0), $\Theta^{(t)} \in (-\pi, \pi]$.*

(a) *If $|\Theta^{(t)}| \le \pi/K$, then $g(t, \omega_0) > 0$ and $(\delta_{\omega_0}, 0)$ is feasible at t. At $t = 0$ the converse holds.*

(b) *If $(g, s)(0, \omega_0) \ne (0, 0)$ and $|\Theta(\omega_0)| < \pi/K$, then $(\delta_{\omega_0}, 0)$ is feasible at some $t > 0$, with $0 = |C| < \bar{g}$. Consequently $\rho_{\mathcal{F}} \ge e^t > 1$, attained by a trajectory in $\mathbb{C} \cong \mathbb{R}^2$ with $S_{k+1} = e^t U S_k$ (Theorem 4.7(d)), and for a stable tuning of $\mathcal{M}$ also $\rho^0_{\mathcal{F}} \ge e^t > 1$ (Lemma C.11).*

Proof. At $t = 0$, the criterion is [32, Proposition D.9]. (a) For $\nu = \delta_{\omega_0}$ and $C = 0$, $\bar{g} = g$, $g = |g - 2is| \cos \Theta^{(t)}$, and $2s = |g - 2is| \sin \Theta^{(t)}$, so

$$\ell_n = g(\cosh nt - 1) + |g - 2is|(\cos \Theta^{(t)} - \cos(n\omega_0 - \Theta^{(t)})),$$
$$\upsilon_n = g(\cosh nt - 1) + |g - 2is|(\cos \Theta^{(t)} - \cos(n\omega_0 + \Theta^{(t)})),$$

Since $\gcd(j, K) = 1$, the residues $m = nj \bmod K$ run through $0, \dots, K - 1$. The cosine differences above equal $2 \sin(\pi m/K) \sin(\pi m/K \mp \Theta^{(t)})$. For $|\Theta^{(t)}| \le \pi/K$, these are nonnegative, $g > 0 = |C|$, and $\cosh nt \ge 1$; hence the pair is feasible. At $t = 0$, the terms $g(\cosh nt - 1)$ vanish, and $|\Theta| > \pi/K$ makes the lower or upper slack at residue $m = 1$ negative. This proves the converse. (b) The values g, s at (t, ω_0) are continuous in t, and $g - 2is \ne 0$ at $t = 0$ with $\Theta(\omega_0) \in (-\pi/K, \pi/K)$, away from the branch point $\pm\pi$; so $\Theta^{(t)}$ is continuous near $t = 0$, and $|\Theta^{(t)}| < \pi/K$ for small $t > 0$. By (a), $(\delta_{\omega_0}, 0)$ is feasible at such t, with $\bar{g} = g > 0 = |C|$. Theorem 4.7(d) realizes it by a trajectory with $S_{k+1} = e^t U S_k$ in $\mathbb{C} \cong \mathbb{R}^2$, so $\Gamma(n) \ge e^{nt}$ and $\rho_{\mathcal{F}} \ge e^t$; for a stable tuning of $\mathcal{M}$, Lemma C.11 applies, as $|C| < \bar{g}$, and gives $\rho^0_{\mathcal{F}} \ge e^t$. □

Proof of Proposition 4.10. On the unit circle $u = N_\tau \ne 0$ (its root $\tau/(1 + \tau)$ lies inside the disk), and, by (15)–(16), $g - 2is = |g - 2is|e^{-i\Theta(\omega)}$ with $\Theta \in (0, \pi)$ on $(0, \pi)$. (a) By Lemma C.14(a), $(\delta_{\omega_0}, 0)$ is feasible at $t = 0$, and Lemma C.11 with one atom gives $f \in \mathcal{F}_{\mu,L}(\mathbb{C} \oplus \mathbb{R}) = \mathcal{F}_{\mu,L}(\mathbb{R}^3)$ and x_0. The trajectory is $x_k = (z^{k+1}, P_k Y)$, $k \ge 0$, with $z = e^{i\omega_0}$ and $z^K = 1$: its first component rotates by ω_0 in the plane $\mathbb{C}$ and is K-periodic, and $P_k \to 0$. Hence the trajectory approaches the K-periodic orbit $(z^{k+1}, 0)$, and $\|x_N\| \ge 1 = (1 + Y^2)^{-1/2}\|x_0\|$. (b) This is Lemma C.14(b), since $\Theta(\omega_0) \in (0, \pi/K)$. □

Proposition C.15 (Lyapunov form)**.** *Let a stable tuning of $\mathcal{M}$ admit a strict certificate at $t = \log r$, $r \in (r_c, 1]$, with lags at most $M \ge 1$. For any trajectory on $f \in \mathcal{F}_{\mu,L}(H)$, put*

$$\sigma_k = \nabla\psi(\tilde{x}_k - x^\star), \qquad \mathbf{X}_k = (x_{k-1} - x^\star, x_k - x^\star, \sigma_k, \dots, \sigma_{k+M-1}).$$

There is a function V_k, quadratic in $\mathbf{X}_k$ minus a linear combination of $\psi(\tilde{x}_{k+j} - x^\star)$, $0 \le j < M$, such that

$$c_-\|\mathbf{X}_k\|^2 \le V_k \le c_+\|\mathbf{X}_k\|^2, \qquad V_{k+1} \le r^2 V_k - \eta\|\mathbf{X}_k\|^2.$$

Its coefficients and the constants $c_-, c_+, \eta > 0$ depend only on the tuning, μ, L, M, and certificate, and work for every H, f and trajectory. From rest, $\|x_N - x^\star\| \le K_V r^N \|x_0 - x^\star\|$, with the same dependence for K_V. For the lag-one certificate of Proposition 4.9(a), $M = 1$, and V_k is quadratic in $\mathbf{X}_k$ plus

$$\frac{4}{L - \mu}\left(f(\tilde{x}_k) - f(x^\star) - \frac{L + \mu}{4}\|\tilde{x}_k - x^\star\|^2\right).$$

Proof of Proposition C.15. State space. Write $X_k = x_k - x^\star$ and $e_k = \tilde{x}_k - x^\star$, so that $\sigma_k = \nabla\psi(e_k)$ and $\mathbf{X}_k = (X_{k-1}, X_k, \sigma_k, \dots, \sigma_{k+M-1})$. Let $\lambda_+ = \frac{L+\mu}{2}$ and $\lambda_- = \frac{L-\mu}{2}$, so that $e_k = (1 + \tau)X_k - \tau X_{k-1}$

and, by (10), (1) becomes $X_{k+1} = S_+X_k - Q_+X_{k-1} + h\lambda_-\sigma_k$ with $S_+ = 1+\beta-h\lambda_+(1+\tau)$ and $Q_+ = \beta - h\lambda_+\tau$. With the input $d_k = \sigma_{k+M}$, $\mathbf{X}_{k+1} = \tilde{A}\mathbf{X}_k + \tilde{B}d_k$, where $\tilde{A}$ updates (X_{k-1}, X_k) by this recursion and shifts the gradients, and $\tilde{B}$ feeds d_k into the last slot. The matrix $\tilde{A}$ is block triangular; its eigenvalues are 0 (from the shift) and the roots of $z^2 - S_+z + Q_+ = \chi_{\lambda_+}(z)$, of modulus at most $r_c < r$. The pair $(\tilde{A}, \tilde{B})$ is controllable: $\tilde{A}^j\tilde{B}$, $0 \le j < M$, are the unit vectors of the gradient slots, $\tilde{A}^M\tilde{B} = (0, h\lambda_-, 0, \dots)$ and $\tilde{A}^{M+1}\tilde{B} = (h\lambda_-, S_+h\lambda_-, 0, \dots)$, and these $M+2$ vectors are linearly independent. Each e_{k+j}, $0 \le j \le M$, is a linear function of $\mathbf{X}_k$: X_{k+j} depends on X_{k-1}, X_k and $\sigma_k, \dots, \sigma_{k+j-1}$.

Supply and telescoping. Let $\mathcal{Q}_k$ be the combination

$$\mathcal{Q}_k = \sum_n 2r^{-n}\big(\alpha_n(R+T)_{k,k+n} + \beta_n(R-T)_{k,k+n}\big) + 4\gamma_+(R+T)_{\star k} + 4\gamma_-(R-T)_{\star k} \ \ge 0$$

of interpolation inequalities in the window $\{\star, k, \dots, k+M\}$. Its function values are $\sum_j \mathfrak{a}_j\psi(e_{k+j})$ with $\mathfrak{a}_j = 2r^{-j}(\alpha_j - \beta_j)$ for $1 \le j \le M$ and $\mathfrak{a}_0 = 4(\gamma_+ - \gamma_-) - \sum_{j\ge1}\mathfrak{a}_j$, and $\sum_j \mathfrak{a}_j r^{2j} = 4[\sum_n(\alpha_n - \beta_n)\sinh nt + \gamma_+ - \gamma_-] = 0$, as in the proof of Lemma C.6. So $\sum_j \mathfrak{a}_j\zeta^j$ is divisible by $\zeta - r^2$; define real numbers $\mathfrak{b}_0, \dots, \mathfrak{b}_{M-1}$ by $\sum_j \mathfrak{a}_j\zeta^j = (\zeta - r^2)\sum_{j<M}\mathfrak{b}_j\zeta^j$, and comparing coefficients ($\mathfrak{a}_0 = -r^2\mathfrak{b}_0$, $\mathfrak{a}_j = \mathfrak{b}_{j-1} - r^2\mathfrak{b}_j$, $\mathfrak{a}_M = \mathfrak{b}_{M-1}$) gives $\sum_j \mathfrak{a}_j\psi(e_{k+j}) = \Psi_{k+1} - r^2\Psi_k$ with $\Psi_k = \sum_{j<M}\mathfrak{b}_j\psi(e_{k+j})$. Let Σ_r be the symmetric matrix of the remaining quadratic part of $\mathcal{Q}_k$ as a form in $\mathbf{Z}_k = (\mathbf{X}_k, d_k)$, so that $\mathcal{Q}_k = \langle \mathbf{Z}_k, \Sigma_r\mathbf{Z}_k\rangle + \Psi_{k+1} - r^2\Psi_k$.

Frequency inequality. Let $\mathcal{L}(\mathsf{W}) = [\tilde{A}\ \tilde{B}]^\top\mathsf{W}[\tilde{A}\ \tilde{B}] - r^2E^\top\mathsf{W}E + \Sigma_r$, $E = [I\ 0]$. On a complex solution $\mathbf{Z}_k = z^k\mathbf{Z}$, $|z| = r$, of the state equation, the vector $\mathbf{Z}$ is determined by d (as z is not an eigenvalue of $\tilde{A}$), and since $p(z) \ne 0$, the latent sequence is a wave cz^k, $c \in \mathbb{C}$, with $\|\mathbf{Z}\| \le K|c|$. The computation in the proof of Lemma C.1, with the function values removed, gives $\mathbf{Z}^*\Sigma_r\mathbf{Z} = \Phi_t(|\omega|)|c|^2 \le -\epsilon|c|^2$; the terms in $\mathrm{Re}(p\bar{u})$ cancel by the second condition of (20). Hence the strict frequency inequality holds on $|z| = r$, and it persists with Σ_r replaced by $\Sigma_r + \eta I$ for $\eta < \epsilon/K^2$. The discrete-time Kalman–Yakubovich–Popov lemma on the circle $|z| = r$ [13, Lemma 2.4] (see also [36]), applicable because $(\tilde{A}, \tilde{B})$ is controllable, gives a real symmetric $(M+2)\times(M+2)$ matrix W with $\mathcal{L}(\mathsf{W}) \preceq -\eta I$.

Dissipation. With $V_k = \langle \mathbf{X}_k, \mathsf{W}\mathbf{X}_k\rangle - \Psi_k$ (W acting as $\mathsf{W}\otimes I$ on H^{M+2}),

$$V_{k+1} - r^2V_k = \langle \mathbf{Z}_k, \mathcal{L}(\mathsf{W})\mathbf{Z}_k\rangle - \mathcal{Q}_k \le -\eta\|\mathbf{Z}_k\|^2,$$

because $\langle \mathbf{X}_{k+1}, \mathsf{W}\mathbf{X}_{k+1}\rangle - r^2\langle \mathbf{X}_k, \mathsf{W}\mathbf{X}_k\rangle = \langle \mathbf{Z}_k, (\mathcal{L}(\mathsf{W}) - \Sigma_r)\mathbf{Z}_k\rangle$ and $\mathcal{Q}_k \ge 0$. For lag one, $\mathcal{Q}_k = 2(R+T)_{k,k+1} + 2(1-r^2)(R+T)_{\star k}$ and $\Psi_k = 2\psi(e_k)$, i.e. $\mathfrak{b}_0 = 2$, and $-2\psi(e_k) = \frac{4}{L-\mu}\big(f(\tilde{x}_k) - f(x^\star) - \frac{L+\mu}{4}\|e_k\|^2\big)$ by (10).

Bounds on V. Upper bound: $|\psi(e)| \le \frac12\|e\|^2$ and e_{k+j} is linear in $\mathbf{X}_k$, so $|V_k| \le \eta^+\|\mathbf{X}_k\|^2$ with η^+ depending only on W, $\mathfrak{b}$ and the tuning. Lower bound: by Theorem 4.7(c), $\rho_{\mathcal{F}} < r$, so for some $r' < r$ every trajectory satisfies $\|\mathbf{X}_k\| \le c'r'^k\|\mathbf{X}_0\|$ (the gradients in $\mathbf{X}_k$ are bounded by the queries, hence by the states $S_k, \dots, S_{k+M}$). Summing $r^{-2(k+1)}V_{k+1} \le r^{-2k}V_k - \eta r^{-2(k+1)}\|\mathbf{X}_k\|^2$ over $k < n$, $r^{-2n}V_n \le V_0 - \eta r^{-2}\|\mathbf{X}_0\|^2$, and $r^{-2n}|V_n| \le \eta^+c'^2(r'/r)^{2n}\|\mathbf{X}_0\|^2 \to 0$. Hence $V_0 \ge \eta r^{-2}\|\mathbf{X}_0\|^2$, and the same holds at every k by restarting. From rest, $\|\mathbf{X}_0\| \le K''\|X_0\|$, because $\|\sigma_j\| \le \|e_j\|$ and $\|S_j\| \le \Gamma(j)\|S_0\|$; so $\frac{\eta}{r^2}\|X_N\|^2 \le V_N \le r^{2N}V_0 \le \eta^+K''^2r^{2N}\|X_0\|^2$. □

Remark C.16 (Tongues as phase locking of a Hill recursion). On the planar orbit of Proposition 4.10(a), the gradient field acts by a rotating curvature. With R the rotation by ω_0, the query-normalized gradient amplitude $\hat{p} = p(z)/u(z)$ ($|\hat{p}| \le 1$ by feasibility), and a symmetric matrix H_0 of norm at most 1 with first column $(\mathrm{Re}\,\hat{p}, \mathrm{Im}\,\hat{p})$, for instance $\left(\begin{smallmatrix}\mathrm{Re}\,\hat{p} & \mathrm{Im}\,\hat{p}\\ \mathrm{Im}\,\hat{p} & -\mathrm{Re}\,\hat{p}\end{smallmatrix}\right)$, of norm $|\hat{p}|$, in a basis in which $\tilde{x}_0$ is real and positive the orbit satisfies $\sigma_n = \mathsf{H}_n\tilde{x}_n$ with $\mathsf{H}_n = \mathsf{R}^n\mathsf{H}_0\mathsf{R}^{-n}$, a symmetric matrix of norm at most 1 whose trace-free part rotates by $2\omega_0$ per step. In the plane of the orbit the method is therefore a linear recursion with periodic symmetric coefficients (period dividing K), a discrete Hill equation. In the rotating frame $\mathsf{R}^{-n}x_n$, the orbit is a fixed point. Thus a K-tongue is a resonance region for periodically

rotating curvatures that one function of the class can realize. One may take H_0, and then every H_n, scalar if and only if $\operatorname{Im}\hat{p}=0$, i.e. for a quadratic mode.

Spread spectra at a half step. The next lemma constructs spectra for a method with phase lead close to half a step on a frequency band. It gives strict margins for both minimizer constraints, all upper constraints and all lower constraints with lag at least two; lag one is checked separately. We use it in Appendices E and F. For $t<0$, put $C=-(1-\eta)\bar g$, $\eta\in(0,1)$, and

$$\mathcal{E}_\pm(x)=\cosh x\pm(1-\eta)\sinh x\qquad(x\ge 0),\qquad \Phi_t^{(1)}(\omega)=\operatorname{Re}\big[(1-e^{t+i\omega})(g-2is)(t,\omega)\big].$$

Since $\ell_n=\bar g\cosh nt-\operatorname{Re}\int(g-2is)e^{in\omega}d\nu+C\sinh nt$ and $\upsilon_n=\bar g\cosh nt-\operatorname{Re}\int(g+2is)e^{in\omega}d\nu-C\sinh nt$, one has, for $C=-(1-\eta)\bar g$,

$$\ell_n=\bar g\,\mathcal{E}_+(n|t|)-\operatorname{Re}\int(g-2is)e^{in\omega}\,d\nu,\qquad \upsilon_n=\bar g\,\mathcal{E}_-(n|t|)-\operatorname{Re}\int(g+2is)e^{in\omega}\,d\nu. \tag{43}$$

For $n=1$, $\mathcal{E}_+(|t|)=e^{-t}-\eta\sinh|t|$ and $e^{-t}g-\operatorname{Re}[(g-2is)e^{i\omega}]=e^{-t}\Phi_t^{(1)}$, so

$$\ell_1=e^{-t}\int\Phi_t^{(1)}\,d\nu-\eta\sinh|t|\,\bar g. \tag{44}$$

Thus the lower lag-one slack is the average of the certificate symbol $\Phi_t^{(1)}$, up to its explicit positive factor and the correction in η. For $\mathcal{M}$, $\Phi_t^{(1)}=-4H_r/(h^2(L-\mu)^2)$, $r=e^t$, with H_r as in Lemma E.1.

Choose once and for all an even smooth cutoff $\mathsf{b}_0\ge 0$, equal to 1 on $[-\frac18,\frac18]$ and zero for $|u|\ge\frac14$, and set $\mathsf{b}=\mathsf{b}_0/\int\mathsf{b}_0$. Thus $\mathsf{b}\in C_c^\infty((-\frac12,\frac12))$, $\mathsf{b}\ge 0$ and $\int\mathsf{b}=1$. Fix this choice throughout and put $\mathsf{k}=\mathsf{b}*\mathsf{b}$, supported in $[-1,1]$, and $\hat{\mathsf{k}}(x)=\int e^{ixu}\mathsf{k}(u)\,du=\hat{\mathsf{b}}(x)^2$.

Lemma C.17 (Spread spectra at a half step)**.** *There are $\mathsf{s}_1\in(0,\frac16]$, $c_\mathsf{k}>0$ and $C_\mathsf{k}<\infty$, depending only on b, with the following property. Let (u,v) be a stationary linear method, $t<0$, $K\ge 3$, $\omega_0=2\pi/K$, $\mathsf{s}\in(0,\mathsf{s}_1]$, $\mathsf{w}=\mathsf{s}\omega_0$ and $B=[\omega_0-\mathsf{w},\omega_0+\mathsf{w}]$. Suppose that $(g,s)(t,\omega)\neq(0,0)$ on B, write $g-2is=Re^{-i\Theta}$ with $R>0$ and Θ continuous on B, and put $\mathsf{d}=\max_B|\omega-2\Theta(\omega)|$, the phase defect from a half-step lead. Let $\eta\in(0,1)$, $\mathsf{m}=\min(\frac14,\frac{\omega_0^2}{6},c_\mathsf{k}\mathsf{s}^2)$, and assume*

$$\mathsf{d}\le\frac{\mathsf{m}}{2},\qquad C_\mathsf{k}\Big(\frac{|t|}{\mathsf{m}\,\mathsf{w}}\Big)^2+\mathsf{d}<\frac{\sqrt{\eta}}{8}. \tag{45}$$

Let ν be the probability measure on B with density proportional to $\mathsf{k}((\omega-\omega_0)/\mathsf{w})/R(\omega)$, and $C=-(1-\eta)\bar g$. Then $\bar g>0$, $|C|<\bar g$, $\upsilon_n>0$ for every $n\ge1$, and $\ell_n>0$ for every $n\ge2$. If moreover

$$\int_B\Phi_t^{(1)}\,d\nu>\eta\,e^t\sinh|t|\,\bar g, \tag{46}$$

then (ν,C) is feasible at t, and some pair (ν',C') with ν' carried by finitely many points of B and $|C'|<\bar g'$ is feasible at t.

Proof. The kernel. $\hat{\mathsf{b}}$ is real and even, $\hat{\mathsf{b}}(0)=1$, $|\hat{\mathsf{b}}(x)|<1$ for $x\neq0$ (the support of b has interior points), $\hat{\mathsf{b}}(x)=1-\frac{m_\mathsf{b}}{2}x^2+O(x^4)$ with $m_\mathsf{b}=\int u^2\mathsf{b}>0$, and $|\hat{\mathsf{b}}(x)|\le C_1/|x|$. Hence $\hat{\mathsf{k}}=\hat{\mathsf{b}}^2\ge0$, $\hat{\mathsf{k}}(x)=1-m_\mathsf{b}x^2+O(x^4)$ and $\hat{\mathsf{k}}(x)\le C_1^2/x^2$. Choose $x_\circ\in(0,1]$ with $1-\frac32 m_\mathsf{b}x^2\le\hat{\mathsf{k}}(x)\le1-\frac12 m_\mathsf{b}x^2$ on $[0,x_\circ]$, and let $M_0=\max_{x\ge x_\circ}\hat{\mathsf{k}}(x)<1$. Put $a_0=\min(\frac{x_\circ}{3},((1-M_0)/(3m_\mathsf{b}))^{1/2},(3m_\mathsf{b})^{-1/2})$, $c_\mathsf{k}=\min(\frac{m_\mathsf{b}}{3},\frac{1-M_0}{2})$, $\mathsf{s}_1=\min(\frac16,\frac{3a_0}{\pi})$ and $C_\mathsf{k}=16C_1^2$. For $0<a\le a_0$ and $x\ge3a$,

$$\hat{\mathsf{k}}(a)\ge\tfrac12,\qquad \hat{\mathsf{k}}(x)\le\hat{\mathsf{k}}(a)-c_\mathsf{k}\min(x^2,1). \tag{47}$$

Indeed $\hat{\mathsf{k}}(a)\ge1-\frac32 m_\mathsf{b}a^2\ge\frac12$. If $x\le x_\circ$, then $\hat{\mathsf{k}}(a)-\hat{\mathsf{k}}(x)\ge\frac12 m_\mathsf{b}x^2-\frac32 m_\mathsf{b}a^2\ge\frac13 m_\mathsf{b}x^2$, because $a^2\le x^2/9$. If $x\ge x_\circ$, then $\hat{\mathsf{k}}(a)-\hat{\mathsf{k}}(x)\ge1-\frac32 m_\mathsf{b}a^2-M_0\ge\frac12(1-M_0)$.

Reduction. Let $\varrho(\omega) = \mathsf{w}^{-1}\mathsf{k}((\omega-\omega_0)/\mathsf{w})$, so that $\hat\varrho(\zeta) := \int e^{i\zeta\omega}\varrho(\omega)\,d\omega = e^{i\zeta\omega_0}\hat{\mathsf{k}}(\zeta\mathsf{w})$, and $d\nu = \varrho\,d\omega/(ZR)$ with $Z = \int \varrho/R$. Since $g \pm 2is = Re^{\pm i\Theta}$, multiplying (43) by Z shows that $\ell_n > 0$ and $\upsilon_n > 0$ are equivalent to

$$\operatorname{Re}\int e^{i(n\omega-\Theta)}\varrho\,d\omega < \mathcal{E}_+(n|t|)\int\cos\Theta\,\varrho\,d\omega, \qquad \operatorname{Re}\int e^{i(n\omega+\Theta)}\varrho\,d\omega < \mathcal{E}_-(n|t|)\int\cos\Theta\,\varrho\,d\omega.$$

Write $\Theta = \frac{\omega}{2} - \frac{\delta}{2}$ with $|\delta| \le \mathsf{d}$ on B. Since $|e^{ix}-1| \le |x|$,

$$\Big|\int e^{i(n\omega\mp\Theta)}\varrho\,d\omega - \hat\varrho(n\mp\tfrac12)\Big| \le \frac{\mathsf{d}}{2}, \qquad \int\cos\Theta\,\varrho\,d\omega \ge G_0 - \frac{\mathsf{d}}{2}, \quad G_0 := \operatorname{Re}\hat\varrho(\tfrac12) = \cos\tfrac{\omega_0}{2}\,\hat{\mathsf{k}}(\tfrac{\mathsf{w}}{2}).$$

As $\omega_0 \le \frac{2\pi}{3}$ and $\frac{\mathsf{w}}{2} \le \frac{\pi\mathsf{s}}{3} \le a_0$, $G_0 \ge \frac14$. Also $\mathsf{d} \le \omega_0^2/12 < (5/6)\omega_0 \le \omega$ on B, so $\Theta > 0$. Finally $\mathsf{d} \le \frac{\mathsf{m}}{2} \le \frac18$ and $\omega_0 + \mathsf{w} \le \frac{7\pi}{9}$, so $0 < \Theta \le \frac12(\frac{7\pi}{9} + \frac18) < \frac{5\pi}{12}$ on B; hence $\cos\Theta > \frac14$, $\bar g = \int R\cos\Theta\,d\nu > 0$, and $|C| = (1-\eta)\bar g < \bar g$.

The margin. For $n \ge 1$ let $\mathfrak{m}(n) = G_0 - \operatorname{Re}\hat\varrho(n+\frac12) = G_0 - \cos((n+\frac12)\omega_0)\,\hat{\mathsf{k}}((n+\frac12)\mathsf{w})$. We claim $\mathfrak{m}(n) \ge \mathsf{m}$. Put $a = \frac{\mathsf{w}}{2}$ and $x = (n+\frac12)\mathsf{w} \ge 3a$. If $\cos((n+\frac12)\omega_0) \le 0$, then $\mathfrak{m}(n) \ge G_0 \ge \frac14$, since $\hat{\mathsf{k}} \ge 0$. If $n \equiv 0$ or $n \equiv -1 \pmod K$, then $n \ge K-1$, $\cos((n+\frac12)\omega_0) = \cos\frac{\omega_0}{2}$, and by (47) $\mathfrak{m}(n) = \cos\frac{\omega_0}{2}(\hat{\mathsf{k}}(a) - \hat{\mathsf{k}}(x)) \ge \frac12 c_{\mathsf{k}}\min(x^2,1)$; here $x \ge (K-\frac12)\mathsf{w} = 2\pi\mathsf{s}(1-\frac{1}{2K}) \ge \frac{5\pi}{3}\mathsf{s}$, so $x^2 \ge 27\mathsf{s}^2$, and $27\mathsf{s}^2 \le 1$, whence $\mathfrak{m}(n) \ge c_{\mathsf{k}}\mathsf{s}^2$. Otherwise $(n+\frac12)\omega_0$ is congruent modulo 2π to $(i+\frac12)\omega_0$ with $1 \le i \le K-2$, so $\cos((n+\frac12)\omega_0) \le \cos\frac{3\omega_0}{2}$; if this cosine is positive, then $\hat{\mathsf{k}}(x) \le \hat{\mathsf{k}}(a)$ gives $\mathfrak{m}(n) \ge \hat{\mathsf{k}}(a)(\cos\frac{\omega_0}{2} - \cos\frac{3\omega_0}{2}) \ge \sin\omega_0\sin\frac{\omega_0}{2} \ge \frac{\omega_0^2}{6}$, by concavity of sin on $[0,\pi]$ ($\sin\varphi \ge \frac{3\sqrt3}{4\pi}\varphi$ on $[0,\frac{2\pi}{3}]$ and $\sin\varphi \ge \frac{3\sqrt3}{2\pi}\varphi$ on $[0,\frac{\pi}{3}]$, and $\frac{27}{16\pi^2} > \frac16$).

Upper constraints. Let $n \ge 1$ and $x = n|t|$. If $x \le \frac{\mathsf{m}}{4}$, then $\mathcal{E}_-(x) \ge e^{-x} \ge 1-x$, so the right-hand side is at least $(1-x)(G_0 - \frac{\mathsf{d}}{2}) \ge G_0 - \frac{\mathsf{d}}{2} - x$, while the left-hand side is at most $G_0 - \mathfrak{m}(n) + \frac{\mathsf{d}}{2} \le G_0 - \mathsf{m} + \frac{\mathsf{d}}{2}$; since $\mathsf{m} > \frac{\mathsf{m}}{4} + \mathsf{d}$, $\upsilon_n > 0$. If $x > \frac{\mathsf{m}}{4}$, then $(n+\frac12)\mathsf{w} > \mathsf{m}\mathsf{w}/(4|t|)$, so $|\operatorname{Re}\hat\varrho(n+\frac12)| \le C_1^2(4|t|/(\mathsf{m}\mathsf{w}))^2$, and $\mathcal{E}_-(x) = (1-\frac{\eta}{2})e^{-x} + \frac{\eta}{2}e^x \ge \sqrt{\eta(2-\eta)} \ge \sqrt\eta$; the left-hand side is at most $C_{\mathsf{k}}(|t|/(\mathsf{m}\mathsf{w}))^2 + \frac{\mathsf{d}}{2}$, the right-hand side at least $\sqrt\eta(G_0 - \frac{\mathsf{d}}{2}) \ge \frac{\sqrt\eta}{8}$, and $\upsilon_n > 0$ by (45).

Lower constraints, $n \ge 2$. $\mathcal{E}_+ \ge 1$, so the right-hand side is at least $G_0 - \frac{\mathsf{d}}{2}$, and the left-hand side is at most $\operatorname{Re}\hat\varrho(n-\frac12) + \frac{\mathsf{d}}{2} = G_0 - \mathfrak{m}(n-1) + \frac{\mathsf{d}}{2} \le G_0 - \mathsf{m} + \frac{\mathsf{d}}{2}$; as $\mathsf{m} > \mathsf{d}$, $\ell_n > 0$.

Lag one and feasibility. By (44), $\ell_1 > 0$ is (46). With $|C| < \bar g$, all constraints of (19) hold strictly.

Finitely many atoms. For every probability measure ν' on B and $C' = -(1-\eta)\bar g'$, the identities (43) hold, and $|\operatorname{Re}\int(g \mp 2is)e^{in\omega}d\nu'| \le \int R\,d\nu' < 4\bar g'$ because $\cos\Theta > \frac14$ on B. Since $\mathcal{E}_\pm(x) \ge \frac{\eta}{2}e^x$, all constraints with $n \ge n_2 := \lceil |t|^{-1}\log(8/\eta)\rceil$ hold for every such ν'. The finitely many remaining constraints, ℓ_n, υ_n with $n < n_2$, are integrals of continuous functions on B against ν' and hold strictly for ν; they persist for the Riemann sums ν' of ν on a fine enough grid, which converge weakly to ν. Finally $|C'| = (1-\eta)\bar g' < \bar g'$. □

For $\mathcal{M}$ in the accelerated scaling the hypotheses of the lemma are checked once, in Lemma E.9, where the defect d is read off the mixed symbol of Lemma E.2 through (41).

D. Proofs for Section 5

Throughout, a second-order family with slow data (θ, y) is fixed, $L = 1$, $\mu = \varepsilon^2$, and $\theta_\varepsilon = (1-\beta_\varepsilon)/\varepsilon$, $y_\varepsilon = -1/q_\varepsilon(1)$. For the member ε we write $\phi_\lambda^\varepsilon = \lambda u_\varepsilon - v_\varepsilon$; at $\varepsilon = 0$, $\phi_0^0 = \phi_0$ and $\phi_1^0 = \phi_1$ are the polynomials of the limit method in Section 5.4.

Lemma D.1 (Slow normal form)**.** *For $z = e^{-\varepsilon\varpi}$, uniformly for ϖ in compact subsets of $\mathbb{C}$,*

$$\hat\lambda_\varepsilon(z) := \frac{v_\varepsilon(z)}{u_\varepsilon(z)} = \frac{\varepsilon^2}{y_\varepsilon}\,\varpi(\theta_\varepsilon - \varpi) + O(\varepsilon^3).$$

For small ε the polynomial $\phi^\varepsilon_{\varepsilon^2} = \varepsilon^2 u_\varepsilon - v_\varepsilon$ has two roots $e^{-\varepsilon\varpi_\pm}$ with $\varpi_\pm$ converging to the roots of $\varpi^2 - \theta\varpi + y$. Hence $\rho_q \ge 1 - \varepsilon c_Q(\theta, y) + o(\varepsilon)$.

Proof. With $E(x) = (1-e^{-x})/x$, continuously extended by $E(0) = 1$, $z - 1 = -\varepsilon\varpi E(\varepsilon\varpi)$ and $z - \beta_\varepsilon = \varepsilon(\theta_\varepsilon - \varpi E(\varepsilon\varpi))$. The coefficients are bounded and $|z-1| = O(\varepsilon)$, so $q_\varepsilon(z) = q_\varepsilon(1) + O(\varepsilon)$ and $u_\varepsilon(z) = u_\varepsilon(1) + O(\varepsilon) = 1 + O(\varepsilon)$. Multiplying, $v_\varepsilon(z) = -\varepsilon^2\varpi(\theta_\varepsilon - \varpi)q_\varepsilon(1) + O(\varepsilon^3)$, which gives the first claim. Next, $\varepsilon^{-2}\phi^\varepsilon_{\varepsilon^2}(e^{-\varepsilon\varpi}) = u_\varepsilon(z)(1 - \varepsilon^{-2}\hat\lambda_\varepsilon(z))$ converges, uniformly on compact sets, to $\frac{1}{y}(\varpi^2 - \theta\varpi + y)$; Rouché's theorem, counting multiplicities, gives two zeros converging to the roots of $\varpi^2 - \theta\varpi + y$. The moduli of the corresponding roots of $\phi^\varepsilon_{\varepsilon^2}$ are $e^{-\varepsilon \operatorname{Re}\varpi_\pm}$, and the smaller of $\operatorname{Re}\varpi_\pm$ tends to $c_Q(\theta, y)$ (to ζ_- if the roots are real, to $\theta/2$ otherwise). Since ρ_q is the largest modulus of a root of ϕ^ε_λ, $\lambda \in [\mu, L]$, the last claim follows. □

Slow spectra. Here a *slow spectrum* is supported in $[0, \varepsilon\gamma_{\max}]$ for some $\gamma_{\max}$ independent of ε. The lower bounds of Theorems 2.6 and 5.2(a) use $t = -c\varepsilon$ and $\omega_j = \varepsilon\gamma_j$, with $\Delta_c = c^2 - \theta c + y$. We choose the value constant $C = C_*$ of Lemma D.2. In query normalization, where each wave has query amplitude 1, the lag inequalities have exactly the kernels $\mathcal{A}_\gamma(q), \mathcal{B}_\gamma(q)$ of the flow program at $q = n\varepsilon$; their coefficients converge uniformly to the flow coefficients on bounded frequency sets (Lemma D.3). The uniform margin of Lemma B.4 absorbs these perturbations for all lags simultaneously (Lemma D.5). Conversely, Lemma D.4 shows that every feasible slow spectrum has $C = C_* + O(\varepsilon^3)$, which will exclude a single slow frequency.

Lemma D.2 (Per-atom splitting)**.** *Let $\nu = \sum_j \nu_j\delta_{\omega_j}$ and $t < 0$. Set $C_* = t^{-1}\sum_j 2\nu_j s_j\omega_j$. For $C = C_*$,*

$$\ell_n,\ \upsilon_n = \textstyle\sum_j \nu_j\big[g_j(\cosh nt - \cos n\omega_j) \pm 2s_j\big(\frac{\omega_j \sinh n|t|}{|t|} - \sin n\omega_j\big)\big] \quad (\textit{signs } +, - \textit{ respectively}),$$

and $|C_*| \le \bar g$ if and only if $|\sum_j \nu_j s_j\omega_j| \le \frac{|t|}{2}\bar g$.

Proof. Substitute $C_* \sinh nt = \sum_j \nu_j 2s_j\omega_j \sinh(n|t|)/|t|$ in (19). The two lag identities follow, and $|C_*| \le \bar g$ is exactly $|\sum_j \nu_j 2s_j\omega_j| \le |t|\bar g$. □

For a finitely supported ν with $u(z_j) \ne 0$, the constraints (19) are homogeneous in (ν, C); replacing g, s by $g/|u|^2 = 1 - |\hat p|^2$ and $s/|u|^2 = \operatorname{Im}\hat p$, $\hat p = p/u$, and ν_j by $\nu_j|u(z_j)|^2$ (renormalized, with C divided by the same constant) gives an equivalent system. We call this the *query normalization*. The original amplitudes u, p and spectral weights in (19) give the *latent normalization*. By (14) with $L = 1$, $\mu = \varepsilon^2$, $\hat p = (1 + \varepsilon^2 - 2\hat\lambda)/(1 - \varepsilon^2)$ for every stationary linear method, where $\hat\lambda = v(z)/u(z)$ is the complex curvature of the wave (for $\mathcal{M}$, $\hat\lambda = -(z-1)(z-\beta)/(hN_\tau(z))$). Define $\tilde g, \tilde s$ by

$$1 - |\hat p|^2 = \tfrac{4\varepsilon^2}{y_\varepsilon}\tilde g, \qquad 2\operatorname{Im}\hat p = \tfrac{4\varepsilon^2}{y_\varepsilon}\tilde s. \tag{48}$$

Lemma D.3 (Exact reduction and expansion)**.** *Let $c > 0$, $0 < \varepsilon < 1$ with $y_\varepsilon > 0$, and put $t = -c\varepsilon$, $\omega_j = \varepsilon\gamma_j \in [0, \pi]$ with $\gamma_j \ge 0$. Let ν be a finitely supported probability measure on $\{\omega_j\}$ in the query normalization. The pair (ν, C_*) of Lemma D.2 is feasible at t if and only if, for all $q \in \varepsilon\mathbb{N}$,*

$$\textstyle\sum_j \nu_j\big[\tilde g_j\mathcal{A}_{\gamma_j}(q) \pm \tilde s_j\mathcal{B}_{\gamma_j}(q)\big] \ge 0, \qquad \big|\sum_j \nu_j\gamma_j\tilde s_j\big| \le c\sum_j \nu_j\tilde g_j.$$

Moreover $u_\varepsilon(z_j) \to 1$, so the query normalization is defined for small ε, and $\tilde g \to \gamma^2 - \Delta_c$, $\tilde s \to (\theta - 2c)\gamma$ as $\varepsilon \to 0$, uniformly for (c, γ) in compact subsets of $(0, \infty) \times [0, \infty)$.

Proof. Reduction. With $q = n\varepsilon$, $|t| = c\varepsilon$ and $\omega_j = \varepsilon\gamma_j$ we have, exactly,

$$\cosh nt - \cos n\omega_j = \cosh cq - \cos\gamma_j q = \mathcal{A}_{\gamma_j}(q), \qquad \frac{\omega_j \sinh n|t|}{|t|} - \sin n\omega_j = \frac{\gamma_j \sinh cq}{c} - \sin\gamma_j q = \mathcal{B}_{\gamma_j}(q).$$

In the query normalization $g_j = 1 - |\hat p_j|^2 = \frac{4\varepsilon^2}{y_\varepsilon}\tilde g_j$ and $2s_j = 2\operatorname{Im}\hat p_j = \frac{4\varepsilon^2}{y_\varepsilon}\tilde s_j$. Lemma D.2, divided by $4\varepsilon^2/y_\varepsilon$, gives the lag constraints, and its minimizer condition $|\sum_j \nu_j 2s_j\omega_j| \le c\varepsilon\,\bar g$ becomes $|\sum_j \nu_j\gamma_j\tilde s_j| \le c\sum_j \nu_j\tilde g_j$.

Expansion. Put $\varpi = c - i\gamma$, so that $z = e^{t+i\omega} = e^{-\varepsilon\varpi}$. The coefficients of u_ε are bounded and $|z-1| = O(\varepsilon)$, so $u_\varepsilon(z) = u_\varepsilon(1) + O(\varepsilon) = 1 + O(\varepsilon)$; in particular the query normalization is defined for small ε. By Lemma D.1, the complex curvature is $\hat\lambda = \varepsilon^2\varpi(\theta_\varepsilon - \varpi)/y_\varepsilon + O(\varepsilon^3)$. Writing $\hat p = 1 - 2\varepsilon^2\tilde w$, we get $\tilde w = (\hat\lambda/\varepsilon^2 - 1)/(1-\varepsilon^2) \to \varpi(\theta-\varpi)/y - 1$. Since $1 - |\hat p|^2 = 4\varepsilon^2(\operatorname{Re}\tilde w - \varepsilon^2|\tilde w|^2)$ and $2\operatorname{Im}\hat p = -4\varepsilon^2\operatorname{Im}\tilde w$, (48) gives $\tilde g = y_\varepsilon(\operatorname{Re}\tilde w - \varepsilon^2|\tilde w|^2)$ and $\tilde s = -y_\varepsilon \operatorname{Im}\tilde w$. Finally $\varpi(\theta-\varpi) = (\gamma^2 - c^2 + \theta c) + i(2c-\theta)\gamma$, so $\tilde g \to \gamma^2 - \Delta_c$ and $\tilde s \to (\theta - 2c)\gamma$. All remainders are uniform for (c,γ) in compact sets, as in Lemma D.1. □

Lemma D.4 (The value constant is forced)**.** *Fix $\gamma_{\max} > 0$, a compact set $I_c \subset (0,\infty)$, and a second-order family with slow data (θ, y). There are $\varepsilon_2, K_* > 0$ such that every pair (ν, C) feasible at $t = -c\varepsilon$, $c \in I_c$, $0 < \varepsilon < \varepsilon_2$, in query normalization with support in $[0, \varepsilon\gamma_{\max}]$, satisfies*

$$|C - C_*| \le K_*\varepsilon^3, \qquad C_* := \frac{1}{t}\int 2s\omega\, d\nu = -\frac{4\varepsilon^2}{y_\varepsilon c}\int \gamma\tilde s\, d\nu.$$

Proof. By Lemma D.3, $|g|, |s| \le K_1\varepsilon^2$ on the support of ν, uniformly for $c \in I_c$. The lag-one constraints $\ell_1, \upsilon_1 \ge 0$ read $\bar a_1 - \bar b_1 + C\sinh t \ge 0$ and $\bar a_1 + \bar b_1 - C\sinh t \ge 0$, i.e. $|\bar b_1 - C\sinh t| \le \bar a_1$. Here $\bar a_1 = \int g(\cosh c\varepsilon - \cos\varepsilon\gamma)\, d\nu \le K_2\varepsilon^4$, since the bracket is $O(\varepsilon^2)$. Next,

$$\bar b_1 - C_*\sinh t = \int 2s\Big(\sin\varepsilon\gamma - \varepsilon\gamma\frac{\sinh c\varepsilon}{c\varepsilon}\Big)d\nu,$$

and the bracket is $O(\varepsilon^3)$ uniformly, so $|\bar b_1 - C_*\sinh t| \le K_3\varepsilon^5$. Hence $|C - C_*|\,|\sinh t| \le |\bar b_1 - C\sinh t| + |\bar b_1 - C_*\sinh t| \le K_4\varepsilon^4$, and since $|\sinh t| \ge c\varepsilon$, $|C - C_*| \le (K_4/\min I_c)\varepsilon^3$. The formula for C_* follows from $2s = \frac{4\varepsilon^2}{y_\varepsilon}\tilde s$, $\omega = \varepsilon\gamma$ and $t = -c\varepsilon$. □

Lemma D.5 (Sampled flow spirals)**.** *Let $\theta/3 < c < c_Q(\theta, y)$, and let J, $\gamma_i = \gamma_0\lambda_i$ and $\nu_i = 1/J$ $(1 \le i \le J)$ be given by Lemma B.4 with $m = y$. For all small ε, the spectrum with atoms $\omega_i = \varepsilon\gamma_i$ and weights ν_i in the query normalization, with $C = C_*$, is feasible at $t = -c\varepsilon$ and has $|C_*| < \bar g$. Consequently the corresponding pair in the latent normalization, with weights $\nu_i|u_\varepsilon(z_i)|^{-2}$ renormalized and C rescaled accordingly, is feasible at t, has J atoms and satisfies $|C| < \bar g$.*

Proof. Let ϵ_* be as in Lemma B.4. The quantities $c, \gamma_0, J, \epsilon_*$ are fixed before ε is chosen; take ε small enough that also $\varepsilon\gamma_i \in [0,\pi]$ for every i. By Lemma D.3 choose ε_0 such that, for $\varepsilon < \varepsilon_0$ and all i,

$$|\tilde g_i - (\gamma_i^2 - \Delta_c)| \le \eta', \qquad |\tilde s_i - (\theta - 2c)\gamma_i| \le \eta', \qquad \eta' = \min\big(\tfrac{\epsilon_*}{2}, \tfrac{1}{2(c+2\gamma_0)}\big).$$

Since $\mathcal{A}_\gamma, \mathcal{B}_\gamma \ge 0$, (36) and (37) give, for every $q > 0$,

$$\sum_i \nu_i\big[\tilde g_i\mathcal{A}_{\gamma_i}(q) \pm \tilde s_i\mathcal{B}_{\gamma_i}(q)\big] \ge (\epsilon_* - \eta')\sum_i \nu_i(\mathcal{A}_{\gamma_i}(q) + \mathcal{B}_{\gamma_i}(q)) \ge 0$$

(the minus sign uses (36), the plus sign (37)). For the minimizer condition, using $\gamma_i \le 2\gamma_0$ and (38),

$$\begin{aligned} c\sum_i \nu_i\tilde g_i - \Big|\sum_i \nu_i\gamma_i\tilde s_i\Big| &\ge c\sum_i \nu_i(\gamma_i^2 - \Delta_c) - (\theta - 2c)\sum_i \nu_i\gamma_i^2 - (c + 2\gamma_0)\eta' \\ &= (3c - \theta)\sum_i \nu_i\gamma_i^2 - c\Delta_c - (c + 2\gamma_0)\eta' \;\ge\; \tfrac{1}{2}. \end{aligned}$$

Lemmas D.2 and D.3 now give feasibility and $|C_*| < \bar g$. Homogeneity preserves both conclusions under the inverse change of spectral weights from query to latent normalization. □

Proof of Theorem 5.2. If $c_Q \le \theta/3$, Lemma D.1 and $\rho_{\mathcal{F}} \ge \rho_q$ (Lemma 2.3) give the claim. Let $\theta/3 < c < c_Q$. Lemma D.5 gives J, independent of ε, and for small ε a pair with J atoms that is feasible at $t = -c\varepsilon$. By Theorem 4.7(d), it is realized by a trajectory on $H = L^2(\nu;\mathbb{C}) \cong \mathbb{R}^{2J}$, with $S_{k+1} = e^{-c\varepsilon}US_k$. Hence $\Gamma(n) \ge e^{-c\varepsilon n}$, $\rho_{\mathcal{F}} \ge e^{-c\varepsilon}$, and $\limsup\sqrt{\kappa}(1 - \rho_{\mathcal{F}}) \le c$. Letting $c \downarrow \theta/3$ completes the nonlinear branch. □

Proof of Corollary 5.3. (i) By Lemma D.1 the quadratic constant is at most $c_Q(\theta, y)$; apply Theorem 5.2 and Corollary 3.2. (ii) $c_{\mathcal{F}}(2\sqrt{y}, y) = \min(\frac{2}{3}\sqrt{y}, \sqrt{y})$. For $(\frac{4}{\sqrt{3}}, \frac{4}{3})$, $\theta^2 = 4y$, so $c_{\mathcal{F}} = \min(\frac{4}{3\sqrt{3}}, \frac{2}{\sqrt{3}})$. (iii) $c_{\mathcal{F}} \geq 1$ requires $\theta \geq 3$ and $c_Q(\theta, y) \geq 1$. If $\theta^2 \geq 4y$, $\zeta_- \geq 1$ is equivalent to $\theta - 2 \geq \sqrt{\theta^2 - 4y}$, i.e. $y \geq \theta - 1$; if $\theta^2 < 4y$, then $y > \theta^2/4 \geq \theta - 1$. □

Proof of Lemma 5.4. First, the transfer-function form of τ_{eff} in Section 5.4 follows from $v_0 = (z-1)^2 q_0$, $(q_0/u_0)(1) = -1/y$ and $(q_0/u_0)'(1) = q_0'(1) + u_0'(1)/y$. Here $\phi_0\overline{\phi_1} = -v_0\overline{(u_0 - v_0)}$. Since $v_0 = O(\omega^2)$ on the circle near 1, $\phi_0\overline{\phi_1} = -(z-1)^2 q_0(z)\overline{u_0(z)} + O(\omega^4)$, and $\overline{u_0(z)} = u_0(1/z)$ for $|z| = 1$. Put $G(z) = q_0(z)u_0(1/z)$: $G(1) = -1/y$ and $G'(1) = q_0'(1) - q_0(1)u_0'(1) = q_0'(1) + u_0'(1)/y$. With $(z-1)^2 = -\omega^2 - i\omega^3 + O(\omega^4)$ and $G(e^{i\omega}) = G(1) + i\omega G'(1) + O(\omega^2)$,

$$-(z-1)^2 G(z) = \omega^2 G(1) + i\omega^3 (G(1) + G'(1)) + O(\omega^4),$$

and $G(1) + G'(1) = \tau_{\text{eff}}/y$. For $\mathcal{S}_0$, multiply by $1 - e^{i\omega} = -i\omega + \frac{\omega^2}{2} + O(\omega^3)$: the real part is $\frac{\tau_{\text{eff}}}{y}\omega^4 - \frac{\omega^4}{2y} + O(\omega^5)$, and $\mathcal{S}_0$ is even in ω, since the coefficients are real. □

Lemma D.6 (Two-scale expansion)**.** *Fix $c \in \mathbb{R}$, put $r = 1 - c\varepsilon$, $z = re^{i\omega}$, and $\mathcal{H}(\varepsilon, \omega) = \operatorname{Re}\big[(1 - z)\phi^{\varepsilon}_{\varepsilon^2}(z)\overline{\phi^{\varepsilon}_1(z)}\big]$. For every second-order family, $\mathcal{H}$ is even in ω, and, as $(\varepsilon, \omega) \to (0^+, 0)$,*

$$\mathcal{H}(\varepsilon, \omega) = \frac{\varepsilon}{y}\big(c\Delta_c\, \varepsilon^2 + (\theta - 3c)\, \omega^2\big) + o\big((\varepsilon + |\omega|)^3\big).$$

If the family is of class C^3, then $\mathcal{H}$ is also of class C^3 on the half-neighbourhood $\varepsilon \geq 0$.

Proof. Write $s = \varepsilon + |\omega|$. The factorization following Definition 5.1 and coefficient continuity give

$$z - 1 = -c\varepsilon + i\omega + O(s^2), \qquad z - \beta_\varepsilon = (\theta - c)\varepsilon + i\omega + o(s),$$
$$q_\varepsilon(z) = -\frac{1}{y} + o(1), \qquad u_\varepsilon(z) = 1 + o(1).$$

Consequently, with $a = c\varepsilon$ and $b = (\theta - c)\varepsilon$,

$$\phi^{\varepsilon}_{\varepsilon^2}(z) = \varepsilon^2 + \tfrac{1}{y}(i\omega - a)(i\omega + b) + o(s^2), \qquad \phi^{\varepsilon}_1(z) = 1 + o(1).$$

The other factor is $1 - z = a - i\omega + O(s^2)$, so their product has real part

$$a\varepsilon^2 - \frac{a^2 b}{y} + \frac{\omega^2(b - 2a)}{y} + o(s^3) = \frac{\varepsilon}{y}(c\Delta_c\varepsilon^2 + (\theta - 3c)\omega^2) + o(s^3).$$

This proves the joint expansion for every relative scale of ω and ε. Replacing ω by $-\omega$ conjugates the bracket, proving evenness. Finally, $\mathcal{H}$ is a polynomial in $r\cos\omega$, $r\sin\omega$ and the real coefficients; hence C^3 coefficient regularity implies C^3 regularity of $\mathcal{H}$. □

Proof of Theorem 5.5. (a) Let $0 < c < c_{\mathcal{F}}$; then $\Delta_c > 0$ and $\theta - 3c > 0$. Here $\mathcal{S}_0 > 0$ near 0, so $s_{2\mathfrak{p}} > 0$. The family is of class $C^{2\mathfrak{p}}$, so the polynomial dependence on its coefficients makes $\mathcal{H}$ of class $C^{2\mathfrak{p}}$, and by Taylor's formula at $(0, 0)$, $\mathcal{H} = \sum_{i+j \leq 2\mathfrak{p}} \mathfrak{h}_{ij}\varepsilon^i\omega^j + R$ with $|R| \leq \eta(\varepsilon + |\omega|)^{2\mathfrak{p}}$ for $\varepsilon + |\omega| \leq \rho_\eta$, for every $\eta > 0$. Here $\mathfrak{h}_{ij} = 0$ for odd j, since $\mathcal{H}$ is even in ω; $\mathfrak{h}_{0j}$ are the Taylor coefficients of $\mathcal{H}(0, \omega) = \mathcal{S}_0(\omega)$, so $\mathfrak{h}_{0j} = 0$ for $j < 2\mathfrak{p}$ and $\mathfrak{h}_{0,2\mathfrak{p}} = s_{2\mathfrak{p}}$; and by Lemma D.6 there are no terms of degree below 3, and the cubic ones are $\frac{\varepsilon}{y}(c\Delta_c\varepsilon^2 + (\theta - 3c)\omega^2)$. Every other monomial has $i \geq 1$ and either $j = 0$, $i \geq 4$, or $j \geq 2$, $i + j \geq 4$; for $\varepsilon + |\omega| \leq 1$ it is bounded by $\varepsilon^3(\varepsilon + |\omega|)$ in the first case and by $\varepsilon\omega^2(\varepsilon + |\omega|)$ in the second. With M_0 the sum of the absolute values of their coefficients, $\eta = s_{2\mathfrak{p}}2^{-2\mathfrak{p}}$ and $(\varepsilon + |\omega|)^{2\mathfrak{p}} \leq 2^{2\mathfrak{p}-1}(\varepsilon^4 + \omega^{2\mathfrak{p}})$ (as $2\mathfrak{p} \geq 4$, $\varepsilon \leq 1$), for $\varepsilon + |\omega| \leq \rho \leq \min(\rho_\eta, 1)$,

$$\mathcal{H} \geq \varepsilon^3\Big(\frac{c\Delta_c}{y} - (M_0 + \tfrac{s_{2\mathfrak{p}}}{2})\rho\Big) + \varepsilon\omega^2\Big(\frac{\theta - 3c}{y} - M_0\rho\Big) + \frac{s_{2\mathfrak{p}}}{2}\,\omega^{2\mathfrak{p}}.$$

Fix ρ so small that both brackets are positive; then $\mathcal{H} > 0$ for $0 < \varepsilon \le \rho/2$, $|\omega| \le \rho/2$. On $\rho/2 \le |\omega| \le \pi$, $\mathcal{H}(\varepsilon,\omega) \to \mathcal{S}_0(\omega)$ uniformly, and $\min_{[\rho/2,\pi]} \mathcal{S}_0 > 0$; so $\mathcal{H} > 0$ there for small ε. Hence the lag-one symbol is positive on $|z| = 1 - c\varepsilon$. By (14) the normalized polynomial is proportional to $(1+\varepsilon^2)u_\varepsilon - 2v_\varepsilon$, whose roots converge to those of $u_0 - 2v_0$ (the leading coefficient is -2 times that of v_ε, which is nonzero at $\varepsilon = 0$); so $r_c < 1 - c\varepsilon$ for small ε, and the lag-one criterion gives $\rho_{\mathcal{F}} < 1 - c\varepsilon$. With Theorem 5.2 the limit follows.

A criterion for (b). At $t = 0$ let $\Xi = \phi^\varepsilon_{\varepsilon^2}\overline{\phi^\varepsilon_1}$ on the unit circle, let $\mathcal{H} = \operatorname{Re}[(1 - e^{i\omega})\Xi] = (1 - \cos\omega)\operatorname{Re}\Xi + \sin\omega\,\operatorname{Im}\Xi$ be the lag-one symbol, and let $\check{\mathcal{H}} = \operatorname{Re}[(1 - e^{-i\omega})\Xi] = \mathcal{H} - 2\sin\omega\,\operatorname{Im}\Xi$ be the symbol of the backward multiplier $1 - \bar{z}$. By (22), $g - 2is$ is a positive multiple of $-\Xi$, so by Lemma C.2, where $\Xi \ne 0$, $\mathcal{H}$ is a positive multiple of $\sin(\Theta - \frac{\omega}{2})$ and $\check{\mathcal{H}}$ a positive multiple of $-\sin(\Theta + \frac{\omega}{2})$. If $\mathcal{H} < 0$ and $\check{\mathcal{H}} < 0$ at some $\omega \in (0,\pi)$, then, as $\Theta \in (-\pi,\pi]$, $\Theta \in (\frac{\omega}{2} - \pi, \frac{\omega}{2}) \cap (-\frac{\omega}{2}, \pi - \frac{\omega}{2})$, i.e. $|\Theta(\omega)| < \omega/2$.

(b) At $\varepsilon = 0$ the symbols of the criterion are those of the limit method, $\mathcal{H} = \mathcal{S}_0$ and $\check{\mathcal{H}} = \check{\mathcal{H}}_0 := \mathcal{S}_0 - 2\sin\omega\,\operatorname{Im}\Xi_0$ with $\Xi_0 = [\phi_0\overline{\phi_1}](e^{i\omega})$, and $-\Xi_0 = |\Xi_0|\,e^{-i\Theta_0}$ by the definition of Θ_0. By Lemma 5.4, $\operatorname{Im}\Xi_0 = \frac{\tau_{\text{eff}}}{y}\omega^3 + O(\omega^5)$ (it is odd in ω), so

$$\mathcal{S}_0 = \frac{2\tau_{\text{eff}} - 1}{2y}\,\omega^4 + O(\omega^6), \qquad \check{\mathcal{H}}_0 = -\frac{1 + 2\tau_{\text{eff}}}{2y}\,\omega^4 + O(\omega^6).$$

If $|\tau_{\text{eff}}| < \frac{1}{2}$, both coefficients are negative. If $\tau_{\text{eff}} = \frac{1}{2}$ and $s_{2\mathfrak{p}} < 0$, then $\mathcal{S}_0 \le \frac{1}{2}s_{2\mathfrak{p}}\omega^{2\mathfrak{p}}$ and $\check{\mathcal{H}}_0 \le -\frac{1}{2y}\omega^4$ near 0. In both cases there is $\omega_1 \in (0, \frac{2\pi}{3}]$ with $\mathcal{S}_0 < 0$ and $\check{\mathcal{H}}_0 < 0$ on $(0,\omega_1]$, and the criterion gives $|\Theta_0(\omega)| < \omega/2$ there. With $K_0 = \lceil 2\pi/\omega_1 \rceil \ge 3$, every $K \ge K_0$ has $2\pi/K \le \omega_1$, so $|\Theta_0(2\pi/K)| < \pi/K$; as $\mathcal{S}_0(2\pi/K) \ne 0$, also $\phi_0\phi_1(e^{2\pi i/K}) \ne 0$, and (d), with $j = 1$, gives the feasible spectra.

(c) Let $\mathcal{R} = v_0/u_0$, holomorphic near $z = 1$ as $u_0(1) = 1$. By the transfer-function form of τ_{eff}, $\mathcal{R}(1+w) = -\frac{w^2}{y}(1 - (1 + \tau_{\text{eff}})w + O(w^2))$. The function $G(\zeta,\delta) = \delta^{-2}\mathcal{R}(1 + \delta\zeta) - 1$ is holomorphic near $(\zeta,\delta) = (i\sqrt{y}, 0)$, with $G(\zeta,0) = -\zeta^2/y - 1$ and $\partial_\zeta G(i\sqrt{y},0) = -2i/\sqrt{y} \ne 0$. The implicit function theorem gives a holomorphic $\zeta(\delta)$ with $G(\zeta(\delta),\delta) = 0$ and $\zeta(0) = i\sqrt{y}$, and the terms of order δ give $\zeta'(0) = \frac{1+\tau_{\text{eff}}}{2}\zeta(0)^2 = -\frac{(1+\tau_{\text{eff}})y}{2}$. So for small $\lambda = \delta^2 > 0$, $z = 1 + \delta\zeta(\delta)$ is a root of $\lambda u_0 - v_0$ with

$$|z|^2 = 1 + 2\delta\operatorname{Re}\zeta(\delta) + \delta^2|\zeta(\delta)|^2 = 1 - \tau_{\text{eff}}\,\lambda y + O(\lambda^{3/2}).$$

If $\tau_{\text{eff}} < 0$, fix such a $\lambda_1 \in (0,1]$ with a root of modulus $r_1 > 1$. For small ε, $\lambda_1 \in [\varepsilon^2, 1]$, and the roots of $\lambda_1 u_\varepsilon - v_\varepsilon$ converge to those of $\lambda_1 u_0 - v_0$, since its leading coefficient $-v_D$ is nonzero at $\varepsilon = 0$; so $\rho_q \ge (1 + r_1)/2$ for small ε, and the root gives growing solutions on $\frac{\lambda_1}{2}x^2$ on $\mathbb{R}$. By Lemma 2.3, $\rho_{\mathcal{F}} \ge \rho_q > 1$.

(d) At $t = 0$ and $\omega_0 = 2\pi j/K$ the values g, s depend continuously on ε, with a nonzero limit by (22) and $\phi_0\phi_1(e^{i\omega_0}) \ne 0$. Hence, for small ε, $(g,s)(0,\omega_0) \ne (0,0)$ and $|\Theta(\omega_0)| < \pi/K$ (as $\Theta_0(\omega_0)$ lies away from the branch point $\pm\pi$), and Lemma C.14(b) gives a pair feasible at some $t > 0$ and $\rho_{\mathcal{F}} > 1$. □

Proof of Corollary 5.6. If $\tau_{\text{eff}} < 0$, Theorem 5.5(c) gives $\rho_q > 1$ for small ε. If $\rho_{\mathcal{F}} < 1$ for all small ε, then $\tau_{\text{eff}} \ge 0$ by Lemma 2.3 and the first claim, and $\tau_{\text{eff}} \ge \frac{1}{2}$ by Theorem 5.5(b), whose conclusion is $\rho_{\mathcal{F}} > 1$ whenever $|\tau_{\text{eff}}| < \frac{1}{2}$. □

Proof of Theorem 5.8. (a) Put

$$Q(z) = -\frac{z(1+z)\,q_0(z)}{\phi_1(z)} = \frac{z(1+z)\,\phi_0(z)}{(z-1)^2\phi_1(z)}.$$

Since $\deg u_0 < \deg v_0$, ϕ_1 and $z(1+z)q_0$ have the same degree and opposite leading coefficients, so $Q(\infty) = 1$; and $Q(1) = -2q_0(1)/\phi_1(1) = 2/y$, as $\phi_1(1) = u_0(1) = 1$. By the hypothesis, Q is holomorphic in $|z| > 1$ and continuous on $|z| \ge 1$, the simple root of ϕ_1 at -1, if any, being cancelled

by $1+z$. On the unit circle, $z/(z-1)^2 = -1/(4\sin^2\frac{\omega}{2})$ and $(1+z)/(1-z) = i\cot\frac{\omega}{2}$, so

$$Q(e^{i\omega}) = -\frac{i\cot\frac{\omega}{2}}{4\sin^2\frac{\omega}{2}} \cdot \frac{(1-z)\phi_0(z)\overline{\phi_1(z)}}{|\phi_1(z)|^2}, \qquad \operatorname{Im} Q(e^{i\omega}) = -\frac{\cos\frac{\omega}{2}\ \mathcal{S}_0(\omega)}{4\sin^3\frac{\omega}{2}\ |\phi_1(e^{i\omega})|^2}$$

for $0 < \omega < \pi$. Let $F(\zeta) = Q(1/\zeta) - 1$: it is holomorphic in the unit disk, continuous on its closure, has real Taylor coefficients and $F(0) = 0$. By the Schwarz formula, $F(\zeta) = \frac{i}{2\pi}\int_{-\pi}^{\pi} \operatorname{Im} F(e^{it})\,\frac{e^{it}+\zeta}{e^{it}-\zeta}\,dt$, and since $\operatorname{Im} F(e^{-it}) = -\operatorname{Im} F(e^{it})$, pairing t with $-t$ gives

$$F(\zeta) = \frac{2}{\pi}\int_0^{\pi} \operatorname{Im} F(e^{it})\,\frac{\zeta\sin t}{1-2\zeta\cos t+\zeta^2}\,dt \qquad (|\zeta|<1).$$

Here $\operatorname{Im} F(e^{it}) = \operatorname{Im} Q(e^{-it}) = -\operatorname{Im} Q(e^{it})$. Let $\zeta = x \uparrow 1$. As $1-2x\cos t+x^2 \geq 2x(1-\cos t)$, the kernel is at most $\frac{1}{2}\cot\frac{t}{2}$ and tends to it, while $\operatorname{Im} F(e^{it})\cot\frac{t}{2} = \cos^2\frac{t}{2}\,\mathcal{S}_0(t)/(4\sin^4\frac{t}{2}\,|\phi_1|^2)$ is bounded: near $t=0$ because $\mathcal{S}_0 = O(t^4)$ (Lemma 5.4), and near $t=\pi$ because Q is continuous at -1. By dominated convergence and the continuity of F at 1,

$$\frac{2}{y} - 1 = F(1) = \frac{1}{\pi}\int_0^{\pi} \frac{\cos^2\frac{t}{2}\ \mathcal{S}_0(t)}{4\sin^4\frac{t}{2}\ |\phi_1(e^{it})|^2}\,dt,$$

which is (24), since $\sin^2\frac{t}{2} = \frac{\xi}{2}$ and $\cos^2\frac{t}{2} = \frac{2-\xi}{2}$. If $\mathcal{S}_0 \equiv 0$, then $\operatorname{Im} F = 0$ on the circle, so the harmonic function $\operatorname{Im} F$ vanishes, F is a real constant, and $F = F(0) = 0$: $Q \equiv 1$. Then $u_0 - (z-1)^2 q_0 = \phi_1 = -z(1+z)q_0$, i.e. $u_0 = (1-3z)q_0$, and $v_0/u_0 = -(z-1)^2/(3z-1)$. Conversely, if $v_0/u_0 = -(z-1)^2/(3z-1)$, then $u_0 = (1-3z)q_0$, $\phi_0/\phi_1 = (v_0/u_0)/(v_0/u_0-1) = (z-1)^2/(z(z+1))$, and on the circle $(1-z)\phi_0/\phi_1 = -(z-1)^3/(z(z+1)) = 4i\sin^2\frac{\omega}{2}\tan\frac{\omega}{2}$ is imaginary, so $\mathcal{S}_0 = |\phi_1|^2\operatorname{Re}[(1-z)\phi_0/\phi_1] \equiv 0$; moreover $q_0(1) = -\frac{1}{2}u_0(1) = -\frac{1}{2}$, so $y = 2$, and $u_0'(1) = -2q_0'(1) + \frac{3}{2}$, so $\tau_{\text{eff}} = \frac{1}{2}$.

(b) Divide u_0 and v_0 by their greatest common divisor r, normalized by $r(1) = 1$ (possible since $u_0(1) = 1$). This preserves v_0/u_0 and y, keeps the roots of $\lambda u_0 - v_0$ among the old ones, and gives a reduced symbol $\mathcal{S}_0^{\text{red}}$ with $\mathcal{S}_0 = |r|^2\mathcal{S}_0^{\text{red}}$. Thus $\mathcal{S}_0^{\text{red}} \geq 0$ away from the finitely many roots of r, and everywhere by continuity. We may therefore assume u_0 and v_0 coprime. Put $\phi_\lambda = \lambda u_0 - v_0 = \lambda\phi_1 + (1-\lambda)\phi_0$. On the circle

$$\operatorname{Re}\big[(1-z)\phi_0\overline{\phi_\lambda}\big] = \lambda\,\mathcal{S}_0 + (1-\lambda)\,\xi\,|\phi_0|^2. \tag{49}$$

Let $\lambda \in (\lambda_0, 1)$. If $\phi_\lambda(e^{i\alpha}) = 0$, then $\alpha \neq 0$, since $\phi_\lambda(1) = \lambda$, and (49) vanishes at α; both terms are nonnegative ($\mathcal{S}_0$ is even and continuous), so $\phi_0(e^{i\alpha}) = 0$, and then $u_0(e^{i\alpha}) = 0$, against coprimality. So the roots of ϕ_λ lie in the open unit disk, and (a) applies to the method $(u_0, v_0/\lambda)$, whose step is λy and whose ϕ_0, ϕ_1 are ϕ_0/λ, ϕ_λ/λ:

$$\frac{2}{\lambda y} - 1 = \frac{1}{2\pi}\int_0^{\pi} \frac{(2-\xi)\big(\lambda\mathcal{S}_0 + (1-\lambda)\xi|\phi_0|^2\big)}{\xi^2\,|\phi_\lambda(e^{i\omega})|^2}\,d\omega\ \geq 0. \tag{50}$$

For the same strictly stable curvatures, put

$$Q_\lambda(z) = -\frac{z(1+z)q_0(z)}{\phi_\lambda(z)}, \qquad F_\lambda(\zeta) = Q_\lambda(1/\zeta) - 1.$$

The paired Schwarz representation in (a) has a nonnegative density by (49). For $0 < \zeta < 1$, its kernel is nonnegative and at most its limiting value at $\zeta = 1$. Thus, for every real $x > 1$,

$$0 \leq Q_\lambda(x) - 1 \leq F_\lambda(1) = \frac{2}{\lambda y} - 1.$$

Letting $\lambda \uparrow 1$ gives $y \leq 2$. If $y = 2$, fix $x > 1$ with $\phi_1(x) \neq 0$. Continuity in λ and the displayed bounds give $Q_1(x) = 1$, hence $-x(1+x)q_0(x) = \phi_1(x)$. Since $\phi_1(1) = 1$, only finitely many real x are excluded. The two polynomials agree at infinitely many points, so $\phi_1(z) = -z(1+z)q_0(z)$, or $u_0 = (1-3z)q_0$. Therefore $v_0/u_0 = -(z-1)^2/(3z-1)$. Conversely, this transfer function has $y = 2$, as computed in (a). □

Proof of Corollary 5.9. The limit method satisfies the hypotheses of Theorem 5.8(b): for small ε the roots of $\lambda u_\varepsilon - v_\varepsilon$ lie in the open unit disk for $\lambda \in [\varepsilon^2, 1]$, and they depend continuously on ε ($\deg v_0 = D$), so for $\lambda \in (0,1)$ the roots of $\lambda u_0 - v_0$ lie in the closed unit disk. (i) $y \le 2$ by Theorem 5.8(b), and Corollary 5.3(i) gives $\limsup \sqrt{\kappa}(1-\rho_{\mathcal{F}}) \le \sqrt{y/2} \le 1$ and $\limsup \sqrt{\kappa}(1-\rho_q) \le \sqrt{y} \le \sqrt{2}$, since the first coordinates $\zeta_-(\theta', y)$, $\theta' \ge 2\sqrt{y}$, of the arc of Corollary 3.2 are at most $\zeta_-(2\sqrt{y}, y) = \sqrt{y}$. (ii) By Corollary 5.3(iii), $\theta \ge 3$ and $y \ge \theta - 1 \ge 2$; with (i), $y = 2$ and $\theta = 3$, and the equality case of Theorem 5.8(b) gives the transfer function, for which $\tau_{\text{eff}} = \frac{1}{2}$ and $\mathcal{S}_0 \equiv 0$ by Theorem 5.8(a). (iii) By Theorem 5.5(a), $\rho_{\mathcal{F}} < 1$ for small ε, so the family is stable on quadratics (Lemma 2.3), and $\mathcal{S}_0 > 0$ on $(0, \pi]$. So $y \le 2$, and $y \ne 2$, since $y = 2$ would force $\mathcal{S}_0 \equiv 0$. The constant is $c_{\mathcal{F}}(\theta, y)$ by Theorem 5.5(a), and $c_{\mathcal{F}}(\theta, y) \le \sqrt{y/2}$ by Corollary 3.2. □

E. Two-Step Rates and Boundary Layers

Write $\chi_\mu(z) = z^2 - S_\mu z + Q_\mu$ and $\chi_L(z) = z^2 - S_L z + Q_L$ with $S_\mu = 1 + \beta - y_\mu(1+\tau)$, $Q_\mu = \beta - y_\mu \tau$, $S_L = 1 + \beta - y(1+\tau)$, $Q_L = \beta - y\tau$.

Lemma E.1 (Lag-one cubic)**.** *For $z = re^{i\omega}$, let $H_r(\omega) = \operatorname{Re}[(1-z)\chi_\mu(z)\overline{\chi_L(z)}]$, $E_0 = r^4 + S_\mu S_L r^2 + Q_\mu Q_L$, $E_1 = S_L r^2 + S_\mu Q_L$ and $E_2 = S_\mu r^2 + Q_\mu S_L$. Then*

$$H_r = E_0 + r^2 E_2 - r(E_0 + E_1 + E_2 + Q_\mu r^2)\cos\omega + r^2(E_1 + Q_\mu + Q_L)\cos 2\omega - Q_L r^3 \cos 3\omega.$$

In $\xi = 1 - \cos\omega$, this becomes $H_r = \mathsf{P}_0 + \mathsf{P}_1\xi + \mathsf{P}_2\xi^2 + \mathsf{P}_3\xi^3$, where $\mathsf{P}_0 = (1-r)\chi_\mu(r)\chi_L(r)$, $\mathsf{P}_3 = 4Q_L r^3$, $\mathsf{P}_2 = 2r^2(E_1 + Q_\mu + Q_L) - 12Q_L r^3$ and $\mathsf{P}_1 = r(E_0 + E_1 + E_2 + Q_\mu r^2) - 4r^2(E_1 + Q_\mu + Q_L) + 9Q_L r^3$. At $\omega = \pi$ ($\xi = 2$), $H_r = (1+r)\chi_\mu(-r)\chi_L(-r)$.

Proof. Using $z\bar{z} = r^2$, expand

$$\chi_\mu \overline{\chi_L} = E_0 - E_1 z - E_2 \bar{z} + Q_L z^2 + Q_\mu \bar{z}^2.$$

Multiplication by $1 - z$, followed by $\operatorname{Re} z^n = \operatorname{Re} \bar{z}^n = r^n \cos n\omega$, gives the trigonometric polynomial. Substitute $\cos\omega = 1 - \xi$, $\cos 2\omega = 1 - 4\xi + 2\xi^2$ and $\cos 3\omega = 1 - 9\xi + 12\xi^2 - 4\xi^3$ and collect powers of ξ. The evaluations at $z = r$ and $z = -r$ give P_0 and the endpoint formula. □

Lemma E.2 (Mixed test)**.** *Let (β, τ, h) be a tuning of $\mathcal{M}$, $r > 0$, and for $z = re^{i\omega}$ put $H_r^{\text{mix}}(\omega) = \operatorname{Re}[(1 - e^{i\omega})\chi_\mu(z)\overline{\chi_L(z)}]$. Then*

$$\operatorname{Im}\big(\chi_\mu(z)\overline{\chi_L(z)}\big) = (y - y_\mu)\, r \sin\omega \,\big[(1+\tau)(r^2 - \beta) + \tau(1+\beta) - 2\tau r\cos\omega\big],$$

and $H_r^{\text{mix}} = \xi\, \mathsf{M}_r(\xi)$, $\xi = 1 - \cos\omega$, with the quadratic $\mathsf{M}_r(\xi) = E_0 - (E_1 + E_2)r(1-\xi) + (Q_\mu + Q_L)r^2(1 - 4\xi + 2\xi^2) + (2 - \xi)(y - y_\mu)r[(1+\tau)(r^2 - \beta) + \tau(1+\beta) - 2\tau r(1-\xi)]$ (notation of Lemma E.1). If $K \ge 3$, $\omega_0 = 2\pi/K$, $\operatorname{Im}(\chi_\mu\overline{\chi_L})(re^{i\omega_0}) > 0$ and $H_r^{\text{mix}}(\omega_0) \le 0$, then $(\delta_{\omega_0}, 0)$ is feasible at $t = \log r$.

Proof. Writing $\chi_\lambda = \Omega + y_\lambda N_\tau$ gives $\operatorname{Im}(\chi_\mu\overline{\chi_L}) = (y - y_\mu)\operatorname{Im}(\Omega\overline{N_\tau})$. Substitution of $\Omega = (z-1)(z-\beta)$, $N_\tau = (1+\tau)z - \tau$ and $z\bar{z} = r^2$ gives the stated imaginary part. The product expansion in Lemma E.1 gives the real part, so

$$H_r^{\text{mix}} = \xi \operatorname{Re}(\chi_\mu\overline{\chi_L}) + \sin\omega \operatorname{Im}(\chi_\mu\overline{\chi_L}) = \xi \mathsf{M}_r,$$

using $\sin^2\omega = \xi(2 - \xi)$. At $t = \log r$, (15) gives $g - 2is = -\mathfrak{c}\chi_\mu\overline{\chi_L}$, $\mathfrak{c} > 0$. Thus $s > 0$, and $g - 2is = Re^{-i\Theta_r}$ with $R > 0$, $\Theta_r \in (0, \pi)$. By Lemma C.2 and the sign hypothesis,

$$0 \le -\mathfrak{c} H_r^{\text{mix}}(\omega_0) = 2R \sin\frac{\omega_0}{2} \sin\frac{\omega_0 - 2\Theta_r}{2}.$$

Since $(\omega_0 - 2\Theta_r)/2 \in (-\pi, \pi/2)$, this implies $\Theta_r \le \omega_0/2 = \pi/K$. Lemma C.14(a), with $j = 1$, gives the claimed feasible atom. □

Proof of Proposition 6.1(a). The coefficients of L_τ are

$$\begin{aligned} \mathsf{A}_0 &= 2(1-\beta)(y-y_\mu)+y_\mu y, \qquad \mathsf{A}_2 = 4(\beta-\tau y),\\ \mathsf{A}_1 &= 2(1-\beta)^2 - 2y_\mu(\beta+\beta\tau+\tau) + y(6\tau-2-2\beta\tau) + 2\tau(1+\tau)y_\mu y; \end{aligned} \tag{51}$$

at $\beta = 1$, $y_\mu = 0$ they are 0, $2y(2\tau-1)$ and $4(1-\tau y)$. On $|z| = 1$, put $\mathcal{R}_0 = \mathrm{Re}(\chi_\mu \overline{\chi_L})$ and $q = (y-y_\mu)[(1-\beta)+2\tau\xi]$. By (16),

$$H_1 = \xi\mathcal{R}_0 + q\sin^2\omega = \xi\big[\mathcal{R}_0 + (2-\xi)q\big], \qquad L_\tau = \mathcal{R}_0 + (2-\xi)q.$$

Substituting $\chi_\lambda = (z-1)(z-\beta) + y_\lambda((1+\tau)z-\tau)$, $z\bar z = 1$ and $\mathrm{Re}\, z = 1-\xi$ and collecting powers of ξ gives (51).

Stability gives $\chi_\mu(-1)\chi_L(-1) > 0$ (Lemma 2.1), hence $L_\tau(2) = \chi_\mu(-1)\chi_L(-1) > 0$. Also $L_\tau(0) = \mathsf{A}_0 > 0$. For the angle, let $\omega \in (0,\pi]$ with $g - 2is \neq 0$ at $(0,\omega)$ (for $\omega < \pi$ this holds by (16), and at $\omega = \pi$ for stable tunings). By (15) and (16), $2s = \mathfrak{c} q \sin\omega \geq 0$ with $\mathfrak{c} = 4/(h^2(L-\mu)^2)$, so $\Theta \in [0,\pi]$ and $\Theta - \frac{\omega}{2} \in [-\frac{\pi}{2}, \pi)$. By Lemma C.2, $H_1 = \xi L_\tau$ is a positive multiple of $\sin(\Theta - \frac{\omega}{2})$; hence $L_\tau(\xi) > 0$ if and only if $\Theta > \omega/2$, and $L_\tau(\xi) = 0$ if and only if $\Theta = \omega/2$. The last claim is Proposition 4.9(b), since $H_1 = \xi L_\tau$. □

Lemma E.3 (Resonance on a band)**.**

(a) *Let* $\mathsf{s} \in [0,1)$ *and* $0 < \omega_a < \omega_b \leq \frac{2\pi}{3}$ *with* $2\pi(1-\mathsf{s})/\omega_a - 2\pi(1+\mathsf{s})/\omega_b \geq 1$. *Then there is an integer* $K \geq 2\pi(1+\mathsf{s})/\omega_b \geq 3$ *with* $[(1-\mathsf{s})\frac{2\pi}{K}, (1+\mathsf{s})\frac{2\pi}{K}] \subset [\omega_a, \omega_b]$.

(b) *Let a tuning of* $\mathcal{M}$ *be stable,* $K \geq 3$, *and* $L_\tau(1-\cos\frac{2\pi}{K}) < 0$. *Then* $\Theta(2\pi/K) < \pi/K$, *the method has trajectories from rest that approach* K*-periodic orbits in* $\mathbb{R}^3$, *and* $\rho^0_{\mathcal{F}} > 1$. *In particular this holds for some* $K \geq 2\pi/\omega_b$ *if* $L_\tau(1-\cos\omega) < 0$ *for all* $\omega \in [\omega_a, \omega_b]$, *with* ω_a, ω_b *as in (a) for* $\mathsf{s} = 0$.

Proof. (a) The interval $[2\pi(1+\mathsf{s})/\omega_b,\, 2\pi(1-\mathsf{s})/\omega_a]$ has length at least 1, so it contains an integer K; then $(1+\mathsf{s})2\pi/K \leq \omega_b$, $(1-\mathsf{s})2\pi/K \geq \omega_a$, and $K \geq 2\pi/\omega_b \geq 3$. (b) By Proposition 6.1(a), $\Theta(2\pi/K) < \pi/K$, and Proposition 4.10, with $j = 1$, gives the orbits and $\rho^0_{\mathcal{F}} > 1$. For the last claim, (a) with $\mathsf{s} = 0$ gives K with $2\pi/K \in [\omega_a, \omega_b]$. □

Lemma E.4 (Accelerated tunings)**.** *Let a stable tuning of* $\mathcal{M}$ *on* $[\mu, L]$, $L = 1$, $\mu = \varepsilon^2$, $0 < \varepsilon < 1$, *have* $\rho_q \leq 1 - c\varepsilon$, $c > 0$.

(a) $y_\mu \geq c^2\varepsilon^2$, $c^2 \leq y < 4$, $\tau < 2/c^2$ *and* $1 - \beta < \max(4, 8/c)\,\varepsilon$.

(b) *If such tunings are given along a sequence* $\varepsilon \to 0^+$, *with parameters* $\beta = 1 - \theta_\varepsilon\varepsilon$, $h = y_\varepsilon$ *and* τ_ε, *a subsequence has* $(\theta_\varepsilon, y_\varepsilon, \tau_\varepsilon) \to (\theta, y, \tau)$ *with* $\theta \geq 2c$, $y \geq c^2$, $\tau \geq 0$ *and* $y(1+2\tau) \leq 4$*; along it the tunings form a family* (3).

Proof. (a) As in the proof of Theorem 2.7, $y_\mu \geq (1-\rho_q)^2 \geq c^2\varepsilon^2$ and $y = \kappa y_\mu \geq c^2$. Stability gives $y(1+2\tau) < 2(1+\beta) < 4$, so $\tau < (4-y)/(2y) < 2/c^2$. Put $\delta = 1-\beta$. For $\eta \geq 0$, $\chi_\mu(1-\eta) = \eta^2 - \eta(\delta + y_\mu(1+\tau)) + y_\mu \leq \eta^2 - \eta\delta + y_\mu$. If $y_\mu < \delta^2/4$, then $\chi_\mu(1 - 2y_\mu/\delta) < 0 < \chi_\mu(1)$, so χ_μ has a root in $(1 - 2y_\mu/\delta, 1)$, and $c\varepsilon < 2y_\mu/\delta$, i.e. $\delta < 2y_\mu/(c\varepsilon) < 8\varepsilon/c$. Otherwise $\delta \leq 2\sqrt{y_\mu} < 4\varepsilon$. (b) The roots of χ_μ have moduli at most $\rho_q \leq 1 - c\varepsilon$, and their product is $Q_\mu = \beta - y_\mu\tau$. Hence $1 - \theta_\varepsilon\varepsilon - y_\varepsilon\tau\varepsilon^2 = Q_\mu \leq (1-c\varepsilon)^2$, and $y_\varepsilon\tau < 8/c^2$ by (a); so $\theta_\varepsilon \geq 2c - (c^2 + 8/c^2)\varepsilon$. By (a), $(\theta_\varepsilon, y_\varepsilon, \tau_\varepsilon)$ stays in a bounded set, so a subsequence converges; the limits satisfy $\theta \geq 2c > 0$, $y \geq c^2 > 0$, $\tau \geq 0$, and $y(1+2\tau) \leq 4$ by the stability condition $y_\varepsilon(1+2\tau_\varepsilon) < 2(1+\beta)$ (Lemma 2.1). □

Proof of Theorem 6.2. If the first claim failed, Lemma E.4 would give a subsequence with $\beta \to 1$, $y_\mu \to 0$, $y_\varepsilon \to y > 0$, and $\tau_\varepsilon \to \tau < 1/2$. By Proposition 6.1, its lag-one polynomials converge coefficientwise to

$$L_0(\xi) = 2y(2\tau-1)\xi + 4(1-\tau y)\xi^2.$$

The negative linear coefficient makes $L_0 < 0$ on a sufficiently small interval $(0, \xi_0)$. Choose one integer $K \geq 6$ with $1 - \cos(2\pi/K) < \xi_0$. Coefficientwise convergence gives $L_{\tau_\varepsilon}(1 - \cos(2\pi/K)) < 0$ for all sufficiently small ε on the subsequence. Propositions 4.10 and 6.1 then give the periodic orbits and $\rho^0_{\mathcal{F}} > 1$, contradicting the choice of that subsequence.

If an accelerated family on $\mathcal{F}_{\mu,L}$ had $\liminf \tau_\kappa < 1/2$, a subsequence with $\limsup \tau_\kappa < 1/2$ would satisfy the first claim, contrary to $\rho^0_{\mathcal{F}} \leq \rho_{\mathcal{F}} < 1$. □

Proposition E.5 (Range of resonant periods)**.** *Let $\tau \in [0, \frac{1}{2})$, $\theta, y > 0$, $y(1 + 2\tau) < 4$, and consider $\beta = 1 - \theta\varepsilon$, $h = y$. Then $\tau y < 1$. Put $\xi_* = \frac{y(1-2\tau)}{2(1-\tau y)} \in (0, 2)$ and $\xi_{\min} = \frac{\theta\varepsilon}{1-2\tau}$. For every $\delta \in (0, \frac{1}{2})$ and all small ε, $L_\tau < 0$ on $[(1 + \delta)\xi_{\min}, (1 - \delta)\xi_*]$, while $L_\tau > 0$ on $(0, (1 - \delta)\xi_{\min}]$ and on $[(1 + \delta)\xi_*, 2]$. Consequently the method has K-periodic orbits for every K with $1-\cos(2\pi/K) \in [(1+\delta)\xi_{\min}, (1-\delta)\xi_*]$; the largest such K is $\pi\big(2(1 - 2\tau)/((1 + \delta)\theta)\big)^{1/2}\varepsilon^{-1/2}(1 + o(1))$, and the number of such K grows like $\varepsilon^{-1/2}$.*

Proof of Proposition E.5. Stability implies $\tau y < 4\tau/(1 + 2\tau) < 1$, and $\xi_* < 2$. The coefficients of Proposition 6.1 satisfy

$$\mathsf{A}_0 = 2\theta y\varepsilon + O(\varepsilon^2), \quad \mathsf{A}_1 = -2y(1 - 2\tau) + O(\varepsilon), \quad \mathsf{A}_2 = 4(1 - \tau y) + O(\varepsilon) > 0.$$

The limit polynomial has the two simple roots 0 and ξ_*. Its perturbed roots are therefore

$$\xi_1 = \frac{\theta\varepsilon}{1 - 2\tau} + O(\varepsilon^2) = \xi_{\min} + O(\varepsilon^2), \qquad \xi_2 = \xi_* + O(\varepsilon).$$

For small ε, the polynomial is negative between these roots and positive outside; the claimed intervals follow for every fixed $\delta > 0$. Lemma E.3(b) gives the periodic orbits. Finally, $1 - \cos(2\pi/K) = 2\pi^2 K^{-2}(1 + O(K^{-2}))$ gives the largest admitted period and the asserted count. □

For the remainder of this appendix, use (3) with $L = 1$, $\mu = \varepsilon^2$, and $\Delta^\varepsilon_c = c^2 - \theta_\varepsilon c + y_\varepsilon$.

Lemma E.6 (Lag-one and mixed coefficients in the accelerated scaling)**.** *Let $r = 1 - c\varepsilon$ with $c \in \mathbb{R}$. The following expansions hold uniformly for c and the parameters in compact sets.*

(a) *The coefficients of Lemma E.1 satisfy $\mathsf{P}_0 = c\varepsilon\,\chi_\mu(r)\chi_L(r)$ with $\chi_\mu(r) = \varepsilon^2(\Delta^\varepsilon_c - y_\varepsilon(1 + \tau_\varepsilon)c\varepsilon)$ and $\chi_L(r) = y_\varepsilon + O(\varepsilon)$, and*

$$\mathsf{P}_1 = 2y_\varepsilon(\theta_\varepsilon - 3c)\varepsilon + O(\varepsilon^2), \qquad \mathsf{P}_2 = 2y_\varepsilon(2\tau_\varepsilon - 1) + O(\varepsilon), \qquad \mathsf{P}_3 = 4(1 - \tau_\varepsilon y_\varepsilon) + O(\varepsilon).$$

(b) *For $z = re^{i\omega}$, $\operatorname{Im}(\chi_\mu\overline{\chi_L}) = (y_\varepsilon - y_\mu)r\sin\omega\,[\varepsilon(\theta_\varepsilon - 2c) + (1 + \tau_\varepsilon)c^2\varepsilon^2 + 2\tau_\varepsilon r\xi]$, and the quadratic of Lemma E.2 is $\mathsf{M}_r = \mathsf{Q}_0 + \mathsf{Q}_1\xi + \mathsf{Q}_2\xi^2$ with $\mathsf{Q}_0 = 2y_\varepsilon(\theta_\varepsilon - 2c)\varepsilon + O(\varepsilon^2)$, $\mathsf{Q}_1 = 2y_\varepsilon(2\tau_\varepsilon - 1) + O(\varepsilon)$ and $\mathsf{Q}_2 = 4(1 - \tau_\varepsilon y_\varepsilon) + O(\varepsilon)$.*

(c) *Regard $(\theta', y', \tau') = (\theta_\varepsilon, y_\varepsilon, \tau_\varepsilon)$ as independent parameters. Then*

$$\mathsf{P}_1 = \varepsilon\dot{\mathsf{P}}_1 + O(\varepsilon^2), \quad \mathsf{P}_2 = 2y'(2\tau' - 1) + \varepsilon\dot{\mathsf{P}}_2 + O(\varepsilon^2), \quad \mathsf{P}_3 = 4(1 - \tau' y') + \varepsilon\dot{\mathsf{P}}_3 + O(\varepsilon^2),$$
$$\mathsf{Q}_0 = \varepsilon\dot{\mathsf{Q}}_0 + O(\varepsilon^2), \quad \mathsf{Q}_1 = 2y'(2\tau' - 1) + \varepsilon\dot{\mathsf{Q}}_1 + O(\varepsilon^2), \quad \mathsf{Q}_2 = 4(1 - \tau' y') + \varepsilon\dot{\mathsf{Q}}_2 + O(\varepsilon^2),$$

where $\dot{\mathsf{P}}_k, \dot{\mathsf{Q}}_k$ are polynomials in (c, θ', y', τ'), and at $(y', \tau') = (2, \frac{1}{2})$

$$\dot{\mathsf{P}}_1 = 4(\theta' - 3c), \quad \dot{\mathsf{P}}_2 = 2(\theta' + 2c), \quad \dot{\mathsf{P}}_3 = -4\theta', \qquad \dot{\mathsf{Q}}_0 = 4(\theta' - 2c), \quad \dot{\mathsf{Q}}_1 = 2(\theta' + c), \quad \dot{\mathsf{Q}}_2 = -4\theta'.$$

Proof. Treat $p = (c, \theta', y', \tau')$ as independent parameters and substitute $r = 1 - c\varepsilon$, $\beta = 1 - \theta'\varepsilon$ and $y_\mu = y'\varepsilon^2$ in the coefficient formulas of Lemmas E.1 and E.2. Each of the six coefficients F is polynomial, so

$$F(\varepsilon, p) = F(0, p) + \varepsilon\partial_\varepsilon F(0, p) + \varepsilon^2 R_F(\varepsilon, p),$$

with R_F polynomial. Collecting the constant and linear powers of ε gives the displayed coefficients in (a)–(c); setting $(y', \tau') = (2, 1/2)$ in the linear coefficients gives the six displayed derivative values. For

every compact parameter set K, both the first coefficients and R_F are bounded on $[-1,1]\times K$, proving all the stated uniform remainders.

The exact characteristic formulas in (a) follow from $r-1=-c\varepsilon$, $r-\beta=(\theta'-c)\varepsilon$ and $N_{\tau'}(r)=1-(1+\tau')c\varepsilon$. For (b), the imaginary-part bracket of Lemma E.2 reduces exactly to $\varepsilon(\theta'-2c)+(1+\tau')c^2\varepsilon^2+2\tau' r\xi$. The same coefficient extraction then gives the assertions about M_r. □

Lemma E.7 (Lag-one certificates in the accelerated scaling)**.** *Let tunings of $\mathcal{M}$ with $L=1$, $\mu=\varepsilon^2$, $\beta=1-\theta_\varepsilon\varepsilon$, $h=y_\varepsilon$, $\tau=\tau_\varepsilon$ and $(\theta_\varepsilon,y_\varepsilon,\tau_\varepsilon)\to(\theta,y,\tau)$, $\theta,y>0$, be given for small ε (or along a sequence $\varepsilon\to 0$). Let $0<c<c_Q(\theta,y)$ and $r=1-c\varepsilon$. Then, for small ε, $\mathsf{P}_0>0$ and the roots of χ_μ lie in $|z|<r$. If moreover $\mathsf{P}_1+\mathsf{P}_2\xi+\mathsf{P}_3\xi^2>0$ on $[0,2]$ for small ε, then $H_r>0$ on $[0,2]$, and $\rho_q<r$, $r_c<r$ and $\rho_{\mathcal{F}}<r$; the last bound is certified by the lag-one criterion.*

Proof. Since $c<c_Q$, Lemma B.1(a) gives $\Delta_c>0$, and Lemma E.6(a) yields $\mathsf{P}_0=c\varepsilon^3(\Delta_c+o(1))(y+o(1))>0$. Exactly,

$$\chi_\mu(1-\varepsilon\zeta)=\varepsilon^2(\zeta^2-\theta''_\varepsilon\zeta+y_\varepsilon),\qquad \theta''_\varepsilon=\theta_\varepsilon+y_\varepsilon(1+\tau_\varepsilon)\varepsilon\to\theta.$$

The two ζ-roots are bounded, and their minimum real part tends to $c_Q>c$. Hence, for some $\eta>0$ and $K<\infty$, $|1-\varepsilon\zeta|^2\le 1-2(c+\eta)\varepsilon+K^2\varepsilon^2<(1-c\varepsilon)^2$ for small ε. This puts the roots of χ_μ inside $|z|<r$. The additional coefficient hypothesis gives $H_r=\mathsf{P}_0+\xi(\mathsf{P}_1+\mathsf{P}_2\xi+\mathsf{P}_3\xi^2)>0$ on $[0,2]$; Lemma C.13 then gives all three rate bounds. □

Lemma E.8 (The sign-alternating mode)**.**

(a) *Let tunings of $\mathcal{M}$ with $L=1$ be given along $\varepsilon\to 0$, with $\chi_L(-1)>0$, $\chi_L(-1)\to 0$ and $S_L\to S_\infty\in(-2,0)$. Then, for small ε, χ_L has the real roots $z_L=-1+\delta$ and $z_2=1+S_L-\delta$, with $\delta=\frac{\chi_L(-1)}{2+S_L}(1+O(\chi_L(-1)))>0$ and $z_2\to 1+S_\infty\in(-1,1)$; so $|z_2|<|z_L|=1-\delta$, and $\rho_q\ge 1-\delta$. From rest with $x_0=1$ on $\frac12 x^2$ on $\mathbb{R}$, $\limsup_N|x_N|^{1/N}=1-\delta$, so $\rho^0_{\mathcal{F}}\ge 1-\delta$; the dominant mode z_L^k alternates in sign.*

(b) *Let a family (3) have $\tau\ge\frac12$, $y(1+2\tau)=4$ and $\varsigma_\varepsilon\to\varsigma\in[2\theta,\infty]$ (notation (25)), and be stable for small ε. Then (a) applies, with $S_\infty=-\frac{2}{1+2\tau}$, $z_2\to\frac{2\tau-1}{1+2\tau}\in[0,1)$ and $\delta/\varepsilon\to(\varsigma-2\theta)\frac{1+2\tau}{4\tau}$, which is c_π if $\varsigma>2\theta$ (∞ if $\varsigma=\infty$) and 0 if $\varsigma=2\theta$.*

Proof. (a) Put $e=\chi_L(-1)$ and $\sigma=2+S_L\to 2+S_\infty\in(0,2)$. Then $\chi_L(-1+\delta)=e-\sigma\delta+\delta^2$, whose roots for $e<\sigma^2/4$ are $\delta=\frac{2e}{\sigma+\sqrt{\sigma^2-4e}}=\frac{e}{\sigma}(1+O(e))>0$ and $\sigma-\delta$. So $z_L=-1+\delta$ and $z_2=-1+\sigma-\delta=1+S_L-\delta\to 1+S_\infty\in(-1,1)$, and $|z_2|<1-\delta=|z_L|$ for small ε. Hence $\rho(\chi_L)=1-\delta$, and Lemma C.12 at $\lambda=L$ gives $\rho^0_{\mathcal{F}}\ge 1-\delta$, with dominant mode z_L^k. (b) By (25), $\chi_L(-1)=(\varsigma_\varepsilon-2\theta_\varepsilon)\varepsilon$, which is positive by stability (Lemma 2.1) and tends to 0, since $\varsigma_\varepsilon\varepsilon=4-y_\varepsilon(1+2\tau_\varepsilon)\to 0$. Further $S_L=1+\beta-y_\varepsilon(1+\tau_\varepsilon)\to 2-\frac{4(1+\tau)}{1+2\tau}=-\frac{2}{1+2\tau}\in[-1,0)$, so $2+S_L\to\frac{4\tau}{1+2\tau}$, $z_2\to 1+S_\infty=\frac{2\tau-1}{1+2\tau}$, and $\delta/\varepsilon=\frac{\varsigma_\varepsilon-2\theta_\varepsilon}{2+S_L}(1+o(1))\to(\varsigma-2\theta)\frac{1+2\tau}{4\tau}$, also when $\varsigma=\infty$. □

Lemma E.9 (Spread witnesses for $\mathcal{M}$)**.** *Let stable tunings of $\mathcal{M}$ with $L=1$, $\mu=\varepsilon^2$, $\beta=1-\theta_\varepsilon\varepsilon$, $h=y_\varepsilon$ and $\tau=\tau_\varepsilon$ be given along a sequence $\varepsilon\to 0$, with $(\theta_\varepsilon,y_\varepsilon,\tau_\varepsilon)\to(\theta,y,\tau)$, $\theta,y,\tau>0$. Let $0<c<\theta/2$, $r=1-c\varepsilon$, $t=\log r$, and fix $\mathsf{s}\in(0,\mathsf{s}_1]$ (Lemma C.17). For each ε let $K\ge 3$, $\omega_0=2\pi/K$, $B=[(1-\mathsf{s})\omega_0,(1+\mathsf{s})\omega_0]$, $\xi_{\max}=1-\cos((1+\mathsf{s})\omega_0)$, and $\Xi_B=\max_B|\chi_\mu\overline{\chi_L}|$ on $|z|=r$. Suppose that, as $\varepsilon\to 0$,*

$$\varepsilon=o(\xi_{\max}^{3/2}),\qquad \max_B|\mathsf{M}_r|=o(\xi_{\max}^{3/2}),\qquad H_r\le-\lambda_0\varepsilon\Xi_B\ \text{ on } B,$$

with a constant $\lambda_0>0$. Then, for all small ε, there is a pair with finitely many atoms in B and $|C|<\bar g$ feasible at t. For some finite $J=J_\varepsilon$, it is realized from rest by $f\in\mathcal{F}_{\varepsilon^2,1}(\mathbb{R}^{2J+1})$ and $x_0\ne 0$, with $\|x_N\|\ge c_0(1-c\varepsilon)^N\|x_0\|$ for every $N\ge 0$, where $c_0>0$ may depend on ε; in particular $\rho^0_{\mathcal{F}}\ge 1-c\varepsilon$.

Proof. Put $\mathfrak{c} = 4/(y_\varepsilon^2(1-\varepsilon^2)^2)$; by (15), $g - 2is = -\mathfrak{c}\chi_\mu\overline{\chi_L}$ at t. On B, $\omega \le \frac{7}{6}\cdot\frac{2\pi}{3} < \pi$, so $\sin\omega > 0$ and $2-\xi \ge \frac{1}{5}$, and $\xi \ge \frac{1}{2}\xi_{\max}$, because $\sin\frac{(1-\mathsf{s})\omega_0}{2} \ge \frac{1-\mathsf{s}}{1+\mathsf{s}}\sin\frac{(1+\mathsf{s})\omega_0}{2}$ and $(\frac{1-\mathsf{s}}{1+\mathsf{s}})^2 \ge \frac{1}{2}$.

Signs. By Lemma E.6(b), $\mathrm{Im}(\chi_\mu\overline{\chi_L}) = (y_\varepsilon - y_\mu)r\sin\omega\,\iota_\varepsilon$ with $\iota_\varepsilon = \varepsilon(\theta_\varepsilon - 2c) + (1+\tau_\varepsilon)c^2\varepsilon^2 + 2\tau_\varepsilon r\xi$, and $\iota_\varepsilon > 0$ for small ε, as $\theta > 2c$. By Lemma E.2, $\xi\mathsf{M}_r = \xi\,\mathrm{Re}(\chi_\mu\overline{\chi_L}) + \sin\omega\,\mathrm{Im}(\chi_\mu\overline{\chi_L})$, and $\sin^2\omega = \xi(2-\xi)$; hence

$$\mathrm{Re}(\chi_\mu\overline{\chi_L}) = \mathsf{M}_r - (2-\xi)(y_\varepsilon - y_\mu)r\,\iota_\varepsilon \le |\mathsf{M}_r| - 2(2-\xi)(y_\varepsilon - y_\mu)\tau_\varepsilon r^2\xi \le -\mathsf{a}\xi, \qquad \mathsf{a} = \tfrac{y\tau}{10},$$

for small ε, because $2(2-\xi)(y_\varepsilon - y_\mu)\tau_\varepsilon r^2 \to 2(2-\xi)y\tau \ge \frac{2}{5}y\tau$ uniformly on B, and $|\mathsf{M}_r| = o(\xi_{\max}) \le \frac{y\tau}{10}\xi_{\max} \le \frac{y\tau}{5}\xi$. Thus $g = -\mathfrak{c}\,\mathrm{Re}(\chi_\mu\overline{\chi_L}) > 0$ and $2s = \mathfrak{c}\,\mathrm{Im}(\chi_\mu\overline{\chi_L}) > 0$ on B: $g - 2is = Re^{-i\Theta}$ with $\Theta \in (0, \frac{\pi}{2})$ continuous and $R = \mathfrak{c}|\chi_\mu\overline{\chi_L}| \ge \mathfrak{c}\mathsf{a}\xi$.

Defect. By Lemmas C.2 and E.2, $2R\sin\frac{\omega}{2}\sin\frac{\omega-2\Theta}{2} = \mathrm{Re}[(1-e^{i\omega})(g-2is)] = -\mathfrak{c}\xi\mathsf{M}_r$. As $\sin\frac{\omega}{2} \ge \frac{\omega}{\pi}$ and $|\omega - 2\Theta| < \pi$, $|\omega - 2\Theta| \le \pi|\sin\frac{\omega-2\Theta}{2}| \le \pi^2|\mathsf{M}_r|/(2\mathsf{a}\omega)$, so $\mathsf{d} \le \pi^2\max_B|\mathsf{M}_r|/(\mathsf{a}\omega_0)$.

Hypotheses of Lemma C.17. Since $c_\mathsf{k} \le \frac{1}{2}$, $\mathsf{s} \le \frac{1}{6}$ and $\omega_0 \le \frac{2\pi}{3}$, $\mathsf{m} = \min(\frac{1}{4}, \frac{\omega_0^2}{6}, c_\mathsf{k}\mathsf{s}^2) \ge m_\mathsf{s}\omega_0^2$ with $m_\mathsf{s} = 9c_\mathsf{k}\mathsf{s}^2/(4\pi^2)$. Moreover $\xi_{\max} \le \frac{1}{2}(1+\mathsf{s})^2\omega_0^2 \le \omega_0^2$, so $\mathsf{m} \ge m_\mathsf{s}\xi_{\max}$ and $\mathsf{w} = \mathsf{s}\omega_0 \ge \mathsf{s}\xi_{\max}^{1/2}$. Hence $\mathsf{d} \le \pi^2\max_B|\mathsf{M}_r|/(\mathsf{a}\xi_{\max}^{1/2}) = o(\xi_{\max})$, so $\mathsf{d} \le \mathsf{m}/2$ and $\mathsf{d} \to 0$; and $|t| \le 2c\varepsilon$, so $|t|/(\mathsf{mw}) \le 2c\varepsilon/(m_\mathsf{s}\mathsf{s}\xi_{\max}^{3/2}) \to 0$. Thus (45) holds for every fixed $\eta \in (0,1)$ and small ε.

The lag-one constraint. On B, $\Phi_t^{(1)} = -\mathfrak{c}H_r \ge \mathfrak{c}\lambda_0\varepsilon\Xi_B$, so $\int_B \Phi_t^{(1)}\,d\nu \ge \mathfrak{c}\lambda_0\varepsilon\Xi_B$ for every probability measure ν on B; while $\bar{g} \le \max_B g \le \mathfrak{c}\Xi_B$ and $e^t\sinh|t| = \frac{1}{2}(1-r^2) \le c\varepsilon$. With $\eta = \min(\frac{1}{2}, \frac{\lambda_0}{2c})$, $\eta\,e^t\sinh|t|\,\bar{g} \le \frac{1}{2}\lambda_0\varepsilon\mathfrak{c}\Xi_B < \mathfrak{c}\lambda_0\varepsilon\Xi_B$, as $\Xi_B \ge \mathsf{a}\xi_{\max}/2 > 0$; so (46) holds. Lemma C.17 gives a pair with finitely many atoms in B, feasible at t, with $|C'| < \bar{g}'$, and Lemma C.11 gives $f \in \mathcal{F}_{\varepsilon^2,1}(\mathbb{C}^J \oplus \mathbb{R})$ and x_0 with $\|x_N\| \ge c_0 r^N\|x_0\|$; so $\rho_\mathcal{F}^0 \ge 1 - c\varepsilon$. □

Proof of Proposition 6.1(b). $u_0 = N_\tau$ gives $u_0'(1) = 1+\tau$, and $q_0 = -1/y$ is constant, so $\tau_{\mathrm{eff}} = \tau$. Since $\phi_\lambda = \chi_\lambda/h$, the lag-one symbol here is that of Section 4 divided by y_ε^2; at $\varepsilon = 0$, Lemma E.6(a) gives $\mathsf{P}_0 = \mathsf{P}_1 = 0$, $\mathsf{P}_2 = 2y(2\tau-1)$, $\mathsf{P}_3 = 4(1-\tau y)$, so $\mathcal{S}_0 = \xi^2(\mathsf{P}_2 + \mathsf{P}_3\xi)/y^2$. The affine function $\mathsf{P}_2 + \mathsf{P}_3\xi$ is positive on $(0,2]$ if and only if $\mathsf{P}_2 \ge 0$ and $\mathsf{P}_2 + 2\mathsf{P}_3 = 8 - 2y(1+2\tau) > 0$, and nonnegative on $[0,2]$ if and only if $\mathsf{P}_2 \ge 0$ and $\mathsf{P}_2 + 2\mathsf{P}_3 \ge 0$. Finally $u_0 - 2v_0 = \frac{2}{y}(z^2 - (2 - \frac{y}{2}(1+\tau))z + 1 - \frac{y\tau}{2})$, whose roots lie in the open unit disk by the Schur–Cohn conditions $|1 - \frac{y\tau}{2}| < 1$, i.e. $0 < y\tau < 4$, and $|2 - \frac{y}{2}(1+\tau)| < 2 - \frac{y\tau}{2}$, i.e. $\frac{y}{2} > 0$ and $4 - \frac{y}{2}(1+2\tau) > 0$, all implied by $y(1+2\tau) < 4$ and $\tau \ge \frac{1}{2}$. □

Lemma E.10 (Quadratic constant up to the boundary)**.** *Let a family* (3) *have* $\tau \ge \frac{1}{2}$, $y(1+2\tau) = 4$ *and* $\varsigma_\varepsilon \to \varsigma \in (2\theta, \infty]$, *with* ς_ε *and* c_π *as in* (25)–(26). *Then* $\sqrt{\kappa}\,(1-\rho_q) \to \min(c_Q(\theta, y), c_\pi)$.

Proof. Write $\ell = h\lambda \in [y_\mu, y_\varepsilon]$ and $P_\ell(z) = (z-1)(z-\beta) + \ell N_{\tau_\varepsilon}(z)$, so $\chi_\lambda = P_{h\lambda}$. The proof of Lemma 2.5 already bounds the positive real roots by the slow endpoint contribution $1 - \varepsilon c_Q(\theta, y) + o(\varepsilon)$, and the complex roots by $\sqrt{\beta} = 1 - \theta\varepsilon/2 + o(\varepsilon)$. Both estimates hold up to the limiting stability boundary.

For a negative real root w of P_ℓ,

$$\chi_L(w) = P_\ell(w) + (y_\varepsilon - \ell)N_{\tau_\varepsilon}(w) \le 0,$$

because $N_{\tau_\varepsilon}(w) < 0$. By Lemma E.8(b), χ_L has two real roots for small ε, the smaller being $z_L = -1 + \delta_\varepsilon$. A monic quadratic is nonpositive only between its roots, so $w \ge z_L$. Thus the largest modulus of a negative real mode is attained at $\lambda = L$, and $\delta_\varepsilon/\varepsilon \to c_\pi$, also when $c_\pi = \infty$. Since $c_Q \le \theta/2$, combining the three types of roots gives the claimed minimum. □

Proof of Theorem 2.6. (a) The claim for some δ implies it for every larger δ, so δ may be taken small. The family is a second-order family with slow data (θ, y) (Section 5). *Case* $\theta/3 < c_Q$. Let $0 < \delta < c_Q - \theta/3$ and $c = \theta/3 + \delta = c_\mathcal{F} + \delta$. Lemma D.5 gives J, depending on (θ, y, δ) only, and for small ε a pair with J atoms that is feasible at $t = -c\varepsilon$ and has $|C| < \bar{g}$. Lemma C.11 with $t < 0$ gives $f \in \mathcal{F}_{\varepsilon^2,1}(\mathbb{C}^J \oplus \mathbb{R}) = \mathcal{F}_{\varepsilon^2,1}(\mathbb{R}^{2J+1})$ and x_0 with $\|x_N\| \ge c_0 e^{-c\varepsilon N}\|x_0\|$.

Case $c_Q \le \theta/3$. Then $\theta^2 > 4y$, and the roots $a_\varepsilon > b_\varepsilon$ of χ_μ lie in $(0,1)$ for small ε, with $a_\varepsilon = 1 - c_Q\varepsilon + o(\varepsilon)$. On $f(x) = \mu x^2/2$, normalize the rest initialization to $x_{-1} = x_0 = 1$. The exact solution is

$$x_N = Aa_\varepsilon^N + Bb_\varepsilon^N, \qquad A = \frac{a_\varepsilon(1-b_\varepsilon)}{a_\varepsilon - b_\varepsilon} \ge 1, \qquad B = 1 - A \le 0.$$

Hence $x_N/a_\varepsilon^N = A + B(b_\varepsilon/a_\varepsilon)^N \ge A + B = 1$. For each $\delta > 0$, small ε gives $a_\varepsilon \ge e^{-(c_Q+\delta)\varepsilon}$, proving the required bound for every N, with $c_0 = 1$. Embed this scalar example in $\mathbb{R}^3$ to match the stated dimension.

(b) Let $0 < c < c_{\mathcal{F}}$ and $r = 1 - c\varepsilon$. Then $c < c_Q$ and $\theta - 3c > 0$. By Lemma E.6(a), $\mathsf{P}_1 = 2y(\theta - 3c)\varepsilon + o(\varepsilon) > 0$ for small ε, $\mathsf{P}_2 \to 2y(2\tau - 1) > 0$, because $\tau > \frac{1}{2}$, and $\mathsf{P}_2 + 2\mathsf{P}_3 \to 2y(2\tau-1) + 8(1-\tau y) = 2(4 - y(1+2\tau)) > 0$; so the affine function $\xi \mapsto \mathsf{P}_2 + \mathsf{P}_3\xi$ is positive at $\xi = 0$ and $\xi = 2$, hence on $[0,2]$, for small ε, and $\mathsf{P}_1 + \mathsf{P}_2\xi + \mathsf{P}_3\xi^2 > 0$ on $[0,2]$. Lemma E.7 gives $\rho_{\mathcal{F}} < r$.

(c) By (a), $\rho_{\mathcal{F}} \ge \rho_{\mathcal{F}}^0 \ge e^{-(c_{\mathcal{F}}+\delta)\varepsilon}$ for small ε (Lemma 2.3), so $\limsup \sqrt{\kappa}(1 - \rho_{\mathcal{F}}^0) \le c_{\mathcal{F}} + \delta$ for every $\delta > 0$, and the same for $\rho_{\mathcal{F}}$. By (b), $\liminf \sqrt{\kappa}(1 - \rho_{\mathcal{F}}) \ge c$ for every $c < c_{\mathcal{F}}$, and the same for $\rho_{\mathcal{F}}^0 \le \rho_{\mathcal{F}}$. □

Proof of Proposition 6.4. Let $0 < c < \min(c_{\mathcal{F}}(\theta, y), c_\pi)$, put $r = 1 - c\varepsilon$, and let $\Psi(\xi) = \mathsf{P}_1 + \mathsf{P}_2\xi + \mathsf{P}_3\xi^2$. By Lemma E.6(a), $\Psi(0) = 2y(\theta - 3c)\varepsilon + o(\varepsilon) > 0$ and $\mathsf{P}_3 \to -4(2\tau - 1)/(1 + 2\tau) < 0$. Thus Ψ is concave, and it remains to check $\Psi(2) > 0$. By Lemma E.1,

$$2\Psi(2) = (1 + r)\chi_\mu(-r)\chi_L(-r) - \mathsf{P}_0.$$

Here $\chi_\mu(-r) \to 4$ and $\mathsf{P}_0 = O(\varepsilon^3)$. With the roots of Lemma E.8(b),

$$\chi_L(-r) = (\delta_\varepsilon - c\varepsilon)(r + z_2), \qquad r + z_2 \to \frac{4\tau}{1 + 2\tau} > 0.$$

Since $\delta_\varepsilon/\varepsilon \to c_\pi > c$, this product is bounded below by a positive multiple of ε, including the case $c_\pi = \infty$. Hence $\Psi(2) > 0$, so concavity gives $\Psi > 0$ on $[0,2]$. Lemma E.7 proves $\rho_{\mathcal{F}} < 1 - c\varepsilon$, certified by lag one.

For the limit, Theorem 2.6(a) gives $\limsup \sqrt{\kappa}(1 - \rho_{\mathcal{F}}^0) \le c_{\mathcal{F}}$, and the same for $\rho_{\mathcal{F}}$. If $c_\pi < \infty$, Lemma E.8(b) gives, from rest on $f(x) = \frac{1}{2}x^2$ on $\mathbb{R}$, $\rho_{\mathcal{F}}^0 \ge 1 - \delta_\varepsilon = 1 - c_\pi\varepsilon + o(\varepsilon)$. By the first part, $\liminf \sqrt{\kappa}(1 - \rho_{\mathcal{F}}) \ge c$ for every $c < \min(c_{\mathcal{F}}, c_\pi)$, and the same for $\rho_{\mathcal{F}}^0 \le \rho_{\mathcal{F}}$. □

Proof of Corollary 6.6. (i) $1 - \beta = 2/(\sqrt{\kappa} + 1)$, so $\theta_\varepsilon = 2/(1 + \varepsilon) \to 2$, $y = 1$, $\tau_\varepsilon \to 1$; the tunings are stable since $1 + 2\beta < 2(1 + \beta)$, and $y(1 + 2\tau) = 3 < 4$. As $\theta^2 = 4y$, $c_Q = 1$ and $c_{\mathcal{F}} = \frac{2}{3}$. Squaring the distance doubles the constant. (ii) Substitution in (3) and (25) gives the stated slow data and margin, with $c_Q = c_\pi = 2/\sqrt{3}$. Proposition 6.4 gives the constant $\min(4/(3\sqrt{3}), 2/\sqrt{3}, 2/\sqrt{3}) = 4/(3\sqrt{3})$. (iii) By Remark 4.4, QHM with $(\alpha, \beta, \nu_{\mathrm{qh}})$ has the query representation (1) with $h = \alpha(1 - \beta)$, the same β and $\tau = \tau_{\mathrm{qh}}$; its iterates are the queries of (1) from rest, and their rates satisfy $\rho_{\tilde{x}}^0 \le \rho_{\mathcal{F}}^0 \le \max(\rho_{\tilde{x}}^0, z_\tau)$, $z_\tau = \tau_{\mathrm{qh}}/(1 + \tau_{\mathrm{qh}}) < 1$. If $\beta \ne z_\tau$, then on each $\frac{\lambda}{2}x^2$, $\lambda \in [\mu, L]$, their asymptotic root factor is $\rho(\chi_\lambda)$ (Lemma C.12). (a) Let the iterates accelerate on quadratics with constant c. Then $\beta \ne z_\tau$ for large κ: if $\beta = z_\tau$, then $h_2 = \beta h_1$ in the gradient-correction form, so $e_k = \tilde{x}_{k+1} - \tilde{x}_k + h_1\nabla f(\tilde{x}_k)$ satisfies $e_k = \beta e_{k-1}$; from rest $e_0 = 0$, hence $e_k = 0$ for all k, and the iterates are those of gradient descent with step h_1, whose factor on quadratics is at least $(\kappa - 1)/(\kappa + 1)$. So $\rho_q \le 1 - c\kappa^{-1/2}$, and Lemma E.4 gives $\tau_{\mathrm{qh}} < 2/c^2$ and $1 - \beta < \max(4, 8/c)\kappa^{-1/2}$, hence $1 - \nu_{\mathrm{qh}} = \tau_{\mathrm{qh}}(1 - \beta)/\beta = O(\kappa^{-1/2})$. If $\limsup \tau_{\mathrm{qh}} < \frac{1}{2}$, Theorem 6.2 gives $\rho_{\mathcal{F}}^0 > 1$, hence $\rho_{\tilde{x}}^0 > 1$, for all large κ; so acceleration on $\mathcal{F}_{\mu,L}$ forces $\liminf \tau_{\mathrm{qh}} \ge \frac{1}{2}$. (b) This is Theorem 2.6(c) with $hL = \alpha L(1 - \beta) \to y$; as τ_{qh} is bounded, z_τ stays below 1 while $\rho_{\mathcal{F}}^0 \to 1$, so $\rho_{\tilde{x}}^0 = \rho_{\mathcal{F}}^0$ for large κ. (iv) The gradient-correction dictionary gives $h = \gamma$, $\tau = 1$ and the same β. The tuning is stable by Lemma 2.1 and has slow data $(2, 1, 1)$, so Theorem 2.6(c) gives the constant $2/3$.

(v) Write α for the HNAG time step, B for its gradient coefficient, and γ for the coefficient of $y_k - x_{k+1}$ in its first update ($\gamma = 1$ for HNAG^{++}, $\gamma = 2$ for HNAG^{+}). With $\alpha B = 1/L$, eliminating

the auxiliary state gives the gradient-correction parameters

$$D = (1+\alpha)(1+\gamma\alpha), \qquad \beta = D^{-1},$$
$$hL = \frac{\gamma\alpha^2\kappa + \alpha}{D}, \qquad \tau = \frac{1}{\gamma\alpha^2\kappa + \alpha}.$$

The first update recovers the auxiliary state from two consecutive iterates. The resulting state maps have norm bounds uniform over the function class, so they preserve the uniform rate. For HNAG^{++}, $\alpha = \sqrt{2}\varepsilon = \eta$, giving the parameters in (v) and

$$(\theta, a, b) = \left(2\sqrt{2}, \frac{\sqrt{2}}{4}, 3\sqrt{2}\right).$$

These coordinates satisfy the flow-region inequalities of Remark 7.3, so $c = \theta/3 = 2\sqrt{2}/3$. The quadratic constant is $c_Q(2\sqrt{2}, 2) = \sqrt{2}$; the alternating constant $c_\pi = 3\sqrt{2}$ is larger (Lemma E.10). For HNAG^{+}, $\alpha = \varepsilon/(1-\varepsilon)$, giving exactly $\beta = (1-\varepsilon)^2/(1+\varepsilon)$, $hL = 2-\varepsilon$ and $\tau = (1-\varepsilon)^2/((2-\varepsilon)(1+\varepsilon))$, the TMM parameters of Corollary 7.1. □

Corollary E.11 (Ceilings at fixed extrapolation)**.**

(i) *Every family in $\mathcal{M}$ with $\tau \to \tau_\infty$ has quadratic constant* $\limsup \sqrt{\kappa}(1-\rho_q) \le 2/\sqrt{1+2\tau_\infty}$.

(ii) *If a family in $\mathcal{M}$ is accelerated on $\mathcal{F}_{\mu,L}$, its quadratic constant* $\limsup \sqrt{\kappa}(1-\rho_q)$ *is at most* $\sqrt{2}$.

(iii) *Fix $\tau > \frac{1}{2}$ and $y_\tau = 4/(1+2\tau)$. The exact constants of the families of Theorem 2.6(c) with this limit τ have supremum $\sqrt{y_\tau/2} = \sqrt{2/(1+2\tau)} < 1$, and none of them attains it. It is attained on the stability boundary by the families of Proposition 6.4 with $\theta = 3\sqrt{y_\tau/2}$ and $c_\pi \ge \theta/3$, i.e. $\varsigma \ge 2\theta + \frac{4\tau}{1+2\tau}\cdot\frac{\theta}{3}$.*

(iv) *Fix $\tau \ge \frac{1}{2}$ and $y_\tau = 4/(1+2\tau)$. For every family (3) with this limit τ (so that $y \le y_\tau$), the pair formed by* $\limsup \sqrt{\kappa}(1-\rho_q)$ *and* $\limsup \sqrt{\kappa}(1-\rho_{\mathcal{F}}^0)$ *is dominated componentwise by a point of the arc $\mathcal{C}_\tau = \{(\zeta_-(\theta, y_\tau), \theta/3) : 2\sqrt{y_\tau} \le \theta \le 3\sqrt{y_\tau/2}\}$. For $\tau > \frac{1}{2}$, every point of $\mathcal{C}_\tau$ is attained by a family on the stability boundary; for $\tau = \frac{1}{2}$, see Corollary 7.4.*

(v) *If a family of stable tunings of $\mathcal{M}$ satisfies* $\liminf_{\kappa\to\infty} \sqrt{\kappa}(1-\rho_{\mathcal{F}}) \ge 1$, *then, in the notation of (3), $(\theta_\varepsilon, y_\varepsilon, \tau_\varepsilon) \to (3, 2, \frac{1}{2})$.*

Proof of Corollary 6.7. Put $\varepsilon = \kappa^{-1/2}$ and $\theta = \sqrt{6}$. For the proposed tuning, $y = 4/3$, $\tau_\varepsilon = \beta = 1 - \theta\varepsilon$, and

$$2(1+\beta) - y(1+2\tau) = \frac{2\theta}{3}\varepsilon > 0, \qquad \varsigma_\varepsilon = \frac{8\theta}{3}.$$

Thus $c_\pi = \theta/2$ and $c_Q(\theta, 4/3) = \theta/3 = \sqrt{2/3}$. For $\kappa \ge 6$, $\beta \in [0,1)$ and $\tau = \beta$, so Theorem 8.1 also gives linear convergence at each such condition number. Proposition 6.4 gives the two asymptotic expansions.

For optimality, suppose a family with $\tau = \beta$ has a larger positive asymptotic constant. It is accelerated on quadratics, so Lemma E.4 gives, along a subsequence, convergent slow data (θ, y, τ) with $\beta \to 1$, hence $\tau = 1$. Stability forces $y \le 4/3$, while Theorem 2.6(a) and Corollary 3.2 bound the constant by $c_{\mathcal{F}}(\theta, y) \le \sqrt{y/2} \le \sqrt{2/3}$, a contradiction. □

Proof of Corollary E.11. (i) By Theorem 2.7, $1 - \rho_q \le 2/(\sqrt{\kappa_\tau} + 1)$, and $\sqrt{\kappa}/\sqrt{\kappa_\tau} = (1 + 2\tau - 2\tau/\kappa)^{-1/2} \to (1+2\tau_\infty)^{-1/2}$. (ii) By Theorem 6.2 and Lemma E.4, τ_κ is bounded with $\liminf \tau_\kappa \ge \frac{1}{2}$; apply (i) along subsequences. (iii) By Corollary 3.2 with $m = y$, $c_{\mathcal{F}}(\theta, y) \le \sqrt{y/2}$, with equality only at $\theta = 3\sqrt{y/2}$. The families of Theorem 2.6(c) with limit τ have $y < y_\tau$, so their constants are at most $\sqrt{y/2} < \sqrt{y_\tau/2}$, and the families with constant parameters $\theta = 3\sqrt{y/2}$, $y \uparrow y_\tau$, show that the supremum is $\sqrt{y_\tau/2}$; it is below 1 since $y_\tau < 2$ for $\tau > \frac{1}{2}$. On the stability boundary, Proposition 6.4

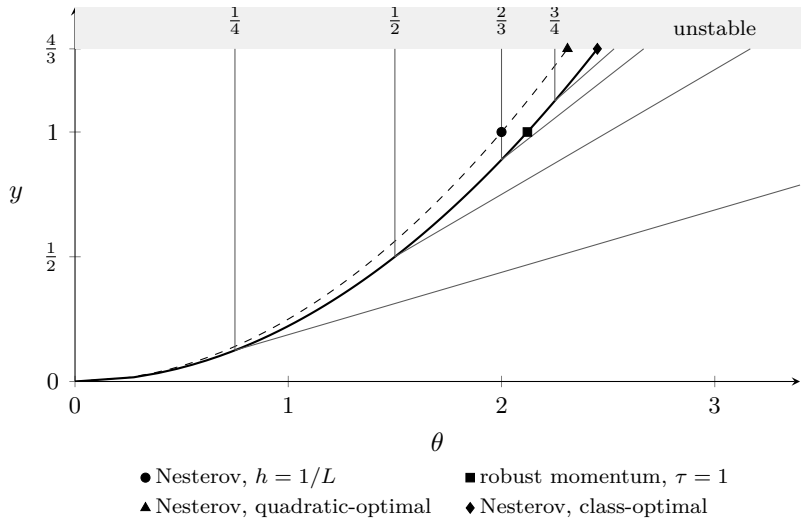


Figure 7: Design level sets $c_{\mathcal{F}} = c$ at $\tau = 1$, for $c = \frac{1}{4}, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}$ (labels at the top). Each joins $\theta = 3c$, $y \geq 2c^2$, to $y = c\theta - c^2$, $\theta \geq 3c$, on the balance curve $\theta = 3\sqrt{y/2}$ (solid). The dashed curve is critical damping $\theta = 2\sqrt{y}$; the shaded region is unstable. At $y = 4/3$, the alternating mode also constrains the constant (Proposition 6.4). The marked tunings are described in Corollaries 6.6 and 6.7 and Section 7.1.

gives the constant $\min(\theta/3, c_Q(\theta, y_\tau), c_\pi)$. At $\theta = 3\sqrt{y_\tau/2}$, $\theta/3 = c_Q(\theta, y_\tau) = \sqrt{y_\tau/2}$ by Corollary 3.2, so the constant equals $\sqrt{y_\tau/2}$ if and only if $c_\pi \geq \theta/3$, i.e. $(\varsigma - 2\theta)(1 + 2\tau)/(4\tau) \geq \theta/3$.

(iv) *Domination.* Lemma 2.5 and Theorem 2.6(a) bound the two constants componentwise by $(c_Q, c_{\mathcal{F}})(\theta, y)$, and stability gives $y \leq y_\tau$. By Lemma B.1(b), this pair is $\alpha(c_Q, c_{\mathcal{F}})(\theta/\alpha, y_\tau)$, where $\alpha = \sqrt{y/y_\tau} \leq 1$. Corollary 3.2 now gives domination by a point of $\mathcal{C}_\tau$.

Attainment. Let $\tau > \frac{1}{2}$ and $(\zeta_-(\theta, y_\tau), \theta/3) \in \mathcal{C}_\tau$; on this arc $\theta/3 \leq \zeta_-(\theta, y_\tau)$. Take $\theta_\varepsilon = \theta$, $\tau_\varepsilon = \tau$ and $y_\varepsilon = y_\tau - \varepsilon^{1/2}$, so that $\varsigma_\varepsilon = (1 + 2\tau)\varepsilon^{-1/2} \to \infty$ and $c_\pi = \infty$; Proposition 6.4 gives the constant $c_{\mathcal{F}}(\theta, y_\tau) = \theta/3$ and Lemma E.10 the quadratic constant $\zeta_-(\theta, y_\tau)$.

(v) Let a family of stable tunings (with $L = 1$, $\mu = \varepsilon^2$, $\theta_\varepsilon = (1 - \beta)/\varepsilon$, $y_\varepsilon = h$) satisfy $\liminf \sqrt{\kappa}(1 - \rho_{\mathcal{F}}) \geq 1$, and suppose that along a sequence $\varepsilon \to 0$ the triple $(\theta_\varepsilon, y_\varepsilon, \tau_\varepsilon)$ stays away from $(3, 2, \frac{1}{2})$. Fix $c \in (0, 1)$. For small ε, $\rho_q \leq \rho_{\mathcal{F}} \leq 1 - c\varepsilon$, so by Lemma E.4(b) a subsequence converges to some (θ, y, τ) with $\theta \geq 2c > 0$ and is a family (3). By Theorem 6.2 (applied along the subsequence, which is accelerated on quadratics), $\tau \geq \frac{1}{2}$; by stability, $y(1 + 2\tau) \leq 4$, so $y \leq 2$. Theorem 2.6(a) gives $\limsup \sqrt{\kappa}(1 - \rho^0_{\mathcal{F}}) \leq c_{\mathcal{F}}(\theta, y) \leq \sqrt{y/2} \leq 1$. Since $\rho^0_{\mathcal{F}} \leq \rho_{\mathcal{F}}$, the assumption forces $\sqrt{y/2} = 1$, i.e. $y = 2$ and then $\tau = \frac{1}{2}$, and $c_{\mathcal{F}}(\theta, 2) = 1$, i.e. $\theta = 3$. This contradicts the choice of the sequence. □

Proof of Proposition 6.3. Suppose $\varepsilon_k \to 0$, $c_k \in (\theta/3, c_\sharp]$ and $\gamma_k \in [0, \gamma_{\max}]$ with $(\delta_{\varepsilon_k \gamma_k}, C_k)$ feasible at $-c_k\varepsilon_k$. Pass to a subsequence with $c_k \to c_\infty$, $\gamma_k \to \gamma_\infty$. By Lemma D.4, $C_k = C_{*,k} + O(\varepsilon_k^3)$; in the normalization of (48) the lag constraints become $\tilde{g}\mathcal{A}_{\gamma_k}(q) \pm \tilde{s}\mathcal{B}_{\gamma_k}(q) + O(\varepsilon_k)\sinh(c_k q) \geq 0$ for $q \in \varepsilon_k\mathbb{N}$, and the minimizer constraint becomes $|\gamma_k \tilde{s}| \leq c_k\tilde{g} + O(\varepsilon_k)$. For fixed $q > 0$ take $q_k = \varepsilon_k \lfloor q/\varepsilon_k \rfloor \to q$; by Lemma D.3 the limits show that γ_∞ satisfies (34) at c_∞, and $|(\theta - 2c_\infty)\gamma_\infty^2| \leq c_\infty(\gamma_\infty^2 - \Delta_{c_\infty})$, which is (35) (with $m = y$). If $c_\infty > \theta/3$ this contradicts Proposition 3.3; if $c_\infty = \theta/3$, (35) reads $0 \geq \frac{\theta}{3}\Delta_{\theta/3} > 0$, a contradiction. □

Proof of Theorem 6.5. Throughout, $2\tau_\varepsilon - 1 = -2A_\varepsilon\varepsilon^{1/2}$, $\tau_\varepsilon \to \frac{1}{2}$, and $4(1 - \tau_\varepsilon y_\varepsilon) \to 2(2 - y) > 0$. We use repeatedly that $\varepsilon^{1/2}\xi \leq \frac{1}{2}(\varepsilon + \xi^2)$ and $\varepsilon\xi \leq \frac{1}{2}\varepsilon^{1/2}(\varepsilon + \xi^2)$.

(a) It suffices to take $0 < c < c_{\text{layer}}(A)$; put $r = 1 - c\varepsilon$. Then $c < c_Q$ and $3c < \Lambda(A)$. By Lemma E.6(a), uniformly for $\xi \in [0, 2]$,

$$\mathsf{P}_1 + \mathsf{P}_2\xi + \mathsf{P}_3\xi^2 \;\geq\; 2y(\theta - 3c)\varepsilon - 4yA_+\varepsilon^{1/2}\xi + 2(2 - y)\xi^2 - o(\varepsilon + \xi^2),$$

because $\mathsf{P}_2 \geq -4y_\varepsilon(A_\varepsilon)_+\varepsilon^{1/2} - O(\varepsilon)$ and $y_\varepsilon(A_\varepsilon)_+ \to yA_+$ (also for $A = -\infty$). The binary form $2y(\theta-3c)a^2-4yA_+ab+2(2-y)b^2$ is positive definite, since $3c < \Lambda(A)$ is $yA_+^2 < (\theta-3c)(2-y)$; evaluated at $(\varepsilon^{1/2}, \xi)$ it is at least $m_1(\varepsilon+\xi^2)$ with $m_1 > 0$. Hence $\mathsf{P}_1 + \mathsf{P}_2\xi + \mathsf{P}_3\xi^2 \geq (m_1 - o(1))(\varepsilon+\xi^2) > 0$ on $[0,2]$ for small ε, and Lemma E.7 gives $\rho_{\mathcal{F}} < r$.

(b) Let $A \in (0,\infty)$, and let $c > \Lambda(A)_+/3$; we may assume $c < \theta/2$ (if c works, so does every larger c). Put $r = 1 - c\varepsilon$, $\varphi_* = Ay/(2-y)$ and $m_3(\varphi) = 2y(\theta-3c) - 4Ay\varphi + 2(2-y)\varphi^2$. Then $m_3(\varphi_*) = \frac{2y}{2-y}\big[(\theta-3c)(2-y) - yA^2\big] < 0$, because $3c > \Lambda(A)$. Put $m_* = -\frac{1}{2}m_3(\varphi_*) > 0$, and choose $\mathsf{s} \in (0, \mathsf{s}_1]$ (Lemma C.17) so small that $m_3 \leq -m_*$ on $I_\varphi = [\varphi_*(1-2\mathsf{s})^2, \varphi_*(1+2\mathsf{s})^2]$.

The band. Let $\omega_* = \arccos(1-\varepsilon^{1/2}\varphi_*)$; as $\varepsilon \to 0$, $\omega_* \to 0$ and $\omega_*^2/(2\varepsilon^{1/2}\varphi_*) \to 1$. For $[\omega_a, \omega_b] = [(1-\frac{3}{2}\mathsf{s})\omega_*, (1+\frac{3}{2}\mathsf{s})\omega_*]$, $2\pi(1-\mathsf{s})/\omega_a - 2\pi(1+\mathsf{s})/\omega_b = \frac{2\pi}{\omega_*}\big(\frac{1-\mathsf{s}}{1-3\mathsf{s}/2} - \frac{1+\mathsf{s}}{1+3\mathsf{s}/2}\big) \to \infty$, so Lemma E.3(a) gives $K \geq 3$ with $B = [(1-\mathsf{s})\omega_0, (1+\mathsf{s})\omega_0] \subset [\omega_a, \omega_b]$, $\omega_0 = 2\pi/K$. Since $\xi = 1 - \cos\omega = \frac{\omega^2}{2}(1+O(\omega^2))$, every $\omega \in B$ has $\xi = \varepsilon^{1/2}\varphi$ with $\varphi \in I_\varphi$, for small ε.

Symbols on the band. By Lemma E.6(a), $\mathsf{P}_0 = O(\varepsilon^3)$, $\mathsf{P}_1 = 2y(\theta-3c)\varepsilon + o(\varepsilon)$, $\mathsf{P}_2 = -4y_\varepsilon A_\varepsilon\varepsilon^{1/2} + O(\varepsilon)$ and $\mathsf{P}_3 = 2(2-y) + o(1)$, so on B

$$H_r = \varepsilon^{3/2}\varphi\big[m_3(\varphi) + o(1)\big] + O(\varepsilon^3) \leq -\tfrac{1}{4}\varepsilon^{3/2}\varphi_* m_*.$$

By Lemma E.6(b), $\mathsf{M}_r = O(\varepsilon)$ on B (its coefficients are $O(\varepsilon)$, $-4y_\varepsilon A_\varepsilon\varepsilon^{1/2} + O(\varepsilon)$ and $O(1)$), and $|\chi_\mu\overline{\chi_L}| \leq C_2\varepsilon^{1/2}$ on B, as $|\chi_\mu| = O(\xi)$ and $|\chi_L| = O(1)$ there. Since $\xi \asymp \varepsilon^{1/2}$ on B, the hypotheses of Lemma E.9 hold, with $\lambda_0 = \varphi_* m_*/(4C_2)$: ε and $\max_B|\mathsf{M}_r|$ are $O(\varepsilon) = o(\varepsilon^{3/4})$. It gives $\rho_{\mathcal{F}}^0 \geq 1 - c\varepsilon$.

(c) By Proposition 6.1, $L_\tau = \mathsf{A}_0 + \mathsf{A}_1\xi + \mathsf{A}_2\xi^2$ with $\mathsf{A}_0 = 2\theta_\varepsilon y_\varepsilon\varepsilon + O(\varepsilon^2) > 0$, $\mathsf{A}_1 = 2y_\varepsilon(2\tau_\varepsilon - 1) + O(\varepsilon) = -4y_\varepsilon A_\varepsilon\varepsilon^{1/2}(1+o(1)) < 0$ and $\mathsf{A}_2 = 4(\beta - \tau_\varepsilon y_\varepsilon) \to 2(2-y)$; the $o(1)$ is uniform because $\liminf A_\varepsilon > 0$. Hence $\mathsf{A}_1^2/(4\mathsf{A}_0\mathsf{A}_2) = \frac{y_\varepsilon A_\varepsilon^2}{\theta_\varepsilon(2-y_\varepsilon)}(1+o(1)) \geq (A'/A_*)^2(1+o(1))$ for some $A' > A_*$, so L_τ has two roots $0 < \xi_1 < \xi_2$ with $\xi_2/\xi_1 = (1+\sqrt{d})/(1-\sqrt{d})$, $d = 1 - 4\mathsf{A}_0\mathsf{A}_2/\mathsf{A}_1^2 \geq d_0 > 0$, and $L_\tau < 0$ on (ξ_1, ξ_2). Moreover $\xi_2 < -\mathsf{A}_1/\mathsf{A}_2 \to 0$. With $\omega_i = \arccos(1-\xi_i)$, therefore $\omega_2 \to 0$ and $\omega_2/\omega_1 \geq 1 + \eta'$ for small ε, with $\eta' > 0$; so $2\pi/\omega_1 - 2\pi/\omega_2 \geq 2\pi\eta'/\omega_2 \to \infty$, and $(2\pi/\omega_2, 2\pi/\omega_1)$ contains an integer $K \geq 3$, at which $L_\tau(1 - \cos\frac{2\pi}{K}) < 0$, and Lemma E.3(b) gives the orbits and $\rho_{\mathcal{F}}^0 > 1$. If $A_\varepsilon \to A < \infty$, then $\xi_1\xi_2 = \mathsf{A}_0/\mathsf{A}_2 \asymp \varepsilon$ and ξ_2/ξ_1 is bounded, so $\xi_i \asymp \varepsilon^{1/2}$, $\omega_i \asymp \varepsilon^{1/4}$ and $K \asymp \varepsilon^{-1/4} = \kappa^{1/8}$.

Consequences. Let $A_\varepsilon \to A \in [-\infty, A_*)$. By (a), $\liminf\sqrt{\kappa}(1-\rho_{\mathcal{F}}) \geq c_{\mathrm{layer}}(A)$. Theorem 2.6(a), which holds for every τ, gives $\limsup\sqrt{\kappa}(1-\rho_{\mathcal{F}}^0) \leq c_{\mathcal{F}}(\theta, y)$, and for $A > 0$, (b) gives $\limsup\sqrt{\kappa}(1-\rho_{\mathcal{F}}^0) \leq \Lambda(A)/3$. Since $\Lambda(A) \leq \theta$, with equality for $A \leq 0$, $\min(c_{\mathcal{F}}, \Lambda(A)/3) = c_{\mathrm{layer}}(A)$ in both cases, and $\rho_{\mathcal{F}}^0 \leq \rho_{\mathcal{F}}$ closes the chain. If $A_\varepsilon \to A \in [A_*, \infty)$, (b) applies to every $c > 0 = \Lambda(A)_+/3$, and if $A = \infty$, (c) applies; in both cases the family is not accelerated. Finally c_{layer} is nonincreasing in A and positive on $[-\infty, A_*)$; applying the above along subsequences gives the last claim. □

Remark E.12 (One-atom spectra in the layer). For a family as in Theorem 6.5, let $A_\varepsilon \to A \in (0,\infty)$, $c > \Lambda(A)_+/2$ and $r = 1 - c\varepsilon$. By Lemma E.6(b), with $\xi = \varepsilon^{1/2}\varphi$,

$$\varepsilon^{-1}\mathsf{M}_r = 2y(\theta-2c) - 4Ay\varphi + 2(2-y)\varphi^2 + o(1)$$

uniformly for φ in compact sets. The quadratic in φ is minimal at $\varphi_* = Ay/(2-y)$, with the value $\frac{2y}{2-y}\big[(\theta-2c)(2-y) - yA^2\big]$, which is negative because $2c > \Lambda(A)$. The points $\xi_K = 1 - \cos\frac{2\pi}{K}$ with $K \asymp \varepsilon^{-1/4}$ have consecutive ratios tending to 1, so for small ε some $\xi_K = \varepsilon^{1/2}\varphi$ has φ close to φ_*; there $H_r^{\mathrm{mix}} = \xi_K\mathsf{M}_r < 0$, while $\mathrm{Im}(\chi_\mu\overline{\chi_L}) > 0$ by Lemma E.6(b). Lemma E.2 shows that $(\delta_{2\pi/K}, 0)$ is feasible at $t = \log r$, and Lemma C.11 gives $\rho_{\mathcal{F}}^0 \geq 1 - c\varepsilon$. So the one-atom spectra of Lemma E.2 give $\limsup\sqrt{\kappa}(1-\rho_{\mathcal{F}}^0) \leq \Lambda(A)_+/2$, and with Theorem 2.6(a) the bound $\min(c_{\mathcal{F}}, \Lambda(A)_+/2)$ drawn dashed in Figure 4.

F. Proofs for Section 7

F.1. The exact robust-momentum certificate

Robust momentum [8] is (1) with

$$hL = \kappa(1-\varrho)^2(1+\varrho), \qquad \beta = \frac{\kappa\varrho^3}{\kappa-1}, \qquad \tau = \frac{\varrho^3}{(\kappa-1)(1-\varrho)^2(1+\varrho)}, \qquad \varrho \in [1-\kappa^{-1/2},\, 1-\kappa^{-1}], \tag{52}$$

where ϱ is its design rate; for $\varrho = 1-\kappa^{-1/2}$ it is TMM, and for $\varrho = 1-\kappa^{-1}$ its queries follow gradient descent with step $1/L$ [8]. For this family the lag-one certificate can be computed in closed form at every condition number.

Theorem F.1 (Robust momentum)**.** *Let $\kappa > 1$ and $\varrho \in [1-\kappa^{-1/2}, 1-\kappa^{-1}]$, and consider the tuning* (52)*. Then*

$$\chi_\mu(z) = (z-\varrho)(z-\varrho^2), \qquad \chi_L(z) = z(z-z_L), \qquad z_L = 1-\kappa(1-\varrho)^2(1+\varrho),$$

and on the circle $|z| = \varrho$, with $\xi = 1-\cos\omega$,

$$\mathrm{Re}\big[(1-z)\chi_\mu(z)\overline{\chi_L(z)}\big] = \varrho^3(1+\varrho)\big(1-\kappa(1-\varrho)^2\big)\,\xi\,\big((1-\varrho)^2+2\varrho\xi\big) \;\geq 0.$$

Moreover, on every circle $|z| = r$ with $\varrho < r < 1$,

$$\mathrm{Re}\big[(1-z)\chi_\mu(z)\overline{\chi_L(z)}\big] > 0.$$

The tuning is stable and has exact rates

$$\rho_{\mathcal{F}} = \rho^0_{\mathcal{F}} = \rho_q = \varrho.$$

At TMM, $\varrho = 1-\kappa^{-1/2}$, and the lag-one symbol vanishes on $|z| = \varrho$.

Proof of Theorem F.1. Characteristic polynomials. From (52), $hL\tau = \kappa\varrho^3/(\kappa-1) = \beta$ and $h\mu\tau = \beta/\kappa$. Hence $Q_L = \beta - hL\tau = 0$ and $S_L = 1+\beta-hL(1+\tau) = 1-hL = z_L$, so $\chi_L(z) = z(z-z_L)$. Further $Q_\mu = \beta(1-\frac{1}{\kappa}) = \varrho^3$ and $S_\mu = 1+\beta-h\mu-\frac{\beta}{\kappa} = 1+\varrho^3-(1-\varrho)^2(1+\varrho) = \varrho+\varrho^2$, so $\chi_\mu(z) = (z-\varrho)(z-\varrho^2)$.

Admissibility and stability. Since $0 < \varrho \leq 1-\kappa^{-1}$, we have $\varrho^3 < \varrho$ and $\beta \in [0,1)$; also $\tau \geq 0$, $h > 0$. Since $\varrho \geq 1-\kappa^{-1/2}$, $\kappa(1-\varrho)^2 \leq 1$ and $hL \leq 1+\varrho < 2$. As $hL\tau = \beta$, $y(1+2\tau)-2(1+\beta) = hL-2 < 0$, and the tuning is stable by Lemma 2.1. Moreover $-\varrho \leq z_L \leq \varrho$: indeed

$$\varrho + z_L = (1+\varrho)\big(1-\kappa(1-\varrho)^2\big) \geq 0, \qquad \varrho - z_L = (1-\varrho)\big(\kappa(1-\varrho^2)-1\big) \geq 0,$$

since $\kappa(1-\varrho)^2 \leq 1$ and $1-\varrho^2 \geq 1-\varrho \geq \kappa^{-1}$.

The symbol. Let $z = \varrho e^{i\omega}$. Since $1-\varrho e^{i\omega} = e^{i\omega}\,\overline{(e^{i\omega}-\varrho)}$,

$$(1-z)(z-\varrho^2) = \varrho(1-\varrho e^{i\omega})(e^{i\omega}-\varrho) = \varrho\, e^{i\omega}\,|e^{i\omega}-\varrho|^2, \qquad (z-\varrho)\bar{z} = \varrho^2(1-e^{-i\omega}).$$

Hence

$$(1-z)\chi_\mu(z)\overline{\chi_L(z)} = \varrho^3|e^{i\omega}-\varrho|^2\,(e^{i\omega}-1)(\varrho e^{-i\omega}-z_L),$$

and $\mathrm{Re}[(e^{i\omega}-1)(\varrho e^{-i\omega}-z_L)] = (\varrho+z_L)(1-\cos\omega) = (\varrho+z_L)\xi$. With $|e^{i\omega}-\varrho|^2 = (1-\varrho)^2+2\varrho\xi$ and $\varrho+z_L = (1+\varrho)(1-\kappa(1-\varrho)^2)$ this is the stated formula.

Lower bounds. ϱ is a root of χ_μ, so $\rho_{\mathcal{F}} \geq \rho_q \geq \varrho$ (Lemma 2.3), and $\rho^0_{\mathcal{F}} \geq \varrho$ by Lemma C.12 with $\lambda = \mu$, since the roots of χ_μ are ϱ and ϱ^2.

Positive circles and the upper bound. Fix $\varrho < r < 1$ and $\xi \in [0,2]$. Let $H_r(s;\xi)$ be the circle symbol with $\chi_\mu = (z-\varrho)(z-\varrho^2)$ and $\chi_L = z(z-s)$. It is affine in s, so

$$2\varrho H_r(z_L;\xi) = (\varrho - z_L)H_r(-\varrho;\xi) + (\varrho+z_L)H_r(\varrho;\xi).$$

At the two endpoints, polynomial expansion gives

$$H_r(-\varrho;\xi) = r(r^2-\varrho^2)\big[(r-\varrho^2)(1-r) + \xi\big(r^2+\varrho^2+2r\varrho(2-\xi)\big)\big],$$
$$H_r(\varrho;\xi) = r\big[(r-\varrho)^2+2r\varrho\xi\big]\big[(r-\varrho^2)(1-r)+(r^2+\varrho^2)\xi\big].$$

Both are strictly positive: $(r-\varrho^2)(1-r)>0$, and all remaining bracketed terms are nonnegative, with $(r-\varrho)^2>0$. Since $-\varrho\le z_L\le\varrho$, the interpolation weights are nonnegative and sum to $2\varrho>0$. Therefore $H_r(z_L;\xi)>0$ for every $\varrho<r<1$. The roots ϱ,ϱ^2 of χ_μ lie in $|z|<r$, so Lemma C.13 gives $\rho_q<r$ and $\rho_{\mathcal F}<r$. Letting $r\downarrow\varrho$ proves $\rho_{\mathcal F}\le\varrho$. With the lower bounds and Lemma 2.3, $\varrho\le\rho_q\le\rho_{\mathcal F}\le\varrho$ and $\varrho\le\rho^0_{\mathcal F}\le\rho_{\mathcal F}$. At $\varrho=1-\kappa^{-1/2}$, $\kappa(1-\varrho)^2=1$, so the symbol at radius ϱ vanishes identically. □

Proof of Corollary 7.1. TMM [48] is (1) with $\beta=\varrho^2/(2-\varrho)$, $\tau=\varrho^2/((1+\varrho)(2-\varrho))$ and $hL=1+\varrho$, where $\varrho=1-\kappa^{-1/2}$. (It reports the point $(1+\delta_{\mathrm T})x_k-\delta_{\mathrm T}x_{k-1}$, $\delta_{\mathrm T}=\varrho^2/(1-\varrho^2)$; as in Remark 4.4, this point has the same rate as x_k, because $\delta_{\mathrm T}/(1+\delta_{\mathrm T})=\varrho^2<\varrho$.) This is (52) at $\varrho=1-\kappa^{-1/2}$: then $\kappa(1-\varrho)^2=1$ and $\kappa-1=\varrho(2-\varrho)/(1-\varrho)^2$, so $\kappa(1-\varrho)^2(1+\varrho)=1+\varrho$, $\kappa\varrho^3/(\kappa-1)=\varrho^2/(2-\varrho)$ and $\varrho^3/((\kappa-1)(1-\varrho)^2(1+\varrho))=\varrho^2/((2-\varrho)(1+\varrho))$. Theorem F.1 gives $\rho_{\mathcal F}=\rho^0_{\mathcal F}=\rho_q=\varrho$. With $\varepsilon=1-\varrho$, $1-\beta=\varepsilon(3-\varepsilon)/(1+\varepsilon)$, $2-hL=\varepsilon$, and

$$\tfrac12-\tau=\frac{(1-\varrho)(2+3\varrho)}{2(2+\varrho-\varrho^2)}=\tfrac54\varepsilon\,(1+O(\varepsilon));$$

in the notation of (28), $(\theta_\varepsilon,a_\varepsilon,b_\varepsilon)\to(3,\frac54,1)$. In particular $\tau<\frac12$ for every $\varrho\in(0,1)$. Finally $\zeta_-(3,2)=1=\theta/3$, and optimality among deterministic first-order methods with value and gradient access follows from [10]. □

Lemma F.2 (Expansions at the corner)**.** *Under* (28)*, let $r=1-c\varepsilon$, $c\in\mathbb R$, and write $\doteq$ for equality up to $o(\varepsilon)$, uniformly for $\xi\in[0,2]$ and for c in compact sets. (With $\theta_\varepsilon,a_\varepsilon,b_\varepsilon$ in place of their limits the remainders are $O(\varepsilon^2)$.) Then the coefficients of Lemma E.1 and Lemma E.2 satisfy*

$$\mathsf P_1\doteq4(\theta-3c)\varepsilon,\quad \mathsf P_2\doteq2(\theta+2c-4a)\varepsilon,\quad \mathsf P_3\doteq2(4a+b-2\theta)\varepsilon,$$

$$\mathsf P_1+\mathsf P_2\xi+\mathsf P_3\xi^2\doteq\varepsilon\big[\mathsf N(\xi)-4c(3-\xi)\big],\qquad \mathsf M_r(\xi)\doteq\varepsilon\big[\mathsf N(\xi)-2c(4-\xi)\big].$$

Moreover $\mathsf P_0=c\varepsilon\chi_\mu(r)\chi_L(r)$ with $\chi_\mu(r)=\varepsilon^2(c^2-\theta_\varepsilon c+2+O(\varepsilon))$ and $\chi_L(r)=2+O(\varepsilon)$, $\chi_L(-1)=c_\pi\varepsilon+o(\varepsilon)$, $S_L\to-1$, and χ_L has a real root $z_L=-1+c_\pi\varepsilon+o(\varepsilon)$.

Proof. Substitute $y_\varepsilon=2-b_\varepsilon\varepsilon$ and $\tau_\varepsilon=1/2-a_\varepsilon\varepsilon$ in Lemma E.6. The corner coordinates are bounded, so collecting the polynomial coefficients gives $O(\varepsilon^2)$ remainders with the member parameters; replacing them by their limits costs $o(\varepsilon)$, uniformly on the stated compact sets. The characteristic formulas at r also follow from Lemma E.6(a); retaining θ_ε gives the asserted $O(\varepsilon)$ remainder inside $\chi_\mu(r)/\varepsilon^2$. Finally,

$$\chi_L(-1)=(4a_\varepsilon+2b_\varepsilon-2\theta_\varepsilon)\varepsilon-2a_\varepsilon b_\varepsilon\varepsilon^2=c_\pi\varepsilon+o(\varepsilon),\qquad \chi_L'(-1)=-1+O(\varepsilon).$$

The simple real root near -1 therefore satisfies $z_L=-1+\chi_L(-1)(1+O(\varepsilon))=-1+c_\pi\varepsilon+o(\varepsilon)$. The expansion holds for every sign of c_π. □

Proof of Theorem 7.2. The tunings are stable for small ε: by Lemma 2.1, stability is $\chi_L(-1)>0$, and $\chi_L(-1)=c_\pi\varepsilon+o(\varepsilon)$ with $c_\pi>0$.

(a) Let $0<c<\min(c_Q(\theta,2),c_{\mathrm{lag}})$ and $r=1-c\varepsilon$. Since $3-\xi\ge1$ on $[0,2]$,

$$\mathsf N(\xi)-4c(3-\xi)=4(3-\xi)\big(\mathsf q(\xi)-c\big)\ge4(c_{\mathrm{lag}}-c)=:m_1>0,$$

so by Lemma F.2 $\mathsf P_1+\mathsf P_2\xi+\mathsf P_3\xi^2\ge\varepsilon m_1-o(\varepsilon)>0$ on $[0,2]$ for small ε. The family has $(\theta_\varepsilon,y_\varepsilon,\tau_\varepsilon)\to(\theta,2,\frac12)$, and Lemma E.7 gives $\rho_{\mathcal F}<r$.

(b) For a requested constant $\widehat c$, it suffices to construct a witness at some $0<c<\widehat c$, since $\rho^0_{\mathcal F}\ge1-c\varepsilon>1-\widehat c\varepsilon$. We treat the terms in the minimum separately and choose c strictly between the selected term (or zero, if larger) and $\widehat c$. The family (28) is a family (3) with $y=2$ and $\tau=\frac12$, so Theorem 2.6(a) gives $\rho^0_{\mathcal F}\ge e^{-(c_{\mathcal F}(\theta,2)+\delta)\varepsilon}$ for every $\delta>0$ and small ε, with the witnesses described there; note that $c_{\mathcal F}(\theta,2)=\min(\theta/3,c_Q(\theta,2))$.

For c_π, Lemma E.8(a) applies, since $\chi_L(-1) = c_\pi\varepsilon + o(\varepsilon) > 0$ and $S_L \to -1$ (Lemma F.2): from rest on $f(x) = \frac{1}{2}x^2$ on $\mathbb{R}$ (here $L = 1$), $\rho^0_{\mathcal{F}} \geq |z_L| = 1 - c_\pi\varepsilon + o(\varepsilon)$, and the dominant mode z_L^k alternates in sign.

For $\mathsf{q}(\xi_K)$, let $K \geq 3$ with $\mathsf{q}(\xi_K) < \theta/3$ (otherwise every $c > \mathsf{q}(\xi_K)$ exceeds $c_{\mathcal{F}}(\theta, 2)$, and the first term applies), $\omega_0 = 2\pi/K$, and $\max(0, \mathsf{q}(\xi_K)) < c < \theta/3$. Then $\mathsf{N}(\xi_K) - 4c(3 - \xi_K) < 0$; fix $\mathsf{s} \in (0, \mathsf{s}_1]$ such that $\mathsf{N}(\xi) - 4c(3 - \xi) \leq -m_* < 0$ for $\omega \in B = [(1 - \mathsf{s})\omega_0, (1 + \mathsf{s})\omega_0]$, and put $r = 1 - c\varepsilon$. On B, $\xi \geq \xi_{\min} > 0$, and by Lemma F.2 $H_r = \mathsf{P}_0 + \xi\varepsilon[\mathsf{N}(\xi) - 4c(3-\xi) + o(1)] \leq -\frac{1}{2}\xi_{\min} m_* \varepsilon$ and $\mathsf{M}_r = O(\varepsilon)$ for small ε, while $|\chi_\mu \overline{\chi_L}| \leq C_3$ on B. Since B is fixed across ε, Lemma E.9 applies with $\lambda_0 = \xi_{\min} m_*/(2C_3)$ and gives $\rho^0_{\mathcal{F}} \geq 1 - c\varepsilon$.

(c) The upper bound on lim sup is (b). If $c_{\mathrm{lag}} > 0$, (a) gives $\liminf \sqrt{\kappa}(1 - \rho_{\mathcal{F}}) \geq \min(c_Q(\theta, 2), c_{\mathrm{lag}})$, and $\rho^0_{\mathcal{F}} \leq \rho_{\mathcal{F}}$ gives the middle inequality. If q attains its minimum at a point of $\{0, 2\} \cup \{\xi_K\}$, then $c_{\mathrm{lag}} = c_{\mathrm{disc}}$; if $c_Q(\theta, 2) \leq c_{\mathrm{lag}}$, both bounds equal $c_Q(\theta, 2)$. If $\mathsf{N}(\xi) \geq 4c_\angle(3 - \xi)$ on $[0, 2]$, then $c_{\mathrm{lag}} \geq c_\angle$, while $c_{\mathrm{disc}} \leq \min(\mathsf{q}(0), \mathsf{q}(2)) = \min(\theta/3, c_\pi)$; so both bounds equal $c_\angle$. If $4a + b \leq 2\theta$, then N is concave, so $\mathsf{N}(\xi) - 4c_\angle(3 - \xi)$ is concave on $[0, 2]$; at $\xi = 0$ it equals $4(\theta - 3c_\angle) \geq 0$, and at $\xi = 2$ it equals $8(2a + b - \theta) - 4c_\angle = 4(c_\pi - c_\angle) \geq 0$. Hence it is nonnegative on $[0, 2]$.

Explicit regions (Remark 7.3). For $\theta \leq 3$, $c_Q(\theta, 2) \geq \theta/3$ by Corollary 3.2, so $c_\angle = \min(\theta/3, c_\pi)$. The identities

$$\mathsf{N}(\xi) - \tfrac{4\theta}{3}(3 - \xi) = 2\xi\Big[(\tfrac{5\theta}{3} - 4a) + (4a + b - 2\theta)\xi\Big],$$

$$\mathsf{N}(\xi) - 4c_\pi(3 - \xi) = 2(2 - \xi)\Big[(7\theta - 12a - 6b) - (4a + b - 2\theta)\xi\Big],$$

reduce the criterion of (c) to the signs of affine functions at $\xi = 0$ and $\xi = 2$. If $a \leq \frac{5\theta}{12}$ and $2a + b \geq \frac{7\theta}{6}$, the first bracket is nonnegative at both ends (its value at $\xi = 2$ is $2(2a+b) - \frac{7\theta}{3}$), and $c_\pi = 2(2a+b-\theta) \geq \theta/3$, so $c_\angle = \theta/3$ and (c) applies. If $2a + b < \frac{7\theta}{6}$, then $c_\pi < \theta/3$, so $c_\angle = c_\pi$; the second bracket is positive at $\xi = 0$ and equals $11\theta - 20a - 8b \geq 0$ at $\xi = 2$ when $5a + 2b \leq \frac{11\theta}{4}$, and (c) applies.

(d) Let $\theta = 3$ and $a \leq \frac{5}{4}$, $2a + b \geq \frac{7}{2}$. By Remark 7.3, the constant is $\theta/3 = 1$. Conversely, if $\theta \neq 3$, then $\min(\theta/3, c_Q(\theta, 2)) = c_{\mathcal{F}}(\theta, 2) < 1$ by Corollary 3.2 (with $m = 2$, the maximum $\sqrt{m/2} = 1$ of $c_{\mathcal{F}}(\cdot, 2)$ is attained only at $\theta = 3$), and (b) gives $\limsup \sqrt{\kappa}(1 - \rho^0_{\mathcal{F}}) < 1$. Let $\theta = 3$. If $2a + b < \frac{7}{2}$, then $c_\pi < 1$. If $a > \frac{5}{4}$, then $\mathsf{q}(\xi) - 1 = 2\xi[(5 - 4a) + (4a + b - 6)\xi]/(4(3 - \xi)) < 0$ for small $\xi > 0$, hence $\mathsf{q}(\xi_K) < 1$ for all large K, as $\xi_K \to 0$. In both cases $c_{\mathrm{disc}} < 1$, and (b) again gives $\limsup \sqrt{\kappa}(1 - \rho_{\mathcal{F}}) \leq \limsup \sqrt{\kappa}(1 - \rho^0_{\mathcal{F}}) < 1$. □

Proof of Corollary 7.4. (i) These tunings form a family (28) with $\theta = 2\sqrt{2}$, $a = 0$, $b = 4$, so $2a+b = 4 > \theta$, $4a + b = 4 \leq 2\theta$ and $c_\pi = 2(4 - 2\sqrt{2}) > \sqrt{2}$. Since $\theta^2 = 8 = 4y$, $c_Q(\theta, 2) = \sqrt{2}$, and Theorem 7.2(c) gives the constant $c_\angle = \min(\frac{2\sqrt{2}}{3}, \sqrt{2}, c_\pi) = \frac{2\sqrt{2}}{3}$ on $\mathcal{F}_{\mu,L}$. By Lemma E.10 the quadratic constant is $\min(c_Q(\theta, 2), c_\pi) = \sqrt{2}$. (ii) The arc $\mathcal{C}_{1/2}$ has $y_{1/2} = 2$ and $2\sqrt{2} \leq \theta \leq 3$. For such θ take the family (28) with $a = 0$ and $b = 2\theta$: then $2a + b > \theta$, $4a + b = 2\theta$, $c_\pi = 2\theta > \zeta_-(\theta, 2)$, and Theorem 7.2(c) and Lemma E.10 give the pair $(\zeta_-(\theta, 2), \theta/3)$. □

Lemma F.3 (Near the corner)**.** *Let a family* (3) *satisfy* $(\theta_\varepsilon, y_\varepsilon, \tau_\varepsilon) \to (\theta, 2, \frac{1}{2})$, $\theta > 0$, *with* $\alpha_\varepsilon, \varsigma_\varepsilon, \mathsf{N}_\varepsilon$ *as in* (30)*, allowing unbounded* α_ε *and* ς_ε*. Let* $r = 1 - c\varepsilon$ *with* c *in a compact set. Then, uniformly for* $\xi \in [0, 2]$ *and for* c *in that set,*

$$\mathsf{P}_1 + \mathsf{P}_2\xi + \mathsf{P}_3\xi^2 = \varepsilon\big[\mathsf{N}_\varepsilon(\xi) - 4c(3 - \xi) + o(1)\big], \qquad \mathsf{M}_r(\xi) = \varepsilon\big[\mathsf{N}_\varepsilon(\xi) - 2c(4 - \xi) + o(1)\big],$$

with P_k *as in Lemma E.1 and* M_r *as in Lemma E.2. Moreover* $\mathsf{P}_0 = c\varepsilon\,\chi_\mu(r)\chi_L(r)$ *with* $\chi_\mu(r) = \varepsilon^2(c^2 - \theta c + 2 + o(1))$ *and* $\chi_L(r) = 2 + o(1)$*, and* $\chi_L(-1) = (\varsigma_\varepsilon - 2\theta_\varepsilon)\varepsilon$*,* $\chi_L'(-1) \to -1$.

Proof. Apply Lemma E.6(c) to the physical parameters $(\theta_\varepsilon, y_\varepsilon, \tau_\varepsilon)$, which stay in a compact set. The terms involving the possibly unbounded corner coordinates cancel exactly:

$$2y_\varepsilon(2\tau_\varepsilon - 1) = -8\alpha_\varepsilon\varepsilon, \qquad 4(1 - \tau_\varepsilon y_\varepsilon) = (\varsigma_\varepsilon + 4\alpha_\varepsilon)\varepsilon.$$

The first-order coefficient polynomials converge to their values at $(\theta, 2, 1/2)$, uniformly for c in the prescribed compact set. Combining them with the two exact identities gives both symbol expansions, with remainders uniform on $\xi \in [0,2]$, including unbounded α_ε and ς_ε. The formulas at r follow from Lemma E.6(a), while $\chi_L(-1) = 2(1+\beta) - y_\varepsilon(1+2\tau_\varepsilon) = (\varsigma_\varepsilon - 2\theta_\varepsilon)\varepsilon$ and $\chi_L'(-1) = -2 - S_L \to -1$. □

Proof of Theorem 7.5. Reformulation. Suppose $(\theta_\varepsilon, y_\varepsilon, \tau_\varepsilon) \to (3, 2, 1/2)$, and let

$$c_{\mathrm{lag},\varepsilon} = \min_{0 \le \xi \le 2} \frac{\mathsf{N}_\varepsilon(\xi)}{4(3-\xi)}.$$

Define

$$\mathsf{B}_\varepsilon(\xi) = (5 - 4\alpha_\varepsilon) + \mathsf{B}'_\varepsilon \xi, \qquad \mathsf{B}'_\varepsilon = \tfrac{1}{2}(\varsigma_\varepsilon + 4\alpha_\varepsilon - 12).$$

Direct expansion gives

$$\mathsf{N}_\varepsilon(\xi) - 4(3-\xi) = 2\xi\mathsf{B}_\varepsilon(\xi) + (\theta_\varepsilon - 3)(4 + 2\xi - 4\xi^2), \qquad \mathsf{B}_\varepsilon(2) = \varsigma_\varepsilon - 7.$$

Since $4 \le 4(3-\xi) \le 12$ and $\xi\mathsf{B}_\varepsilon(\xi) = 0$ at $\xi = 0$, the condition $c_{\mathrm{lag},\varepsilon} \to 1$ is equivalent to $\min_{[0,2]} \xi\mathsf{B}_\varepsilon(\xi) \to 0$. It requires $\liminf \varsigma_\varepsilon \ge 7$, by evaluation at $\xi = 2$. Assume this inequality, and put $d_\varepsilon = (4\alpha_\varepsilon - 5)_+$. When $d_\varepsilon = 0$, both endpoint values of the affine B_ε are at least $-o(1)$. More generally, if $d_\varepsilon \to 0$, then $\mathsf{B}_\varepsilon(0) \ge -d_\varepsilon$ and $\mathsf{B}_\varepsilon(2) \ge -o(1)$, so $\min_{[0,2]} \xi\mathsf{B}_\varepsilon(\xi) \ge -2d_\varepsilon - o(1) \to 0$. Along any sequence with $d_\varepsilon \ge d_* > 0$, the vertex

$$\xi_v = \frac{d_\varepsilon}{\varsigma_\varepsilon + 5 + d_\varepsilon - 12}$$

lies in $(0, 1 + o(1)]$, and

$$\min_{[0,2]} \xi\mathsf{B}_\varepsilon(\xi) = -\frac{d_\varepsilon^2}{2(\varsigma_\varepsilon + 5 + d_\varepsilon - 12)}.$$

The quantity $\varsigma_\varepsilon + 5 + d_\varepsilon - 12$ is at least $d_* - o(1)$. Adding 12 therefore changes the ratio by a bounded factor, proving equivalence with $d_\varepsilon^2/(\varsigma_\varepsilon + 5 + d_\varepsilon) \to 0$. If this ratio fails to vanish, a subsequence has $d_\varepsilon \ge d_* > 0$ and $m_\varepsilon := d_\varepsilon^2/(2\mathsf{B}'_\varepsilon)$ bounded below by a positive constant. We may pass to further subsequences in the cases below whenever a bounded/unbounded alternative is used.

Sufficiency. Let the conditions hold and $0 < c < 1$. By Lemma F.3 with $c_{\mathrm{lag},\varepsilon} \to 1$, $\mathsf{P}_1 + \mathsf{P}_2\xi + \mathsf{P}_3\xi^2 = \varepsilon[\mathsf{N}_\varepsilon(\xi) - 4c(3-\xi) + o(1)] \ge \varepsilon[4(3-\xi)(c_{\mathrm{lag},\varepsilon} - c) + o(1)] \ge 2(1-c)\varepsilon$ for small ε. Since $c < 1 = c_Q(3,2)$, Lemma E.7 gives $\rho_{\mathcal{F}} < 1 - c\varepsilon$. Hence $\liminf \sqrt{\kappa}(1 - \rho_{\mathcal{F}}) \ge 1$. Conversely, Theorem 2.6(a), applied to the family (3) with limit $(3, 2, \frac{1}{2})$, gives $\limsup \sqrt{\kappa}(1 - \rho^0_{\mathcal{F}}) \le c_{\mathcal{F}}(3,2) = 1$, and $\rho^0_{\mathcal{F}} \le \rho_{\mathcal{F}}$.

Necessity. Suppose $\sqrt{\kappa}(1 - \rho_{\mathcal{F}}) \to 1$. By Corollary E.11(v), the physical parameters tend to $(3, 2, 1/2)$. First, $\liminf \varsigma_\varepsilon \ge 7$: otherwise a subsequence has $\varsigma_\varepsilon \to s < 7$, with $s \ge 6$ by stability. By Lemma F.3, $\chi_L(-1) = (\varsigma_\varepsilon - 2\theta_\varepsilon)\varepsilon > 0$ and $\chi_L'(-1) \to -1$. The alternating quadratic mode of Lemma E.8(a) then gives $\rho^0_{\mathcal{F}} \ge 1 - (s - 6 + o(1))\varepsilon$, contradicting $\rho^0_{\mathcal{F}} \le \rho_{\mathcal{F}}$ and $s - 6 < 1$.

If the ratio criterion fails, pass to a sequence on which

$$D_\varepsilon = \varsigma_\varepsilon + 5 + d_\varepsilon, \qquad R_\varepsilon = \frac{d_\varepsilon^2}{D_\varepsilon} \ge q, \qquad 0 < q \le 1.$$

We separate bounded and unbounded deficits, passing to subsequences as needed.

Bounded deficit. If $d_\varepsilon \le M$, stability and $qD_\varepsilon \le d_\varepsilon^2$ bound ς_ε above and keep d_ε away from zero. By compactness, $d_\varepsilon \to d_* > 0$ and $\varsigma_\varepsilon \to s \ge 7$ along a subsequence. Put $a = (d_* + 5)/4$ and $b = (s - 4a)/2$. The exact identities

$$\frac{1/2 - \tau_\varepsilon}{\varepsilon} = \frac{2\alpha_\varepsilon}{y_\varepsilon} \to a, \qquad \frac{2 - y_\varepsilon}{\varepsilon} = \frac{\varsigma_\varepsilon - 4\alpha_\varepsilon}{2} \to b$$

place this subsequence in Theorem 7.2, since $2a + b = s/2 \ge 7/2 > 3$. Part (d) requires $a \le 5/4$, contradicting $d_* > 0$.

It remains to consider $d_\varepsilon \to \infty$. Then $D_\varepsilon \to \infty$, $\mathsf{B}'_\varepsilon = (D_\varepsilon - 12)/2 > 0$, and

$$m_\varepsilon = \frac{d_\varepsilon^2}{D_\varepsilon - 12} = R_\varepsilon \frac{D_\varepsilon}{D_\varepsilon - 12}, \qquad \frac{m_\varepsilon}{R_\varepsilon} \to 1.$$

Thus the bounded and divergent depth alternatives are the same for m_ε and R_ε. Set $\xi_{\mathsf{B}} = d_\varepsilon/\mathsf{B}'_\varepsilon$; since $\varsigma_\varepsilon \ge 7 - o(1)$, $\xi_{\mathsf{B}}/2 \le 1 + o(1)$. For $0 < \xi \le \xi_{\mathsf{B}}/2$, $\mathsf{B}_\varepsilon(\xi) \le -d_\varepsilon/2$, so the reformulation gives

$$\mathsf{N}_\varepsilon(\xi) - 4c(3-\xi) \le -d_\varepsilon \xi + 12(1-c) + o(1) \qquad (0 < \xi \le \tfrac{\xi_{\mathsf{B}}}{2},\ 0 \le c \le 1). \tag{53}$$

The error is uniform on these ranges.

Moderate dip. Suppose $q \le R_\varepsilon \le M$. Let $\xi_0 = q/d_\varepsilon$ and $c' = 1 - q/48$. Since $m_\varepsilon \ge R_\varepsilon \ge q$, $\xi_0 \le \xi_{\mathsf{B}}/2$. On $[\xi_0/2, \xi_0]$, (53) gives

$$\mathsf{N}_\varepsilon(\xi) - 4c'(3-\xi) \le -\frac{q}{2} + \frac{q}{4} + o(1) \le -\frac{q}{8}, \qquad |\mathsf{N}_\varepsilon(\xi)| \le 15$$

for small ε; the second bound follows from $-d_\varepsilon \le \mathsf{B}_\varepsilon(\xi) \le 0$ and $d_\varepsilon \xi \le q$. By (30), $\varepsilon D_\varepsilon \to 0$. Consequently $\varepsilon d_\varepsilon^2 \le M \varepsilon D_\varepsilon \to 0$, and hence $\varepsilon \xi_0^{-3/2} \to 0$.

Choose $\mathsf{s} = \min(\mathsf{s}_1, 1/20)$. The frequencies corresponding to $[\xi_0/2, \xi_0]$ have ratio tending to $\sqrt{2}$ and tend to zero. Thus Lemma E.3(a) supplies $K \ge 3$ and a band $B = [(1-\mathsf{s})2\pi/K, (1+\mathsf{s})2\pi/K]$ inside that interval of frequencies. At $r = 1 - c'\varepsilon$, Lemma F.3 gives, on B,

$$H_r \le -\frac{q}{64}\xi_0 \varepsilon, \qquad |\mathsf{M}_r| \le 24\varepsilon, \qquad |\chi_\mu \overline{\chi_L}| \le C\xi_0.$$

Here $\mathsf{P}_0 = O(\varepsilon^3) = o(\varepsilon\xi_0)$; the last estimate follows from $\chi_\mu = (z-1)(z-\beta) + y_\varepsilon \varepsilon^2 N_{\tau_\varepsilon}(z) = O(\xi_0 + \varepsilon\sqrt{\xi_0} + \varepsilon^2) = O(\xi_0)$ and uniform boundedness of χ_L. Since $\xi_{\max} \in [\xi_0/2, \xi_0]$ and $\varepsilon\xi_0^{-3/2} \to 0$, Lemma E.9 applies with $\lambda_0 = q/(64C)$, giving $\rho_{\mathcal{F}}^0 \ge 1 - c'\varepsilon$, a contradiction.

Deep dip. Otherwise pass to a subsequence with $R_\varepsilon \to \infty$. Take integers $K_\varepsilon \sim 2\pi\sqrt{d_\varepsilon/24}$ and put $\xi_\varepsilon = 1 - \cos(2\pi/K_\varepsilon)$. Then $K_\varepsilon \ge 3$, $\xi_\varepsilon \to 0$ and $d_\varepsilon \xi_\varepsilon \to 12$. The exact reformulation yields

$$\mathsf{N}_\varepsilon(\xi_\varepsilon) = 12 - 4\xi_\varepsilon - 2d_\varepsilon\xi_\varepsilon + \frac{(d_\varepsilon \xi_\varepsilon)^2}{R_\varepsilon} - 12\xi_\varepsilon^2 + o(1) \to -12.$$

By Lemma F.3 at $c = 0$, $L_\tau(\xi_\varepsilon) = \varepsilon[\mathsf{N}_\varepsilon(\xi_\varepsilon) + o(1)] < 0$. Lemma E.3(b) gives K_ε-periodic witnesses and $\rho_{\mathcal{F}}^0 > 1$, again a contradiction. Thus the ratio criterion is necessary. □

G. Proofs for Section 8

Lemma G.1 (Structure of L_τ)**.** *L_τ is affine in y_μ, and at $y_\mu = y$ it equals $|\chi_L(e^{i\omega})|^2$. Hence, for $y > 0$ and $0 \le y_\mu \le y$, $L_\tau(\xi; y_\mu) = (1 - \frac{y_\mu}{y})L_\tau(\xi; 0) + \frac{y_\mu}{y}|\chi_L(e^{i\omega})|^2$, where $L_\tau(\xi;0) = 2(1-\beta)y + \xi[2(1-\beta)^2 + y(2\tau(3-\beta) - 2)] + 4\xi^2(\beta - \tau y)$. With $y_{\max} = 2(1+\beta)/(1+2\tau)$,*

$$L_\tau(\xi;0)\big|_{y=0} = 2\xi[(1-\beta)^2 + 2\beta\xi], \qquad L_\tau(\xi;0)\big|_{y=y_{\max}} = \frac{2(2-\xi)}{1+2\tau}[(1-\beta^2) + 2\xi(2\tau - \beta)].$$

Proof. The coefficients in (51) are affine in y_μ, and $\chi_\mu = \chi_L$ at $y_\mu = y$, proving the interpolation formula. Setting $y_\mu = 0$ gives the displayed polynomial. At $y = 0$ its value is immediate; substitution of $y_{\max} = 2(1+\beta)/(1+2\tau)$ factors it as $2(2-\xi)[(1-\beta^2) + 2\xi(2\tau - \beta)]/(1+2\tau)$. □

Proof of Theorem 8.1. Use the multiplier weight λ specified in Theorem 8.1, and write $B(\xi) = (1 - \beta^2) + 2\xi(2\tau - \beta)$, so that $L_\tau(\xi;0)|_{y=y_{\max}} = \frac{2(2-\xi)}{1+2\tau}B(\xi)$ (Lemma G.1); note $B(2) = 8(\tau - \tau_*)$ and $B(\frac{3}{2}) = 6(\tau - \tau_3)$.

(a), $\tau \ge \tau_*$. Fix $\xi \in (0, 2]$. $L_\tau(\xi;0)$ is affine in y, positive at $y = 0$, and nonnegative at $y = y_{\max}$, because B is affine with $B(0) = 1 - \beta^2 > 0$ and $B(2) \ge 0$. So $L_\tau(\xi;0) > 0$ for $0 \le y < y_{\max}$, which by Lemma 2.1 is the stability range. For a stable tuning χ_L has no root on the unit circle, and Lemma G.1 gives $L_\tau(\xi; y_\mu) > 0$ for $0 < y_\mu < y$; Proposition 6.1 gives $\rho_{\mathcal{F}} < 1$.

(a), $\tau_3 \le \tau < \tau_*$. Then $\beta - 2\tau > 0$, because $\tau_* \le \beta/2$ (equivalent to $\beta^2 \le 1$), so $\lambda > 0$. On $|z| = 1$ write $\chi_\mu \overline{\chi_L} = \mathcal{R}_0 + iq \sin\omega$ with $q(\xi) = (y - y_\mu)[(1-\beta) + 2\tau\xi]$ by (16), and recall $\mathcal{R}_0 = L_\tau - (2-\xi)q$ from the proof of Proposition 6.1. Since $1 - \bar{z}^2 = 2\sin^2\omega + 2i\sin\omega\cos\omega$,

$$\mathrm{Re}[\chi_\mu \overline{\chi_L}(1-\bar{z}^2)] = 2\sin^2\omega\,(\mathcal{R}_0 - q\cos\omega) = 2\xi(2-\xi)(L_\tau - (3-2\xi)q),$$

and together with $\mathrm{Re}[\chi_\mu \overline{\chi_L}(1-z)] = \xi L_\tau$,

$$\mathrm{Re}\big[\chi_\mu \overline{\chi_L}\,((1-z) + \lambda(1-\bar{z}^2))\big] = \xi F(\xi), \qquad F = (1 + 2\lambda(2-\xi))L_\tau + 2\lambda(2-\xi)(2\xi-3)q.$$

We show $F > 0$ on $(0, 2]$ for every stable tuning. For fixed ξ, F is affine in $y_\mu \in [0, y]$ (both L_τ and q are), and at $y_\mu = 0$ it is affine in $y \in [0, y_{\max}]$. At $y_\mu = y$, $q = 0$ and $F = (1 + 2\lambda(2-\xi))|\chi_L|^2 > 0$, since χ_L has no root on the unit circle. At $y_\mu = y = 0$, $F = (1 + 2\lambda(2-\xi))\,2\xi[(1-\beta)^2 + 2\beta\xi] > 0$. At $y_\mu = 0$, $y = y_{\max}$, using Lemma G.1 and $y_{\max}(1+2\tau)/2 = 1 + \beta$, $F = \frac{2(2-\xi)}{1+2\tau} G(\xi)$ with

$$G(\xi) = (1 + 2\lambda(2-\xi))B(\xi) + 2\lambda(2\xi-3)(1+\beta)\big[(1-\beta) + 2\tau\xi\big].$$

Substitution of the chosen λ and completion of the square give the exact identity

$$G(\xi) = 6(1+\lambda)(\tau - \tau_3) + 4\lambda\beta(1+2\tau)\big(\xi - \tfrac{3}{2}\big)^2 \ \ge 0.$$

So $F(\xi; 0, y)$ is affine in $y \in [0, y_{\max}]$, positive at $y = 0$ and nonnegative at $y = y_{\max}$, hence positive for $y < y_{\max}$; and $F(\xi; y_\mu, y)$, affine in y_μ between the positive values at $y_\mu = 0$ and $y_\mu = y$, is positive. Thus $\mathrm{Re}[\chi_\mu \overline{\chi_L}((1-z) + \lambda(1-\bar{z}^2))] > 0$ for $z \ne 1$, and Proposition 4.9(c) gives $\rho_{\mathcal{F}} < 1$.

(b) If $\tau < \tau_3$, then $L_\tau(\frac{3}{2}; 0)|_{y=y_{\max}} = B(\frac{3}{2})/(1+2\tau) < 0$; by continuity in y and Lemma G.1, $L_\tau(\frac{3}{2}; y_\mu) < 0$ for some $y < y_{\max}$ and small $y_\mu > 0$. Take $h = 1$, $L = y$, $\mu = y_\mu$; the tuning is stable, and Lemma E.3(b) with $K = 3$ ($\xi = \frac{3}{2}$) gives $\Theta(2\pi/3) < \pi/3$, the 3-periodic orbits and $\rho^0_{\mathcal{F}} > 1$.

(c) If $\tau < \tau_*$, $B < 0$ on some $(\xi_1, 2]$, so $L_\tau(\xi; 0) < 0$ at $y = y_{\max}$ for $\xi \in (\xi_1, 2)$, and the same perturbation argument applies.

The special cases follow from $\tau_* - \tau_3 = (1-\beta^2)/24$, $\tau_*(\beta) \le \beta \iff \beta^2 - 4\beta - 1 \le 0$, $\tau_3(\beta) \le 0 \iff \beta \le (\sqrt{13} - 3)/2$ and $\tau_*(\beta) \le 0 \iff \beta \le \sqrt{5} - 2$. □

Proof of Corollary 8.2. Use the query-rate dictionary from the proof of Corollary 6.6(iii). For $\beta \ne z_\tau$, Lemma C.12 identifies stability with convergence of the iterates on every quadratic in $\mathcal{F}_{\mu,L}$. For fixed β, ν_{qh}, the extrapolation τ_{qh} is fixed and varying α covers all $h > 0$. Apply Theorem 8.1: linear convergence of $\mathcal{M}$ gives $\rho^0_{\tilde{x}} \le \rho^0_{\mathcal{F}} \le \rho_{\mathcal{F}} < 1$, and the divergent orbits from rest of Theorem 8.1 (a 3-cycle tongue) have $\rho^0_{\mathcal{F}} > 1$, hence $\rho^0_{\tilde{x}} > 1$ by the second inequality. □

Proof of Proposition 8.3. For any fixed method, inclusion of function classes gives

$$\mu \le \mu' < L \quad \Longrightarrow \quad \Gamma_{\mu',L}(n) \le \Gamma_{\mu,L}(n) \quad \Longrightarrow \quad \rho_{\mathcal{F}_{\mu',L}} \le \rho_{\mathcal{F}_{\mu,L}}.$$

Thus $I = \{\mu \in (0, L) : \rho_{\mathcal{F}} < 1\}$ is an upper interval. It is open: Theorem 4.7(b) supplies a strict certificate at $t = 0$ for each $\mu \in I$, and Lemma C.5 preserves it for nearby μ. Stability, independent of μ by Lemma 2.1, gives $r_c < 1$, so Theorem 4.7(c) applies there.

By Lemma G.1, $L_\tau(\xi; y_\mu)$ is affine in y_μ and is strictly positive at $y_\mu = y$, where it equals $|\chi_L(e^{i\omega})|^2$. Therefore $\{y_\mu : \min_{[0,2]} L_\tau(\xi; y_\mu) > 0\}$ is a nonempty open upper interval with right endpoint y. The lag-one criterion places it inside I. Since $y_\mu = h\mu$, this proves both interval descriptions. The class-inclusion argument also shows that the rate itself is nonincreasing in μ, for any fixed stationary method. □